inguinal ligament and in the femoral triangle (Fig. 6.42). The femoral artery and vein pass inferiorly through the adductor canal and become the popliteal vessels behind the knee where they meet and are distributed with branches of the sciatic nerve, which descends through the posterior thigh from the gluteal region.

From lateral to medial, major structures in the femoral triangle are the femoral nerve, the femoral artery, the femoral vein, and lymphatic vessels. The femoral artery can be palpated in the femoral triangle just inferior to the inguinal ligament and midway between the anterior superior iliac spine and the pubic symphysis.

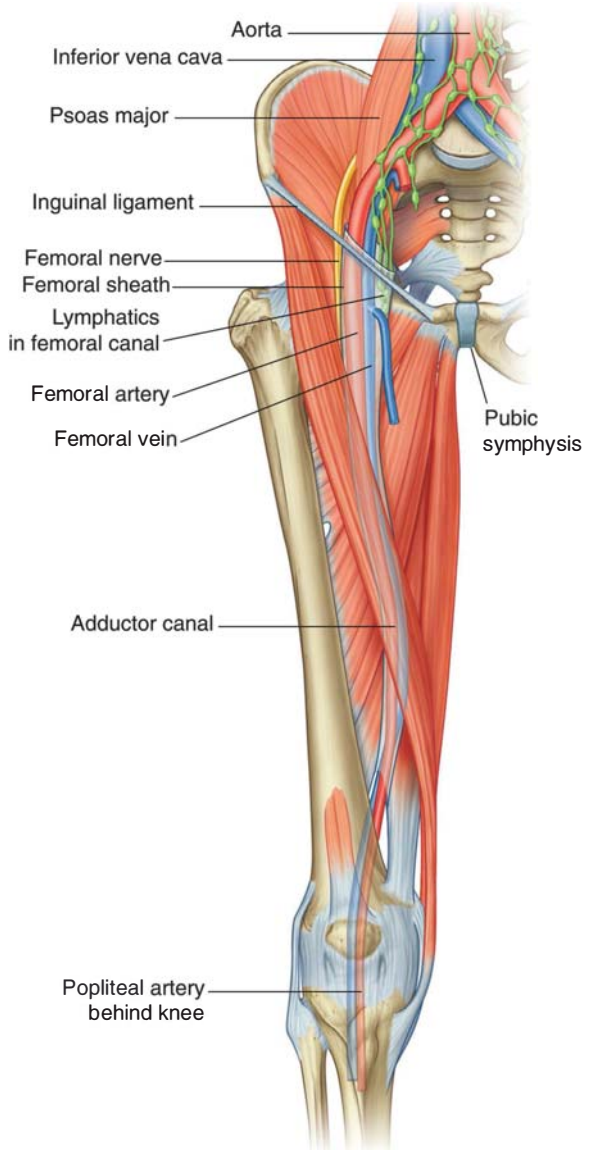


Fig. 6.42 Contents of the femoral triangle.

Femoral sheath

In the femoral triangle, the femoral artery and vein and the associated lymphatic vessels are surrounded by a funnel-shaped sleeve of fascia (the **femoral sheath**). The sheath is continuous superiorly with the transversalis fascia and iliac fascia of the abdomen and merges inferiorly with connective tissue associated with the vessels. Each of the three structures surrounded by the sheath is contained within a separate fascial compartment within the sheath. The most medial compartment (the femoral canal) contains the lymphatic vessels and is conical in shape. The opening of this canal superiorly is potentially a weak point in the lower abdomen and is the site for femoral hernias. The femoral nerve is lateral to and not contained within the femoral sheath.

In the clinic

Vascular access to the lower limb

Deep and inferior to the inguinal ligament are the femoral artery and femoral vein. The femoral artery is palpable as it passes over the femoral head and may be easily demonstrated using ultrasound. If arterial or venous access is needed rapidly, a physician can use the femoral approach to these vessels.

Many radiological procedures involve catheterization of the femoral artery or the femoral vein to obtain access to the contralateral lower limb, the ipsilateral lower limb, the vessels of the thorax and abdomen, and the cerebral vessels.

Cardiologists also use the femoral artery to place catheters in vessels around the arch of the aorta and into the coronary arteries to perform coronary angiography and angioplasty.

Access to the femoral vein permits catheters to be maneuvered into the renal veins, the gonadal veins, the right atrium, and the right side of the heart, including the pulmonary artery and distal vessels of the pulmonary tree. Access to the superior vena cava and the great veins of the neck is also possible.

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GLUTEAL REGION

The gluteal region lies posterolateral to the bony pelvis and proximal end of the femur (Fig. 6.43). Muscles in the region mainly abduct, extend, and laterally rotate the femur relative to the pelvic bone.

The gluteal region communicates anteromedially with the pelvic cavity and perineum through the greater sciatic foramen and lesser sciatic foramen, respectively. Inferiorly, it is continuous with the posterior thigh.

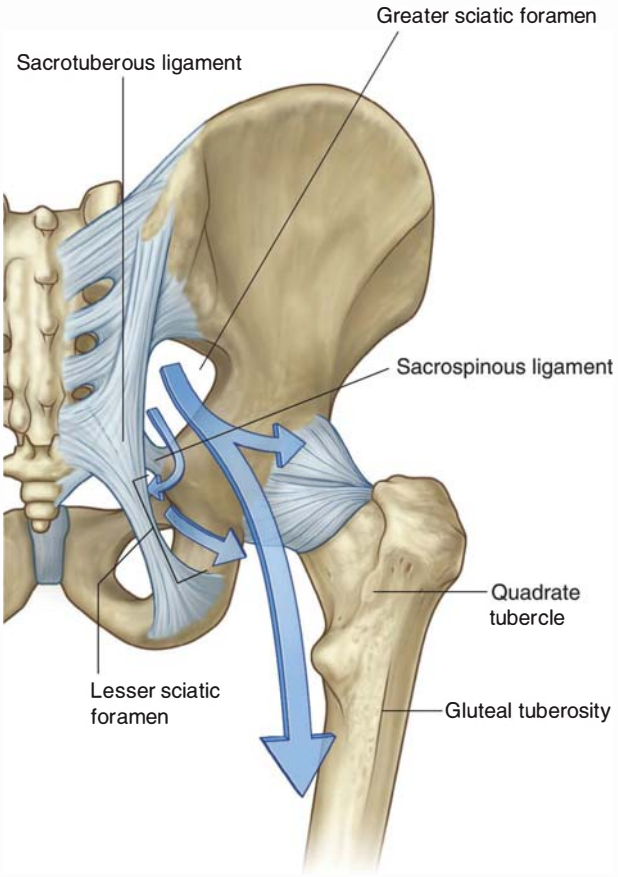


Fig. 6.43 Gluteal region. Posterior view.

The sciatic nerve enters the lower limb from the pelvic cavity by passing through the greater sciatic foramen and descending through the gluteal region into the posterior thigh and then into the leg and foot.

The pudendal nerve and internal pudendal vessels pass between the pelvic cavity and perineum by passing first through the greater sciatic foramen to enter the gluteal region and then immediately passing through the lesser sciatic foramen to enter the perineum. The nerve to the obturator internus and gemellus superior follows a similar course. Other nerves and vessels that pass through the greater sciatic foramen from the pelvic cavity supply structures in the gluteal region itself.

Muscles

Muscles of the gluteal region (Table 6.2) are composed mainly of two groups:

- a deep group of small muscles, which are mainly lateral rotators of the femur at the hip joint and include the piriformis, obturator internus, gemellus superior, gemellus inferior, and quadratus femoris;
- a more superficial group of larger muscles, which mainly abduct and extend the hip and include the gluteus minimus, gluteus medius, and gluteus maximus; an additional muscle in this group, the tensor fasciae latae, stabilizes the knee in extension by acting on a specialized longitudinal band of deep fascia (the iliotibial tract) that passes down the lateral side of the thigh to attach to the proximal end of the tibia in the leg.

Many of the important nerves in the gluteal region are in the plane between the superficial and deep groups of muscles.

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Table 6.2 Muscles of the gluteal region (spinal segments in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Piriformis	Anterior surface of sacrum between anterior sacral foramina	Medial side of superior border of greater trochanter of femur	Branches from S1 and **S2**	Laterally rotates the extended femur at hip joint; abducts flexed femur at hip joint
Obturator internus	Anterolateral wall of true pelvis; deep surface of obturator membrane and surrounding bone	Medial side of greater trochanter of femur	Nerve to obturator internus (L5, **S1**)	Laterally rotates the extended femur at hip joint; abducts flexed femur at hip joint
Gemellus superior	External surface of ischial spine	Along length of superior surface of the obturator internus tendon and into the medial side of greater trochanter of femur with obturator internus tendon	Nerve to obturator internus (L5, **S1**)	Laterally rotates the extended femur at hip joint; abducts flexed femur at hip joint
Gemellus inferior	Upper aspect of ischial tuberosity	Along length of inferior surface of the obturator internus tendon and into the medial side of greater trochanter of femur with obturator internus tendon	Nerve to quadratus femoris (**L5, S1**)	Laterally rotates the extended femur at hip joint; abducts flexed femur at hip joint
Quadratus femoris	Lateral aspect of the ischium just anterior to the ischial tuberosity	Quadrate tubercle on the intertrochanteric crest of the proximal femur	Nerve to quadratus femoris (**L5, S1**)	Laterally rotates femur at hip joint
Gluteus minimus	External surface of ilium between inferior and anterior gluteal lines	Linear facet on the anterolateral aspect of the greater trochanter	Superior gluteal nerve (**L4, L5,** S1)	Abducts femur at hip joint; holds pelvis secure over stance leg and prevents pelvic drop on the opposite swing side during walking; medially rotates thigh
Gluteus medius	External surface of ilium between anterior and posterior gluteal lines	Elongate facet on the lateral surface of the greater trochanter	Superior gluteal nerve (**L4, L5,** S1)	Abducts femur at hip joint; holds pelvis secure over stance leg and prevents pelvic drop on the opposite swing side during walking; medially rotates thigh
Gluteus maximus	Fascia covering gluteus medius, external surface of ilium behind posterior gluteal line, fascia of erector spinae, dorsal surface of lower sacrum, lateral margin of coccyx, external surface of sacrotuberous ligament	Posterior aspect of iliotibial tract of fascia lata and gluteal tuberosity of proximal femur	Inferior gluteal nerve (**L5, S1,** S2)	Powerful extensor of flexed femur at hip joint; lateral stabilizer of hip joint and knee joint; laterally rotates and abducts thigh
Tensor fasciae latae	Lateral aspect of crest of ilium between anterior superior iliac spine and tubercle of the crest	Iliotibial tract of fascia lata	Superior gluteal nerve (**L4, L5,** S1)	Stabilizes the knee in extension

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Deep group

Piriformis

The **piriformis** muscle is the most superior of the deep group of muscles (Fig. 6.44) and is a muscle of the pelvic wall and of the gluteal region (see Chapter 5, p. 451). It originates from between the anterior sacral foramina on the anterolateral surface of the sacrum and passes laterally and inferiorly through the greater sciatic foramen.

In the gluteal region, the piriformis passes posterior to the hip joint and attaches to a facet on the upper margin of the greater trochanter of the femur.

The piriformis externally rotates and abducts the femur at the hip joint and is innervated in the pelvic cavity by the nerve to the piriformis, which originates as branches from S1 and S2 of the sacral plexus (see Chapter 5, p. 487).

In addition to its action on the hip joint, the piriformis is an important landmark because it divides the greater sciatic foramen into two regions, one above and one below the piriformis. Vessels and nerves pass between the pelvis and gluteal region by passing through the greater sciatic foramen either above or below the piriformis.

Obturator internus

The **obturator internus** muscle, like the piriformis muscle, is a muscle of the pelvic wall and of the gluteal region (Fig. 6.44). It is a flat fan-shaped muscle originating from the medial surface of the obturator membrane and adjacent bone of the obturator foramen (see Chapter 5, pp. 450–451). Because the pelvic floor attaches to a thickened band of fascia across the medial surface of the obturator internus, the obturator internus forms:

- the anterolateral wall of the pelvic cavity above the pelvic floor, and
- the lateral wall of the ischio-anal fossa in the perineum below the pelvic floor.

The muscle fibers of the obturator internus converge to form a tendon, which bends 90° around the ischium between the ischial spine and ischial tuberosity and passes through the lesser sciatic foramen to enter the gluteal region. The tendon then passes posteroinferiorly to the hip joint and attaches to the medial surface of the superior

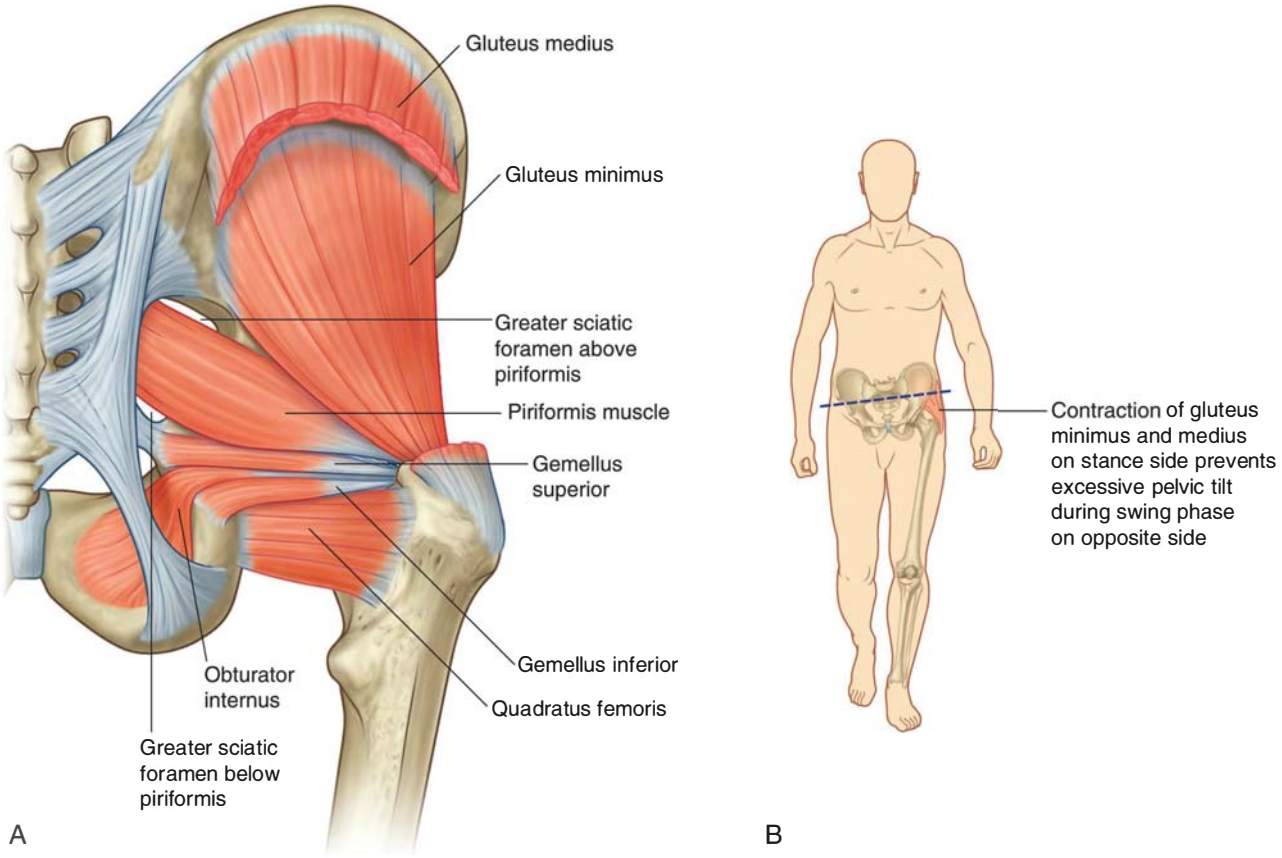


Fig. 6.44 Deep muscles in the gluteal region. **A.** Posterior view. **B.** Function.

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margin of the greater trochanter of the femur just inferior to the attachment of the piriformis muscle.

The obturator internus laterally rotates and abducts the femur at the hip joint and is innervated by the nerve to the obturator internus.

Gemellus superior and inferior

The gemellus superior and inferior (*gemelli* is Latin for "twins") are a pair of triangular muscles associated with the upper and lower margins of the obturator internus tendon (Fig. 6.44):

- The base of the **gemellus superior** originates from the gluteal surface of the ischial spine.
- The base of the **gemellus inferior** originates from the upper gluteal and pelvic surfaces of the ischial tuberosity.

Fibers of the gemellus muscles attach along the length of the obturator internus tendon, and the apices of the two muscles insert with the tendon of the obturator internus on the greater trochanter of the femur.

The gemellus superior is innervated by the nerve to the obturator internus, and the gemellus inferior is innervated by the nerve to the quadratus femoris. The gemellus muscles act with the obturator internus muscle to laterally rotate and abduct the femur at the hip joint.

Quadratus femoris

The **quadratus femoris** muscle is the most inferior of the deep group of muscles in the gluteal region (Fig. 6.44). It is a flat rectangular muscle below the obturator internus muscle and its associated gemellus muscles.

The quadratus femoris is attached at one end to a linear roughening on the lateral aspect of the ischium just anterior to the ischial tuberosity and at the other end to the quadrate tubercle on the intertrochanteric crest of the proximal femur.

The quadratus femoris laterally rotates the femur at the hip joint and is innervated by the nerve to the quadratus femoris.

Superficial group

Gluteus minimus and medius

The gluteus minimus and medius muscles are two muscles of the more superficial group in the gluteal region (Fig. 6.44).

The **gluteus minimus** is a fan-shaped muscle that originates from the external surface of the expanded upper part of the ilium, between the inferior gluteal line and the anterior gluteal line. The muscle fibers converge inferiorly and laterally to form a tendon, which inserts into a broad linear facet on the anterolateral aspect of the greater trochanter.

The **gluteus medius** overlies the gluteus minimus and is also fan shaped. It has a broad origin from the external surface of the ilium between the anterior gluteal line and posterior gluteal line and inserts on an elongate facet on the lateral surface of the greater trochanter.

The gluteus medius and minimus muscles abduct the lower limb at the hip joint and reduce pelvic drop over the opposite swing limb during walking by securing the position of the pelvis on the stance limb (Fig. 6.44B). Both muscles are innervated by the superior gluteal nerve.

In the clinic

Trendelenburg's sign

Trendelenburg's sign occurs in people with weak or paralyzed abductor muscles (gluteus medius and gluteus minimus) of the hip. The sign is demonstrated by asking the patient to stand on one limb. When the patient stands on the affected limb, the pelvis severely drops over the swing limb.

Positive signs are typically found in patients with damage to the superior gluteal nerve. Damage to this nerve may occur with associated pelvic fractures, with space-occupying lesions within the pelvis extending into the greater sciatic foramen, and in some cases relating to hip surgery during which there has been disruption of and subsequent atrophy of the insertion of the gluteus medius and gluteus minimus tendons on the greater trochanter.

In patients with a positive Trendelenburg's sign, gait also is abnormal. Typically during the stance phase of the affected limb, the weakened abductor muscles allow the pelvis to tilt inferiorly over the swing limb. The patient compensates for the pelvic drop by lurching the trunk to the affected side to maintain the level of the pelvis throughout the gait cycle.

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Gluteus maximus

The gluteus maximus is the largest muscle in the gluteal region and overlies most of the other gluteal muscles (Fig. 6.45).

The gluteus maximus is quadrangular in shape and has a broad origin extending from a roughened area of the ilium behind the posterior gluteal line and along the dorsal surface of the lower sacrum and the lateral surface of the coccyx to the external surface of the sacrotuberous ligament. It is also attached to fascia overlying the gluteus medius muscle and, between the ilium and sacrum, to fascia covering the erector spinae muscle, and is often described as being enclosed within two layers of the fascia lata, which covers the thigh and gluteal region.

Laterally, the upper and superficial lower parts of the gluteus maximus insert into the posterior aspect of a tendinous thickening of the fascia lata (the iliotibial tract), which passes over the lateral surface of the greater trochanter and descends down the thigh and into the upper leg. Deep distal parts of the muscle attach to the elongate gluteal tuberosity of the proximal femur.

The gluteus maximus mainly extends the flexed thigh at the hip joint. Through its insertion into the iliotibial tract, it also stabilizes the knee and hip joints. It is innervated by the inferior gluteal nerve.

Tensor fasciae latae

The tensor fasciae latae muscle is the most anterior of the superficial group of muscles in the gluteal region and overlies the gluteus minimus and the anterior part of the gluteus medius (Fig. 6.46).

The tensor fasciae latae originates from the outer margin of the iliac crest from the anterior superior iliac spine to approximately the tuberculum of the iliac crest. The muscle

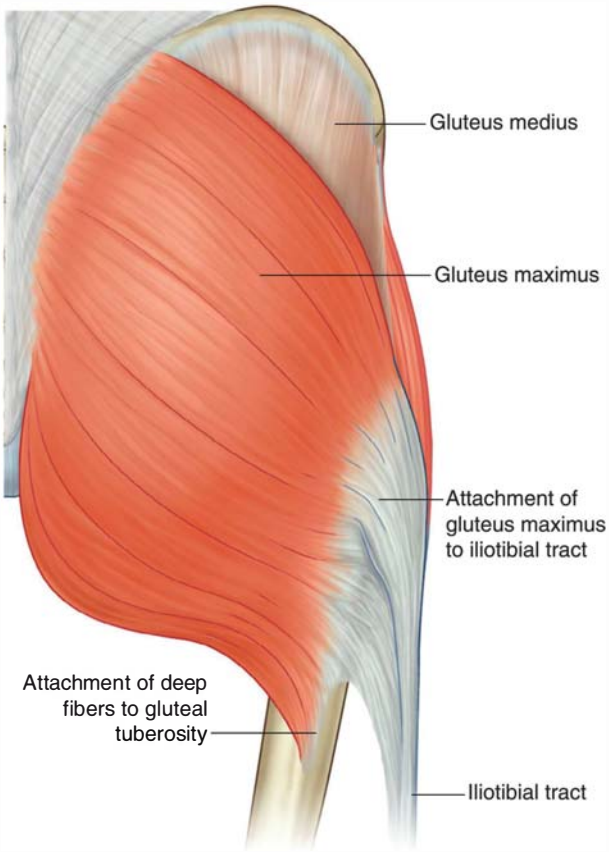


Fig. 6.45 Gluteus maximus muscle. Posterior view.

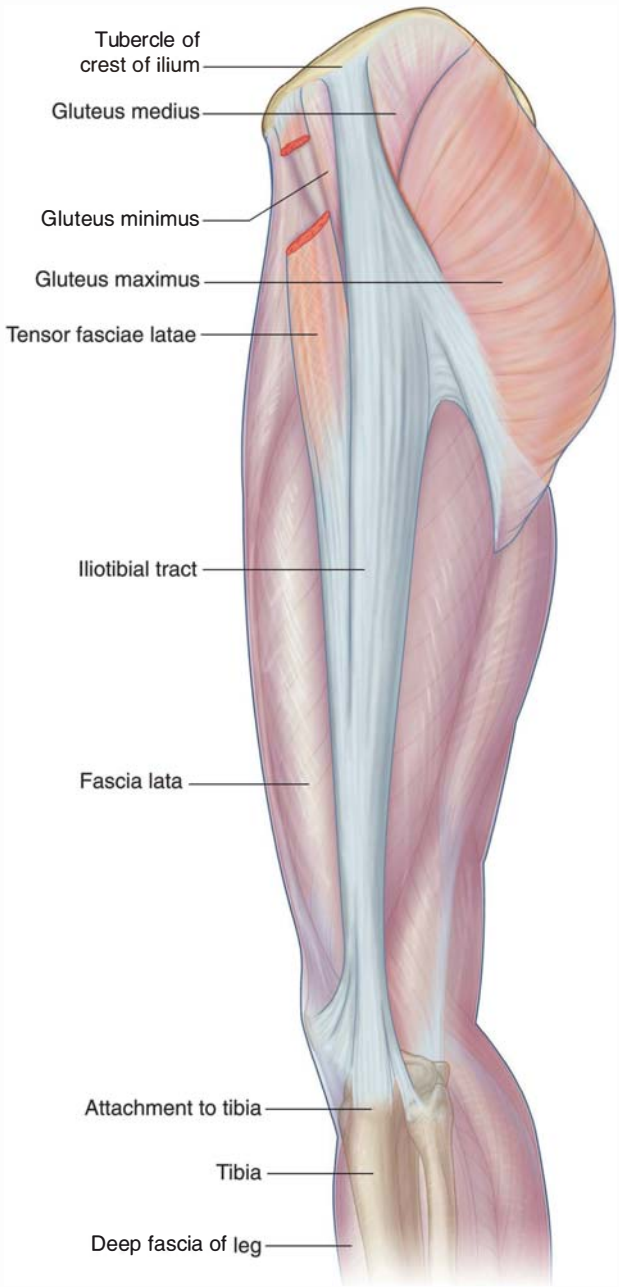


Fig. 6.46 Tensor fasciae latae. Left gluteal region, lateral view.

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fibers descend to insert into the anterior aspect of the iliotibial tract of deep fascia, which runs down the lateral side of the thigh and attaches to the upper tibia. Like the gluteus maximus muscle, the tensor fasciae latae is enclosed within a compartment of the fascia lata.

The tensor fasciae latae stabilizes the knee in extension and, working with the gluteus maximus muscle on the iliotibial tract lateral to the greater trochanter, stabilizes the hip joint by holding the head of the femur in the acetabulum (Fig. 6.46). It is innervated by the superior gluteal nerve.

Nerves

Seven nerves enter the gluteal region from the pelvis through the greater sciatic foramen (Fig. 6.47): the superior gluteal nerve, sciatic nerve, nerve to the quadratus femoris, nerve to the obturator internus, posterior cutaneous nerve of the thigh, pudendal nerve, and inferior gluteal nerve.

An additional nerve, the perforating cutaneous nerve, enters the gluteal region by passing directly through the sacrotuberous ligament.

Some of these nerves, such as the sciatic and pudendal nerves, pass through the gluteal region en route to other areas. Nerves such as the superior and inferior gluteal nerves innervate structures in the gluteal region. Many of the nerves in the gluteal region are in the plane between the superficial and deep groups of muscles.

Superior gluteal nerve

Of all the nerves that pass through the greater sciatic foramen, the superior gluteal nerve is the only one that passes above the piriformis muscle (Fig. 6.47). After entering the gluteal region, the nerve loops up over the inferior margin of the gluteus minimus and travels anteriorly and laterally in the plane between the gluteus minimus and medius muscles.

The superior gluteal nerve supplies branches to the gluteus minimus and medius muscles and terminates by innervating the tensor fasciae latae muscle.

Sciatic nerve

The sciatic nerve enters the gluteal region through the greater sciatic foramen inferior to the piriformis muscle (Fig. 6.47). It descends in the plane between the superficial and deep group of gluteal region muscles, crossing the posterior surfaces of first the obturator internus and associated gemellus muscles and then the quadratus femoris

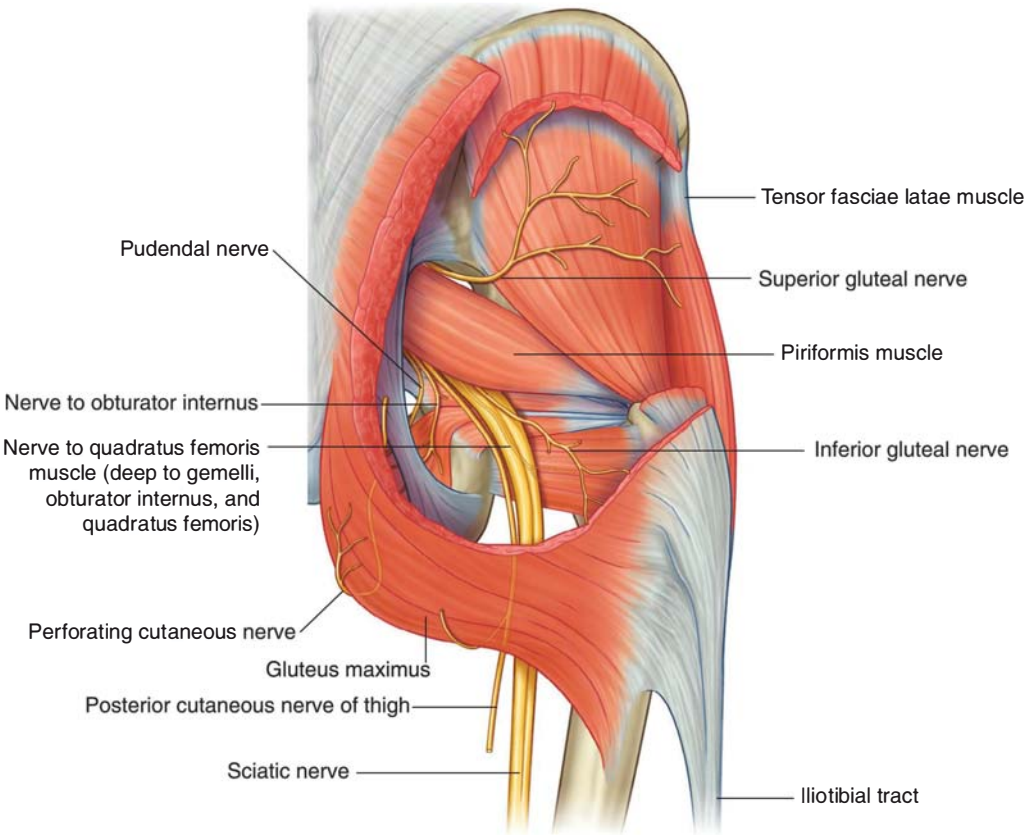


Fig. 6.47 Nerves of the gluteal region. Posterior view.

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muscle. It lies just deep to the gluteus maximus at the midpoint between the ischial tuberosity and the greater trochanter. At the lower margin of the quadratus femoris muscle, the sciatic nerve enters the posterior thigh.

The sciatic nerve is the largest nerve in the body and innervates all muscles in the posterior compartment of the thigh that flex the knee and all muscles that work the ankle and foot. It also innervates a large area of skin in the lower limb.

Nerve to quadratus femoris

The nerve to the quadratus femoris enters the gluteal region through the greater sciatic foramen inferior to the piriformis muscle and deep to the sciatic nerve (Fig. 6.47). Unlike other nerves in the gluteal region, the nerve to the quadratus femoris lies anterior to the plane of the deep muscles.

The nerve to the quadratus femoris descends along the ischium deep to the tendon of the obturator internus muscle and associated gemellus muscles to penetrate and innervate the quadratus femoris. It supplies a small branch to the gemellus inferior.

Nerve to obturator internus

The nerve to the obturator internus enters the gluteal region through the greater sciatic foramen inferior to the piriformis muscle and between the posterior cutaneous nerve of the thigh and the pudendal nerve (Fig. 6.47). It supplies a small branch to the gemellus superior and then passes over the ischial spine and through the lesser sciatic foramen to innervate the obturator internus muscle from the medial surface of the muscle in the perineum.

Posterior cutaneous nerve of the thigh

The posterior cutaneous nerve of the thigh enters the gluteal region through the greater sciatic foramen inferior to the piriformis muscle and immediately medial to the sciatic nerve (Fig. 6.47). It descends through the gluteal region just deep to the gluteus maximus and enters the posterior thigh.

The posterior cutaneous nerve of the thigh has a number of gluteal branches, which loop around the lower margin of the gluteus maximus muscle to innervate skin over the gluteal fold. A small perineal branch passes medially to contribute to the innervation of the skin of the scrotum or labia majora in the perineum. The main trunk of the posterior cutaneous nerve of the thigh passes inferiorly, giving rise to branches that innervate the skin on the posterior thigh and leg.

Pudendal nerve

The pudendal nerve enters the gluteal region through the greater sciatic foramen inferior to the piriformis muscle and medial to the sciatic nerve (Fig. 6.47). It passes over the sacrospinous ligament and immediately passes through the lesser sciatic foramen to enter the perineum. The course of the pudendal nerve in the gluteal region is short and the nerve is often hidden by the overlying upper margin of the sacrotuberous ligament.

The pudendal nerve is the major somatic nerve of the perineum and has no branches in the gluteal region.

Inferior gluteal nerve

The inferior gluteal nerve enters the gluteal region through the greater sciatic foramen inferior to the piriformis muscle and along the posterior surface of the sciatic nerve (Fig. 6.47). It penetrates and supplies the gluteus maximus muscle.

Perforating cutaneous nerve

The perforating cutaneous nerve is the only nerve in the gluteal region that does not enter the area through the greater sciatic foramen. It is a small nerve that leaves the sacral plexus in the pelvic cavity by piercing the sacrotuberous ligament. It then loops around the lower border of the gluteus maximus to supply the skin over the medial aspect of the gluteus maximus (Fig. 6.47).

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In the clinic

Intramuscular injections

From time to time it is necessary to administer drugs intramuscularly, that is, by direct injection into muscles. This procedure must be carried out without injuring neurovascular structures. A typical site for an intramuscular injection is the gluteal region. The sciatic nerve passes through this region and needs to be avoided. The safest place to inject is the upper outer quadrant of either gluteal region.

The gluteal region can be divided into quadrants by two imaginary lines positioned using palpable bony landmarks (Fig. 6.48). One line descends vertically from the highest point of the iliac crest. Another line is horizontal and passes through the first line midway between the highest point of the iliac crest and the horizontal plane through the ischial tuberosity.

It is important to remember that the gluteal region extends as far forward as the anterior superior iliac spine. The sciatic nerve curves through the upper lateral corner of the lower medial quadrant and descends along the medial margin of the lower lateral quadrant.

Occasionally, the sciatic nerve bifurcates into its tibial and common fibular branches in the pelvis, in which case the common fibular nerve passes into the gluteal region through, or even above, the piriformis muscle.

The superior gluteal nerve and vessels normally enter the gluteal region above the piriformis and pass superiorly and forward.

The anterior corner of the upper lateral quadrant is normally used for injections to avoid injuring any part of the sciatic nerve or other nerves and vessels in the gluteal region. A needle placed in this region enters the gluteus medius anterosuperior to the margin of the gluteus maximus.

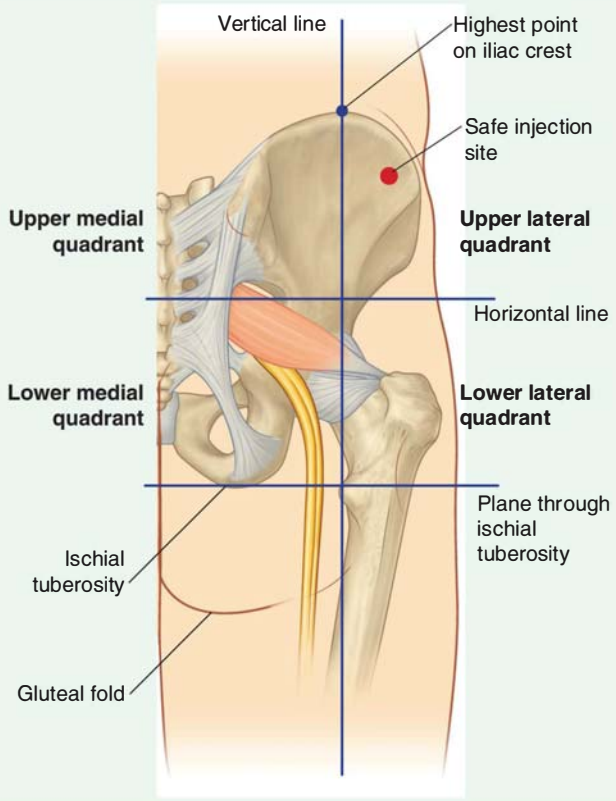


Fig. 6.48 Site for intramuscular injections in the gluteal region.

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Arteries

Two arteries enter the gluteal region from the pelvic cavity through the greater sciatic foramen, the inferior gluteal artery and the superior gluteal artery (Fig. 6.49). They supply structures in the gluteal region and posterior thigh and have important collateral anastomoses with branches of the femoral artery.

Inferior gluteal artery

The inferior gluteal artery originates from the anterior trunk of the internal iliac artery in the pelvic cavity. It leaves the pelvic cavity with the inferior gluteal nerve through the greater sciatic foramen inferior to the piriformis muscle (Fig. 6.49).

The inferior gluteal artery supplies adjacent muscles and descends through the gluteal region and into the posterior thigh where it supplies adjacent structures and anastomoses with perforating branches of the femoral artery. It also supplies a branch to the sciatic nerve.

Superior gluteal artery

The superior gluteal artery originates from the posterior trunk of the internal iliac artery in the pelvic cavity. It leaves the pelvic cavity with the superior gluteal nerve through the greater sciatic foramen above the piriformis muscle (Fig. 6.49). In the gluteal region, it divides into a superficial branch and a deep branch:

- The superficial branch passes onto the deep surface of the gluteus maximus muscle.
- The deep branch passes between the gluteus medius and minimus muscles.

In addition to adjacent muscles, the superior gluteal artery contributes to the supply of the hip joint. Branches of the artery also anastomose with the lateral and medial femoral circumflex arteries from the deep femoral artery in the thigh, and with the inferior gluteal artery (Fig. 6.50).

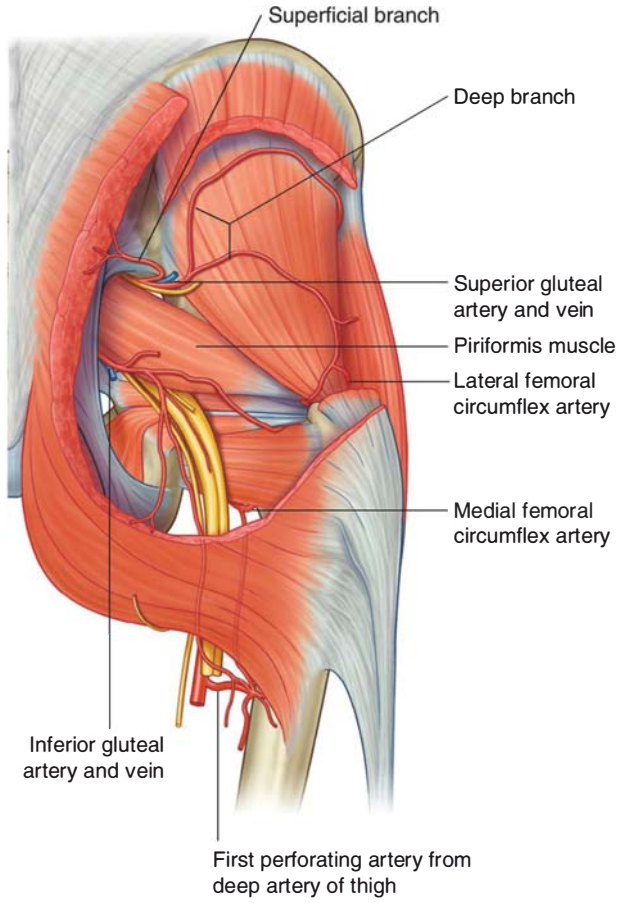


Fig. 6.49 Arteries of the gluteal region.

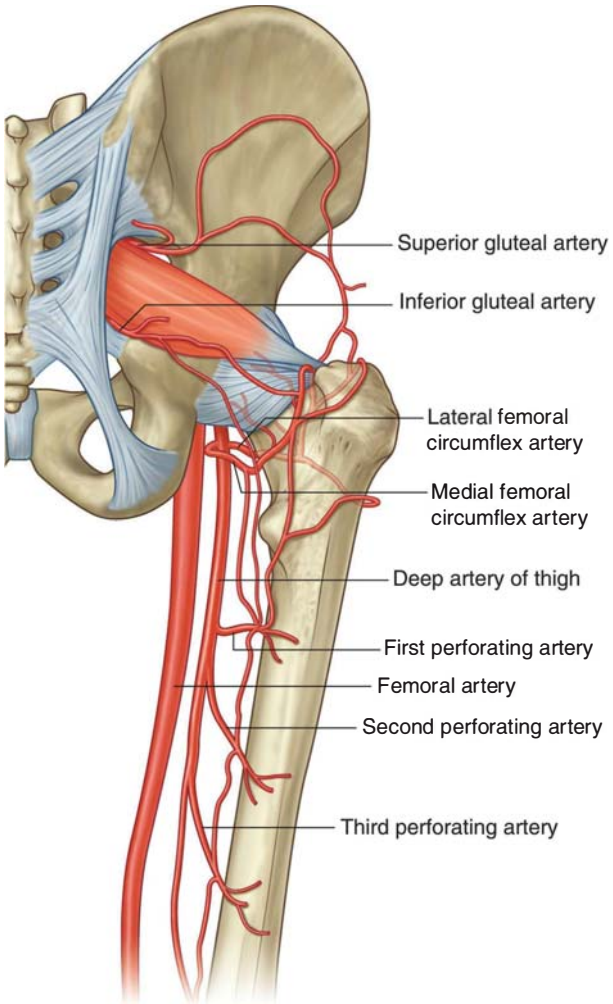


Fig. 6.50 Anastomoses between gluteal arteries and vessels originating from the femoral artery in the thigh. Posterior view.

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Veins

Inferior and superior gluteal veins follow the inferior and superior gluteal arteries into the pelvis where they join the pelvic plexus of veins. Peripherally, the veins anastomose with superficial gluteal veins, which ultimately drain anteriorly into the femoral vein.

Lymphatics

Deep lymphatic vessels of the gluteal region accompany the blood vessels into the pelvic cavity and connect with internal iliac nodes.

Superficial lymphatics drain into the superficial inguinal nodes on the anterior aspect of the thigh.

THIGH

The thigh is the region of the lower limb that is approximately between the hip and knee joints (Fig. 6.51):

- Anteriorly, it is separated from the abdominal wall by the inguinal ligament.
- Posteriorly, it is separated from the gluteal region by the gluteal fold superficially, and by the inferior margins of the gluteus maximus and quadratus femoris on deeper planes.

Structures enter and leave the top of the thigh by three routes:

- Posteriorly, the thigh is continuous with the gluteal region and the major structure passing between the two regions is the sciatic nerve.
- Anteriorly, the thigh communicates with the abdominal cavity through the aperture between the inguinal ligament and pelvic bone, and major structures passing through this aperture are the iliopsoas and pectineus muscles; the femoral nerve, artery, and vein; and lymphatic vessels.

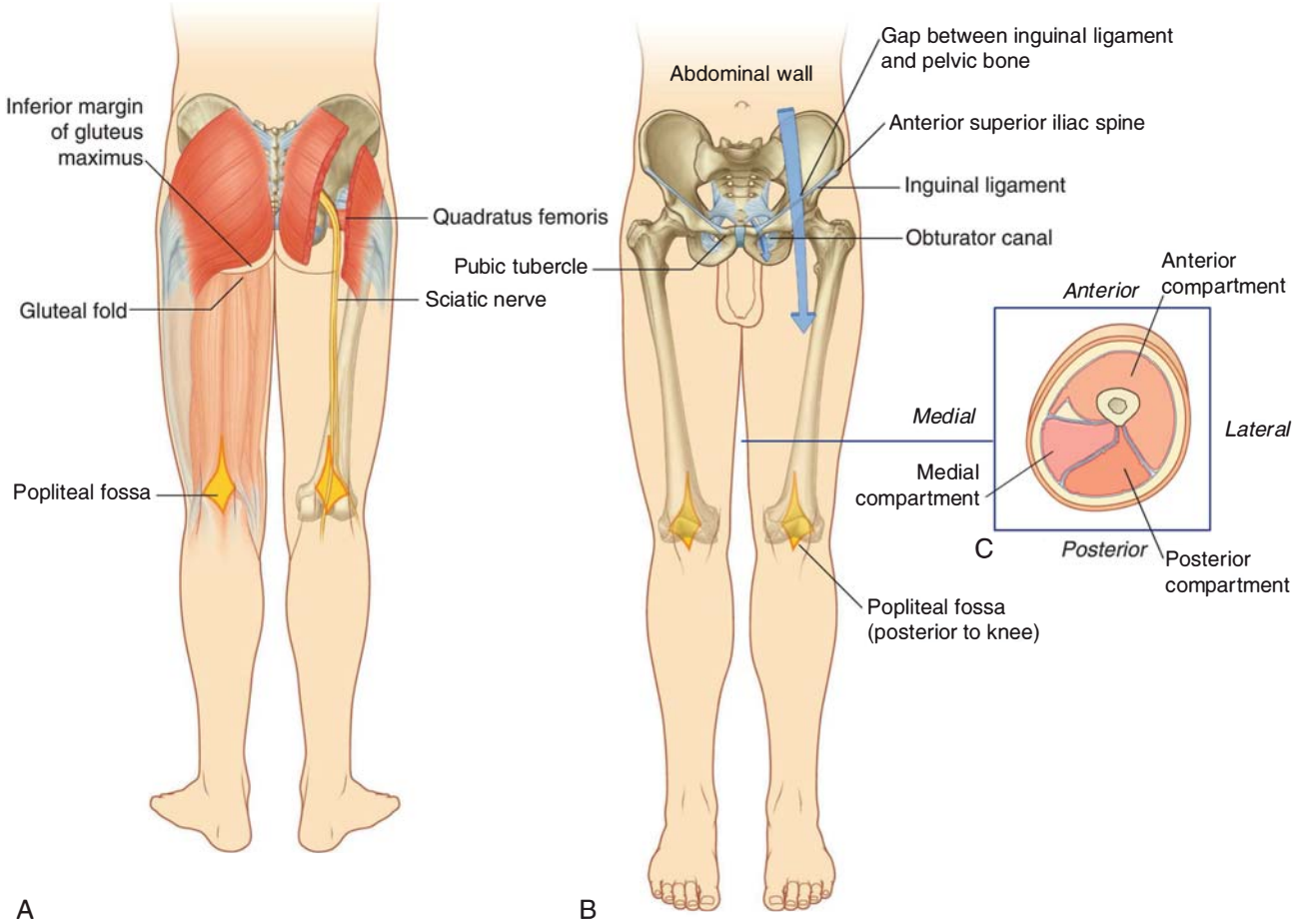


Fig. 6.51 Thigh. **A.** Posterior view. **B.** Anterior view. **C.** Cross section through the midthigh.

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- Medially, structures (including the obturator nerve and associated vessels) pass between the thigh and pelvic cavity through the obturator canal.

The thigh is divided into three compartments by intermuscular septa between the posterior aspect of the femur and the fascia lata (the thick layer of deep fascia that completely surrounds or invests the thigh; Fig. 6.51C):

- The **anterior compartment of the thigh** contains muscles that mainly extend the leg at the knee joint.
- The **posterior compartment of the thigh** contains muscles that mainly extend the thigh at the hip joint and flex the leg at the knee joint.
- The **medial compartment of the thigh** consists of muscles that mainly adduct the thigh at the hip joint.

The sciatic nerve innervates muscles in the posterior compartment of the thigh, the femoral nerve innervates muscles in the anterior compartment of the thigh, and the obturator nerve innervates most muscles in the medial compartment of the thigh.

The major artery, vein, and lymphatic channels enter the thigh anterior to the pelvic bone and pass through the femoral triangle inferior to the inguinal ligament. Vessels and nerves passing between the thigh and leg pass through the popliteal fossa posterior to the knee joint.

Bones

The skeletal support for the thigh is the femur. Most of the large muscles in the thigh insert into the proximal ends of the two bones of the leg (tibia and fibula) and flex and extend the leg at the knee joint. The distal end of the femur provides origin for the gastrocnemius muscles, which are predominantly in the posterior compartment of the leg and plantarflex the foot.

Shaft and distal end of femur

The shaft of the femur is bowed forward and has a n oblique course from the neck of the femur to the distal end (Fig. 6.52). As a consequence of this oblique orientation, the knee is close to the midline under the body's center of gravity.

The middle part of the shaft of the femur is triangular in cross section (Fig. 6.52D). In the middle part of the shaft, the femur has smooth medial (posteromedial), lateral (posterolateral), and anterior surfaces and medial, lateral, and posterior borders. The medial and lateral borders are rounded, whereas the posterior border forms a broad roughened crest—the **linea aspera**.

In proximal and distal regions of the femur, the linea aspera widens to form an additional posterior surface. At the distal end of the femur, this posterior surface forms the floor of the popliteal fossa, and its margins form the **medial** and **lateral supracondylar lines**. The medial supracondylar line terminates at a prominent tubercle (the **adductor tubercle**) on the superior aspect of the **medial condyle** of the distal end. Just lateral to the lower end of the medial supracondylar line is an elongate roughened area of bone for the proximal attachment of the medial head of the gastrocnemius muscle (Fig. 6.51).

The distal end of the femur is characterized by two large condyles, which articulate with the proximal head of the tibia. The condyles are separated posteriorly by an **intercondylar fossa** and are joined anteriorly where they articulate with the patella.

The surfaces of the condyles that articulate with the tibia are rounded posteriorly and become flatter inferiorly. On each condyle, a shallow oblique groove separates the surface that articulates with the tibia from the more anterior surface that articulates with the patella. The surfaces of the medial and lateral condyles that articulate with the patella form a V-shaped trench, which faces anteriorly. The lateral surface of the trench is larger and steeper than the medial surface.

The walls of the intercondylar fossa bear two facets for the superior attachment of the cruciate ligaments, which stabilize the knee joint (Fig. 6.52):

- The wall formed by the lateral surface of the medial condyle has a large oval facet, which covers most of the inferior half of the wall, for attachment of the proximal end of the **posterior cruciate ligament**.
- The wall formed by the medial surface of the lateral condyle has a posterosuperior smaller oval facet for attachment of the proximal end of the **anterior cruciate ligament**.

Epicondyles, for the attachment of collateral ligaments of the knee joint, are bony elevations on the nonarticular outer surfaces of the condyles (Fig. 6.52). Two facets separated by a groove are just posterior to the **lateral epicondyle**:

- The upper facet is for attachment of the lateral head of the gastrocnemius muscle.
- The inferior facet is for attachment of the popliteus muscle.

The tendon of the popliteus muscle lies in the groove separating the two facets.

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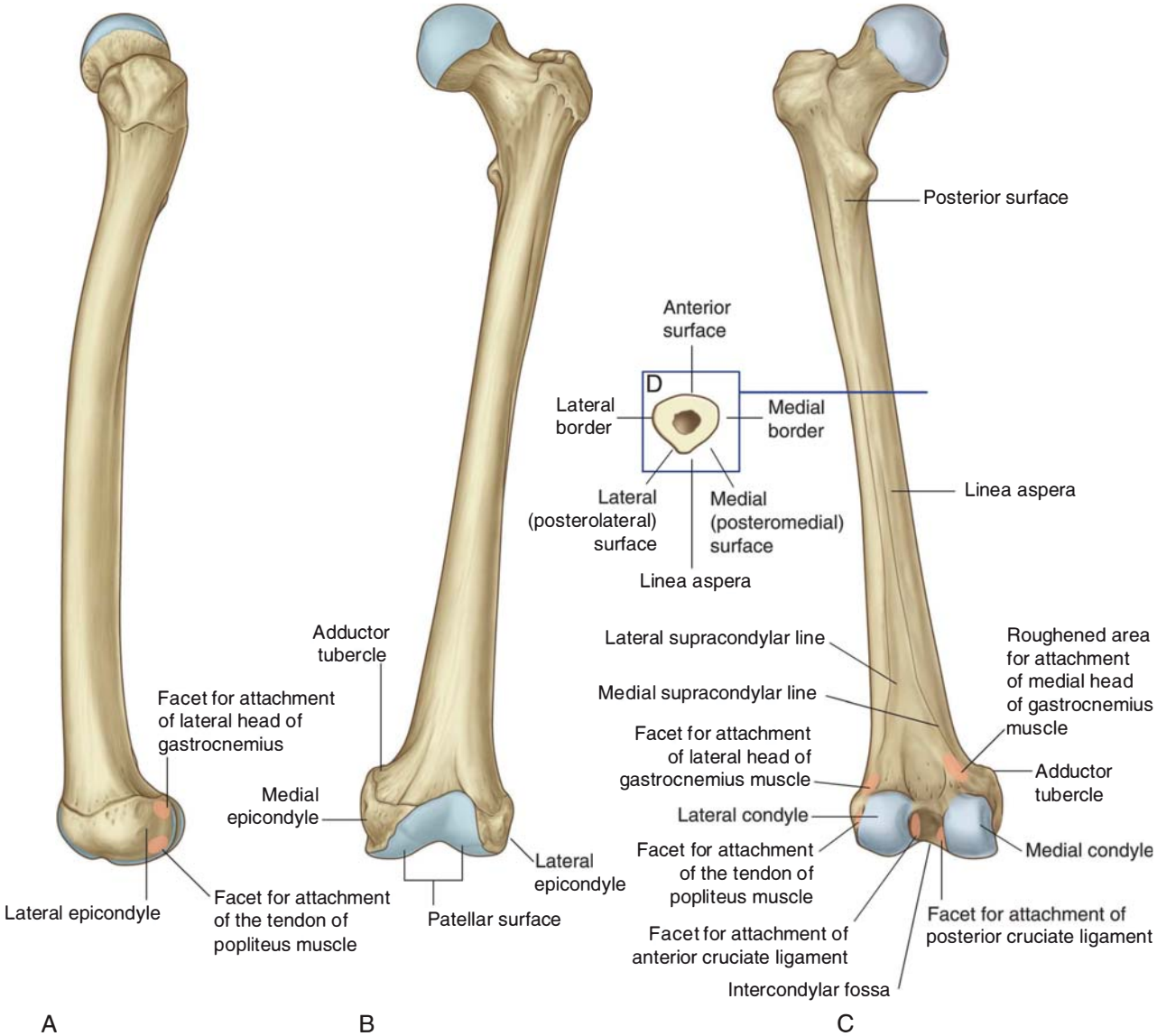


Fig. 6.52 Shaft and distal end of femur. **A.** Lateral view. **B.** Anterior view. **C.** Posterior view. **D.** Cross section through shaft of femur.

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The **medial epicondyle** is a rounded eminence on the medial surface of the medial condyle. Just posterosuperior to the medial epicondyle is the adductor tubercle.

Patella

The patella (knee cap) is the largest sesamoid bone (a bone formed within the tendon of a muscle) in the body and is formed within the tendon of the quadriceps femoris muscle as it crosses anterior to the knee joint to insert on the tibia.

The patella is triangular:

- Its apex is pointed inferiorly for attachment to the patellar ligament, which connects the patella to the tibia (Fig. 6.53).
- Its base is broad and thick for the attachment of the quadriceps femoris muscle from above.
- Its posterior surface articulates with the femur and has medial and lateral facets, which slope away from a raised smooth ridge—the lateral facet is larger than the medial facet for articulation with the larger corresponding surface on the lateral condyle of the femur.

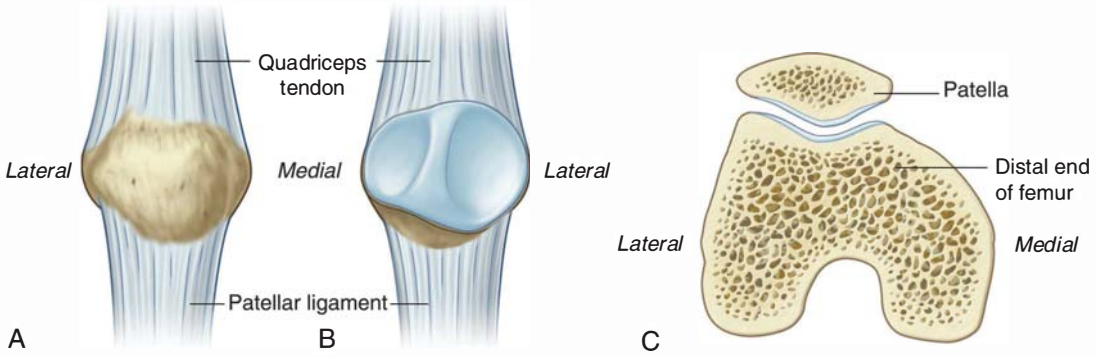


Fig. 6.53 Patella. **A.** Anterior view. **B.** Posterior view. **C.** Superior view.

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Proximal end of tibia

The tibia is the medial and larger of the two bones in the leg, and is the only one that articulates with the femur at the knee joint.

The proximal end of the tibia is expanded in the transverse plane for weight-bearing and consists of a **medial condyle** and a **lateral condyle**, which are both flattened in the horizontal plane and overhang the shaft (Fig. 6.54).

The superior surfaces of the medial and lateral condyles are articular and separated by an intercondylar region, which contains sites of attachment for strong ligaments (cruciate ligaments) and interarticular cartilages (menisci) of the knee joint.

The articular surfaces of the medial and lateral condyles and the intercondylar region together form a "tibial plateau," which articulates with and is anchored to the distal end of the femur. Inferior to the condyles on the proximal part of the shaft is a large **tibial tuberosity** and roughenings for muscle and ligament attachments.

Tibial condyles and intercondylar areas

The tibial condyles are thick horizontal discs of bone attached to the top of the tibial shaft (Fig. 6.54).

The medial condyle is larger than the lateral condyle and is better supported over the shaft of the tibia. Its superior surface is oval for articulation with the medial condyle

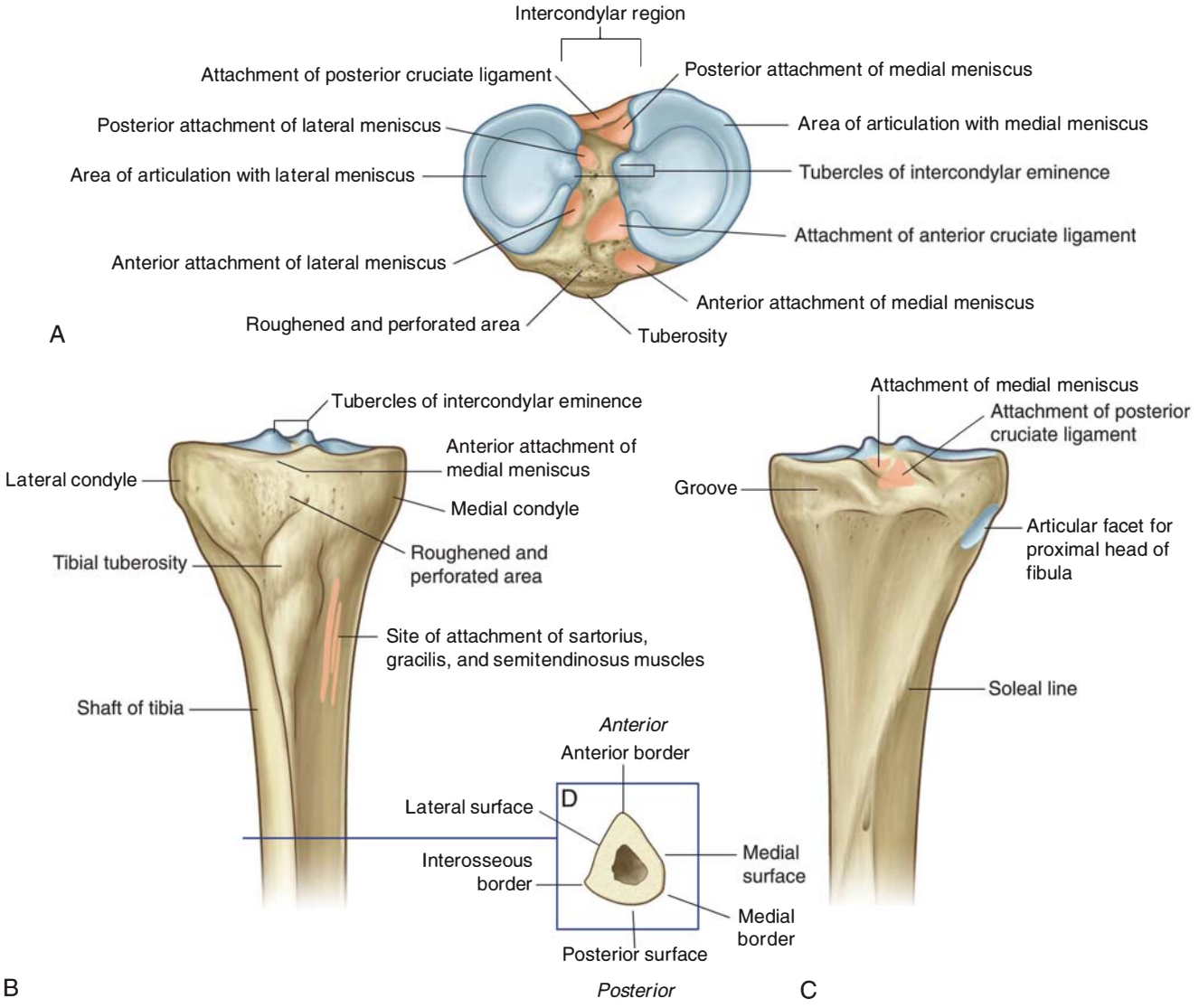


Fig. 6.54 Proximal end of the tibia. **A.** Superior view, tibial plateau. **B.** Anterior view. **C.** Posterior view. **D.** Cross section through the shaft of tibia.

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of the femur. The articular surface extends laterally onto the side of the raised **medial intercondylar tubercle**.

The superior surface of the lateral condyle is circular and articulates above with the lateral condyle of the femur. The medial edge of this surface extends onto the side of the **lateral intercondylar tubercle**.

The superior articular surfaces of both the lateral and medial condyles are concave, particularly centrally. The outer margins of the surfaces are flatter and are the regions in contact with the interarticular discs (menisci) of fibrocartilage in the knee joint.

The nonarticular posterior surface of the medial condyle bears a distinct horizontal groove for part of the attachment of the semimembranosus muscle, and the undersurface of the lateral condyle bears a distinct circular facet for articulation with the proximal head of the fibula.

The intercondylar region of the tibial plateau lies between the articular surfaces of the medial and lateral condyles (Fig. 6.54). It is narrow centrally where it is raised to form the **intercondylar eminence**, the sides of which are elevated further to form medial and lateral intercondylar tubercles.

The intercondylar region bears six distinct facets for the attachment of menisci and cruciate ligaments. The anterior intercondylar area widens anteriorly and bears three facets:

- The most anterior facet is for attachment of the anterior end (horn) of the medial meniscus.
- Immediately posterior to the most anterior facet is a facet for the attachment of the anterior cruciate ligament.
- A small facet for the attachment of the anterior end (horn) of the lateral meniscus is just lateral to the site of attachment of the anterior cruciate ligament.

The posterior intercondylar area also bears three attachment facets:

- The most anterior is for attachment of the posterior horn of the lateral meniscus.
- Posteromedial to the most anterior facet is the site of attachment for the posterior horn of the medial meniscus.
- Behind the site of attachment for the posterior horn of the medial meniscus is a large facet for the attachment of the posterior cruciate ligament.

In addition to these six sites of attachment for menisci and cruciate ligaments, a large anterolateral region of the anterior intercondylar area is roughened and perforated by numerous small nutrient foramina for blood vessels. This region is continuous with a similar surface on the front of the tibia above the tuberosity and lies against infrapatellar connective tissue.

Tibial tuberosity

The **tibial tuberosity** is a palpable inverted triangular area on the anterior aspect of the tibia below the site of junction between the two condyles (Fig. 6.54). It is the site of attachment for the **patellar ligament**, which is a continuation of the quadriceps femoris tendon below the patella.

Shaft of tibia

The shaft of the tibia is triangular in cross section and has three surfaces (posterior, medial, and lateral) and three borders (anterior, interosseous, and medial) (Fig. 6.54D):

- The **anterior border** is sharp and descends from the tibial tuberosity where it is continuous superiorly with a ridge that passes along the lateral margin of the tuberosity and onto the lateral condyle.
- The **interosseous border** is a subtle vertical ridge that descends along the lateral aspect of the tibia from the region of bone anterior and inferior to the articular facet for the head of the fibula.
- The medial border is indistinct superiorly where it begins at the anterior end of the groove on the posterior surface of the medial tibial condyle, but is sharp in midshaft.

The large **medial surface** of the shaft of the tibia, between the anterior and medial borders, is smooth and subcutaneous, and is palpable along almost its entire extent. Medial and somewhat inferior to the tibial tuberosity, this medial surface bears a subtle, slightly roughened elongate elevation. This elevation is the site of the combined attachment of three muscles (sartorius, gracilis, and semitendinosus), which descend from the thigh.

The **posterior surface** of the shaft of the tibia, between the interosseous and medial borders, is widest superiorly where it is crossed by a roughened oblique line (the **soleal line**).

The **lateral surface**, between the anterior and interosseous borders, is smooth and unremarkable.

Proximal end of fibula

The fibula is the lateral bone of the leg and does not take part in formation of the knee joint or in weight-bearing. It is much smaller than the tibia and has a small proximal head, a narrow neck, and a delicate shaft, which ends as the lateral malleolus at the ankle.

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The **head** of the fibula is a globe-shaped expansion at the proximal end of the fibula (Fig. 6.55). A circular facet on the superomedial surface is for articulation above with a similar facet on the inferior aspect of the lateral condyle of the tibia. Just posterolateral to this facet, the bone projects superiorly as a blunt apex (styloid process).

The lateral surface of the head of the fibula bears a large impression for the attachment of the biceps femoris muscle. A depression near the upper margin of this impression is for attachment of the fibular collateral ligament of the knee joint.

The **neck** of the fibula separates the expanded head from the **shaft**. The common fibular nerve lies against the posterolateral aspect of the neck.

Like the tibia, the shaft of the fibula has three borders (anterior, posterior, and interosseous) and three surfaces (lateral, posterior, and medial), which lie between the borders (Fig. 6.55):

- The **anterior border** is sharp midshaft and begins superiorly from the anterior aspect of the head.
- The **posterior border** is rounded and descends from the region of the styloid process of the head.
- The **interosseous border** is medial in position.

The three surfaces of the fibula are associated with the three muscular compartments (lateral, posterior, and anterior) of the leg.

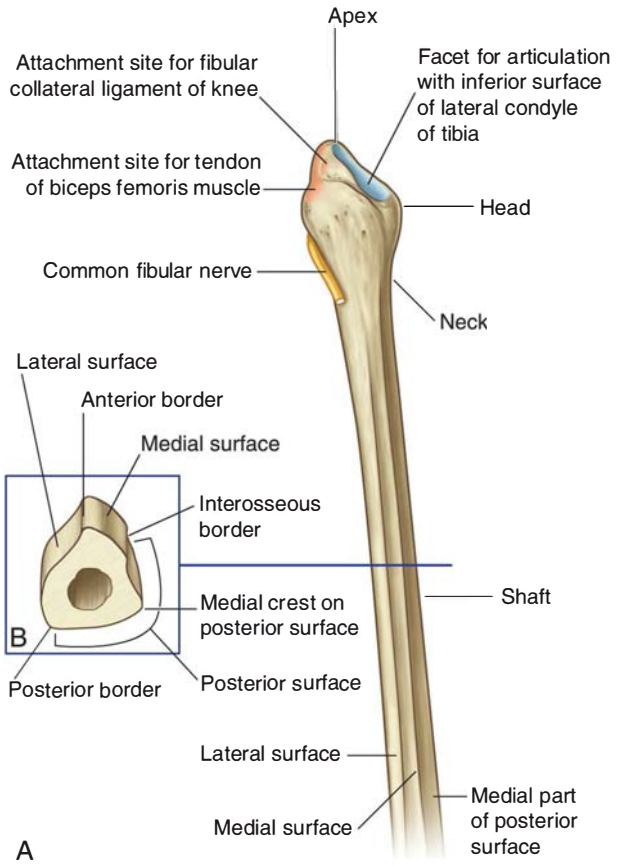


Fig. 6.55 Proximal end of the fibula. **A.** Anterior view. **B.** Cross section through the shaft of fibula.

Muscles

Muscles of the thigh are arranged in three compartments separated by intermuscular septa (Fig. 6.56).

The **anterior compartment of the thigh** contains the sartorius and the four large quadriceps femoris muscles (rectus femoris, vastus lateralis, vastus medialis, and vastus intermedius). All are innervated by the femoral nerve. In addition, the terminal ends of the psoas major and iliacus muscles pass into the upper part of the anterior compartment from sites of origin on the posterior abdominal wall. These muscles are innervated by branches directly from the anterior rami of L1 to L3 (psoas major) or from the femoral nerve (iliacus) as it passes down the abdominal wall.

The **medial compartment of the thigh** contains six muscles (gracilis, pectineus, adductor longus, adductor brevis, adductor magnus, and obturator externus). All except the pectineus, which is innervated by the femoral nerve, and part of the adductor magnus, which is innervated by the sciatic nerve, are innervated by the obturator nerve.

The **posterior compartment of the thigh** contains three large muscles termed the "hamstrings." All are innervated by the sciatic nerve.

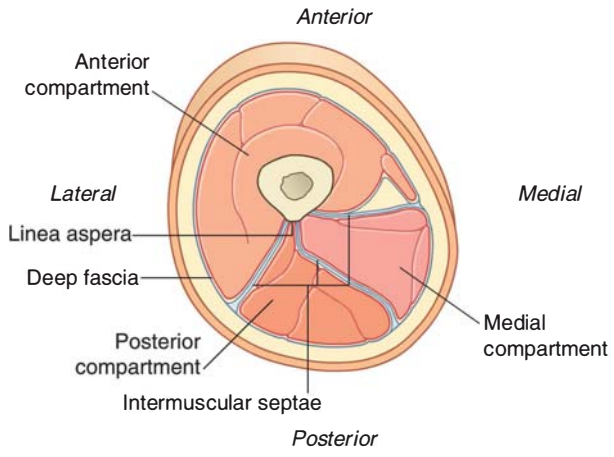


Fig. 6.56 Transverse section through the midthigh.

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In the clinic

Compartment syndrome

Compartment syndrome occurs when there is swelling within a fascial enclosed muscle compartment in the limbs. Typical causes include limb trauma, intracompartment hemorrhage, and limb compression. As pressure within the compartment elevates, capillary blood flow and tissue perfusion is compromised, which can ultimately lead to neuromuscular damage if not treated.

Anterior compartment

Muscles in the anterior compartment (Table 6.3) act on the hip and knee joints:

- the psoas major and iliacus act on the hip joint,
- the sartorius and rectus femoris act on both the hip and knee joints, and
- the vastus muscles act on the knee joint.

Table 6.3 Muscles of the anterior compartment of thigh (spinal segments in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Psoas major	Posterior abdominal wall (lumbar transverse processes, intervertebral discs, and adjacent bodies from TXII to LV and tendinous arches between these points)	Lesser trochanter of femur	Anterior rami (**L1, L2**, L3)	Flexes the thigh at the hip joint
Iliacus	Posterior abdominal wall (iliac fossa)	Lesser trochanter of femur	Femoral nerve (**L2**, L3)	Flexes the thigh at the hip joint
Vastus medialis	Femur—medial part of intertrochanteric line, pectineal line, medial lip of the linea aspera, medial supracondylar line	Quadriceps femoris tendon and medial border of patella	Femoral nerve (L2, **L3, L4**)	Extends the leg at the knee joint
Vastus intermedius	Femur—upper two-thirds of anterior and lateral surfaces	Quadriceps femoris tendon, lateral margin of patella, and lateral condyle of tibia	Femoral nerve (L2, **L3, L4**)	Extends the leg at the knee joint
Vastus lateralis	Femur—lateral part of intertrochanteric line, margin of greater trochanter, lateral margin of gluteal tuberosity, lateral lip of the linea aspera	Quadriceps femoris tendon and lateral margin of patella	Femoral nerve (L2, **L3, L4**)	Extends the leg at the knee joint
Rectus femoris	Straight head originates from the anterior inferior iliac spine; reflected head originates from the ilium just superior to the acetabulum	Quadriceps femoris tendon	Femoral nerve (L2, **L3, L4**)	Flexes the thigh at the hip joint and extends the leg at the knee joint
Sartorius	Anterior superior iliac spine	Medial surface of tibia just inferomedial to tibial tuberosity	Femoral nerve (**L2, L3**)	Flexes the thigh at the hip joint and flexes the leg at the knee joint

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Iliopsoas—psoas major and iliacus

The **psoas major** and **iliacus** muscles originate on the posterior abdominal wall and descend into the upper part of the anterior compartment of the thigh through the lateral half of the gap between the inguinal ligament and the pelvic bone (Fig. 6.57).

Although the iliacus and psoas major originate as separate muscles in the abdomen, both insert by a common tendon onto the lesser trochanter of the femur and together are usually referred to as the **iliopsoas** muscle.

The iliopsoas is a powerful flexor of the thigh at the hip joint and can also contribute to lateral rotation of the thigh. The psoas major is innervated by branches from the anterior rami of L1 to L3 and the iliacus is innervated by branches from the femoral nerve in the abdomen.

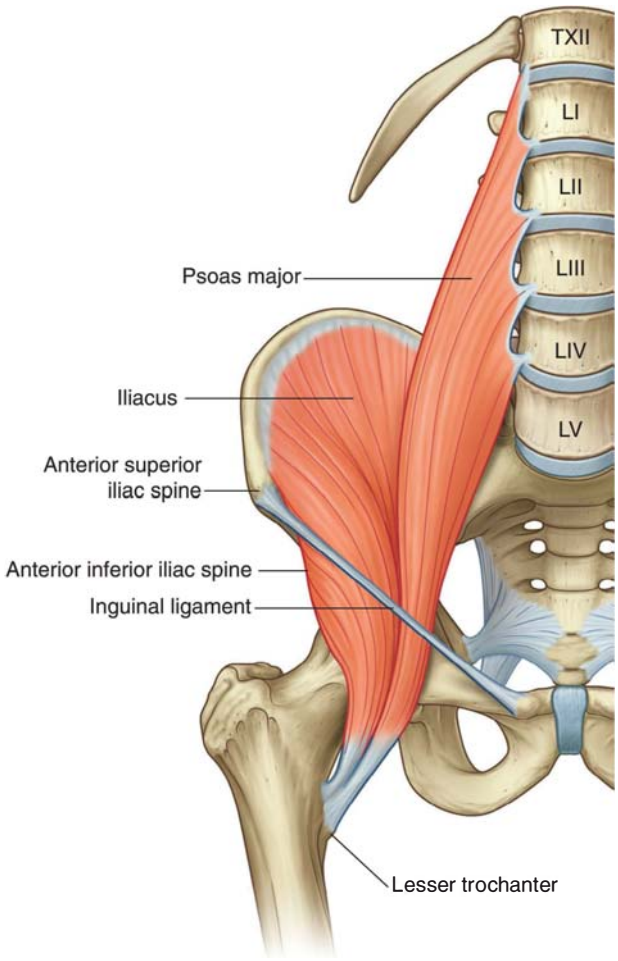


Fig. 6.57 Psoas major and iliacus muscles.

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Quadriceps femoris—vastus medialis, intermedius, and lateralis and rectus femoris

The large **quadriceps femoris** muscle consists of three vastus muscles (vastus medialis, vastus intermedius, and vastus lateralis) and the rectus femoris muscle (Fig. 6.58).

The quadriceps femoris muscle mainly extends the leg at the knee joint, but the rectus femoris component also assists flexion of the thigh at the hip joint. Because the vastus muscles insert into the margins of the patella as well as into the quadriceps femoris tendon, they stabilize the position of the patella during knee joint movement.

The quadriceps femoris is innervated by the femoral nerve with contributions mainly from spinal segments L3 and L4. A tap with a tendon hammer on the patellar ligament therefore tests reflex activity mainly at spinal cord levels L3 and L4.

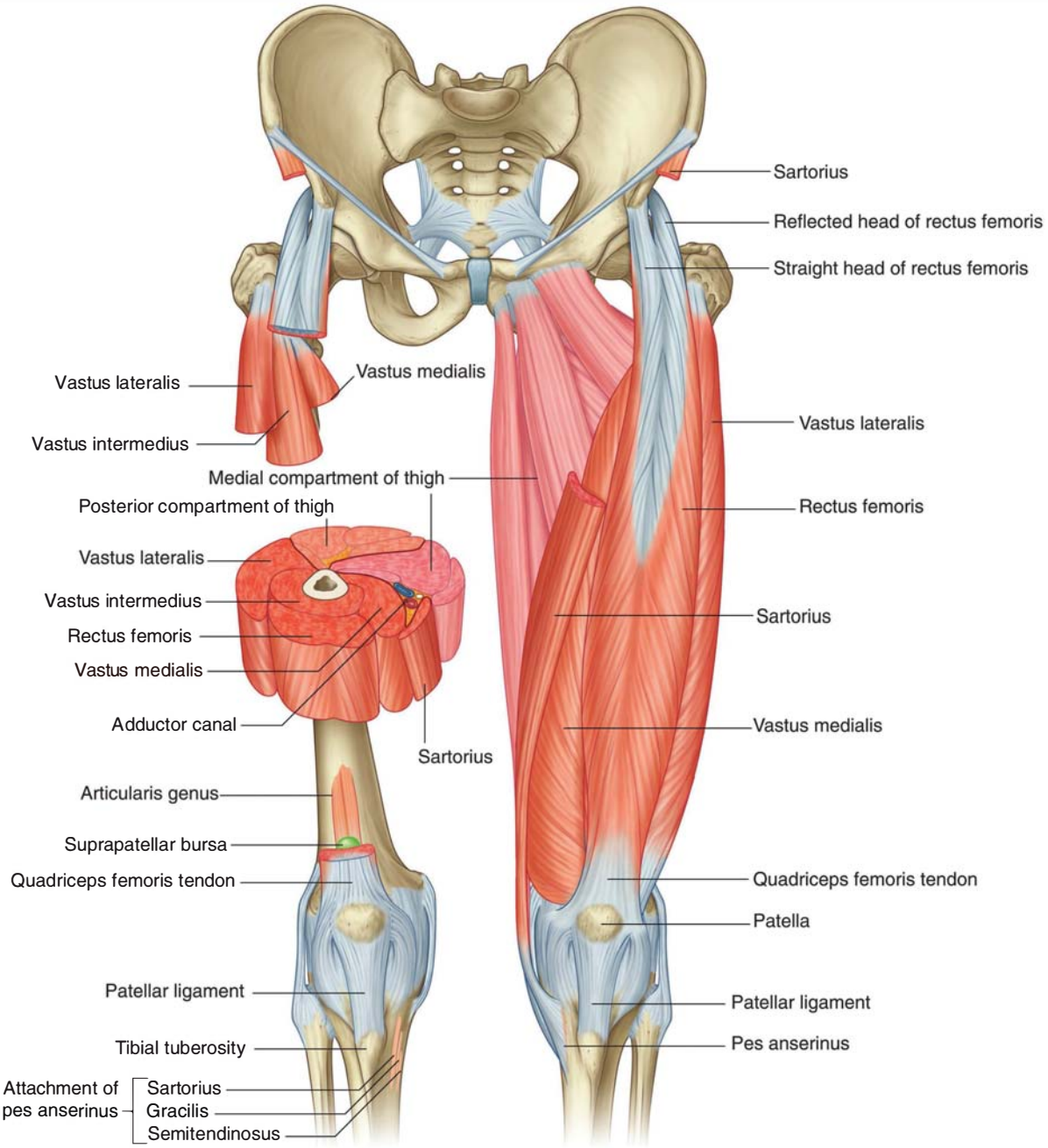


Fig. 6.58 Muscles of the anterior compartment of thigh.

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Vastus muscles

The vastus muscles originate from the femur, whereas the rectus femoris muscle originates from the pelvic bone. All attach first to the patella by the quadriceps femoris tendon and then to the tibia by the **patellar ligament**.

The **vastus medialis** originates from a continuous line of attachment on the femur, which begins anteromedially on the intertrochanteric line and continues posteroinferiorly along the pectineal line and then descends along the medial lip of the linea aspera and onto the medial supracondylar line. The fibers converge onto the medial aspect of the quadriceps femoris tendon and the medial border of the patella (Fig. 6.58).

The **vastus intermedius** originates mainly from the upper two-thirds of the anterior and lateral surfaces of the femur and the adjacent intermuscular septum (Fig. 6.58). It merges into the deep aspect of the quadriceps femoris tendon and also attaches to the lateral margin of the patella and lateral condyle of the tibia.

A tiny muscle (**articularis genus**) originates from the femur just inferior to the origin of the vastus intermedius and inserts into the suprapatellar bursa associated with the knee joint (Fig. 6.58). This articular muscle, which is often part of the vastus intermedius muscle, pulls the bursa away from the knee joint during extension.

The **vastus lateralis** is the largest of the vastus muscles (Fig. 6.58). It originates from a continuous line of attachment, which begins anterolaterally from the superior part of the intertrochanteric line of the femur and then circles laterally around the bone to attach to the lateral margin of the gluteal tuberosity and continues down the upper part of the lateral lip of the linea aspera. Muscle fibers converge mainly onto the quadriceps femoris tendon and the lateral margin of the patella.

Rectus femoris

Unlike the vastus muscles, which cross only the knee joint, the **rectus femoris** muscle crosses both the hip and the knee joints (Fig. 6.58).

The rectus femoris has two tendinous heads of origin from the pelvic bone:

- one from the anterior inferior iliac spine (**straight head**), and
- the other from a roughened area of the ilium immediately superior to the acetabulum (**reflected head**) (Fig. 6.58).

The two heads of the rectus femoris unite to form an elongate muscle belly, which lies anterior to the vastus intermedius muscle and between the vastus lateralis and vastus medialis muscles, to which it is attached on either side. At the distal end, the rectus femoris muscle converges on the quadriceps femoris tendon and inserts on the base of the patella.

Patellar ligament

The patellar ligament is functionally the continuation of the quadriceps femoris tendon below the patella and is attached above to the apex and margins of the patella and below to the tibial tuberosity (Fig. 6.58). The more superficial fibers of the quadriceps femoris tendon and the patellar ligament are continuous over the anterior surface of the patella, and lateral and medial fibers are continuous with the ligament beside the margins of the patella.

Sartorius

The **sartorius** muscle is the most superficial muscle in the anterior compartment of the thigh and is a long strap-like muscle that descends obliquely through the thigh from the anterior superior iliac spine to the medial surface of the proximal shaft of the tibia (Fig. 6.58). Its flat aponeurotic insertion into the tibia is immediately anterior to the insertion of the gracilis and semitendinosus muscles.

The sartorius, gracilis, and semitendinosus muscles attach to the tibia in a three-pronged pattern on the tibia, so their combined tendons of insertion are often termed the **pes anserinus** (Latin for "goose foot").

In the upper one-third of the thigh, the medial margin of the sartorius forms the lateral margin of the femoral triangle.

In the middle one-third of the thigh, the sartorius forms the anterior wall of the adductor canal.

The sartorius muscle assists in flexing the thigh at the hip joint and the leg at the knee joint. It also abducts the thigh and rotates it laterally, as when resting the foot on the opposite knee when sitting.

The sartorius is innervated by the femoral nerve.

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Medial compartment

There are six muscles in the medial compartment of the thigh (Table 6.4): gracilis, pectineus, adductor longus, adductor brevis, adductor magnus, and obturator externus (Fig. 6.59). Collectively, all these muscles except the obturator externus mainly adduct the thigh at the hip joint; the adductor muscles may also medially rotate the thigh. Obturator externus is a lateral rotator of the thigh at the hip joint.

Gracilis

The **gracilis** is the most superficial of the muscles in the medial compartment of thigh and descends almost vertically down the medial side of the thigh (Fig. 6.59). It is attached above to the outer surface of the ischiopubic ramus of the pelvic bone and below to the medial surface of the proximal shaft of the tibia, where it lies sandwiched between the tendon of sartorius in front and the tendon of the semitendinosus behind.

Table 6.4 Muscles of the medial compartment of thigh (spinal segments in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Gracilis	A line on the external surfaces of the body of the pubis, the inferior pubic ramus, and the ramus of the ischium	Medial surface of proximal shaft of tibia	Obturator nerve (**L2**, L3)	Adducts thigh at hip joint and flexes leg at knee joint
Pectineus	Pectineal line (pecten pubis) and adjacent bone of pelvis	Oblique line extending from base of lesser trochanter to linea aspera on posterior surface of proximal femur	Femoral nerve (**L2**, L3)	Adducts and flexes thigh at hip joint
Adductor longus	External surface of body of pubis (triangular depression inferior to pubic crest and lateral to pubic symphysis)	Linea aspera on middle one-third of shaft of femur	Obturator nerve (anterior division) (**L2, L3,** L4)	Adducts and medially rotates thigh at hip joint
Adductor brevis	External surface of body of pubis and inferior pubic ramus	Posterior surface of proximal femur and upper one-third of linea aspera	Obturator nerve (**L2, L3**)	Adducts and medially rotates thigh at hip joint
Adductor magnus	Adductor part—ischiopubic ramus	Posterior surface of proximal femur, linea aspera, medial supracondylar line	Obturator nerve (**L2, L3,** L4)	Adducts and medially rotates thigh at hip joint
	Hamstring part—ischial tuberosity	Adductor tubercle and supracondylar line	Sciatic nerve (tibial division) (**L2, L3,** L4)	
Obturator externus	External surface of obturator membrane and adjacent bone	Trochanteric fossa	Obturator nerve (posterior division) (L3, **L4**)	Laterally rotates thigh at hip joint

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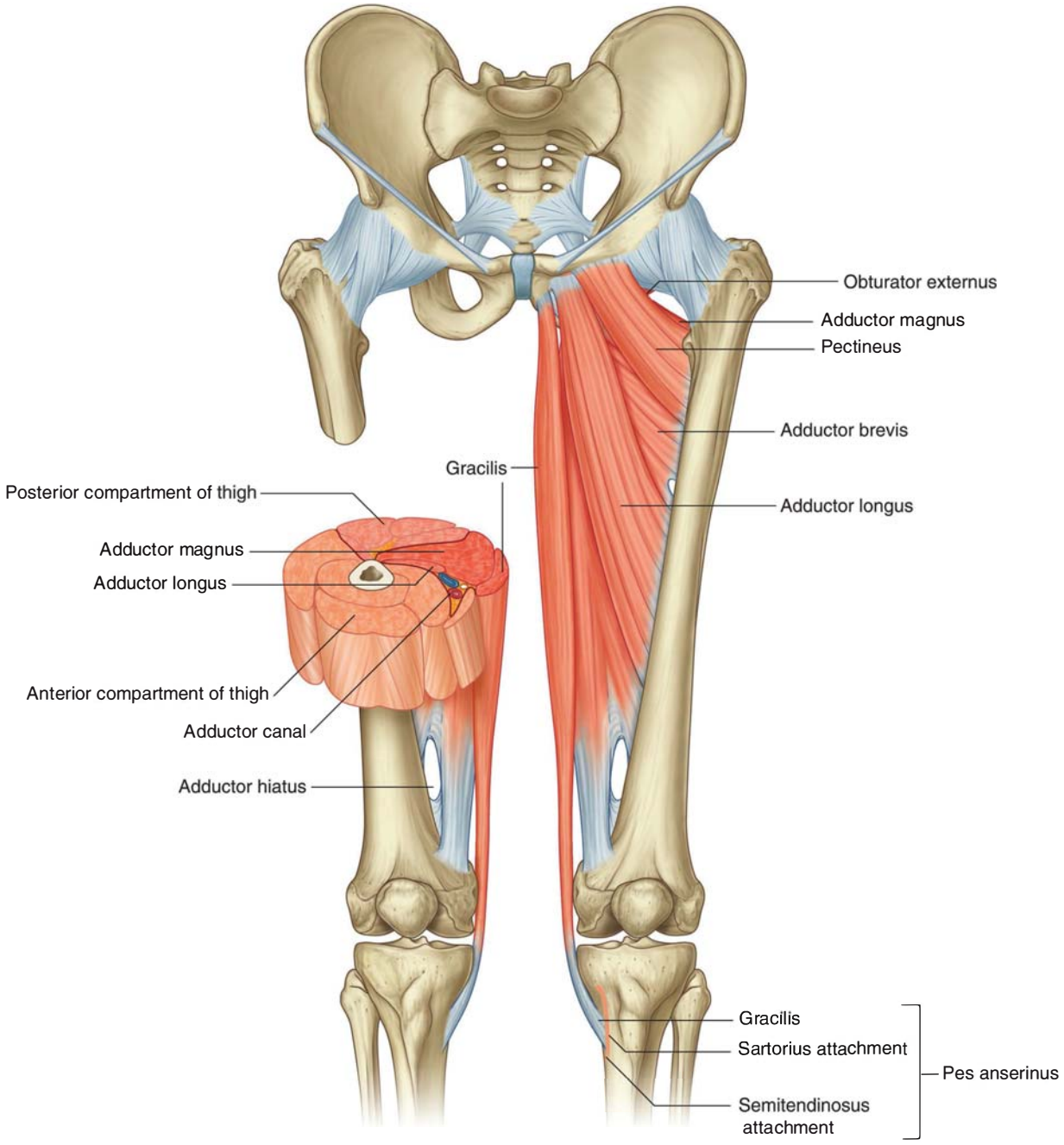


Fig. 6.59 Muscles of the medial compartment of thigh. Anterior view.

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Pectineus

The **pectineus** is a flat quadrangular muscle (Fig. 6.60). It is attached above to the pectineal line of the pelvic bone and adjacent bone, and descends laterally to attach to an oblique line extending from the base of the lesser trochanter to the linea aspera on the posterior surface of the proximal femur.

From its origin on the pelvic bone, the pectineus passes into the thigh below the inguinal ligament and forms part of the floor of the medial half of the femoral triangle.

The pectineus adducts and flexes the thigh at the hip joint and is innervated by the femoral nerve.

Adductor longus

The **adductor longus** is a flat fan-shaped muscle that originates from a small rough triangular area on the external surface of the body of the pubis just inferior to the pubic crest and lateral to the pubic symphysis (Fig. 6.60). It expands as it descends posterolaterally to insert via an aponeurosis into the middle third of the linea aspera.

The adductor longus contributes to the floor of the femoral triangle, and its medial margin forms the medial border of the femoral triangle. The muscle also forms the proximal posterior wall of the adductor canal.

The adductor longus adducts and medially rotates the thigh at the hip joint and is innervated by the anterior division of the obturator nerve.

Adductor brevis

The **adductor brevis** lies posterior to the pectineus and adductor longus. It is a triangular muscle attached at its apex to the body of the pubis and inferior pubic ramus just

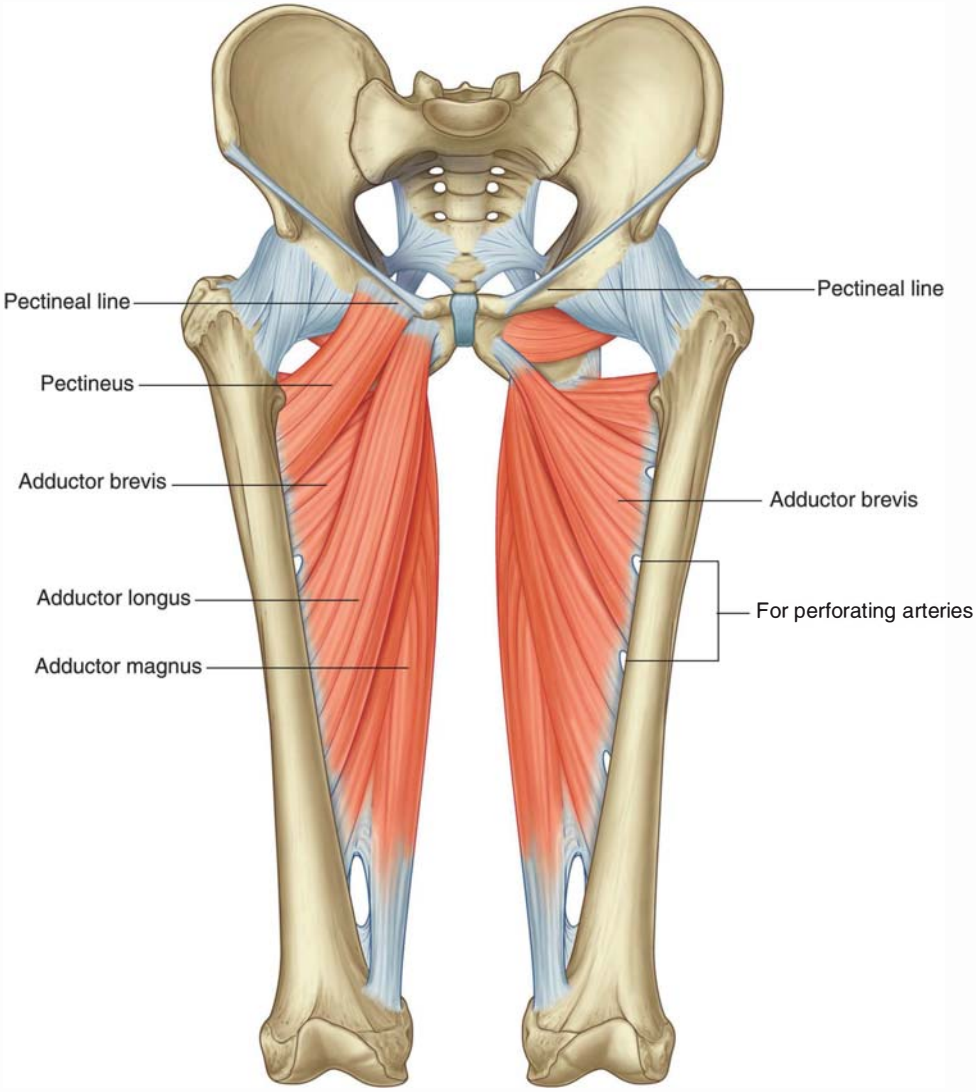


Fig. 6.60 Pectineus, adductor longus, and adductor brevis muscles. Anterior view.

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superior to the origin of the gracilis muscle (Fig. 6.60). The muscle is attached by its expanded base via an aponeurosis to a vertical line extending from lateral to the insertion of the pectineus into the upper aspect of the linea aspera lateral to the attachment of the adductor longus.

The adductor brevis adducts and medially rotates the thigh at the hip joint and is innervated by the obturator nerve.

Adductor magnus

The **adductor magnus** is the largest and deepest of the muscles in the medial compartment of the thigh (Fig. 6.61). The muscle forms the distal posterior wall of the adductor canal. Like the adductor longus and brevis muscles, the adductor magnus is a triangular or fan-shaped muscle anchored by its apex to the pelvis and attached by its expanded base to the femur.

On the pelvis, the adductor magnus is attached along a line that extends from the inferior pubic ramus, above the attachments of the adductor longus and brevis muscles, and along the ramus of the ischium to the ischial tuberosity. The part of the muscle that originates from the ischiopubic ramus expands laterally and inferiorly to insert on the femur along a vertical line of attachment that extends from just inferior to the quadrate tubercle and medial to the gluteal tuberosity, along the linea aspera and onto the medial supracondylar line. This lateral part of the muscle is often termed the "adductor part" of the adductor magnus.

The medial part of the adductor magnus, often called the "hamstring part," originates from the ischial tuberosity of the pelvic bone and descends almost vertically along the thigh to insert via a rounded tendon into the adductor tubercle on the medial condyle of the distal head of the femur. It also inserts via an aponeurosis up onto the medial supracondylar line. A large circular gap inferiorly between the hamstring and adductor parts of the muscle is the **adductor hiatus** (Fig. 6.61), which allows the femoral artery and associated veins to pass between the adductor canal on the anteromedial aspect of the thigh and the popliteal fossa posterior to the knee.

The adductor magnus adducts and medially rotates the thigh at the hip joint. The adductor part of the muscle is innervated by the obturator nerve and the hamstring part is innervated by the tibial division of the sciatic nerve.

Obturator externus

The **obturator externus** is a flat fan-shaped muscle. Its expansive body is attached to the external aspect of the obturator membrane and adjacent bone (Fig. 6.61). The muscle fibers converge posterolaterally to form a tendon, which passes posterior to the hip joint and neck of the femur to insert on an oval depression on the lateral wall of the trochanteric fossa.

The obturator externus externally rotates the thigh at the hip joint and is innervated by the posterior branch of the obturator nerve.

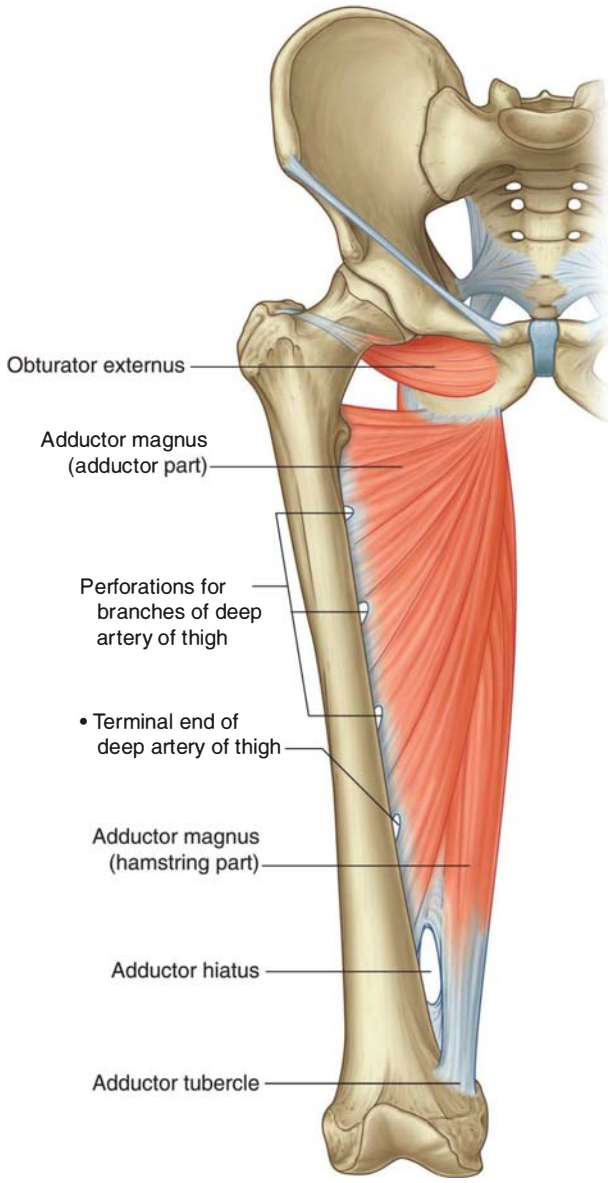


Fig. 6.61 Adductor magnus and obturator externus muscles. Anterior view.

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Posterior compartment

There are three long muscles in the posterior compartment of the thigh: biceps femoris, semitendinosus, and semimembranosus (Table 6.5)—and they are collectively known as the hamstrings (Fig. 6.62). All except the short head of the biceps femoris cross both the hip and knee joints. As a group, the hamstrings flex the leg at the knee joint and extend the thigh at the hip joint. They are also rotators at both joints.

Biceps femoris

The **biceps femoris** muscle is lateral in the posterior compartment of the thigh and has two heads (Fig. 6.62):

- The **long head** originates with the semitendinosus muscle from the inferomedial part of the upper area of the ischial tuberosity.
- The **short head** arises from the lateral lip of the linea aspera on the shaft of the femur.

The muscle belly of the long head crosses the posterior thigh obliquely from medial to lateral and is joined by the short head distally. Together, fibers from the two heads form a tendon, which is palpable on the lateral side of the distal thigh. The main part of the tendon inserts into the lateral surface of the head of the fibula. Extensions from the tendon blend with the fibular collateral ligament and with ligaments associated with the lateral side of the knee joint.

The biceps femoris flexes the leg at the knee joint. The long head also extends and laterally rotates the hip. When the knee is partly flexed, the biceps femoris can laterally rotate the leg at the knee joint.

The long head is innervated by the tibial division of the sciatic nerve and the short head is innervated by the common fibular division of the sciatic nerve.

Semitendinosus

The **semitendinosus** muscle is medial to the biceps femoris muscle in the posterior compartment of the thigh (Fig. 6.62). It originates with the long head of the biceps femoris muscle from the inferomedial part of the upper area of the ischial tuberosity. The spindle-shaped muscle belly ends in the lower half of the thigh and forms a long cord-like tendon, which lies on the semimembranosus muscle and descends to the knee. The tendon curves around the medial condyle of the tibia and inserts into the medial surface of the tibia just posterior to the tendons of the gracilis and sartorius muscles as part of the pes anserinus.

The semitendinosus flexes the leg at the knee joint and extends the thigh at the hip joint. Working with the semimembranosus, it also medially rotates the thigh at the hip joint and medially rotates the leg at the knee joint.

The semitendinosus muscle is innervated by the tibial division of the sciatic nerve.

Table 6.5 Muscles of the posterior compartment of thigh (spinal segments in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Biceps femoris	Long head—inferomedial part of the upper area of the ischial tuberosity; short head—lateral lip of linea aspera	Head of fibula	Sciatic nerve (L5, **S1**, S2)	Flexes leg at knee joint; extends and laterally rotates thigh at hip joint and laterally rotates leg at knee joint
Semitendinosus	Inferomedial part of the upper area of the ischial tuberosity	Medial surface of proximal tibia	Sciatic nerve (L5, **S1**, S2)	Flexes leg at knee joint and extends thigh at hip joint; medially rotates thigh at hip joint and leg at knee joint
Semimembranosus	Superolateral impression on the ischial tuberosity	Groove and adjacent bone on medial and posterior surface of medial tibial condyle	Sciatic nerve (L5, **S1**, S2)	Flexes leg at knee joint and extends thigh at hip joint; medially rotates thigh at hip joint and leg at knee joint

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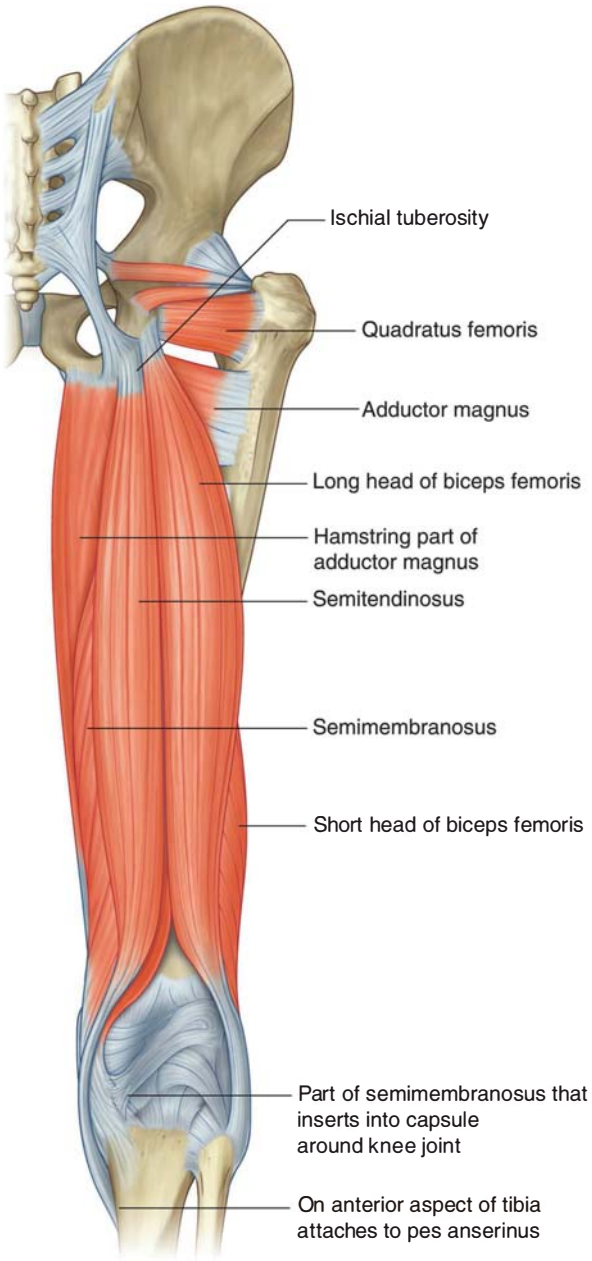


Fig. 6.62 Muscles of the posterior compartment of thigh. Posterior view.

Semimembranosus

The **semimembranosus** muscle lies deep to the semitendinosus muscle in the posterior compartment of the thigh (Fig. 6.62). It is attached above to the superolateral impression on the ischial tuberosity and below mainly to the groove and adjacent bone on the medial and posterior surfaces of the medial tibial condyle. Expansions from the tendon also insert into and contribute to the formation of ligaments and fascia around the knee joint.

The semimembranosus flexes the leg at the knee joint and extends the thigh at the hip joint. Working with the semitendinosus muscle, it medially rotates the thigh at the hip joint and the leg at the knee joint.

The semimembranosus muscle is innervated by the tibial division of the sciatic nerve.

In the clinic

Muscle injuries to the lower limb

Muscle injuries may occur as a result of direct trauma or as part of an overuse syndrome.

Muscle injuries may occur as a minor muscle tear, which may be demonstrated as a focal area of fluid within the muscle. With increasingly severe injuries, further muscle fibers are torn and this may eventually result in a complete muscle tear. The usual muscles in the thigh that tear include the hamstring muscles. Tears in the muscles below the knee typically occur within the soleus muscle, though other muscles may be affected.

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Arteries

Three arteries enter the thigh: the femoral artery, the obturator artery, and the inferior gluteal artery. Of these, the femoral artery is the largest and supplies most of the lower limb. The three arteries contribute to an anastomotic network of vessels around the hip joint.

Femoral artery

The femoral artery is the continuation of the external iliac artery and begins as the external iliac artery passes under the inguinal ligament to enter the femoral triangle on the anterior aspect of the upper thigh (Fig. 6.63). The femoral artery is palpable in the femoral triangle just inferior to the inguinal ligament midway between the anterior superior iliac spine and the pubic symphysis.

The femoral artery passes vertically through the femoral triangle and then continues down the thigh in the adductor canal. It leaves the canal by passing through the adductor hiatus in the adductor magnus muscle and becomes the popliteal artery behind the knee.

A cluster of four small branches—**superficial epigastric artery**, **superficial circumflex iliac artery**, **superficial external pudendal artery**, and **deep external pudendal artery**—originate from the femoral artery in the femoral triangle and supply cutaneous regions of the upper thigh, lower abdomen, and perineum.

Deep artery of thigh

The largest branch of the femoral artery in the thigh is the **deep artery of the thigh** (profunda femoris artery), which originates from the lateral side of the femoral artery in the femoral triangle and is the major source of blood supply to the thigh (Fig. 6.63). The deep artery of the thigh immediately passes:

- posteriorly between the pectineus and adductor longus muscles and then between the adductor longus and adductor brevis muscles, and
- then travels inferiorly between the adductor longus and adductor magnus, eventually penetrating through the adductor magnus to connect with branches of the popliteal artery behind the knee.

The deep artery of the thigh has lateral and medial circumflex femoral branches and three perforating branches.

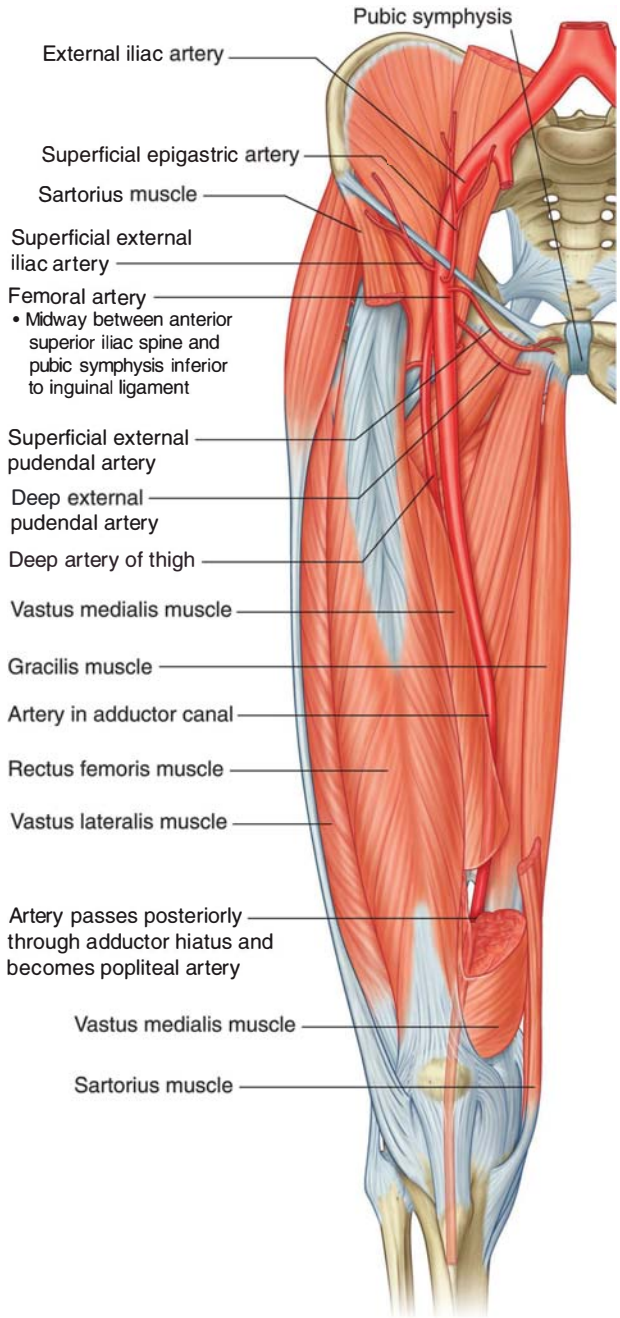


Fig. 6.63 Femoral artery.

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Lateral circumflex femoral artery

The **lateral circumflex femoral artery** normally originates proximally from the lateral side of the deep artery of the thigh, but may arise directly from the femoral artery (Fig. 6.64). It passes deep to the sartorius and rectus femoris and divides into three terminal branches:

- One vessel (**ascending branch**) ascends laterally deep to the tensor fasciae latae muscle and connects with a branch of the medial circumflex femoral artery to form a channel, which circles the neck of the femur and supplies the neck and head of the femur.
- One vessel (**descending branch**) descends deep to the rectus femoris, penetrates the vastus lateralis muscle, and connects with a branch of the popliteal artery near the knee.
- One vessel (**transverse branch**) passes laterally to pierce the vastus lateralis and then circles around the proximal shaft of the femur to anastomose with branches from the medial femoral circumflex artery, the inferior gluteal artery, and the first perforating artery to form the cruciate anastomosis around the hip.

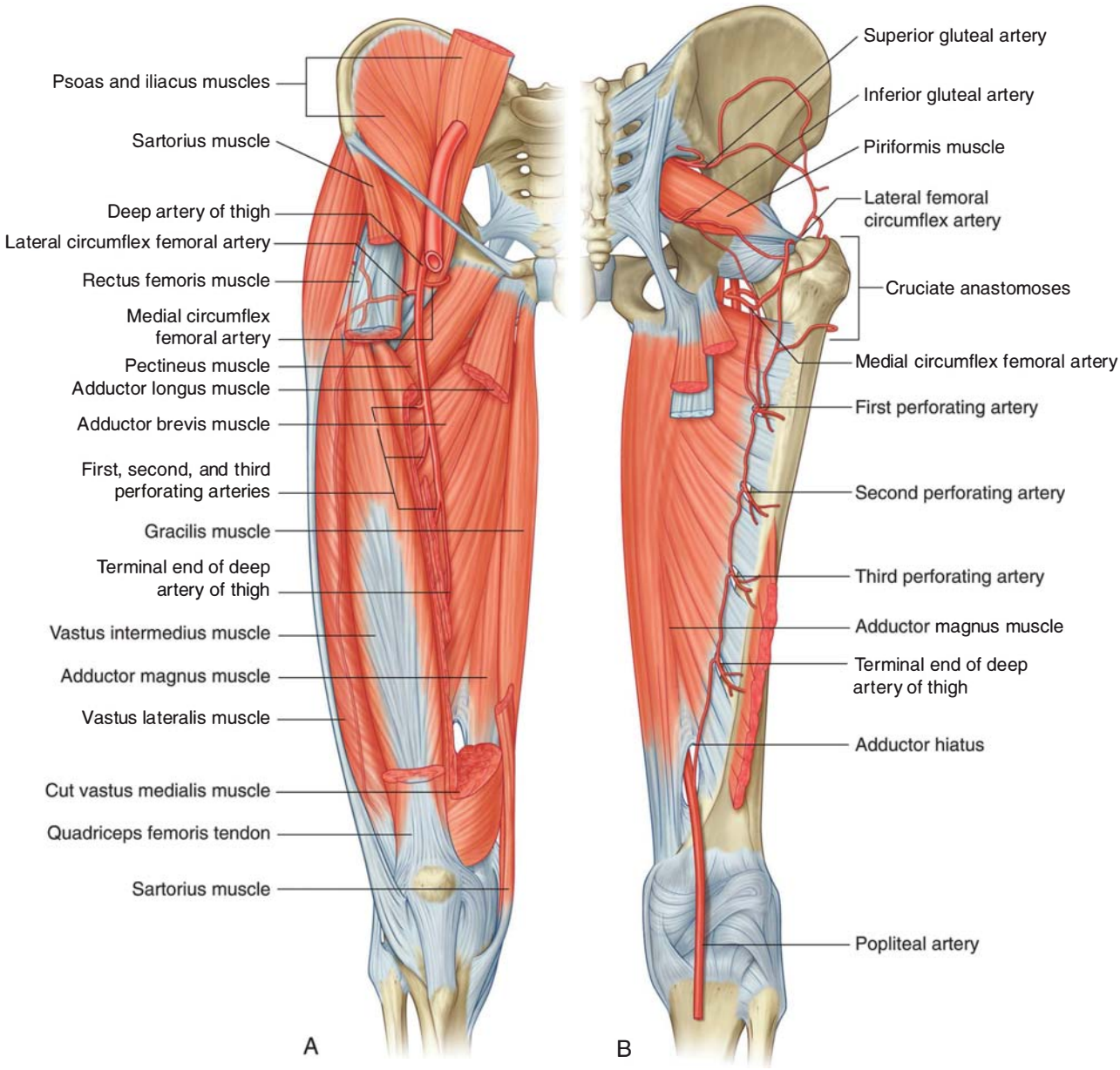


Fig. 6.64 Deep artery of thigh. **A.** Anterior view. **B.** Posterior view.

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Medial circumflex femoral artery

The **medial circumflex femoral artery** normally originates proximally from the posteromedial aspect of the deep artery of the thigh, but may originate from the femoral artery (Fig. 6.64). It passes medially around the shaft of the femur, first between the pectineus and iliopsoas and then between the obturator externus and adductor brevis muscles. Near the margin of the adductor brevis the vessel gives off a small branch, which enters the hip joint through the acetabular notch and anastomoses with the acetabular branch of the obturator artery.

The main trunk of the medial circumflex femoral artery passes over the superior margin of the adductor magnus and divides into two major branches deep to the quadratus femoris muscle:

- One branch ascends to the trochanteric fossa and connects with branches of the gluteal and lateral circumflex femoral arteries.
- The other branch passes laterally to participate with branches from the lateral circumflex femoral artery, the inferior gluteal artery, and the first perforating artery in forming an anastomotic network of vessels around the hip.

Perforating arteries

The three **perforating arteries** branch from the deep artery of the thigh (Fig. 6.64) as it descends anterior to the adductor brevis muscle—the first originates above the muscle, the second originates anterior to the muscle, and the third originates below the muscle. All three penetrate through the adductor magnus near its attachment to the linea aspera to enter and supply the posterior compartment of the thigh. Here, the vessels have ascending and descending branches, which interconnect to form a longitudinal channel, which participates above in forming an anastomotic network of vessels around the hip and inferiorly anastomoses with branches of the popliteal artery behind the knee.

Obturator artery

The **obturator artery** originates as a branch of the internal iliac artery in the pelvic cavity and enters the medial compartment of the thigh through the obturator canal (Fig. 6.65). As it passes through the canal, it bifurcates into an **anterior branch** and a **posterior branch**, which together form a channel that circles the margin of the obturator membrane and lies within the attachment of the obturator externus muscle.

Vessels arising from the anterior and posterior branches supply adjacent muscles and anastomose with the inferior gluteal and medial circumflex femoral arteries. In addition, an acetabular vessel originates from the posterior branch, enters the hip joint through the acetabular notch, and contributes to the supply of the head of the femur.

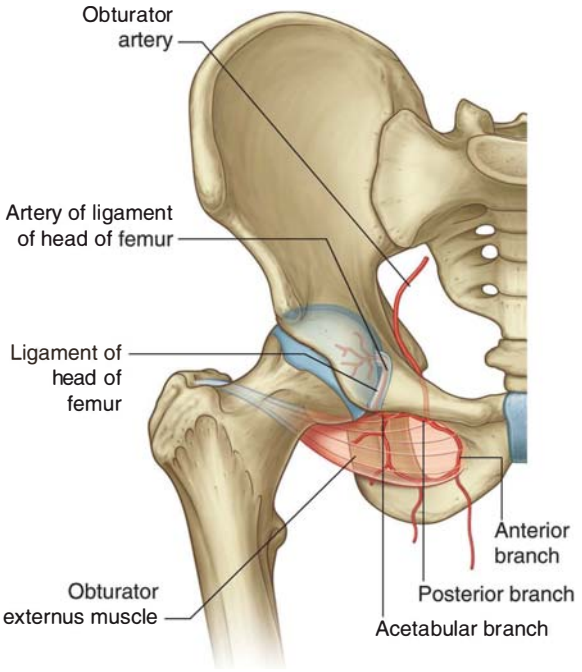


Fig. 6.65 Obturator artery.

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In the clinic

Peripheral vascular disease

Peripheral vascular disease is often characterized by reduced blood flow to the legs. This disorder may be caused by stenoses (narrowing) and/or occlusions (blockages) in the lower aorta and the iliac, femoral, tibial, and fibular vessels. Patients typically have chronic leg ischemia and "acute on chronic" leg ischemia.

Chronic leg ischemia

Chronic leg ischemia is a disorder in which vessels have undergone atheromatous change, and often there is significant luminal narrowing (usually over 50%). Most patients with peripheral arterial disease have widespread arterial disease (including cardiovascular and cerebrovascular disease), which may be clinically asymptomatic. Some of these patients develop such severe ischemia that the viability of the limb is threatened (**critical limb ischemia**).

The commonest symptom of chronic leg ischemia is **intermittent claudication**. Patients typically have a history of pain that develops in the calf muscles (usually associated with occlusions or narrowing in the femoral artery) or the buttocks (usually associated with occlusion or narrowing in the aorto-iliac segments). The pain experienced in these muscles is often cramplike and occurs with walking. The patient rests and is able to continue walking up to the same distance until the pain recurs and stops walking as before.

Acute on chronic ischemia

In some patients with chronic limb ischemia, an acute event blocks the vessels or reduces the blood supply to such a degree that the viability of the limb is threatened.

Occasionally a leg may become acutely ischemic with no evidence of underlying atheromatous disease. In these instances a blood clot is likely to have embolized from the heart. Patients with mitral valve disease and atrial fibrillation are prone to embolic disease.

Critical limb ischemia

Critical limb ischemia occurs when the blood supply to the limb is so poor that the viability of the limb is severely threatened, and in this case many patients develop gangrene, ulceration, and severe rest pain in the foot. These patients require urgent treatment, which may be in the form of surgical reconstruction, radiological angioplasty, or even amputation.

Veins

Veins in the thigh consist of superficial and deep veins. Deep veins generally follow the arteries and have similar names. Superficial veins are in the superficial fascia, interconnect with deep veins, and do not generally accompany arteries. The largest of the superficial veins in the thigh is the great saphenous vein.

Great saphenous vein

The great saphenous vein originates from a venous arch on the dorsal aspect of the foot and ascends along the medial side of the lower limb to the proximal thigh (see p. 568). Here it passes through the saphenous ring in deep fascia covering the anterior thigh to connect with the femoral vein in the femoral triangle (see p. 572).

Nerves

There are three major nerves in the thigh, each associated with one of the three compartments. The femoral nerve is associated with the anterior compartment of the thigh, the obturator nerve is associated with the medial compartment of the thigh, and the sciatic nerve is associated with the posterior compartment of the thigh.

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Femoral nerve

The femoral nerve originates from the lumbar plexus (spinal cord segments L2–L4) on the posterior abdominal wall and enters the femoral triangle of the thigh by passing under the inguinal ligament (Fig. 6.66). In the femoral triangle the femoral nerve lies on the lateral side of the femoral artery and is outside the femoral sheath, which surrounds the vessels.

Before entering the thigh, the femoral nerve supplies branches to the iliacus and pectineus muscles.

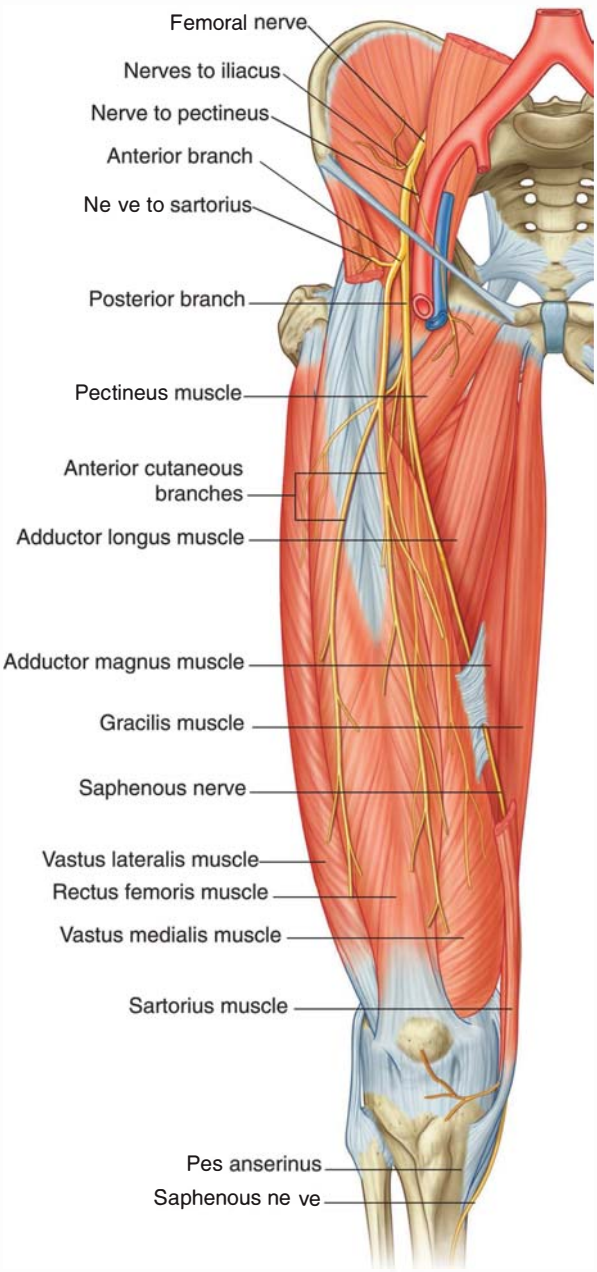


Fig. 6.66 Femoral nerve.

Immediately after passing under the inguinal ligament, the femoral nerve divides into anterior and posterior branches, which supply muscles of the anterior compartment of the thigh and skin on the anterior and medial aspects of the thigh and on the medial sides of the leg and foot.

Branches of the femoral nerve (Fig. 6.66) include:

- anterior cutaneous branches, which penetrate deep fascia to supply skin on the front of the thigh and knee;
- numerous motor nerves, which supply the quadriceps femoris muscles (rectus femoris, vastus lateralis, vastus intermedius, and vastus medialis muscles) and the sartorius muscle; and
- one long cutaneous nerve, the saphenous nerve, which supplies skin as far distally as the medial side of the foot.

The **saphenous nerve** accompanies the femoral artery through the adductor canal, but does not pass through the adductor hiatus with the femoral artery. Rather, the saphenous nerve penetrates directly through connective tissues near the end of the canal to appear between the sartorius and gracilis muscles on the medial side of the knee. Here the saphenous nerve penetrates deep fascia and continues down the medial side of the leg to the foot, and supplies skin on the medial side of the knee, leg, and foot.

Obturator nerve

The obturator nerve is a branch of the lumbar plexus (spinal cord segments L2–L4) on the posterior abdominal wall. It descends in the psoas muscle, and then passes out of the medial margin of the psoas muscle to enter the pelvis (Fig. 6.67). The obturator nerve continues along the lateral pelvic wall and then enters the medial compartment of the thigh by passing through the obturator canal. It supplies most of the adductor muscles and skin on the medial aspect of the thigh. As the obturator nerve enters the thigh, it divides into two branches, an anterior branch and a posterior branch, which are separated by the adductor brevis muscle:

- The **posterior branch** descends behind the adductor brevis muscle and on the anterior surface of the adductor magnus muscle, and supplies the obturator externus and adductor brevis muscles and the part of the adductor magnus that attaches to the linea aspera.
- The **anterior branch** descends on the anterior surface of the adductor brevis muscle and is behind the pectineus and adductor longus muscles—it supplies branches to the adductor longus, gracilis, and adductor brevis muscles, and often contributes to the supply of the pectineus muscle, and cutaneous branches innervate the skin on the medial side of the thigh.

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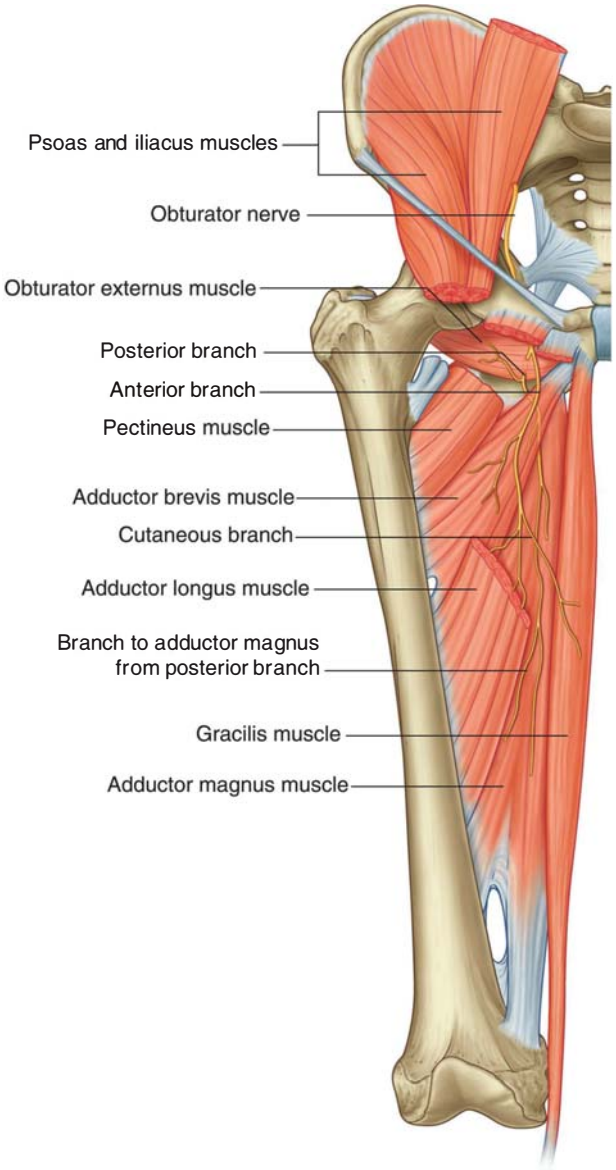


Fig. 6.67 Obturator nerve.

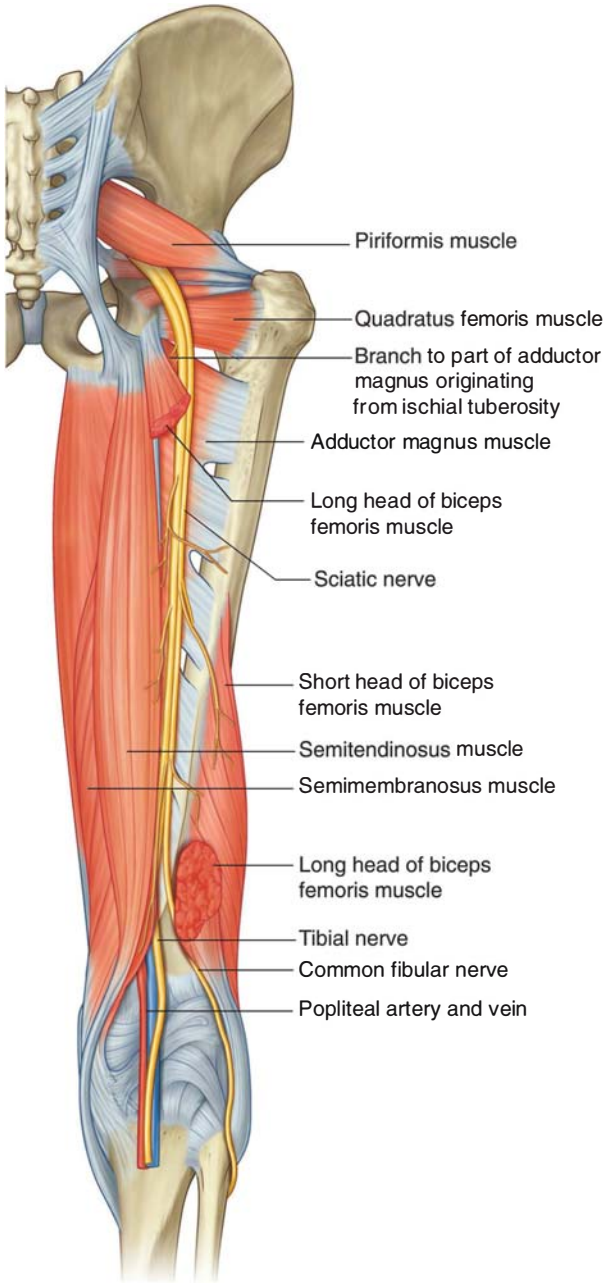


Fig. 6.68 Sciatic nerve.

Sciatic nerve

The sciatic nerve is a branch of the lumbosacral plexus (spinal cord segments L4–S3) and descends into the posterior compartment of the thigh from the gluteal region (Fig. 6.68). It innervates all muscles in the posterior compartment of the thigh and then its branches continue into the leg and foot.

In the posterior compartment of the thigh, the sciatic nerve lies on the adductor magnus muscle and is crossed by the long head of the biceps femoris muscle.

Proximal to the knee, and sometimes within the pelvis, the sciatic nerve divides into its two terminal branches: the **tibial nerve** and the **common fibular nerve**. These nerves travel vertically down the thigh and enter the popliteal fossa posterior to the knee. Here, they meet the popliteal artery and vein.

Tibial nerve

The tibial part of the sciatic nerve, either before or after its separation from the common fibular nerve, supplies branches to all muscles in the posterior compartment of the thigh (long head of biceps femoris, semimembranosus, semitendinosus) except the short head of the biceps femoris, which is innervated by the common fibular part (Fig. 6.68).

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The **tibial nerve** descends through the popliteal fossa, enters the posterior compartment of the leg, and continues into the sole of the foot.

The tibial nerve innervates:

- all muscles in the posterior compartment of the leg,
- all intrinsic muscles in the sole of the foot including the first two dorsal interossei muscles, which also may receive innervation from the deep fibular nerve, and
- skin on the posterolateral side of the lower half of the leg and lateral side of the ankle, foot, and little toe, and skin on the sole of the foot and toes.

Common fibular nerve

The common fibular part of the sciatic nerve innervates the short head of the biceps femoris in the posterior compartment of the thigh and then continues into the lateral and anterior compartments of the leg and onto the foot (Fig. 6.68).

The common fibular nerve innervates:

- all muscles in the anterior and lateral compartments of the leg,
- one muscle (extensor digitorum brevis) on the dorsal aspect of the foot,
- the first two dorsal interossei muscles in the sole of the foot, and
- skin over the lateral aspect of the leg, and ankle, and over the dorsal aspect of the foot and toes.

Knee joint

The knee joint is the largest synovial joint in the body. It consists of:

- the articulation between the femur and tibia, which is weight-bearing, and
- the articulation between the patella and the femur, which allows the pull of the quadriceps femoris muscle to be directed anteriorly over the knee to the tibia without tendon wear (Fig. 6.69).

Two fibrocartilaginous menisci, one on each side, between the femoral condyles and tibia accommodate changes in the shape of the articular surfaces during joint movements.

The detailed movements of the knee joint are complex, but basically the joint is a hinge joint that allows mainly flexion and extension. Like all hinge joints, the knee joint is reinforced by collateral ligaments, one on each side of the joint. In addition, two very strong ligaments (the cruciate ligaments) interconnect the adjacent ends of the femur and tibia and maintain their opposed positions during movement.

Because the knee joint is involved in weight-bearing, it has an efficient "locking" mechanism to reduce the amount of muscle energy required to keep the joint extended when standing.

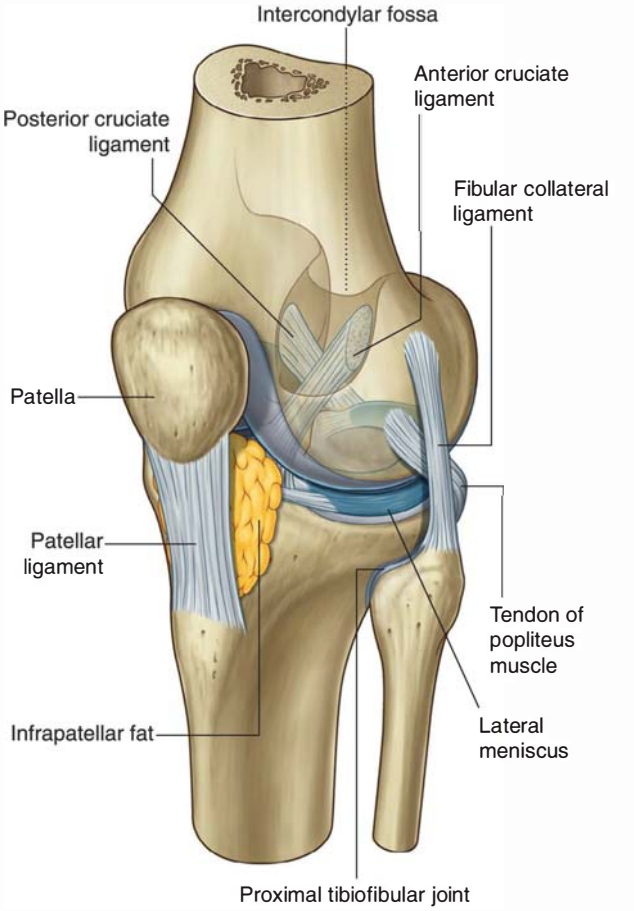


Fig. 6.69 Knee joint. Joint capsule is not shown.

Articular surfaces

The articular surfaces of the bones that contribute to the knee joint are covered by hyaline cartilage. The major surfaces involved include:

- the two femoral condyles, and
- the adjacent surfaces of the superior aspect of the tibial condyles.

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The surfaces of the femoral condyles that articulate with the tibia in flexion of the knee are curved or round, whereas the surfaces that articulate in full extension are flat (Fig. 6.70).

The articular surfaces between the femur and patella are the V-shaped trench on the anterior surface of the distal end of the femur where the two condyles join and the adjacent surfaces on the posterior aspect of the patella. The joint surfaces are all enclosed within a single articular cavity, as are the intraarticular menisci between the femoral and tibial condyles.

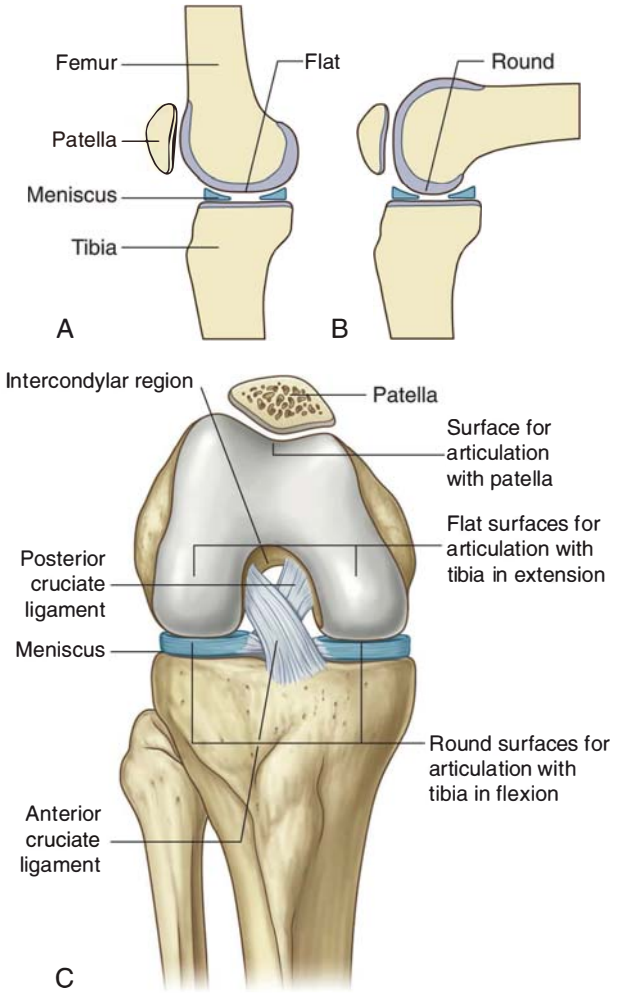


Fig. 6.70 Articular surfaces of the knee joint. **A.** Extended. **B.** Flexed. **C.** Anterior view (flexed).

Menisci

There are two menisci, which are fibrocartilaginous C-shaped cartilages, in the knee joint, one medial (**medial meniscus**) and the other lateral (**lateral meniscus**) (Fig. 6.71). Both are attached at each end to facets in the intercondylar region of the tibial plateau.

The medial meniscus is attached around its margin to the capsule of the joint and to the tibial collateral ligament, whereas the lateral meniscus is unattached to the capsule. Therefore, the lateral meniscus is more mobile than the medial meniscus.

The menisci are interconnected anteriorly by a transverse ligament of the knee. The lateral meniscus is also connected to the tendon of the popliteus muscle, which passes superolaterally between this meniscus and the capsule to insert on the femur.

The menisci improve congruency between the femoral and tibial condyles during joint movements where the surfaces of the femoral condyles articulating with the tibial plateau change from small curved surfaces in flexion to large flat surfaces in extension.

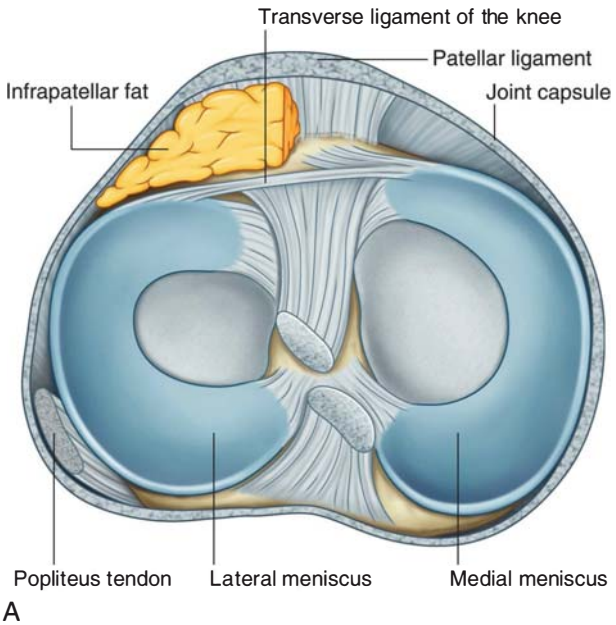


Fig. 6.71 Menisci of the knee joint. **A.** Superior view.

Continued

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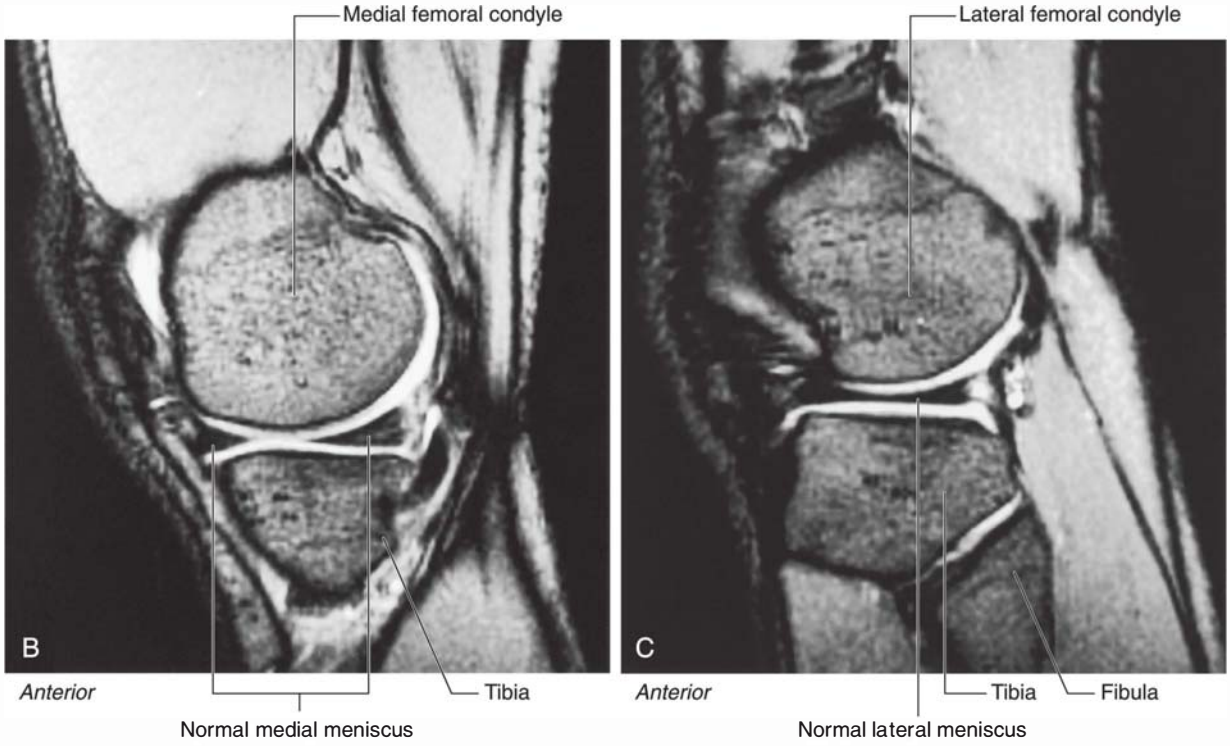


Fig. 6.71, cont'd Menisci of the knee joint. **B.** Normal knee joint showing the medial meniscus. T2-weighted magnetic resonance image in the sagittal plane. **C.** Normal knee joint showing the lateral meniscus. T2-weighted magnetic resonance image in the sagittal plane.

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Synovial membrane

The synovial membrane of the knee joint attaches to the margins of the articular surfaces and to the superior and inferior outer margins of the menisci (Fig. 6.72A). The two cruciate ligaments, which attach in the intercondylar region of the tibia below and the intercondylar fossa of the femur above, are outside the articular cavity, but enclosed within the fibrous membrane of the knee joint.

Posteriorly, the synovial membrane reflects off the fibrous membrane of the joint capsule on either side of the posterior cruciate ligament and loops forward around both ligaments thereby excluding them from the articular cavity.

Anteriorly, the synovial membrane is separated from the patellar ligament by an **infrapatellar fat pad**. On each side of the pad, the synovial membrane forms a fringed margin (an **alar fold**), which projects into the articular cavity. In addition, the synovial membrane covering the lower part of the infrapatellar fat pad is raised into a sharp midline fold directed posteriorly (the **infrapatellar synovial fold**), which attaches to the margin of the intercondylar fossa of the femur.

The synovial membrane of the knee joint forms pouches in two locations to provide low-friction surfaces for the movement of tendons associated with the joint:

- The smallest of these expansions is the **subpopliteal recess** (Fig. 6.72A), which extends posterolaterally from the articular cavity and lies between the lateral meniscus and the tendon of the popliteus muscle, which passes through the joint capsule.
- The second expansion is the **suprapatellar bursa** (Fig. 6.72B), a large bursa that is a continuation of the articular cavity superiorly between the distal end of the shaft of the femur and the quadriceps femoris muscle and tendon—the apex of this bursa is attached to the small articularis genus muscle, which pulls the bursa away from the joint during extension of the knee.

Other bursae associated with the knee but not normally communicating with the articular cavity include the

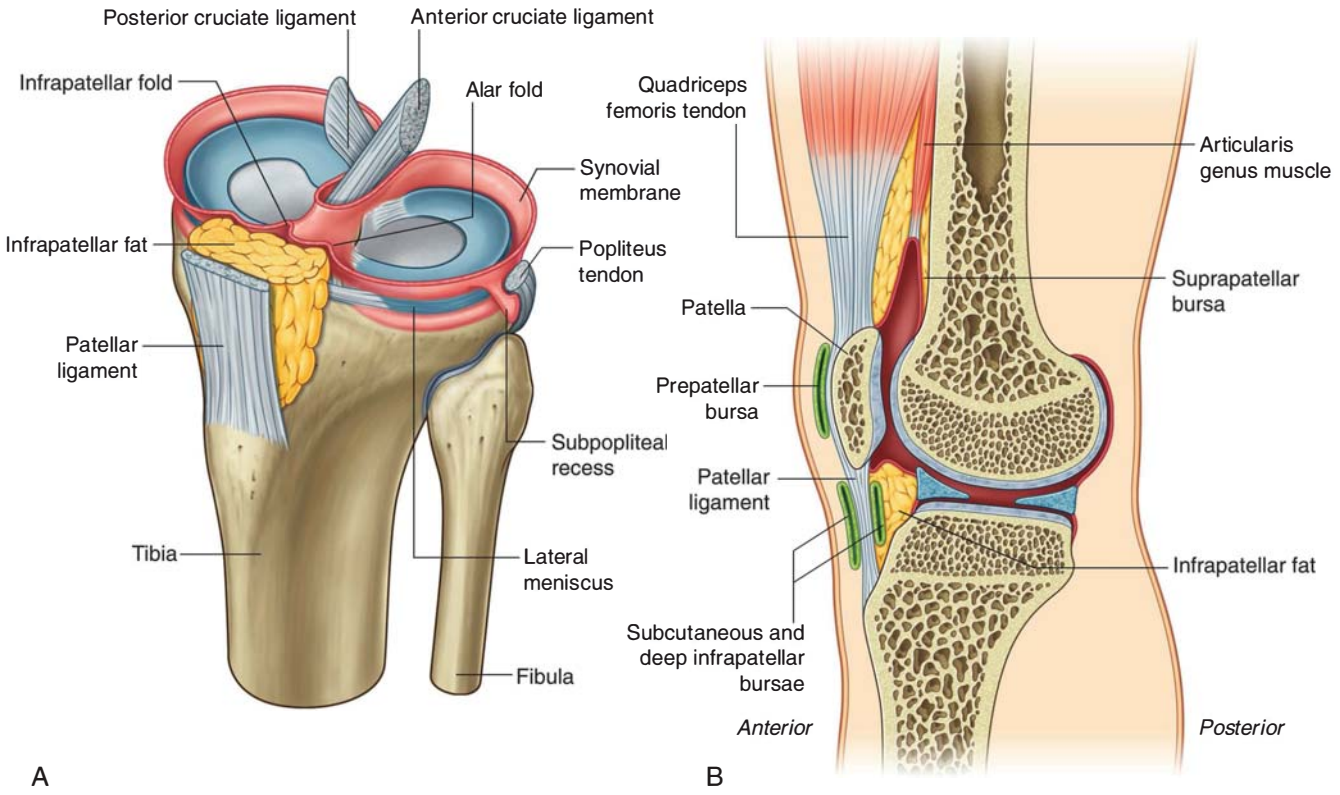


Fig. 6.72 Synovial membrane of the knee joint and associated bursae. **A.** Superolateral view; patella and femur not shown. **B.** Paramedial sagittal section through the knee.

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subcutaneous prepatellar bursa, deep and subcutaneous infrapatellar bursae, and numerous other bursae associated with tendons and ligaments around the joint (Fig. 6.72B).

The prepatellar bursa is subcutaneous and anterior to the patella. The deep and subcutaneous infrapatellar bursae are on the deep and subcutaneous sides of the patellar ligament, respectively.

Fibrous membrane

The fibrous membrane of the knee joint is extensive and is partly formed and reinforced by extensions from tendons of the surrounding muscles (Fig. 6.73). In general, the fibrous membrane encloses the articular cavity and the intercondylar region:

- On the medial side of the knee joint, the fibrous membrane blends with the tibial collateral ligament and is attached on its internal surface to the medial meniscus.
- Laterally, the external surface of the fibrous membrane is separated by a space from the fibular collateral ligament and the internal surface of the fibrous membrane is not attached to the lateral meniscus.
- Anteriorly, the fibrous membrane is attached to the margins of the patella where it is reinforced with tendinous expansions from the vastus lateralis and vastus medialis muscles, which also merge above with the quadriceps femoris tendon and below with the patellar ligament.

The fibrous membrane is reinforced anterolaterally by a fibrous extension from the iliotibial tract and posteromedially by an extension from the tendon of the semimembranosus (the **oblique popliteal ligament**), which reflects superiorly across the back of the fibrous membrane from medial to lateral.

The upper end of the popliteus muscle passes through an aperture in the posterolateral aspect of the fibrous membrane of the knee and is enclosed by the fibrous membrane as its tendon travels around the joint to insert on the lateral aspect of the lateral femoral condyle.

Ligaments

The major ligaments associated with the knee joint are the patellar ligament, the tibial (medial) and fibular (lateral) collateral ligaments, and the anterior and posterior cruciate ligaments.

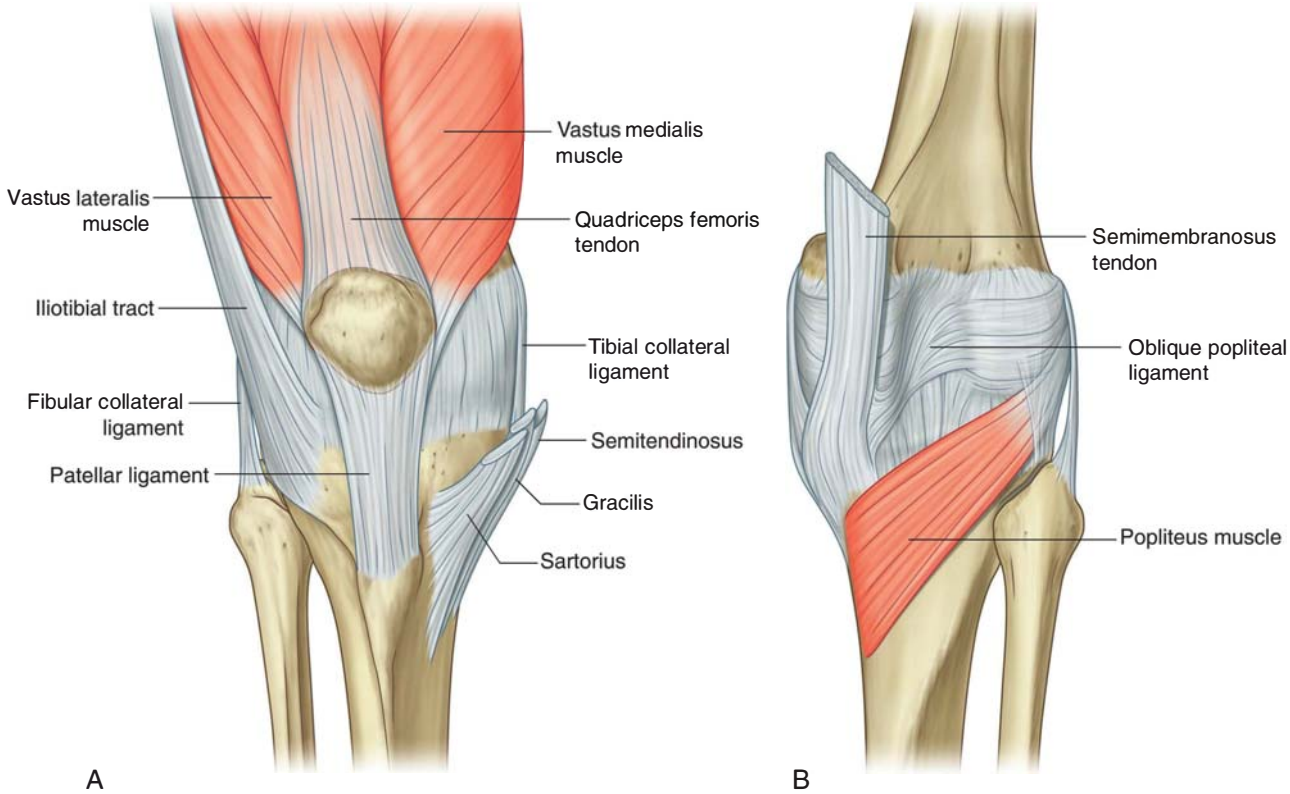


Fig. 6.73 Fibrous membrane of the knee joint capsule. **A.** Anterior view. **B.** Posterior view.

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Patellar ligament

The **patellar ligament** is basically the continuation of the quadriceps femoris tendon inferior to the patella (Fig. 6.73). It is attached above to the margins and apex of the patella and below to the tibial tuberosity.

Collateral ligaments

The collateral ligaments, one on each side of the joint, stabilize the hinge-like motion of the knee (Fig. 6.74).

The cord-like **fibular collateral ligament** is attached superiorly to the lateral femoral epicondyle just above the groove for the popliteus tendon. Inferiorly, it is attached to a depression on the lateral surface of the fibular head. It is separated from the fibrous membrane by a bursa.

The broad and flat **tibial collateral ligament** is attached by much of its deep surface to the underlying fibrous membrane. It is anchored superiorly to the medial femoral epicondyle just inferior to the adductor tubercle and descends anteriorly to attach to the medial margin and

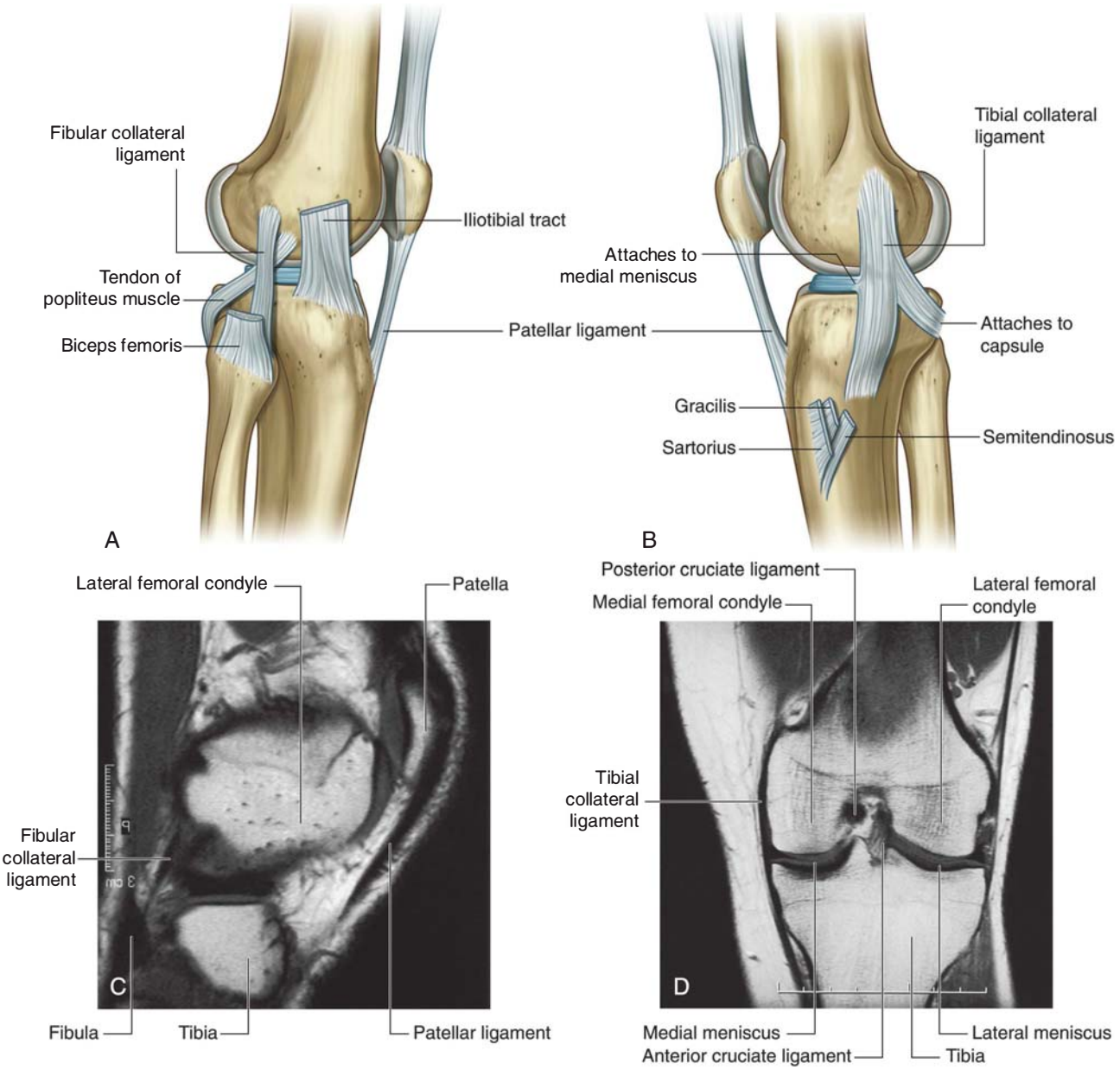


Fig. 6.74 Collateral ligaments of the knee joint. **A.** Lateral view. **B.** Medial view. **C.** Normal knee joint showing the patellar ligament and the fibular collateral ligament. T1-weighted magnetic resonance image in the sagittal plane. **D.** Normal knee joint showing the tibial collateral ligament, the medial and lateral menisci, and the anterior and posterior cruciate ligaments. T1-weighted magnetic resonance image in the coronal plane.

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medial surface of the tibia above and behind the attachment of the sartorius, gracilis, and semitendinosus tendons.

Cruciate ligaments

The two cruciate ligaments are in the intercondylar region of the knee and interconnect the femur and tibia (Figs. 6.74D and 6.75). They are termed "cruciate" (Latin for "shaped like a cross") because they cross each other in the sagittal plane between their femoral and tibial attachments:

- The **anterior cruciate ligament** attaches to a facet on the anterior part of the intercondylar area of the tibia and ascends posteriorly to attach to a facet at the back of the lateral wall of the intercondylar fossa of the femur.
- The **posterior cruciate ligament** attaches to the posterior aspect of the intercondylar area of the tibia and ascends anteriorly to attach to the medial wall of the intercondylar fossa of the femur.

The anterior cruciate ligament crosses lateral to the posterior cruciate ligament as they pass through the intercondylar region.

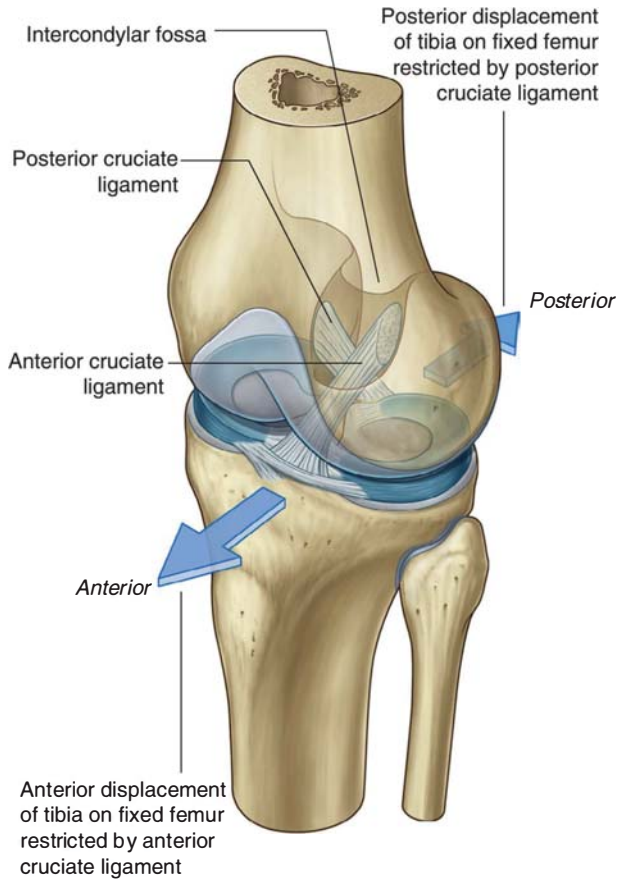


Fig. 6.75 Cruciate ligaments of the knee joint. Superolateral view.

The anterior cruciate ligament prevents anterior displacement of the tibia relative to the femur and the posterior cruciate ligament restricts posterior displacement (Fig. 6.75).

Locking mechanism

When standing, the knee joint is locked into position, thereby reducing the amount of muscle work needed to maintain the standing position (Fig. 6.76).

One component of the locking mechanism is a change in the shape and size of the femoral surfaces that articulate with the tibia:

- In flexion, the surfaces are the curved and rounded areas on the posterior aspects of the femoral condyles.

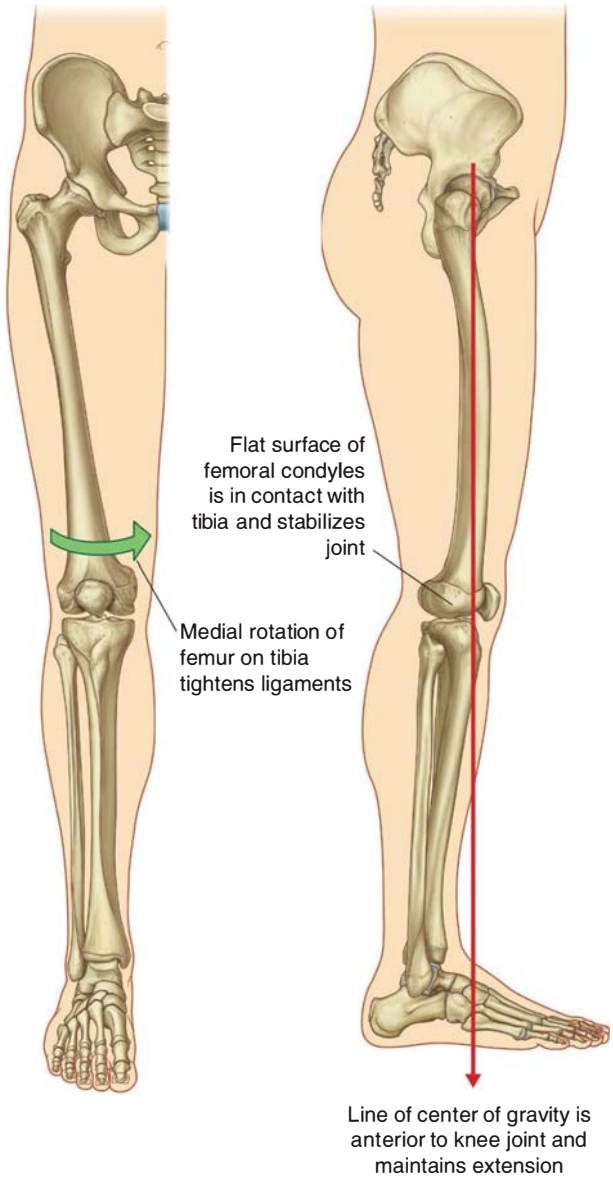


Fig. 6.76 Knee "locking" mechanism.

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- As the knee is extended, the surfaces move to the broad and flat areas on the inferior aspects of the femoral condyles.

Consequently the joint surfaces become larger and more stable in extension.

Another component of the locking mechanism is medial rotation of the femur on the tibia during extension. Medial rotation and full extension tightens all the associated ligaments.

Another feature that keeps the knee extended when standing is that the body's center of gravity is positioned along a vertical line that passes anterior to the knee joint.

The popliteus muscle unlocks the knee by initiating lateral rotation of the femur on the tibia.

Vascular supply and innervation

Vascular supply to the knee joint is predominantly through descending and genicular branches from the femoral, popliteal, and lateral circumflex femoral arteries in the thigh and the circumflex fibular artery and recurrent branches from the anterior tibial artery in the leg. These vessels form an anastomotic network around the joint (Fig. 6.77).

The knee joint is innervated by branches from the obturator, femoral, tibial, and common fibular nerves.

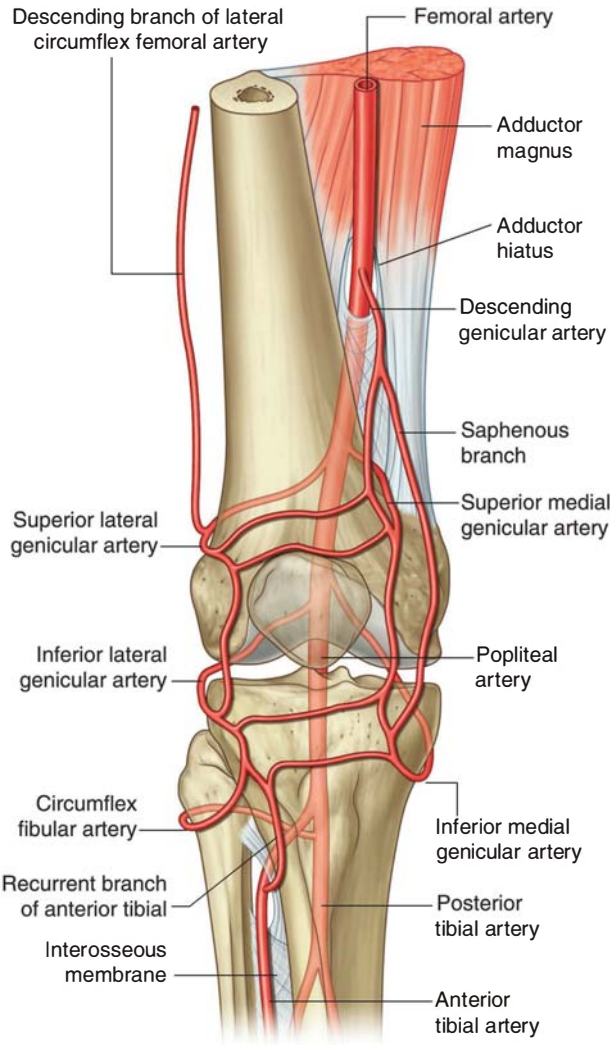


Fig. 6.77 Anastomoses of arteries around the knee. Anterior view.

In the clinic

Soft tissue injuries to the knee

Soft tissue injuries are common at and around the knee joint.

The typical injuries include tears of the anterior and posterior cruciate ligaments, meniscal tears, and trauma to the collateral ligaments. Isolated soft tissue injuries may occur, but it is not uncommon for certain types of injuries to occur together, for example, anterior cruciate ligament disruption, tibial collateral ligament disruption, and tears of the medial meniscus.

Any soft tissue injury at and around the knee joint may involve the neurovascular bundle, and assessment of neurovascular structures is critical in the management of patients with injury to the soft tissues.

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In the clinic

Degenerative joint disease/osteoarthritis

Degenerative joint disease occurs throughout many joints within the body. Articular degeneration may result from an abnormal force across the joint with a normal cartilage or a normal force with abnormal cartilage.

Typically degenerative joint disease occurs in synovial joints and the process is called osteoarthritis. In the joints where osteoarthritis occurs the cartilage and bony tissues are usually involved, with limited change within the synovial membrane. The typical findings include reduction in the joint space, eburnation (joint sclerosis), osteophytosis (small bony outgrowths), and bony cyst formation. As the disease progresses the joint may become malaligned, its movement may become severely limited, and there may be significant pain.

The commonest sites for osteoarthritis include the small joints of the hands and wrist, and in the lower limb, the hip and knee are typically affected, though the tarsometatarsal and metatarsophalangeal articulations may undergo similar change.

The etiology of degenerative joint disease is unclear, but there are some associations, including genetic predisposition, increasing age (males tend to be affected younger than females), overuse or underuse of joints, and nutritional and metabolic abnormalities. Further factors include joint trauma and pre-existing articular disease or deformity.

The histological findings of osteoarthritis consist of degenerative changes within the cartilage and the subchondral bone. Further articular damage worsens these changes, which promote further abnormal stresses upon the joint. As the disease progresses the typical finding is pain, which is usually worse on rising from bed and at the end of a day's activity. Commonly it is aggravated by the extremes of movement or unaccustomed exertion. Stiffness and limitation of movement may ensue.

Treatment in the first instance includes alteration of lifestyle to prevent pain and simple analgesia. As symptoms progress a joint replacement may be necessary, but although joint replacement appears to be the panacea for degenerative joint disease, it is not without risks and complications, which include infection and failure in the short and long term.

In the clinic

Examination of the knee joint

It is important to establish the nature of the patient's complaint before any examination. The history should include information about the complaint, the signs and symptoms, and the patient's lifestyle (level of activity). This history may give a significant clue to the type of injury and the likely findings on clinical examination, for example, if the patient was kicked around the medial aspect of the knee, a valgus deformity injury to the tibial collateral ligament might be suspected.

The examination should include assessment in the erect position, while walking, and on the couch. The affected side must be compared with the unaffected side.

There are many tests and techniques for examining the knee joint, including the following.

Tests for anterior instability

- Lachman's test—the patient lies on the couch. The examiner places one hand around the distal femur and the other around the proximal tibia and then elevates the knee, producing 20° of flexion. The patient's heel rests on the couch. The examiner's thumb must be on the tibial tuberosity. The hand on the tibia applies a brisk anteriorly directed force. If the movement of the tibia on the femur comes to a sudden stop, it is a firm endpoint. If it does not come to a sudden stop, the endpoint is described as soft and is associated with a tear of the anterior cruciate ligament.
- Anterior drawer test—a positive anterior drawer test is when the proximal head of a patient's tibia can be pulled anteriorly on the femur. The patient lies supine on the couch. The knee is flexed to 90° and the heel and sole of the foot are placed on the couch. The examiner sits gently on the patient's foot, which has been placed in a neutral position. The index fingers are used to check that the hamstrings are relaxed while the other fingers encircle the upper end of the tibia and pull the tibia. If the tibia moves forward, the anterior cruciate ligament is torn. Other peripheral structures, such as the medial meniscus or meniscotibial ligaments, must also be damaged to elicit this sign.
- Pivot shift test—there are many variations of this test. The patient's foot is wedged between the examiner's body and elbow. The examiner places one hand flat under the tibia pushing it forward with the knee in extension. The other hand is placed against the patient's thigh pushing it the other way.

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In the clinic—cont'd

The lower limb is taken into slight abduction by the examiner's elbow with the examiner's body acting as a fulcrum to produce the valgus. The examiner maintains the anterior tibial translation and the valgus and initiates flexion of the patient's knee. At about 20°–30° the pivot shift will occur as the lateral tibial plateau reduces. This test demonstrates damage to the posterolateral corner of the knee joint and the anterior cruciate ligament.

Tests for posterior instability

- Posterior drawer test—a positive posterior drawer test occurs when the proximal head of a patient's tibia can be pushed posteriorly on the femur. The patient is placed in a supine position and the knee is flexed to approximately 90° with the foot in the neutral position. The examiner sits gently on the patient's foot placing both thumbs on the tibial tuberosity and pushing the tibia backward. If the tibial plateau moves, the posterior cruciate ligament is torn.

Assessment of other structures of the knee

- Assessment of the tibial collateral ligament can be performed by placing a valgus stress on the knee.
- Assessment of lateral and posterolateral knee structures requires more complex clinical testing.

The knee will also be assessed for:

- joint line tenderness,
- patellofemoral movement and instability,
- presence of an effusion,
- muscle injury, and
- popliteal fossa masses.

Further investigations

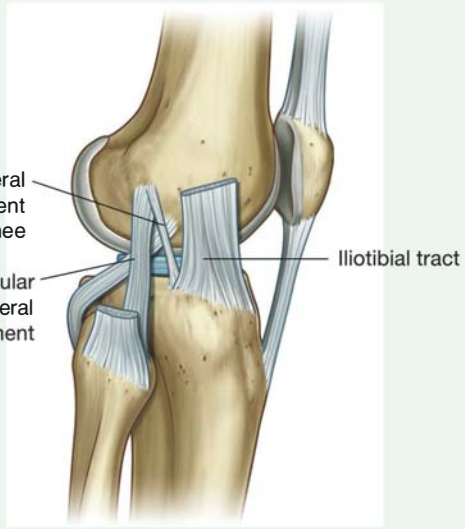
After the clinical examination has been carried out, further investigations usually include **plain radiography** and possibly **magnetic resonance imaging**, which allows the radiologist to assess the menisci, cruciate ligaments, collateral ligaments, bony and cartilaginous surfaces, and soft tissues.

Arthroscopy may be carried out and damage to any internal structures repaired or trimmed. An arthroscope is a small camera that is placed into the knee joint through the anterolateral or anteromedial aspect of the knee joint. The joint is filled with a saline solution and the telescope is manipulated around the knee joint to assess the cruciate ligaments, menisci, and cartilaginous surfaces.

In the clinic

Anterolateral ligament of the knee

A ligament associated at its origin with the fibular collateral ligament of the knee has been described. This ligament (anterolateral ligament of the knee) courses from the lateral femoral epicondyle to the anterolateral region of the proximal end of the tibia and may control internal rotation of the tibia. (*J Anat* 2013;223:321–328)



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Tibiofibular joint

The small proximal tibiofibular joint is synovial in type and allows very little movement (Fig. 6.78). The opposing joint surfaces, on the undersurface of the lateral condyle of the tibia and on the superomedial surface of the head of the fibula, are flat and circular. The capsule is reinforced by anterior and posterior ligaments.

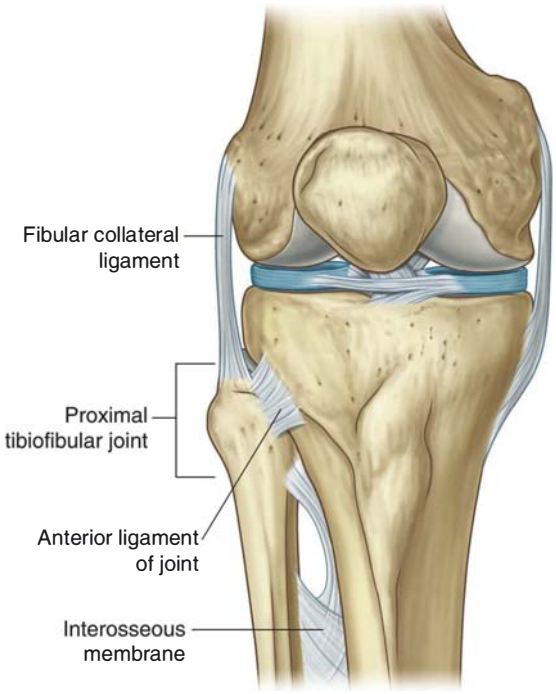


Fig. 6.78 Tibiofibular joint.

Popliteal fossa

The **popliteal fossa** is an important area of transition between the thigh and leg and is the major route by which structures pass from one region to the other.

The popliteal fossa is a diamond-shaped space behind the knee joint formed between muscles in the posterior compartments of the thigh and leg (Fig. 6.79A):

- The margins of the upper part of the diamond are formed medially by the distal ends of the semitendinosus and semimembranosus muscles and laterally by the distal end of the biceps femoris muscle.

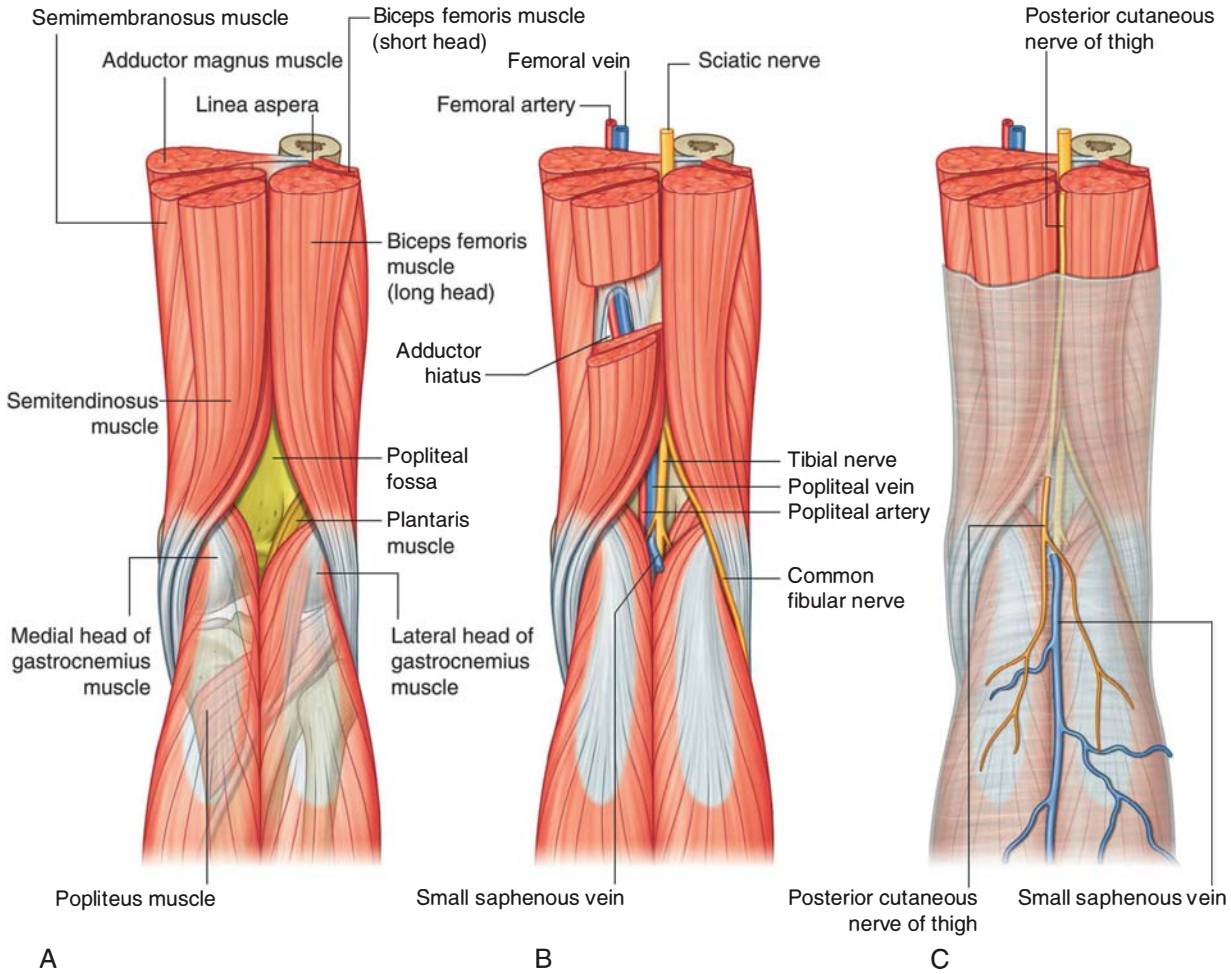


Fig. 6.79 Popliteal fossa. **A.** Boundaries. **B.** Nerves and vessels. **C.** Superficial structures.

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- The margins of the smaller lower part of the space are formed medially by the medial head of the gastrocnemius muscle and laterally by the plantaris muscle and the lateral head of the gastrocnemius muscle.
- The floor of the fossa is formed by the capsule of the knee joint and adjacent surfaces of the femur and tibia, and, more inferiorly, by the popliteus muscle.
- The roof is formed by deep fascia, which is continuous above with the fascia lata of the thigh and below with deep fascia of the leg.

Contents

The major contents of the popliteal fossa are the popliteal artery, the popliteal vein, and the tibial and common fibular nerves (Fig. 6.79B).

Tibial and common fibular nerves

The tibial and common fibular nerves originate proximal to the popliteal fossa as the two major branches of the sciatic nerve. They are the most superficial of the neurovascular structures in the popliteal fossa and enter the region directly from above under the margin of the biceps femoris muscle:

- The tibial nerve descends vertically through the popliteal fossa and exits deep to the margin of the plantaris muscle to enter the posterior compartment of the leg.
- The common fibular nerve exits by following the biceps femoris tendon over the lower lateral margin of the popliteal fossa, and continues to the lateral side of the leg where it swings around the neck of the fibula and enters the lateral compartment of the leg.

Popliteal artery and vein

The popliteal artery is the continuation of the femoral artery in the anterior compartment of the thigh, and begins as the femoral artery passes posteriorly through the adductor hiatus in the adductor magnus muscle.

The popliteal artery appears in the popliteal fossa on the upper medial side under the margin of the semimembranosus muscle. It descends obliquely through the fossa with the tibial nerve and enters the posterior compartment of the leg where it ends just lateral to the midline of the leg by dividing into the anterior and posterior tibial arteries.

The popliteal artery is the deepest of the neurovascular structures in the popliteal fossa and is therefore difficult to palpate; however, a pulse can usually be detected by deep palpation near the midline.

In the popliteal fossa, the popliteal artery gives rise to branches, which supply adjacent muscles, and to a series of geniculate arteries, which contribute to vascular anastomoses around the knee.

The popliteal vein is superficial to and travels with the popliteal artery. It exits the popliteal fossa superiorly to become the femoral vein by passing through the adductor hiatus.

Roof of popliteal fossa

The roof of the popliteal fossa is covered by superficial fascia and skin (Fig. 6.79C). The most important structure in the superficial fascia is the small saphenous vein. This vessel ascends vertically in the superficial fascia on the back of the leg from the lateral side of the dorsal venous arch in the foot. It ascends to the back of the knee where it penetrates deep fascia, which forms the roof of the popliteal fossa, and joins with the popliteal vein.

One other structure that passes through the roof of the fossa is the posterior cutaneous nerve of the thigh, which descends through the thigh superficial to the hamstring muscles, passes through the roof of the popliteal fossa, and then continues inferiorly with the small saphenous vein to innervate skin on the upper half of the back of the leg.

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LEG

The leg is that part of the lower limb between the knee joint and ankle joint (Fig. 6.80):

- Proximally, most major structures pass between the thigh and leg through or in relation to the popliteal fossa behind the knee.
- Distally, structures pass between the leg and foot mainly through the tarsal tunnel on the posteromedial side of the ankle, the exceptions being the anterior tibial artery and the ends of the deep and superficial fibular nerves, which enter the foot anterior to the ankle.

The bony framework of the leg consists of two bones, the tibia and fibula, arranged in parallel.

The **fibula** is much smaller than the tibia and is on the lateral side of the leg. It articulates superiorly with the inferior aspect of the lateral condyle of the proximal tibia, but does not take part in formation of the knee joint. The distal end of the fibula is firmly anchored to the tibia by a fibrous joint and forms the lateral malleolus of the ankle joint.

The **tibia** is the weight-bearing bone of the leg and is therefore much larger than the fibula. Above, it takes part in the formation of the knee joint and below it forms the medial malleolus and most of the bony surface for articulation of the leg with the foot at the ankle joint.

The leg is divided into anterior (extensor), posterior (flexor), and lateral (fibular) compartments by:

- an interosseous membrane, which links adjacent borders of the tibia and fibula along most of their length;
- two intermuscular septa, which pass between the fibula and deep fascia surrounding the limb; and
- direct attachment of the deep fascia to the periosteum of the anterior and medial borders of the tibia (Fig. 6.80).

Muscles in the anterior compartment of the leg dorsiflex the ankle, extend the toes, and invert the foot. Muscles in the posterior compartment plantarflex the ankle, flex the toes, and invert the foot. Muscles in the lateral compartment evert the foot. Major nerves and vessels supply or pass through each compartment.

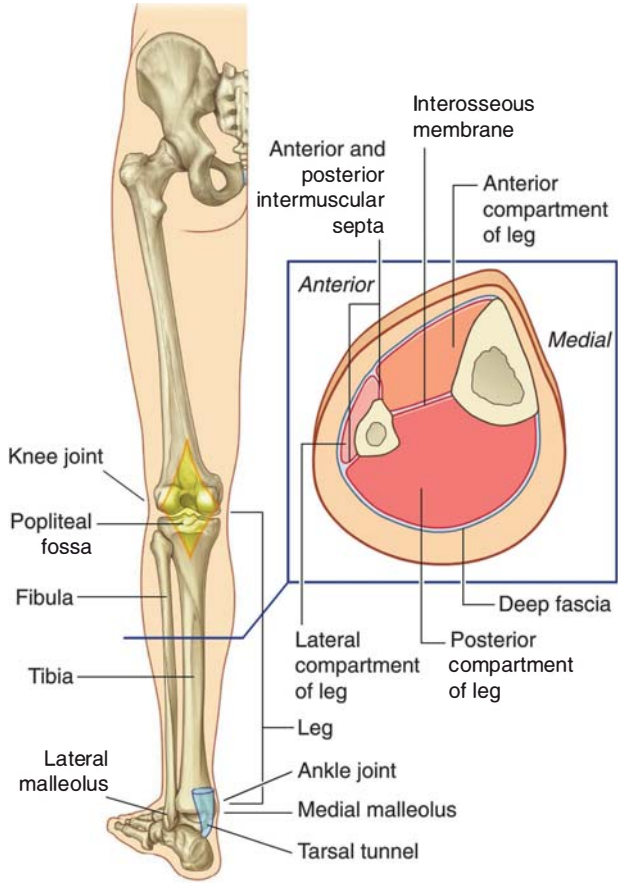


Fig. 6.80 Posterior view of leg; cross section through the left leg (*inset*).

Bones

Shaft and distal end of tibia

The shaft of the tibia is triangular in cross section and has anterior, interosseous, and medial borders and medial, lateral, and posterior surfaces (Fig. 6.81):

- The anterior and medial borders and the entire medial surface are subcutaneous and easily palpable.
- The interosseous border of the tibia is connected, by the interosseous membrane, along its length to the interosseous border of the fibula.
- The posterior surface is marked by an oblique line (the soleal line).

The soleal line descends across the bone from the lateral side to the medial side where it merges with the medial border. In addition, a vertical line descends down the upper part of the posterior surface from the midpoint of the soleal line. It disappears in the lower one-third of the tibia.

The shaft of the tibia expands at both the upper and lower ends to support the body's weight at the knee and ankle joints.

The distal end of the tibia is shaped like a rectangular box with a bony protuberance on the medial side (the **medial malleolus**; Fig. 6.81). The upper part of the box is continuous with the shaft of the tibia while the lower

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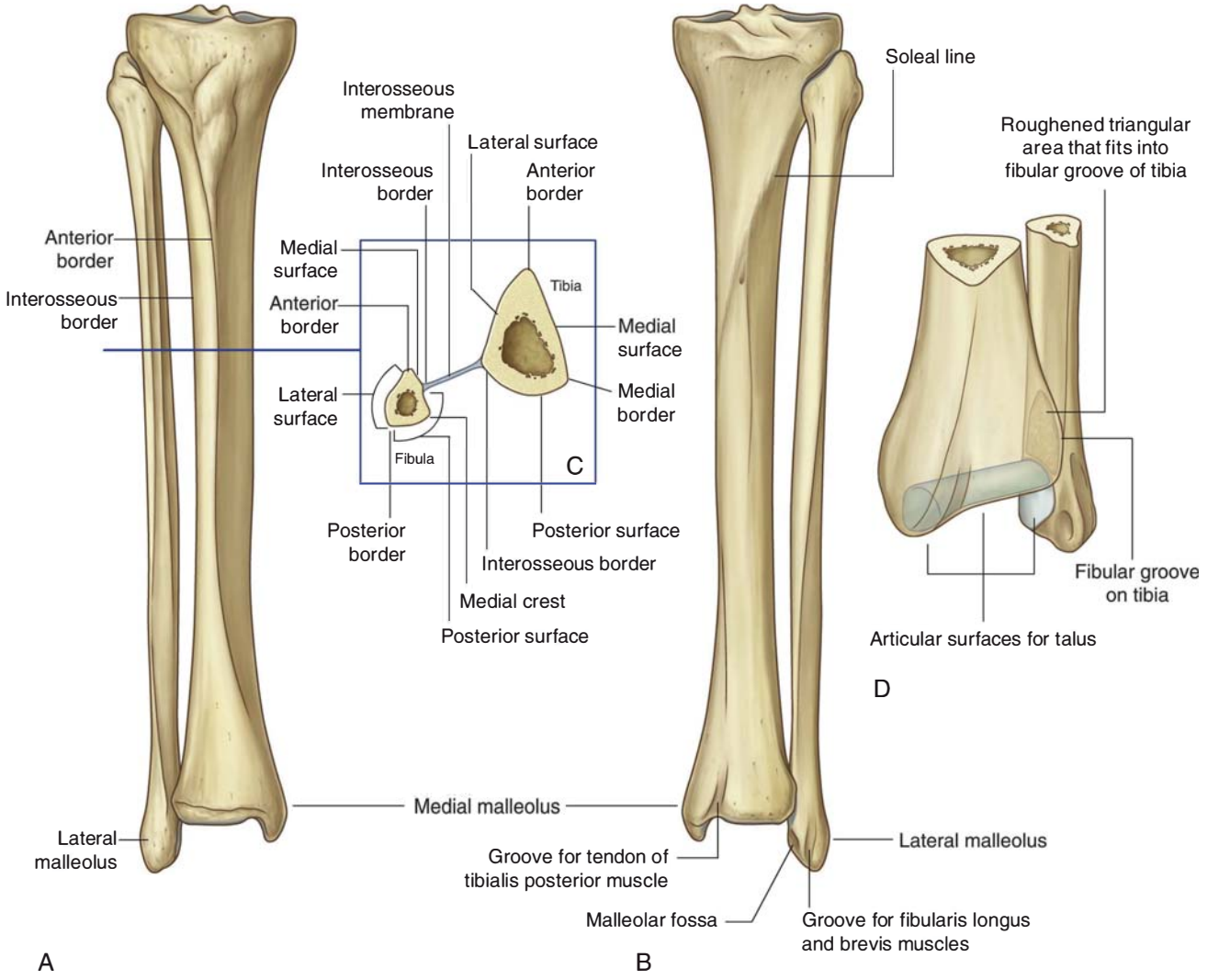


Fig. 6.81 Tibia and fibula. **A.** Anterior view. **B.** Posterior view. **C.** Cross section through shafts. **D.** Posteromedial view of distal ends.

surface and the medial malleolus articulate with one of the tarsal bones (talus) to form a large part of the ankle joint.

The posterior surface of the box-like distal end of the tibia is marked by a vertical groove, which continues inferiorly and medially onto the posterior surface of the medial malleolus. The groove is for the tendon of the tibialis posterior muscle.

The lateral surface of the distal end of the tibia is occupied by a deep triangular notch (the **fibular notch**), to which the distal head of the fibula is anchored by a thickened part of the interosseous membrane.

Shaft and distal end of fibula

The fibula is not involved in weight-bearing. The fibular shaft is therefore much narrower than the shaft of the tibia. Also, and except for the ends, the fibula is enclosed by muscles.

Like the tibia, the shaft of the fibula is triangular in cross section and has three borders and three surfaces for the attachment of muscles, intermuscular septa, and ligaments (Fig. 6.81). The interosseous border of the fibula faces and is attached to the interosseous border of the tibia by the interosseous membrane. Intermuscular septa attach to the anterior and posterior borders. Muscles attach to the three surfaces.

The narrow **medial surface** faces the anterior compartment of the leg, the **lateral surface** faces the lateral compartment of the leg, and the **posterior surface** faces the posterior compartment of the leg.

The posterior surface is marked by a vertical crest (**medial crest**), which divides the posterior surface into two parts each attached to a different deep flexor muscle.

The distal end of the fibula expands to form the spade-shaped **lateral malleolus** (Fig. 6.81).

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The medial surface of the lateral malleolus bears a facet for articulation with the lateral surface of the talus, thereby forming the lateral part of the ankle joint. Just superior to this articular facet is a triangular area, which fits into the fibular notch on the distal end of the tibia. Here the tibia and fibula are joined together by the distal end of the interosseous membrane. Posteroinferior to the facet for articulation with the talus is a pit or fossa (the **malleolar fossa**) for the attachment of the posterior talofibular ligament associated with the ankle joint.

The posterior surface of the lateral malleolus is marked by a shallow groove for the tendons of the fibularis longus and fibularis brevis muscles.

Joints

Interosseous membrane of leg

The interosseous membrane of the leg is a tough fibrous sheet of connective tissue that spans the distance between facing interosseous borders of the tibial and fibular shafts (Fig. 6.82). The collagen fibers descend obliquely from the interosseous border of the tibia to the interosseous border of the fibula, except superiorly where there is a ligamentous band, which ascends from the tibia to fibula.

There are two apertures in the interosseous membrane, one at the top and the other at the bottom, for vessels to pass between the anterior and posterior compartments of the leg.

The interosseous membrane not only links the tibia and fibula together, but also provides an increased surface area for muscle attachment.

The distal ends of the fibula and tibia are held together by the inferior aspect of the interosseous membrane, which spans the narrow space between the fibular notch on the lateral surface of the distal end of the tibia and the corresponding surface on the distal end of the fibula. This expanded end of the interosseous membrane is reinforced by **anterior** and **posterior tibiofibular ligaments**. This firm linking together of the distal ends of the tibia and fibula is essential to produce the skeletal framework for articulation with the foot at the ankle joint.

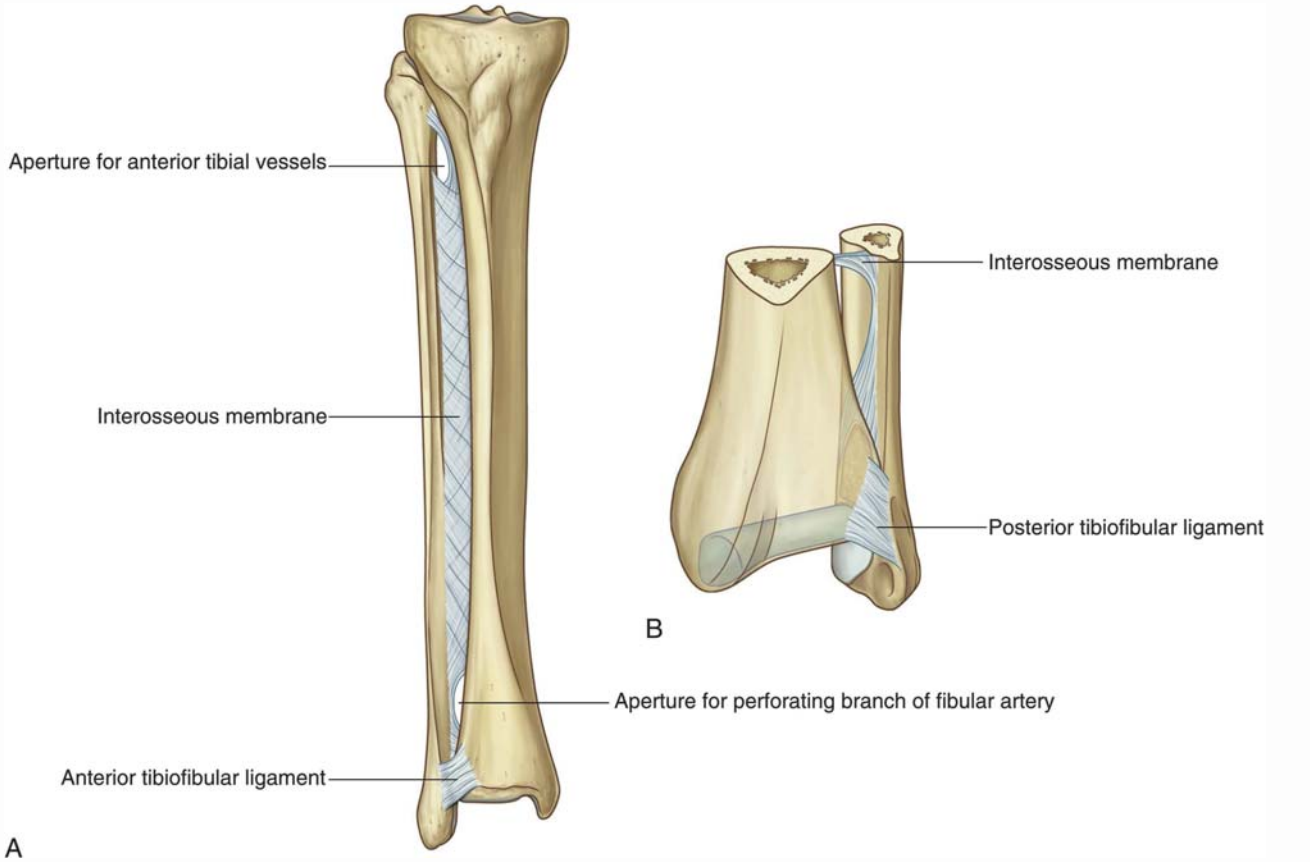


Fig. 6.82 Interosseous membrane. **A.** Anterior view. **B.** Posteromedial view.

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Posterior compartment of leg

Muscles

Muscles in the posterior (flexor) compartment of the leg are organized into two groups, superficial and deep, separated by a layer of deep fascia. Generally, the muscles mainly plantarflex and invert the foot and flex the toes. All are innervated by the tibial nerve.

Superficial group

The superficial group of muscles in the posterior compartment of the leg comprises three muscles—the gastrocnemius, plantaris, and soleus (Table 6.6)—all of which insert onto the heel (calcaneus) of the foot and plantarflex the foot at the ankle joint (Fig. 6.83). As a unit, these muscles are large and powerful because they propel the body forward off the planted foot during walking and can elevate the body upward onto the toes when standing. Two of the muscles (gastrocnemius and plantaris) originate on the distal end of the femur and can also flex the knee.

Gastrocnemius

The **gastrocnemius** muscle is the most superficial of the muscles in the posterior compartment and is one of the largest muscles in the leg (Fig. 6.83). It originates from two heads, one lateral and one medial:

- The **medial head** is attached to an elongate roughening on the posterior aspect of the distal femur just behind the adductor tubercle and above the articular surface of the medial condyle.
- The **lateral head** originates from a distinct facet on the upper lateral surface of the lateral femoral condyle where it joins the lateral supracondylar line.

At the knee, the facing margins of the two heads of the gastrocnemius form the lateral and medial borders of the lower end of the popliteal fossa.

In the upper leg, the heads of the gastrocnemius combine to form a single elongate muscle belly, which forms much of the soft tissue bulge identified as the **calf**.

In the lower leg, the muscle fibers of the gastrocnemius converge with those of the deeper soleus muscle to form the **calcaneal tendon**, which attaches to the calcaneus (heel) of the foot.

The gastrocnemius plantarflexes the foot at the ankle joint and can also flex the leg at the knee joint. It is innervated by the tibial nerve.

Plantaris

The **plantaris** has a small muscle belly proximally and a long thin tendon, which descends through the leg and joins the calcaneal tendon (Fig. 6.82). The muscle takes origin superiorly from the lower part of the lateral supracondylar ridge of the femur and from the oblique popliteal ligament associated with the knee joint.

The short spindle-shaped muscle body of the plantaris descends medially, deep to the lateral head of the gastrocnemius, and forms a thin tendon, which passes between the gastrocnemius and soleus muscles and eventually fuses with the medial side of the calcaneal tendon near its attachment to the calcaneus.

The plantaris contributes to plantarflexion of the foot at the ankle joint and flexion of the leg at the knee joint, and is innervated by the tibial nerve.

Soleus

The **soleus** is a large flat muscle under the gastrocnemius muscle (Fig. 6.83). It is attached to the proximal ends of

Table 6.6 Superficial group of muscles in the posterior compartment of leg (spinal segments in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Gastrocnemius	Medial head—posterior surface of distal femur just superior to medial condyle; lateral head—upper posterolateral surface of lateral femoral condyle	Via calcaneal tendon, to posterior surface of calcaneus	Tibial nerve (**S1, S2**)	Plantarflexes foot and flexes knee
Plantaris	Inferior part of lateral supracondylar line of femur and oblique popliteal ligament of knee	Via calcaneal tendon, to posterior surface of calcaneus	Tibial nerve (**S1, S2**)	Plantarflexes foot and flexes knee
Soleus	Soleal line and medial border of tibia; posterior aspect of fibular head and adjacent surfaces of neck and proximal shaft; tendinous arch between tibial and fibular attachments	Via calcaneal tendon, to posterior surface of calcaneus	Tibial nerve (**S1, S2**)	Plantarflexes the foot

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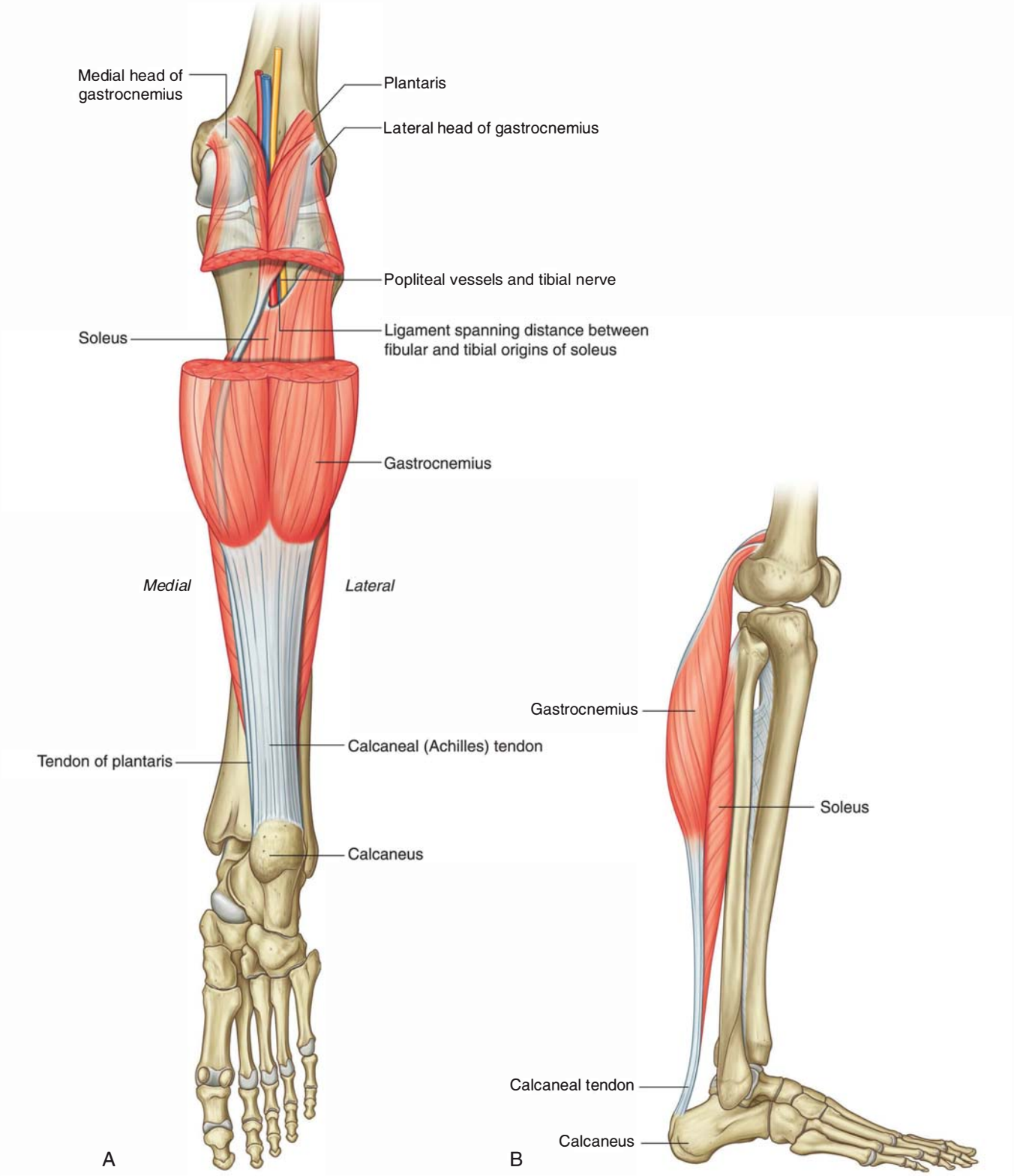


Fig. 6.83 Superficial group of muscles in the posterior compartment of leg. **A.** Posterior view. **B.** Lateral view.

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the fibula and tibia, and to a tendinous ligament, which spans the distance between the two heads of attachment to the fibula and tibia:

- On the proximal end of the fibula, the soleus originates from the posterior aspect of the head and adjacent surface of the neck and upper shaft of the fibula.
- On the tibia, the soleus originates from the soleal line and adjacent medial border.
- The ligament, which spans the distance between the attachments to the tibia and fibula, arches over the popliteal vessels and tibial nerve as they pass from the popliteal fossa into the deep region of the posterior compartment of the leg.

In the lower leg, the soleus muscle narrows to join the calcaneal tendon that attaches to the calcaneus.

The soleus muscle, together with the gastrocnemius and plantaris, plantarflexes the foot at the ankle joint. It is innervated by the tibial nerve.

In the clinic

Achilles tendon rupture

Rupture of the Achilles tendon is often related to sudden or direct trauma. This type of injury frequently occurs in a normal healthy tendon. In addition, there are certain conditions that may predispose the tendon to rupture. Among these conditions are tendinopathy (due to overuse, or to age-related degenerative changes) and previous Achilles tendon interventions such as injections of pharmaceuticals and the use of certain antibiotics (quinolone group). The diagnosis of Achilles tendon rupture is relatively straightforward. The patient typically complains of "being kicked" or "shot" behind the ankle, and clinical examination often reveals a gap in the tendon.

Deep group

There are four muscles in the deep posterior compartment of the leg (Fig. 6.84)—the popliteus, flexor hallucis longus, flexor digitorum longus, and tibialis posterior (Table 6.7). The popliteus muscle acts on the knee, whereas the other three muscles act mainly on the foot.

Popliteus

The **popliteus** is the smallest and most superior of the deep muscles in the posterior compartment of the leg. It unlocks the extended knee at the initiation of flexion and stabilizes the knee by resisting lateral (external) rotation of the tibia on the femur. It is flat and triangular in shape,

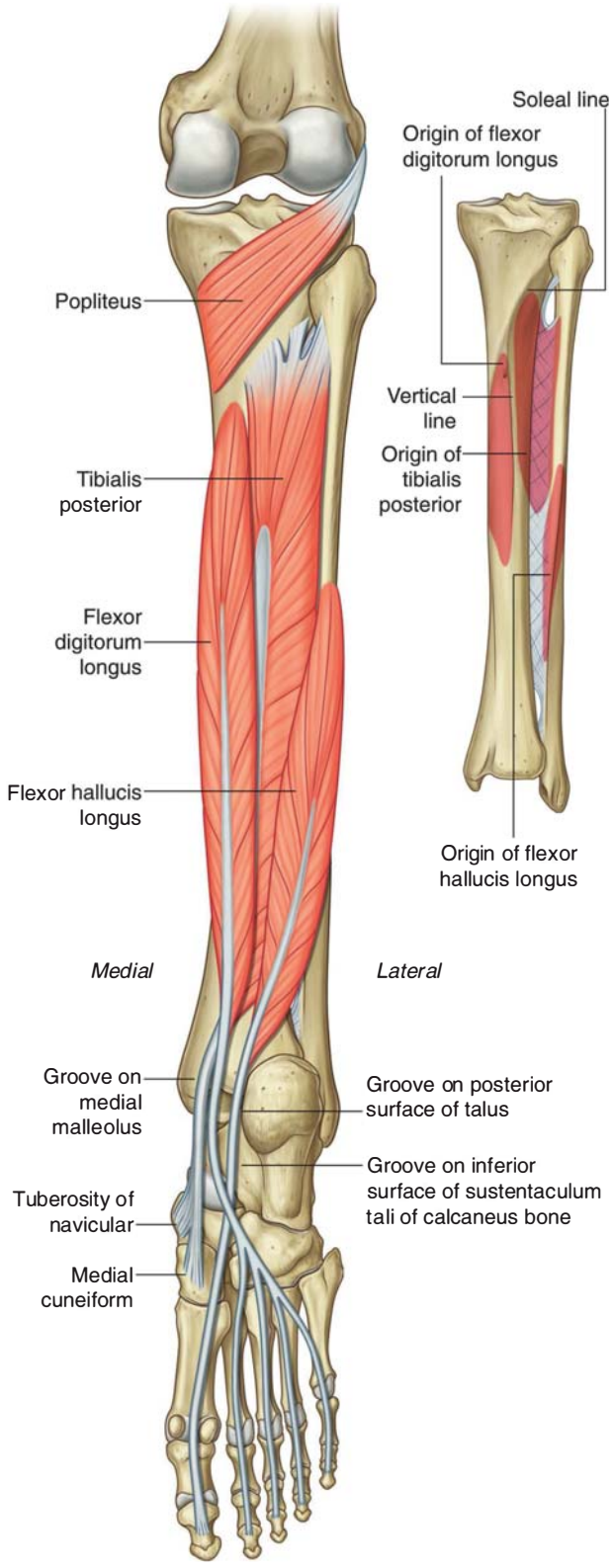


Fig. 6.84 Deep group of muscles in the posterior compartment of leg.

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Table 6.7 Deep group of muscles in the posterior compartment of leg (spinal segments in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Popliteus	Lateral femoral condyle	Posterior surface of proximal tibia	Tibial nerve (L4 to S1)	Stabilizes knee joint (resists lateral rotation of tibia on femur) Unlocks knee joint (laterally rotates femur on fixed tibia)
Flexor hallucis longus	Posterior surface of fibula and adjacent interosseous membrane	Plantar surface of distal phalanx of great toe	Tibial nerve (**S2**, S3)	Flexes great toe
Flexor digitorum longus	Medial side of posterior surface of the tibia	Plantar surfaces of bases of distal phalanges of the lateral four toes	Tibial nerve (**S2**, S3)	Flexes lateral four toes
Tibialis posterior	Posterior surfaces of interosseous membrane and adjacent regions of tibia and fibula	Mainly to tuberosity of navicular and adjacent region of medial cuneiform	Tibial nerve (L4, L5)	Inversion and plantarflexion of foot; support of medial arch of foot during walking

forms part of the floor of the popliteal fossa (Fig. 6.84), and is inserted into a broad triangular region above the soleal line on the posterior surface of the tibia.

The popliteus muscle ascends laterally across the lower aspect of the knee and originates from a tendon, which penetrates the fibrous membrane of the joint capsule of the knee. The tendon ascends laterally around the joint where it passes between the lateral meniscus and the fibrous membrane and then into a groove on the inferolateral aspect of the lateral femoral condyle. The tendon attaches to and originates from a depression at the anterior end of the groove.

When initiating gait from a standing position, contraction of the popliteus laterally rotates the femur on the fixed tibia, unlocking the knee joint. The popliteus muscle is innervated by the tibial nerve.

Flexor hallucis longus

The flexor hallucis longus muscle originates on the lateral side of the posterior compartment of the leg and inserts into the plantar surface of the great toe on the medial side of the foot (Fig. 6.84). It arises mainly from the lower two-thirds of the posterior surface of the fibula and adjacent interosseous membrane.

The muscle fibers of the flexor hallucis longus converge inferiorly to form a large cord-like tendon, which passes behind the distal head of the tibia and then slips into a distinct groove on the posterior surface of the adjacent tarsal bone (talus) of the foot. The tendon curves anteriorly first under the talus and then under a shelf of bone (the sustentaculum tali), which projects medially from the calcaneus, and then continues anteriorly through the sole of the foot to insert on the inferior surface of the base of the distal phalanx of the great toe.

The flexor hallucis longus flexes the great toe. It is particularly active during the toe-off phase of walking when the body is propelled forward off the stance leg and the great toe is the last part of the foot to leave the ground. It can also contribute to plantarflexion of the foot at the ankle joint and is innervated by the tibial nerve.

Flexor digitorum longus

The flexor digitorum longus muscle originates on the medial side of the posterior compartment of the leg and inserts into the lateral four digits of the foot (Fig. 6.84). It arises mainly from the medial side of the posterior surface of the tibia inferior to the soleal line.

The flexor digitorum longus descends in the leg and forms a tendon, which crosses posterior to the tendon of the tibialis posterior muscle near the ankle joint. The tendon continues inferiorly in a shallow groove behind the medial malleolus and then swings forward to enter the sole of the foot. It crosses inferior to the tendon of the flexor hallucis longus muscle to reach the medial side of the foot and then divides into four tendons, which insert on the plantar surfaces of the bases of the distal phalanges of digits II to V.

The flexor digitorum longus flexes the lateral four toes. It is involved with gripping the ground during walking and propelling the body forward off the toes at the end of the stance phase of gait. It is innervated by the tibial nerve.

Tibialis posterior

The tibialis posterior muscle originates from the interosseous membrane and the adjacent posterior surfaces of the tibia and fibula (Fig. 6.84). It lies between and is overlapped by the flexor digitorum longus and the flexor hallucis longus muscles.

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Near the ankle, the tendon of the tibialis posterior is crossed superficially by the tendon of the flexor digitorum longus muscle and lies medial to this tendon in the groove on the posterior surface of the medial malleolus. The tendon curves forward under the medial malleolus and enters the medial side of the foot. It wraps around the medial margin of the foot to attach to the plantar surfaces of the medial tarsal bones, mainly to the tuberosity of the navicular and to the adjacent region of the medial cuneiform.

The tibialis posterior inverts and plantarflexes the foot, and supports the medial arch of the foot during walking. It is innervated by the tibial nerve.

In the clinic

Neurological examination of the legs

Some of the commonest conditions that affect the legs are peripheral neuropathy (particularly associated with diabetes mellitus), lumbar nerve root lesions (associated with pathology of the intervertebral discs), fibular nerve palsy, and spastic paraparesis.

- Look for muscle wasting—loss of muscle mass may indicate loss of or reduced innervation.
- Test the power in muscle groups—hip flexion (L1, L2—iliopsoas—straight leg raise); knee flexion (L5 to S2—hamstrings—the patient tries to bend the knee while the examiner applies force to the leg to hold the knee in extension); knee extension (L3, L4—quadriceps femoris—the patient attempts to keep the leg straight while the examiner applies a force to the leg to flex the knee joint); ankle plantarflexion (S1, S2—the patient pushes the foot down while the examiner applies a force to the plantar surface of the foot to dorsiflex the ankle joint); ankle dorsiflexion (L4, L5—the patient pulls the foot upward while the examiner applies a force to the dorsal aspect of the foot to plantarflex the ankle joint).
- Examine knee and ankle reflexes—a tap with a tendon hammer on the patellar ligament (tendon) tests reflexes at the L3–L4 spinal levels, and tapping the calcaneal tendon tests reflexes at the S1–S2 spinal levels.
- Assess status of general sensory input to lumbar and upper sacral spinal cord levels—test light touch, pin prick, and vibration sense at dermatomes in the lower limb.

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Arteries

Popliteal artery

The **popliteal artery** is the major blood supply to the leg and foot and enters the posterior compartment of the leg from the popliteal fossa behind the knee (Fig. 6.85).

The popliteal artery passes into the posterior compartment of the leg between the gastrocnemius and popliteus muscles. As it continues inferiorly it passes under the tendinous arch formed between the fibular and tibial heads of the soleus muscle and enters the deep region of the posterior compartment of the leg where it immediately divides into an anterior tibial artery and a posterior tibial artery.

Two large sural arteries, one on each side, branch from the popliteal artery to supply the gastrocnemius, soleus, and plantaris muscles (Fig. 6.85). In addition, the popliteal artery gives rise to branches that contribute to a collateral network of vessels around the knee joint (see Fig. 6.77).

Anterior tibial artery

The **anterior tibial artery** passes forward through the aperture in the upper part of the interosseous membrane and enters and supplies the anterior compartment of the leg. It continues inferiorly onto the dorsal aspect of the foot.

Posterior tibial artery

The **posterior tibial artery** supplies the posterior and lateral compartments of the leg and continues into the sole of the foot (Fig. 6.85).

The posterior tibial artery descends through the deep region of the posterior compartment of the leg on the superficial surfaces of the tibialis posterior and flexor digitorum longus muscles. It passes through the tarsal tunnel behind the medial malleolus and into the sole of the foot.

In the leg, the posterior tibial artery supplies adjacent muscles and bone and has two major branches, the circumflex fibular artery and the fibular artery:

- The **circumflex fibular artery** passes laterally through the soleus muscle and around the neck of the fibula to connect with the anastomotic network of vessels surrounding the knee (Fig. 6.85; see also Fig. 6.77).
- The **fibular artery** parallels the course of the tibial artery, but descends along the lateral side of the posterior compartment adjacent to the medial crest on the posterior surface of the fibula, which separates the attachments of the tibialis posterior and flexor hallucis longus muscles.

The fibular artery supplies adjacent muscles and bone in the posterior compartment of the leg and also has

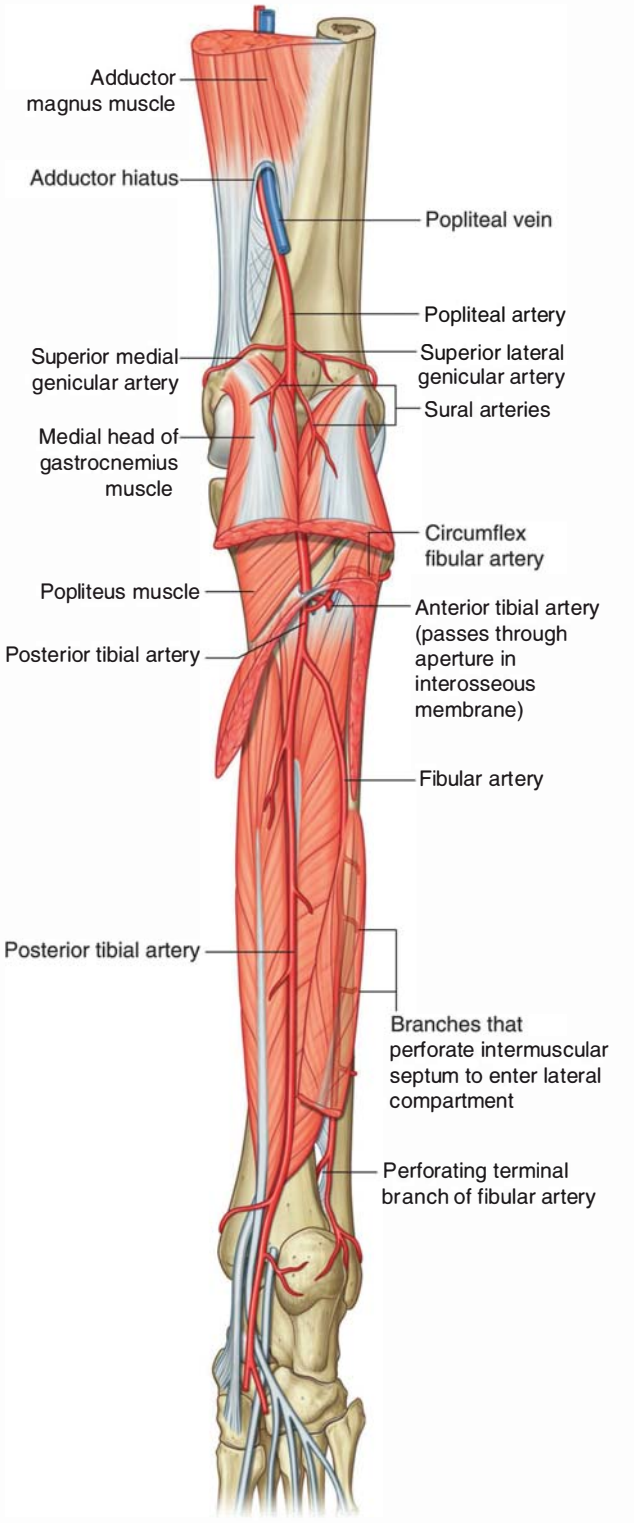


Fig. 6.85 Arteries in the posterior compartment of leg.

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branches that pass laterally through the intermuscular septum to supply the fibularis muscles in the lateral compartment of the leg.

A **perforating branch** that originates from the fibular artery distally in the leg passes anteriorly through the inferior aperture in the interosseous membrane to anastomose with a branch of the anterior tibial artery.

The fibular artery passes behind the attachment between the distal ends of the tibia and fibula and terminates in a network of vessels over the lateral surface of the calcaneus.

Veins

Deep veins in the posterior compartment generally follow the arteries.

Nerves

Tibial nerve

The nerve associated with the posterior compartment of the leg is the tibial nerve (Fig. 6.86), a major branch of the sciatic nerve that descends into the posterior compartment from the popliteal fossa.

The tibial nerve passes under the tendinous arch formed between the fibular and tibial heads of the soleus muscle and passes vertically through the deep region of the posterior compartment of the leg on the surface of the tibialis posterior muscle with the posterior tibial vessels.

The tibial nerve leaves the posterior compartment of the leg at the ankle by passing through the tarsal tunnel behind the medial malleolus. It enters the foot to supply most intrinsic muscles and skin.

In the leg, the tibial nerve gives rise to:

- branches that supply all the muscles in the posterior compartment of the leg, and
- two cutaneous branches, the **sural nerve** and **medial calcaneal nerve**.

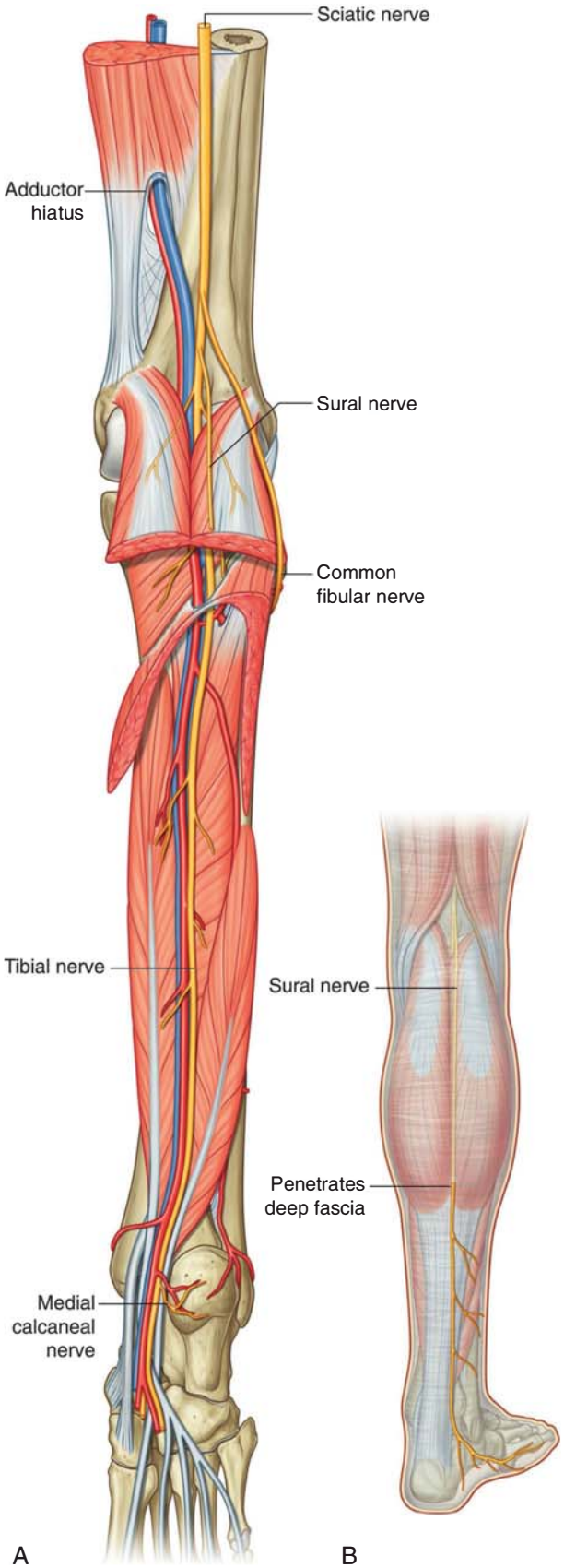


Fig. 6.86 Tibial nerve. **A.** Posterior view. **B.** Sural nerve.

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Branches of the tibial nerve that innervate the superficial group of muscles of the posterior compartment and popliteus muscle of the deep group originate high in the leg between the two heads of the gastrocnemius muscle in the distal region of the popliteal fossa (Fig. 6.87). Branches innervate the gastrocnemius, plantaris, and soleus muscles, and pass more deeply into the popliteus muscle.

Branches to the deep muscles of the posterior compartment originate from the tibial nerve deep to the soleus muscle in the upper half of the leg and innervate the tibialis posterior, flexor hallucis longus, and flexor digitorum longus muscles.

Sural nerve

The sural nerve originates high in the leg between the two heads of the gastrocnemius muscle (Fig. 6.86). It descends superficial to the belly of the gastrocnemius muscle and penetrates through the deep fascia approximately in the middle of the leg where it is joined by a sural communicating branch from the common fibular nerve. It passes down the leg, around the lateral malleolus, and into the foot.

The sural nerve supplies skin on the lower posterolateral surface of the leg and the lateral side of the foot and little toe.

Medial calcaneal nerve

The medial calcaneal nerve is often multiple and originates from the tibial nerve low in the leg near the ankle and descends onto the medial side of the heel.

The medial calcaneal nerve innervates skin on the medial surface and sole of the heel (Fig. 6.86).

Lateral compartment of leg

Muscles

There are two muscles in the lateral compartment of the leg—the fibularis longus and fibularis brevis (Fig. 6.87 and Table 6.8). Both evert the foot (turn the sole outward) and are innervated by the superficial fibular nerve, which is a branch of the common fibular nerve.

Fibularis longus

The **fibularis longus** muscle arises in the lateral compartment of the leg, but its tendon crosses under the foot to attach to bones on the medial side (Fig. 6.87). It originates from both the upper lateral surface of the fibula and from

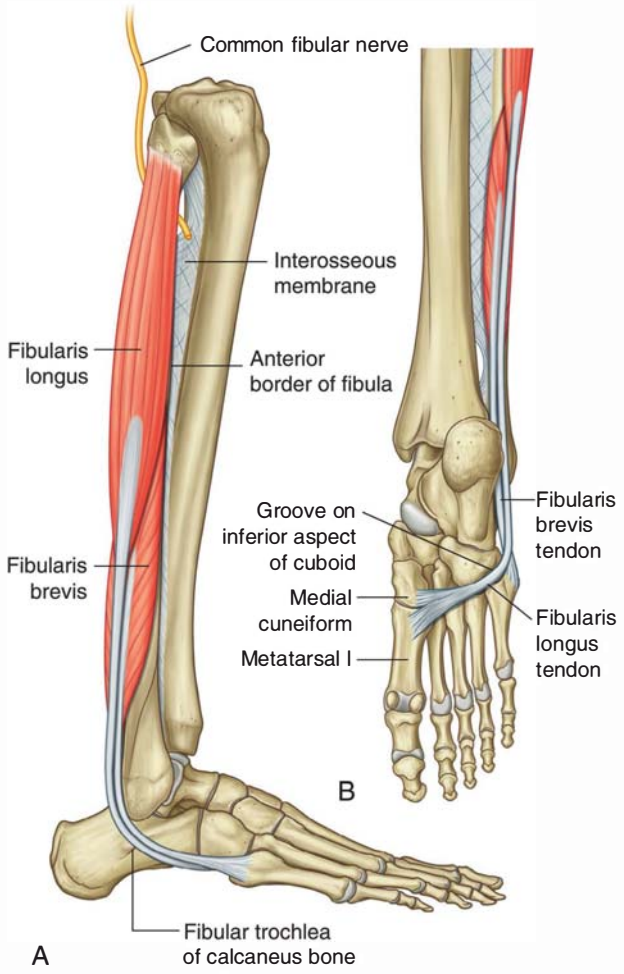


Fig. 6.87 Muscles in the lateral compartment of leg. **A.** Lateral view. **B.** Inferior view of the right foot, with the foot plantarflexed at the ankle.

Table 6.8 Muscles of the lateral compartment of leg (spinal segments in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Fibularis longus	Upper lateral surface of fibula, head of fibula, and occasionally the lateral tibial condyle	Undersurface of lateral sides of distal end of medial cuneiform and base of metatarsal I	Superficial fibular nerve (**L5, S1**, S2)	Eversion and plantarflexion of foot; supports arches of foot
Fibularis brevis	Lower two-thirds of lateral surface of shaft of fibula	Lateral tubercle at base of metatarsal V	Superficial fibular nerve (**L5, S1**, S2)	Eversion of foot

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the anterior aspect of the fibular head and occasionally up onto the adjacent region of the lateral tibial condyle.

The common fibular nerve passes anteriorly around the fibular neck between the attachments of the fibularis longus to the fibular head and shaft.

Distally, the fibularis longus descends in the leg to form a tendon, which, in order:

- passes posterior to the lateral malleolus in a shallow bony groove,
- swings forward to enter the lateral side of the foot,
- descends obliquely down the lateral side of the foot where it curves forward under a bony tubercle (fibular trochlea) of the calcaneus,
- enters a deep groove on the inferior surface of one of the other tarsal bones (the cuboid), and
- swings under the foot to cross the sole and attach to the inferior surfaces of bones on the medial side of the foot (lateral sides of the base of metatarsal I and the distal end of the medial cuneiform).

The fibularis longus everts and plantarflexes the foot. In addition, the fibularis longus, tibialis anterior, and tibialis posterior muscles, which all insert on the undersurfaces of bones on the medial side of the foot, together act as a stirrup to support the arches of the foot. The fibularis longus supports mainly the lateral and transverse arches.

The fibularis longus is innervated by the superficial fibular nerve.

Fibularis brevis

The fibularis brevis muscle is deep to the fibularis longus muscle in the leg and originates from the lower two-thirds of the lateral surface of the shaft of the fibula (Fig. 6.87).

The tendon of the fibularis brevis passes behind the lateral malleolus with the tendon of the fibularis longus muscle and then curves forward across the lateral surface of the calcaneus to attach to a tubercle on the lateral surface of the base of metatarsal V (the metatarsal associated with the little toe).

The fibularis brevis assists in eversion of the foot and is innervated by the superficial fibular nerve.

Arteries

No major artery passes vertically through the lateral compartment of the leg. It is supplied by branches (mainly from the fibular artery in the posterior compartment of the leg) that penetrate into the lateral compartment (Fig. 6.88).

Veins

Deep veins generally follow the arteries.

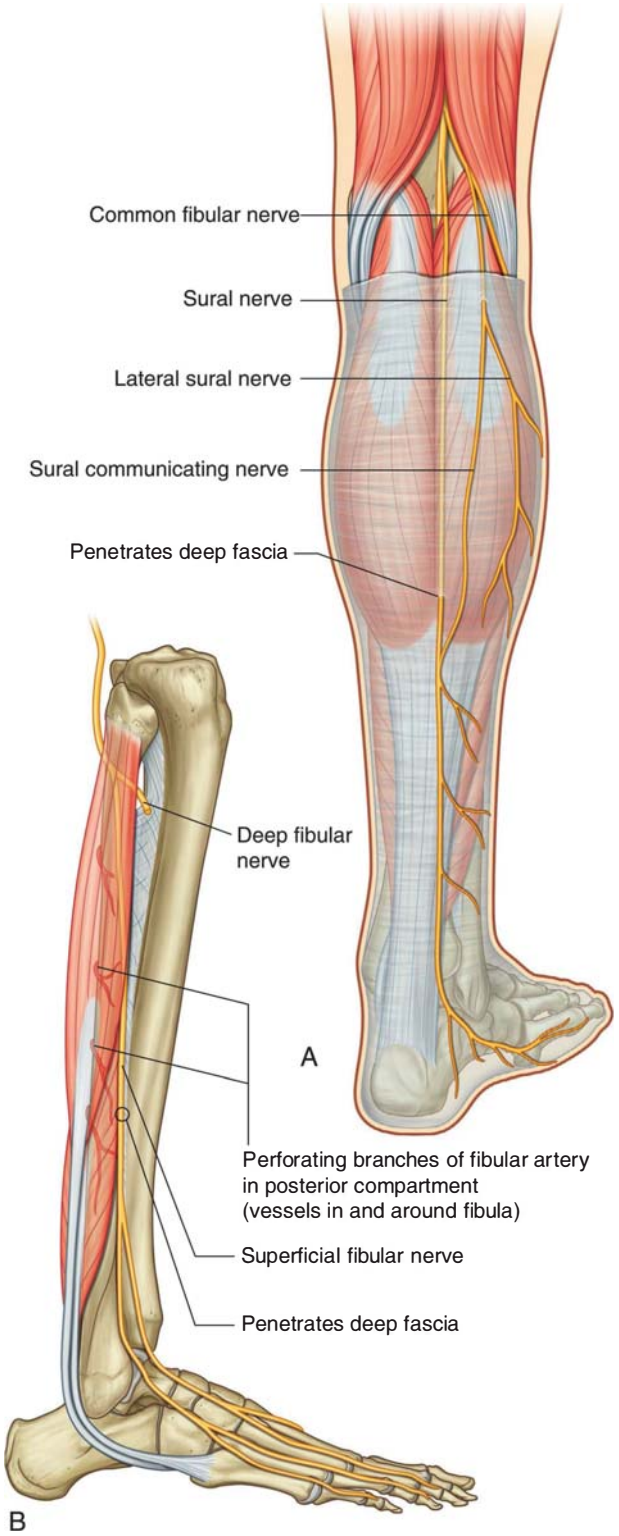


Fig. 6.88 Common fibular nerve, and nerves and arteries of the lateral compartment of leg. **A.** Posterior view, right leg. **B.** Lateral view, right leg.

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Nerves

Superficial fibular nerve

The nerve associated with the lateral compartment of the leg is the **superficial fibular nerve**. This nerve originates as one of the two major branches of the common fibular nerve, which enters the lateral compartment of the leg from the popliteal fossa (Fig. 6.88B).

The common fibular nerve originates from the sciatic nerve in the posterior compartment of the thigh or in the popliteal fossa (Fig. 6.88A), and follows the medial margin of the biceps femoris tendon over the lateral head of the gastrocnemius muscle and toward the fibula. Here it gives origin to two cutaneous branches, which descend in the leg:

- the **sural communicating nerve**, which joins the sural branch of the tibial nerve and contributes to innervation of skin over the lower posterolateral side of the leg; and
- the **lateral sural cutaneous nerve**, which innervates skin over the upper lateral leg.

The common fibular nerve continues around the neck of the fibula and enters the lateral compartment by passing between the attachments of the fibularis longus muscle to the head and shaft of the fibula. Here the common fibular nerve divides into its two terminal branches:

- the superficial fibular nerve, and
- the deep fibular nerve.

The superficial fibular nerve descends in the lateral compartment deep to the fibularis longus and innervates the fibularis longus and fibularis brevis (Fig. 6.87B). It then penetrates deep fascia in the lower leg and enters the foot where it divides into medial and lateral branches, which supply dorsal areas of the foot and toes except for:

- the web space between the great and second toes, which is supplied by the deep fibular nerve; and
- the lateral side of the little toe, which is supplied by the sural branch of the tibial nerve.

The deep fibular nerve passes anteromedially through the intermuscular septum into the anterior compartment of the leg, which it supplies.

Anterior compartment of leg

Muscles

There are four muscles in the anterior compartment of the leg—the tibialis anterior, extensor hallucis longus, extensor digitorum longus, and fibularis tertius (Fig. 6.89 and Table 6.9). Collectively they dorsiflex the foot at the ankle joint, extend the toes, and invert the foot. All are innervated by the deep fibular nerve, which is a branch of the common fibular nerve.

Tibialis anterior

The **tibialis anterior** muscle is the most anterior and medial of the muscles in the anterior compartment of the

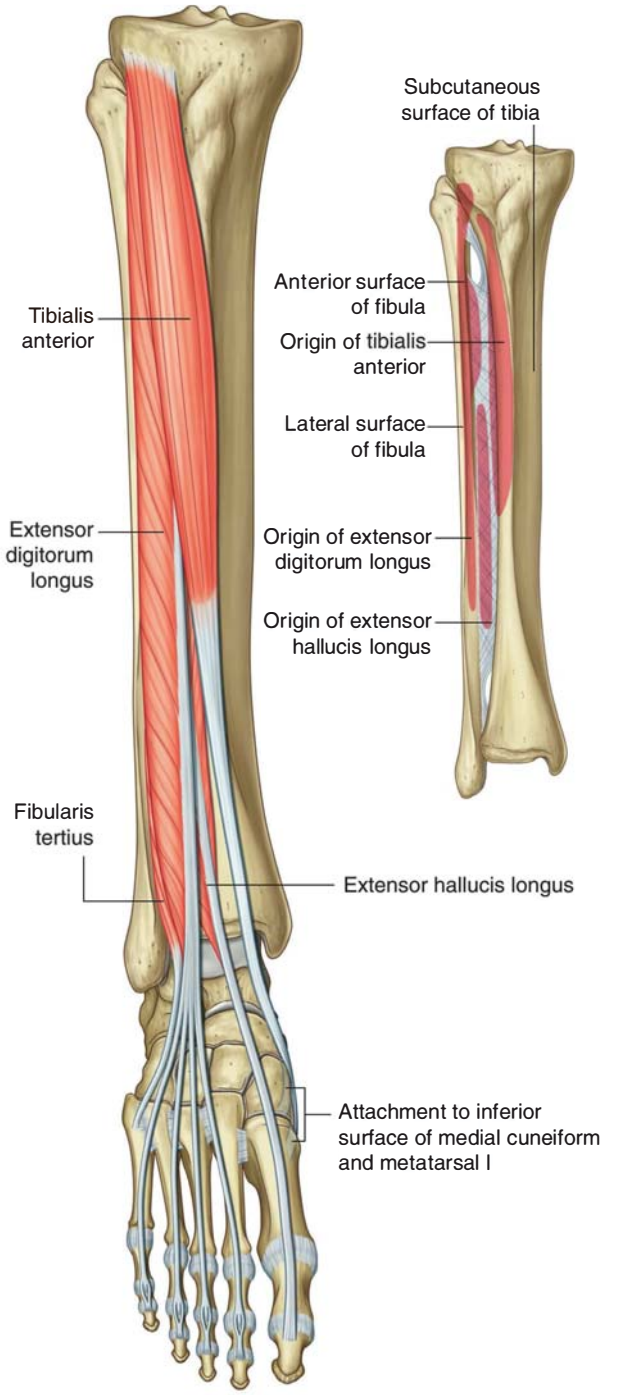


Fig. 6.89 Muscles of the anterior compartment of leg.

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Table 6.9 Muscles of the anterior compartment of leg (spinal segments in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Tibialis anterior	Lateral surface of tibia and adjacent interosseous membrane	Medial and inferior surfaces of medial cuneiform and adjacent surfaces on base of metatarsal I	Deep fibular nerve (**L4**, L5)	Dorsiflexion of foot at ankle joint; inversion of foot; dynamic support of medial arch of foot
Extensor hallucis longus	Middle one-half of medial surface of fibula and adjacent surface of interosseous membrane	Dorsal surface of base of distal phalanx of great toe	Deep fibular nerve (**L5, S1**)	Extension of great toe and dorsiflexion of foot
Extensor digitorum longus	Proximal one-half of medial surface of fibula and related surface of lateral tibial condyle	Via dorsal digital expansions into bases of distal and middle phalanges of lateral four toes	Deep fibular nerve (**L5, S1**)	Extension of lateral four toes and dorsiflexion of foot
Fibularis tertius	Distal part of medial surface of fibula	Dorsomedial surface of base of metatarsal V	Deep fibular nerve (**L5, S1**)	Dorsiflexion and eversion of foot

leg (Fig. 6.89). It originates mainly from the upper two-thirds of the lateral surface of the shaft of the tibia and adjacent surface of the interosseous membrane. It also originates from deep fascia.

The muscle fibers of the tibialis anterior converge in the lower one-third of the leg to form a tendon, which descends into the medial side of the foot, where it attaches to the medial and inferior surfaces of one of the tarsal bones (medial cuneiform) and adjacent parts of metatarsal I associated with the great toe.

The tibialis anterior dorsiflexes the foot at the ankle joint and inverts the foot at the intertarsal joints. During walking, it provides dynamic support for the medial arch of the foot.

The tibialis anterior is innervated by the deep fibular nerve.

Extensor hallucis longus

The **extensor hallucis longus** muscle lies next to and is partly overlapped by the tibialis anterior muscle (Fig. 6.89). It originates from the middle one-half of the medial surface of the fibula and adjacent interosseous membrane.

The tendon of the extensor hallucis longus appears between the tendons of the tibialis anterior and extensor digitorum longus in the lower one-half of the leg and descends into the foot. It continues anteriorly on the medial side of the dorsal surface of the foot to near the end of the great toe where it inserts on the upper surface of the base of the distal phalanx.

The extensor hallucis longus extends the great toe. Because it crosses anterior to the ankle joint, it also dorsiflexes the foot at the ankle joint. Like all muscles in the anterior compartment of the leg, the extensor hallucis longus muscle is innervated by the deep fibular nerve.

Extensor digitorum longus

The **extensor digitorum longus** muscle is the most posterior and lateral of the muscles in the anterior compartment of the leg (Fig. 6.89). It originates mainly from the upper one-half of the medial surface of the fibula lateral to and above the origin of the extensor hallucis longus muscle, and extends superiorly onto the lateral condyle of the tibia. Like the tibialis anterior muscle, it also originates from deep fascia.

The extensor digitorum longus muscle descends to form a tendon, which continues into the dorsal aspect of the foot, where it divides into four tendons, which insert, via dorsal digital expansions, into the dorsal surfaces of the bases of the middle and distal phalanges of the lateral four toes.

The extensor digitorum longus extends the toes and dorsiflexes the foot at the ankle joint, and is innervated by the deep fibular nerve.

Fibularis tertius

The **fibularis tertius** muscle is normally considered part of the extensor digitorum longus (Fig. 6.89). The fibularis tertius originates from the medial surface of the fibula immediately below the origin of the extensor digitorum longus muscle and the two muscles are normally connected.

The tendon of the fibularis tertius descends into the foot with the tendon of the extensor digitorum longus. On the dorsal aspect of the foot, it deviates laterally to insert into the dorsomedial surface of the base of metatarsal V (the metatarsal associated with the little toe).

The fibularis tertius assists in dorsiflexion and possibly eversion of the foot, and is innervated by the deep fibular nerve.

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Arteries

Anterior tibial artery

The artery associated with the anterior compartment of the leg is the **anterior tibial artery**, which originates from the popliteal artery in the posterior compartment of the leg and passes forward into the anterior compartment of the leg through an aperture in the interosseous membrane.

The anterior tibial artery descends through the anterior compartment on the interosseous membrane (Fig. 6.90). In the distal leg, it lies between the tendons of the tibialis anterior and extensor hallucis longus muscles. It leaves the leg by passing anterior to the distal end of the tibia and ankle joint and continues onto the dorsal aspect of the foot as the dorsalis pedis artery.

In the proximal leg, the anterior tibial artery has a recurrent branch, which connects with the anastomotic network of vessels around the knee joint.

Along its course, the anterior tibial artery supplies numerous branches to adjacent muscles and is joined by the perforating branch of the fibular artery, which passes forward through the lower aspect of the interosseous membrane from the posterior compartment of the leg.

Distally, the anterior tibial artery gives rise to an **anterior medial malleolar artery** and an **anterior lateral malleolar artery**, which pass posteriorly around the distal ends of the tibia and fibula, respectively, and connect with vessels from the posterior tibial and fibular arteries to form an anastomotic network around the ankle.

Veins

Deep veins follow the arteries and have similar names.

Nerves

Deep fibular nerve

The nerve associated with the anterior compartment of the leg is the **deep fibular nerve** (Fig. 6.90). This nerve originates in the lateral compartment of the leg as one of the two divisions of the common fibular nerve.

The deep fibular nerve passes anteromedially through the intermuscular septum that separates the lateral from the anterior compartments of the leg and then passes deep to the extensor digitorum longus. It reaches the anterior interosseous membrane where it meets and descends with the anterior tibial artery.

The deep fibular nerve:

- innervates all muscles in the anterior compartment;
- then continues into the dorsal aspect of the foot where it innervates the extensor digitorum brevis, contributes

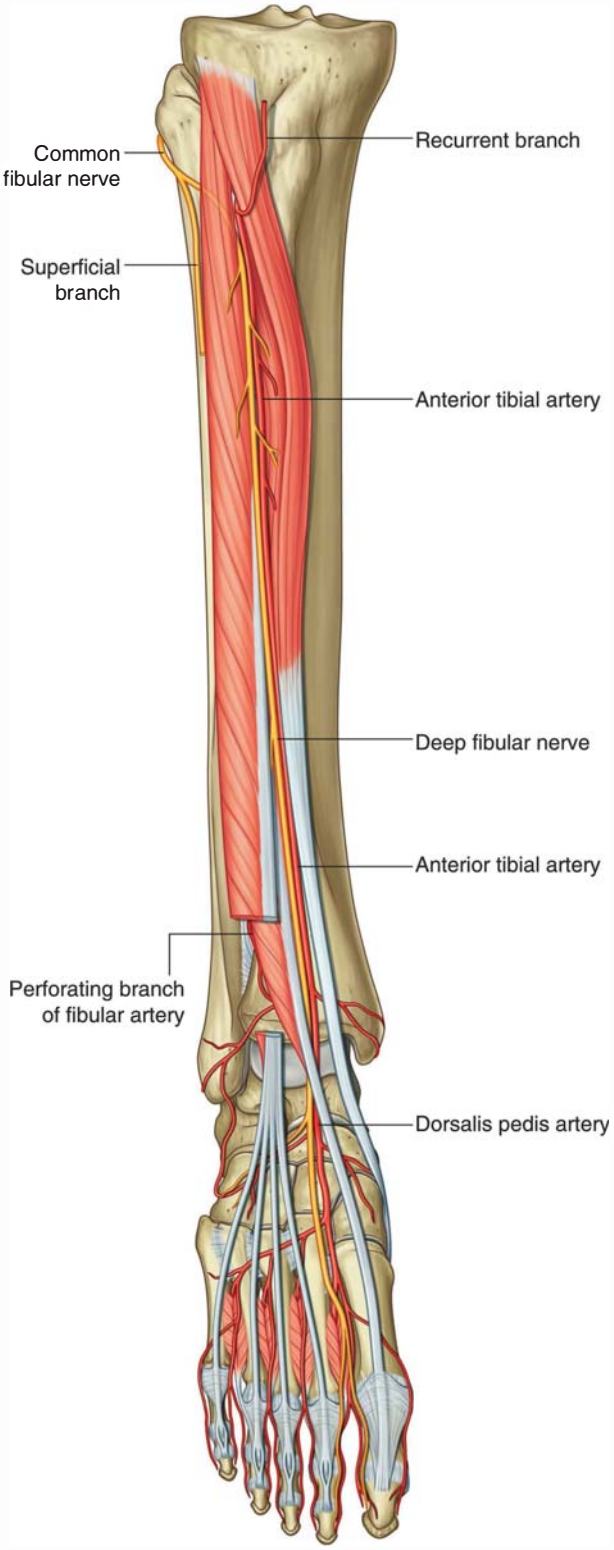


Fig. 6.90 Anterior tibial artery and deep fibular nerve.

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to the innervation of the first two dorsal interossei muscles, and supplies the skin between the great and second toes.

In the clinic

Footdrop

Footdrop is an inability to dorsiflex the foot. Patients with footdrop have a characteristic "steppage" gait. As the patient walks, the knee of the affected limb is elevated to an abnormal height during the swing phase to prevent the foot from dragging. At the end of the swing phase, the foot "slaps" the ground. Also, the unaffected limb often acquires a characteristic tiptoe pattern of gait during the stance phase. A typical cause of footdrop is damage to the common fibular nerve. Other causes include disc protrusion compressing the L5 nerve root, disorders of the sciatic nerve and the lumbosacral plexus, and pathologies of the spinal cord and brain.

FOOT

The foot is the region of the lower limb distal to the ankle joint. It is subdivided into the ankle, the metatarsus, and the digits.

There are five digits consisting of the medially positioned great toe (digit I) and four more laterally placed digits, ending laterally with the little toe (digit V) (Fig. 6.91).

The foot has a superior surface (**dorsum of foot**) and an inferior surface (**sole**; Fig. 6.91).

Abduction and adduction of the toes are defined with respect to the long axis of the second digit. Unlike in the hand, where the thumb is oriented 90° to the other fingers, the great toe is oriented in the same position as the other toes. The foot is the body's point of contact with the ground and provides a stable platform for upright stance. It also levers the body forward during walking.

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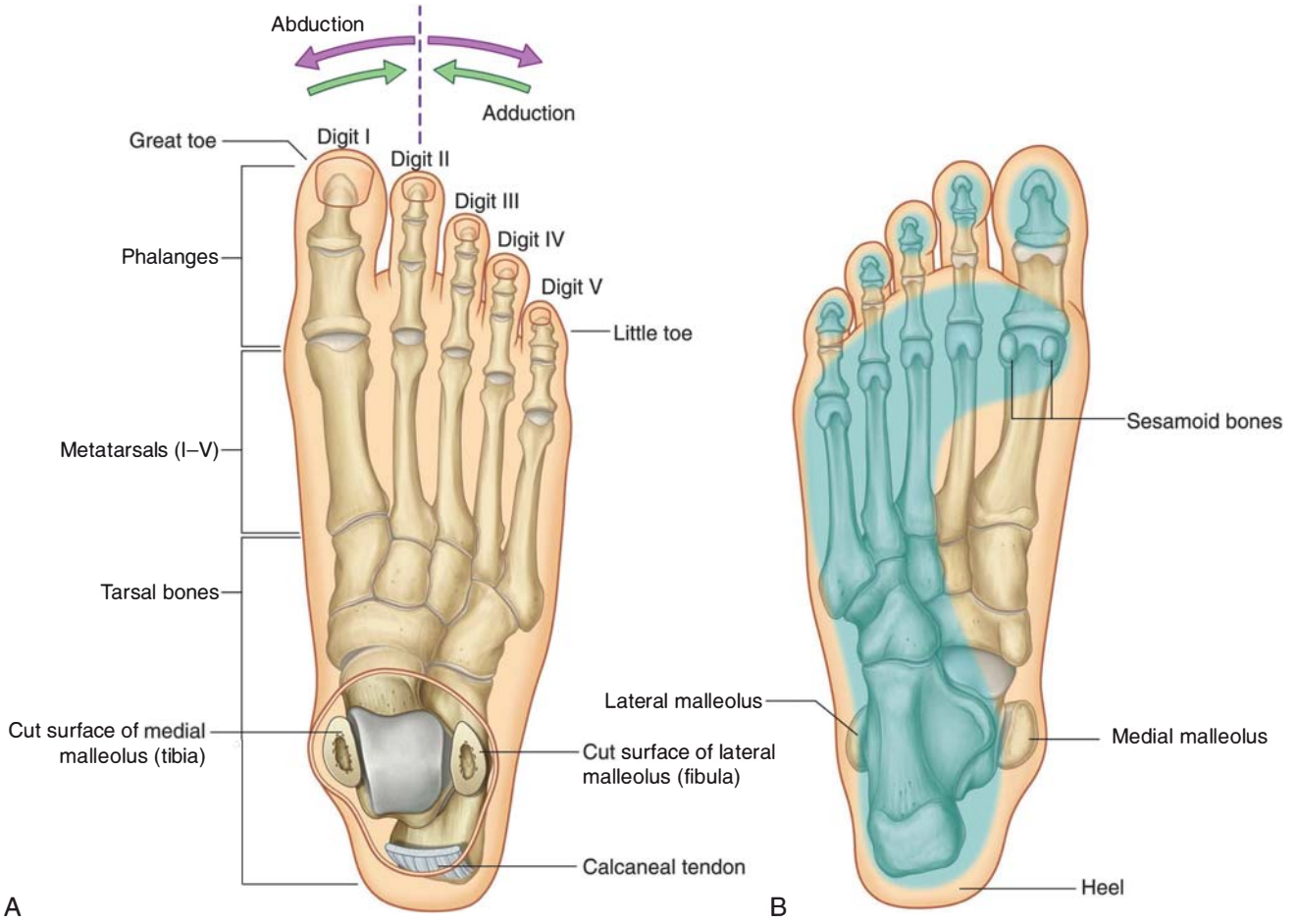


Fig. 6.91 Foot. **A.** Dorsal aspect, right foot. **B.** Plantar aspect, right foot, showing the surface in contact with the ground when standing.

Bones

There are three groups of bones in the foot (Fig. 6.92):

- the seven **tarsal bones**, which form the skeletal framework for the ankle;
- **metatarsals** (**I to V**), which are the bones of the metatarsus; and
- the **phalanges**, which are the bones of the toes—each toe has three phalanges, except for the great toe, which has two.

Tarsal bones

The tarsal bones are arranged in a proximal group and a distal group with an intermediate bone between the two groups on the medial side of the foot (Fig. 6.92A).

Proximal group

The proximal group consists of two large bones, the talus (Latin for "ankle") and the calcaneus (Latin for "heel"):

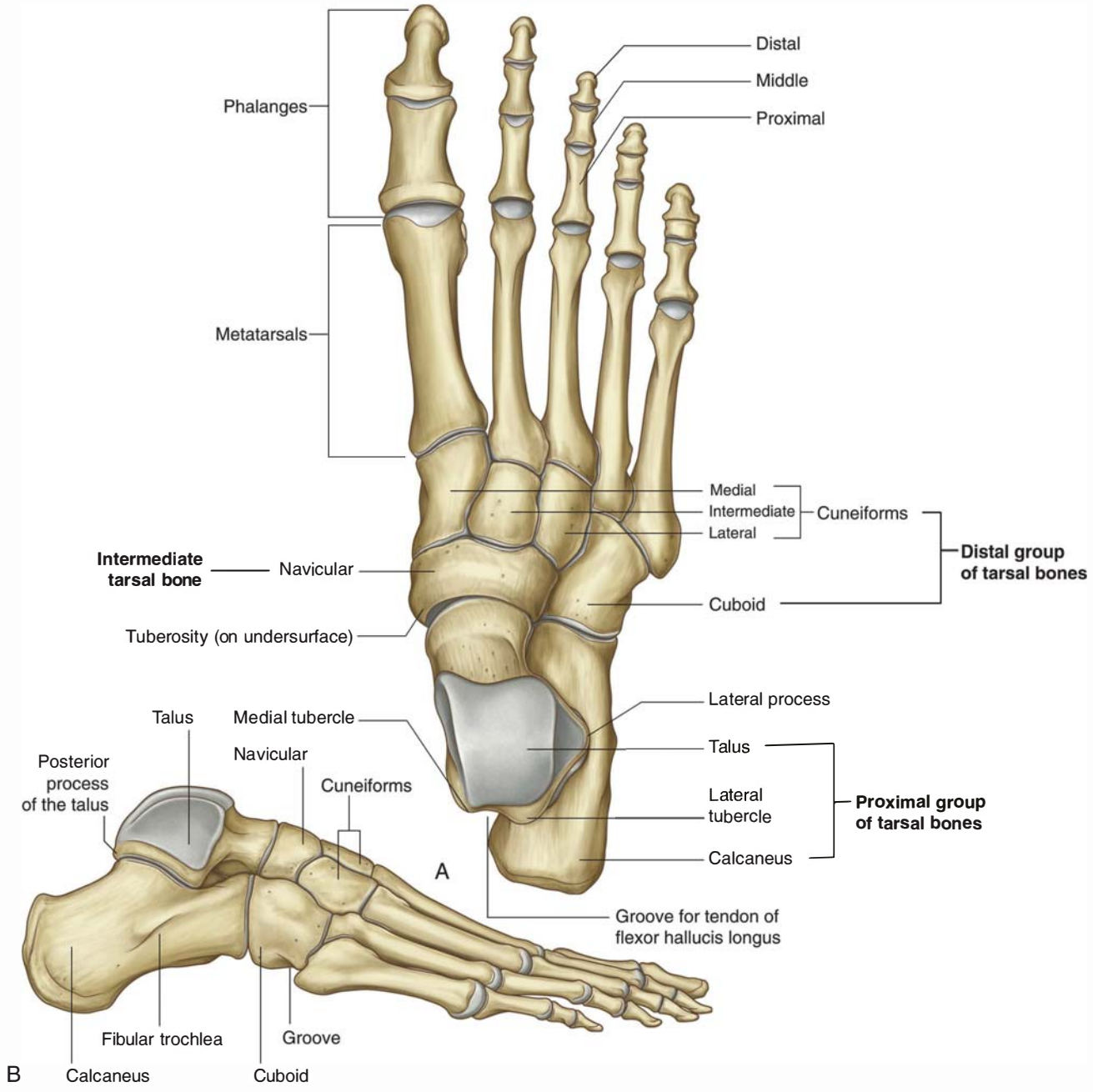


Fig. 6.92 Bones of the foot. **A.** Dorsal view, right foot. **B.** Lateral view, right foot.

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- The **talus** is the most superior bone of the foot and sits on top of and is supported by the calcaneus (Fig. 6.92B)—it articulates above with the tibia and fibula to form the ankle joint and also projects forward to articulate with the intermediate tarsal bone (navicular) on the medial side of the foot.
- The **calcaneus** is the largest of the tarsal bones—posteriorly it forms the bony framework of the heel and anteriorly it projects forward to articulate with one of the distal group of tarsal bones (cuboid) on the lateral side of the foot.

Talus

The talus, when viewed from the medial or lateral sides, is snail shaped (Fig. 6.93A,B). It has a rounded **head**, which is projected forward and medially at the end of a short broad **neck**, which is connected posteriorly to an expanded body.

Anteriorly, the head of the talus is domed for articulation with a corresponding circular depression on the posterior surface of the navicular bone. Inferiorly, this domed articular surface is continuous with an additional three articular facets separated by smooth ridges:

- The anterior and middle facets articulate with adjacent surfaces on the calcaneus bone.
- The other facet, medial to the facets for articulation with the calcaneus, articulates with a ligament—the plantar calcaneonavicular ligament (spring ligament)—which connects the calcaneus to the navicular under the head of the talus.

The neck of the talus is marked by a deep groove (the **sulcus tali**), which passes obliquely forward across the inferior surface from medial to lateral, and expands dramatically on the lateral side. Posterior to the sulcus tali is a large facet (posterior calcaneal surface) for articulation with the calcaneus.

The superior aspect of the body of the talus is elevated to fit into the socket formed by the distal ends of the tibia and fibula to form the ankle joint:

- The upper (trochlear) surface of this elevated region articulates with the inferior end of the tibia.
- The medial surface articulates with the medial malleolus of the tibia.
- The lateral surface articulates with the lateral malleolus of the fibula.

Because the lateral malleolus is larger and projects more inferiorly than the medial malleolus at the ankle joint, the corresponding lateral articular surface on the talus is larger and projects more inferiorly than the medial surface.

The lower part of the lateral surface of the body of the talus, which supports the lower part of the facet for articulation with the fibula, forms a bony projection (the **lateral process**).

The inferior surface of the body of the talus has a large oval concave facet (the **posterior calcaneal articular facet**) for articulation with the calcaneus.

The posterior aspect of the body of the talus consists of a backward and medially facing projection (the **posterior process**). The posterior process is marked on its surface by

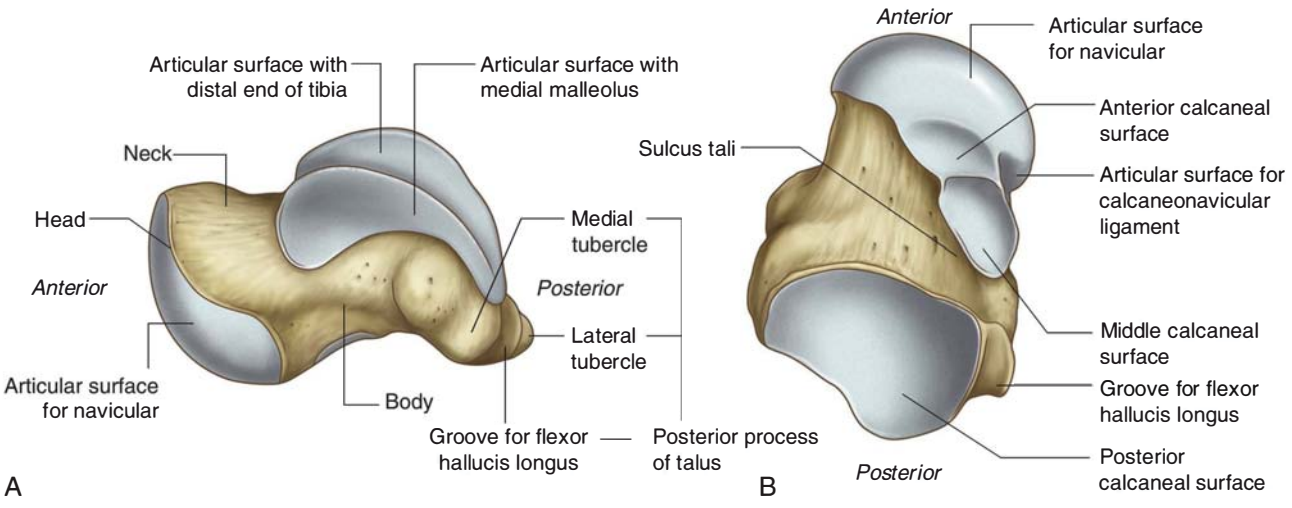


Fig. 6.93 Talus. **A.** Medial view. **B.** Inferior view.

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a lateral tubercle and a medial tubercle, which bracket between them the **groove for the tendon of the flexor hallucis longus** as it passes from the leg into the foot.

Calcaneus

The calcaneus sits under and supports the talus. It is an elongate, irregular, box-shaped bone with its long axis generally oriented along the midline of the foot, but deviating lateral to the midline anteriorly (Fig. 6.94).

The calcaneus projects behind the ankle joint to form the skeletal framework of the heel. The posterior surface of this heel region is circular and divided into upper, middle, and lower parts. The calcaneal tendon (Achilles tendon) attaches to the middle part:

- The upper part is separated from the calcaneal tendon by a bursa.
- The lower part curves forward, is covered by subcutaneous tissue, is the weight-bearing region of the heel, and is continuous onto the plantar surface of the bone as the **calcaneal tuberosity**.

The calcaneal tuberosity projects forward on the plantar surface as a large medial process and a small lateral process separated from each other by a V-shaped notch (Fig. 6.94B). At the anterior end of the plantar surface is a tubercle (the **calcaneal tubercle**) for the posterior attachment of the short plantar ligament of the sole of the foot.

The lateral surface of the calcaneus has a smooth contour except for two slightly raised regions (Fig. 6.94C). One of these raised areas—the **fibular trochlea** (peroneal tubercle)—is anterior to the middle of the surface and often has two shallow grooves, which pass, one above the other, obliquely across its surface. The tendons of the fibularis brevis and longus muscles are bound to the trochlea as they pass over the lateral side of the calcaneus.

Superior and posterior to the fibular trochlea is a second raised area or tubercle for attachment of the calcaneofibular part of the lateral collateral ligament of the ankle joint.

The medial surface of the calcaneus is concave and has one prominent feature associated with its upper margin (the **sustentaculum tali**; Fig. 6.94A), which is a shelf of bone projecting medially and supporting the more posterior part of the head of the talus.

The underside of the sustentaculum tali has a distinct groove running from posterior to anterior and along which the tendon of the flexor hallucis longus muscle travels into the sole of the foot.

The superior surface of the sustentaculum tali has a facet (**middle talar articular surface**) for articulation with the corresponding middle facet on the head of the talus.

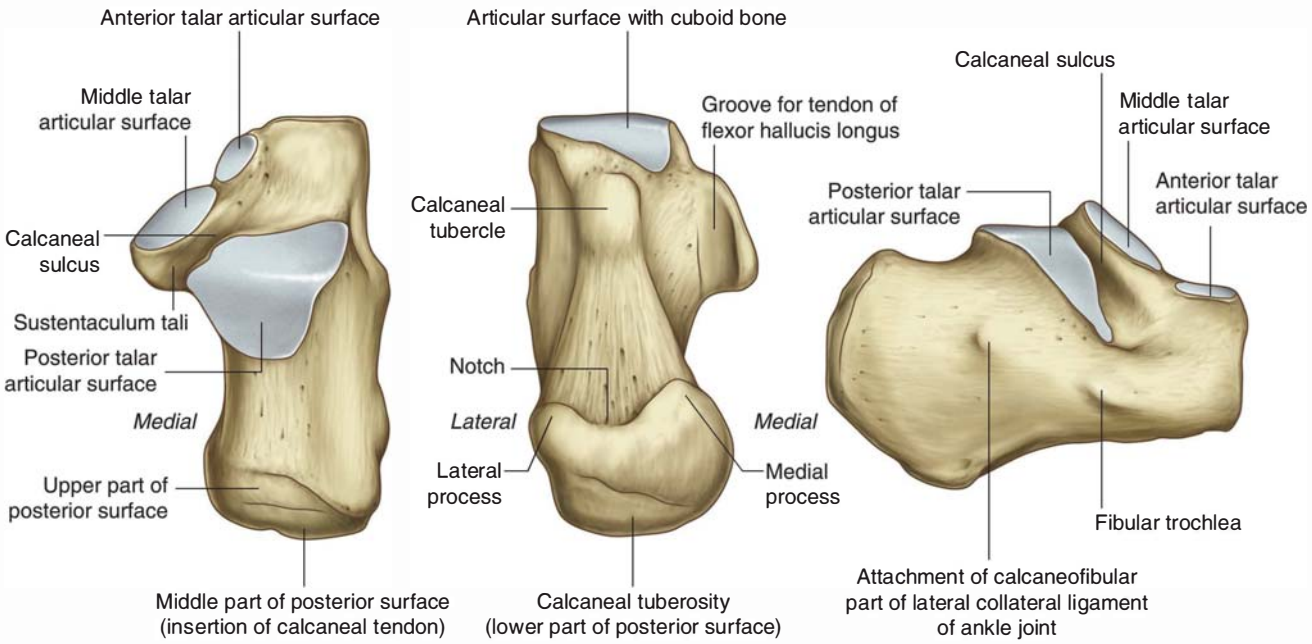


Fig. 6.94 Calcaneus. **A.** Superior view. **B.** Inferior view. **C.** Lateral view.

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Anterior and **posterior talar articular surfaces** are on the superior surface of the calcaneus itself (Fig. 6.94A):

- The anterior talar articular surface is small and articulates with the corresponding anterior facet on the head of the talus.
- The posterior talar articular surface is large and is approximately near the middle of the superior surface of the calcaneus.

Between the posterior talar articular surface, which articulates with the body of the talus, and the other two articular surfaces, which articulate with the head of the talus, is a deep groove (the **calcaneal sulcus**; Fig. 6.94A,C).

The calcaneal sulcus on the superior surface of the calcaneus and the sulcus tali on the inferior surface of the talus together form the **tarsal sinus**, which is a large gap between the anterior ends of the calcaneus and talus that is visible when the skeleton of the foot is viewed from its lateral aspect (Fig. 6.95).

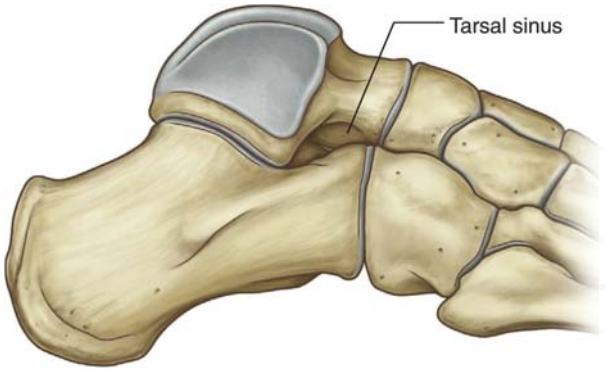


Fig. 6.95 Tarsal sinus. Lateral view, right foot.

Intermediate tarsal bone

The intermediate tarsal bone on the medial side of the foot is the **navicular** (boat shaped) (Fig. 6.92). This bone articulates behind with the talus and articulates in front and on the lateral side with the distal group of tarsal bones.

One distinctive feature of the navicular is a prominent rounded tuberosity for the attachment of the tibialis posterior tendon, which projects inferiorly on the medial side of the plantar surface of the bone.

Distal group

From lateral to medial, the distal group of tarsal bones consists of (Fig. 6.92):

- The **cuboid** (Greek for "cube"), which articulates posteriorly with the calcaneus, medially with the lateral cuneiform, and anteriorly with the bases of the lateral two metatarsals—the tendon of the fibularis longus muscle lies in a prominent groove on the anterior plantar surface, which passes obliquely forward across the bone from lateral to medial.
- Three **cuneiforms** (Latin for "wedge")—the **lateral**, **intermediate**, and **medial** cuneiform bones, in addition to articulating with each other, articulate posteriorly with the navicular bone and anteriorly with the bases of the medial three metatarsals.

Metatarsals

There are five metatarsals in the foot, numbered I to V from medial to lateral (Fig. 6.96). Metatarsal I, associated with the great toe, is shortest and thickest. The second is the longest.

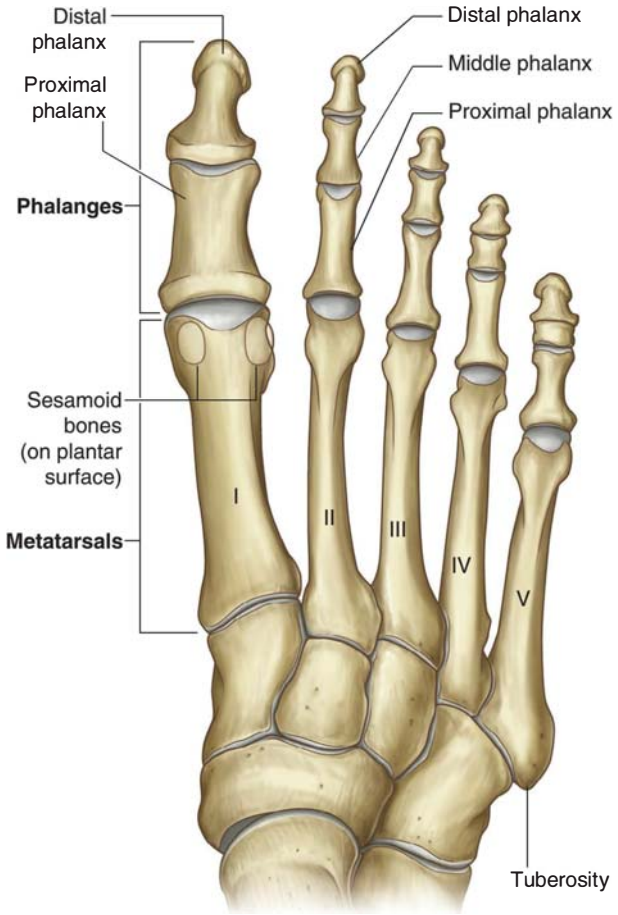


Fig. 6.96 Metatarsals and phalanges. Dorsal view.

Each metatarsal has a **head** at the distal end, an elongate **shaft** in the middle, and a proximal **base**.

The head of each metatarsal articulates with the proximal phalanx of a toe and the base articulates with one or more of the distal group of tarsal bones. The plantar

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surface of the head of metatarsal I also articulates with two sesamoid bones.

The sides of the bases of metatarsals II to V also articulate with each other. The lateral side of the base of metatarsal V has a prominent **tuberosity**, which projects posteriorly and is the attachment site for the tendon of the fibularis brevis muscle.

Phalanges

The phalanges are the bones of the toes (Fig. 6.96). Each toe has three phalanges (**proximal**, **middle**, and **distal**), except for the great toe, which has only two (proximal and distal).

Each phalanx consists of a **base**, a **shaft**, and a distal **head**:

- The base of each proximal phalanx articulates with the head of the related metatarsal.
- The head of each distal phalanx is nonarticular and flattened into a crescent-shaped plantar tuberosity under the plantar pad at the end of the digit.

In each toe, the total length of the phalanges combined is much shorter than the length of the associated metatarsal.

Joints

Ankle joint

The ankle joint is synovial in type and involves the talus of the foot and the tibia and fibula of the leg (Fig. 6.97).

The ankle joint mainly allows hinge-like dorsiflexion and plantarflexion of the foot on the leg.

The distal end of the fibula is firmly anchored to the larger distal end of the tibia by strong ligaments. Together, the fibula and tibia create a deep bracket-shaped socket for the upper expanded part of the body of the talus:

- The roof of the socket is formed by the inferior surface of the distal end of the tibia.
- The medial side of the socket is formed by the medial malleolus of the tibia.
- The longer lateral side of the socket is formed by the lateral malleolus of the fibula.

The articular surfaces are covered by hyaline cartilage.

The articular part of the talus is shaped like a short half-cylinder tipped onto its flat side with one end facing lateral and the other end facing medial. The curved upper surface of the half-cylinder and the two ends are covered by hyaline

In the clinic

Fracture of the talus

The talus is an unusual bone because it ossifies from a single primary ossification center, which initially appears in the neck. The posterior aspect of the talus appears to ossify last, normally after puberty. In up to 50% of people there is a small accessory ossicle (the os trigonum) posterior to the lateral tubercle of the posterior process. Articular cartilage covers approximately 60% of the talar surface and there are no direct tendon or muscle attachments to the bone.

One of the problems with fractures of the talus is that the blood supply to the bone is vulnerable to damage. The main blood supply to the bone enters the talus through the tarsal canal from a branch of the posterior tibial artery. This vessel supplies most of the neck and the body of the talus. Branches of the dorsalis pedis artery enter the superior aspect of the talar neck and supply the dorsal portion of the head and neck, and branches from the fibular artery supply a small portion of the lateral talus.

Fractures of the neck of the talus often interrupt the blood supply to the talus, so making the body and posterior aspect of the talus susceptible to osteonecrosis, which may in turn lead to premature osteoarthritis and require extensive surgery.

Midfoot fractures

Fractures of the midfoot are uncommon. They usually occur when heavy weights have been dropped onto the feet or when the feet have been run over by a vehicle. Plain radiographs are usually sufficient to demonstrate dislocations and fractures.

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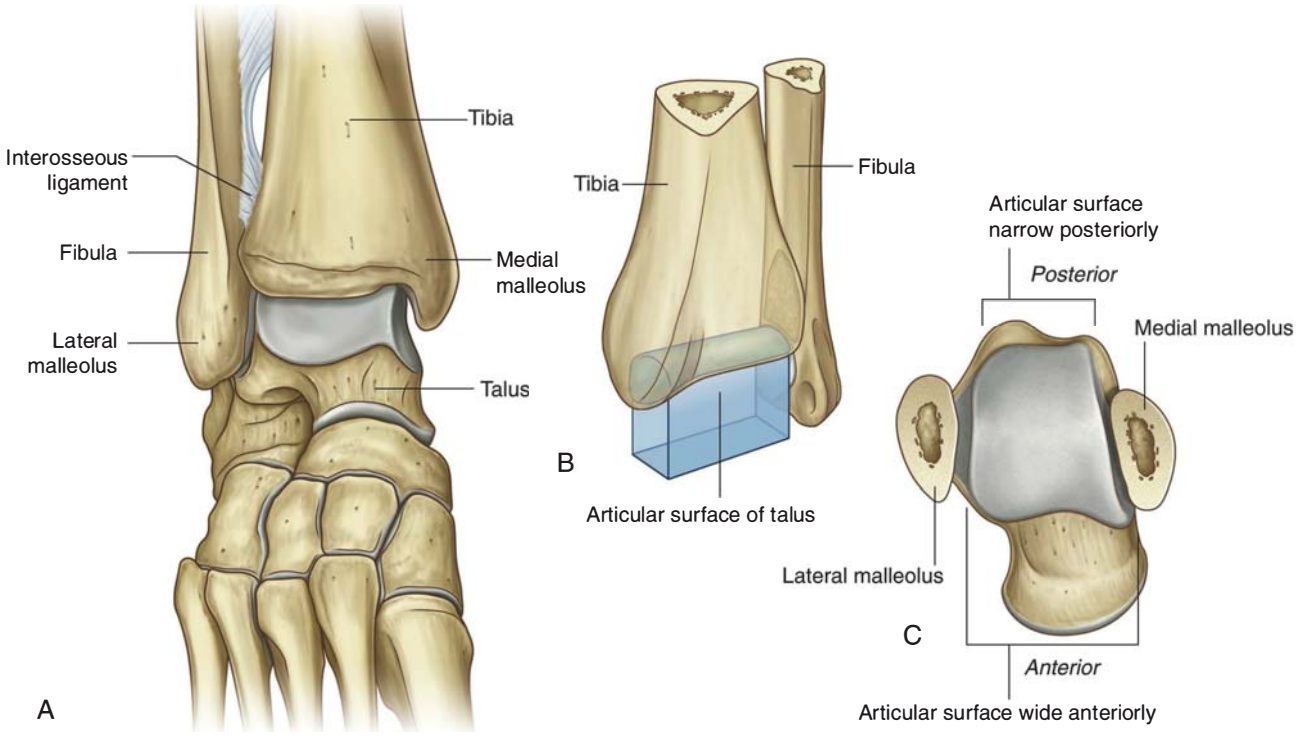


Fig. 6.97 Ankle joint. **A.** Anterior view with foot plantarflexed. **B.** Schematic of joint. **C.** Superior view of the talus to show the shape of the articular surface.

cartilage and fit into the bracket-shaped socket formed by the distal ends of the tibia and fibula.

When viewed from above, the articular surface of the talus is much wider anteriorly than it is posteriorly. As a result, the bone fits tighter into its socket when the foot is dorsiflexed and the wider surface of the talus moves into the ankle joint than when the foot is plantarflexed and the narrower part of the talus is in the joint. The joint is therefore most stable when the foot is dorsiflexed.

The articular cavity is enclosed by a synovial membrane, which attaches around the margins of the articular surfaces, and by a fibrous membrane, which covers the synovial membrane and is also attached to the adjacent bones.

The ankle joint is stabilized by **medial** (deltoid) and **lateral ligaments**.

Medial ligament (deltoid ligament)

The medial (deltoid) ligament is large, strong (Fig. 6.98), and triangular in shape. Its apex is attached above to the medial malleolus and its broad base is attached below to a line that extends from the tuberosity of the navicular bone in front to the medial tubercle of the talus behind.

The medial ligament is subdivided into four parts based on the inferior points of attachment:

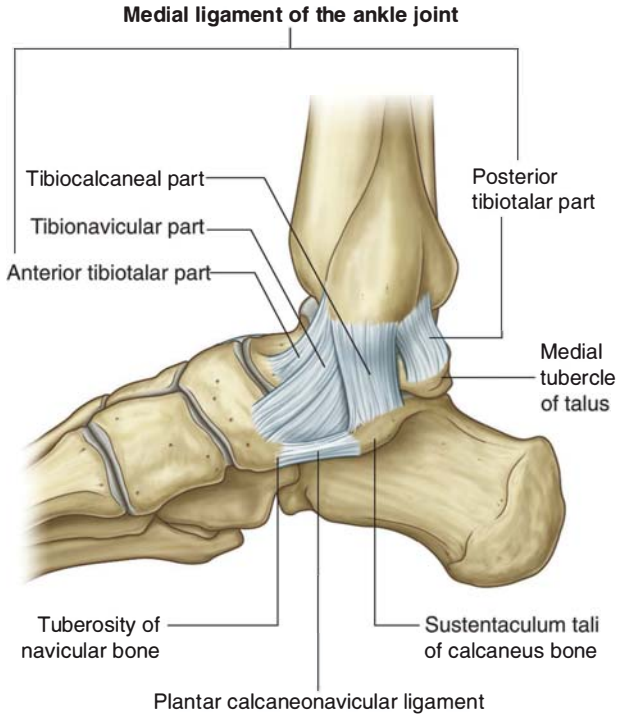


Fig. 6.98 Medial ligament of the ankle joint.

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- The part that attaches in front to the tuberosity of the navicular and the associated margin of the plantar calcaneonavicular ligament (spring ligament), which connects the navicular bone to the sustentaculum tali of the calcaneus bone behind, is the **tibionavicular part** of the medial ligament.
- The **tibiocalcaneal part**, which is more central, attaches to the sustentaculum tali of the calcaneus bone.
- The **posterior tibiotalar part** attaches to the medial side and medial tubercle of the talus.
- The fourth part (the **anterior tibiotalar part**) is deep to the tibionavicular and tibiocalcaneal parts of the medial ligament and attaches to the medial surface of the talus.

Lateral ligament

The lateral ligament of the ankle is composed of three separate ligaments, the anterior talofibular ligament, the posterior talofibular ligament, and the calcaneofibular ligament (Fig. 6.99):

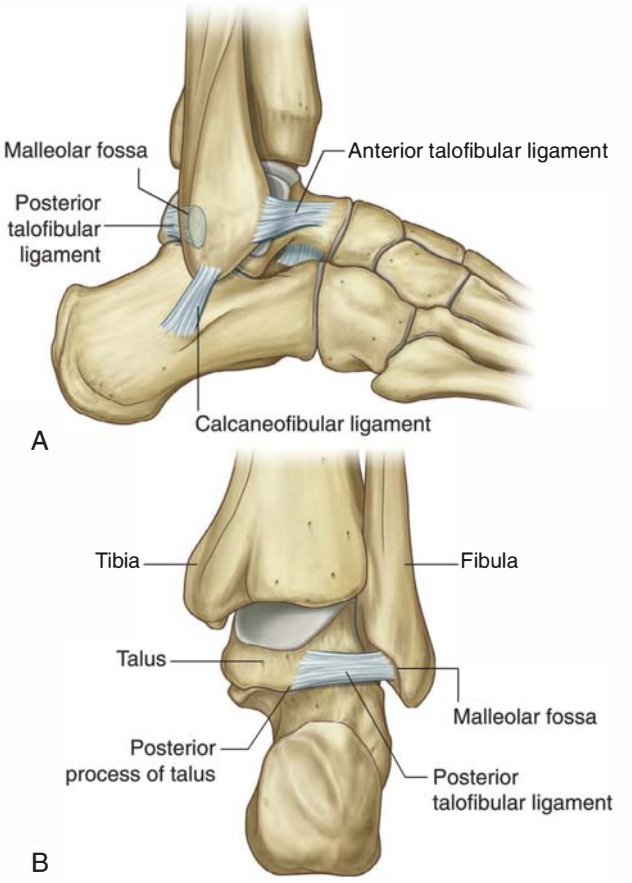


Fig. 6.99 Lateral ligament of the ankle joint. **A.** Lateral view. **B.** Posterior view.

- The **anterior talofibular ligament** is a short ligament, and attaches the anterior margin of the lateral malleolus to the adjacent region of the talus.
- The **posterior talofibular ligament** runs horizontally backward and medially from the malleolar fossa on the medial side of the lateral malleolus to the posterior process of the talus.
- The **calcaneofibular ligament** is attached above to the malleolar fossa on the posteromedial side of the lateral malleolus and passes posteroinferiorly to attach below to a tubercle on the lateral surface of the calcaneus.

Intertarsal joints

The numerous synovial joints between the individual tarsal bones mainly invert, evert, supinate, and pronate the foot:

- Inversion and eversion is turning the whole sole of the foot inward and outward, respectively.
- Pronation is rotating the front of the foot laterally relative to the back of the foot, and supination is the reverse movement.

Pronation and supination allow the foot to maintain normal contact with the ground when in different stances or when standing on irregular surfaces.

The major joints at which movements occur include the subtalar, talocalcaneonavicular, and calcaneocuboid joints (Fig. 6.100). The talocalcaneonavicular and calcaneocuboid joints together form what is often referred to as the **transverse tarsal joint**.

Intertarsal joints between the cuneiforms and between the cuneiforms and the navicular allow only limited movement.

The joint between the cuboid and navicular is normally fibrous.

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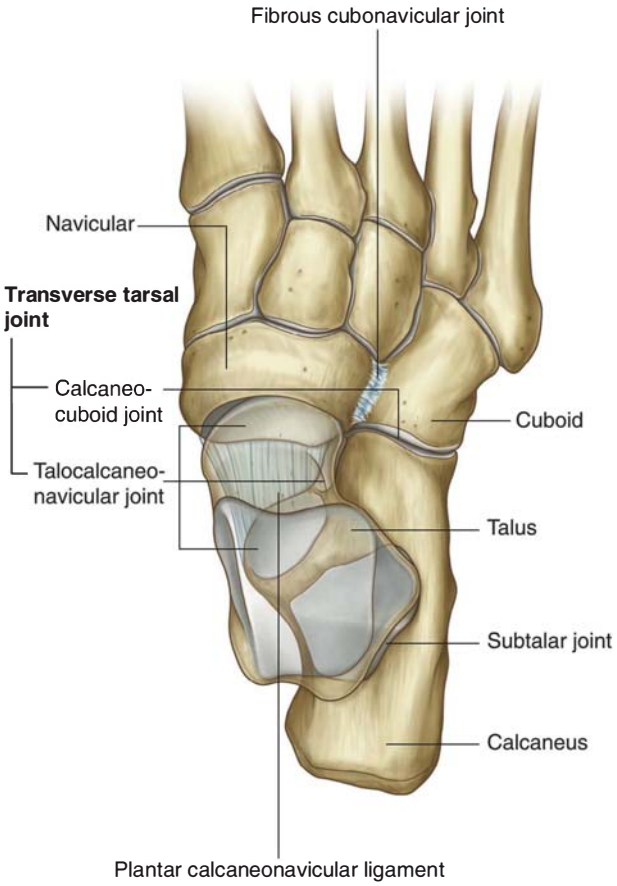


Fig. 6.100 Intertarsal joints.

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In the clinic

Ankle fractures

An appreciation of ankle anatomy is essential to understand the wide variety of fractures that may occur at and around the ankle joint.

The ankle joint and related structures can be regarded as a fibro-osseous ring oriented in the coronal plane.

- The upper part of the ring is formed by the joint between the distal ends of the fibula and tibia and by the ankle joint itself.
- The sides of the ring are formed by the ligaments that connect the medial malleolus and lateral malleolus to the adjacent tarsal bones.
- The bottom of the ring is not part of the ankle joint, but consists of the subtalar joint and the associated ligaments.

Visualizing the ankle joint and surrounding structures as a fibro-osseous ring allows the physician to predict the type of damage likely to result from a particular type of injury. For example, an inversion injury may fracture the medial malleolus and tear ligaments anchoring the lateral malleolus to the tarsal bones.

The ring may be disrupted not only by damage to the bones (which produces fractures), but also by damage to the ligaments. Unlike bone fractures, damage to ligaments is unlikely to be appreciated on plain radiographs. When a fracture is noted on a plain radiograph, the physician must always be aware that there may also be appreciable ligamentous disruption.

Ottawa Ankle Rules

The Ottawa ankle rules were developed to assist clinicians in deciding whether patients with acute ankle injuries require investigation with radiographs in order to avoid unnecessary studies. Named after the hospital where they were developed, the rules are highly sensitive and have reduced the utilization of unwarranted ankle radiographs since their implementation.

(continues)

In the clinic—cont'd

An ankle x-ray series is required if there is ankle pain and any of the following:

- Bone tenderness along the distal 6 cm of the posterior tibia or tip of the medial malleolus
- Bone tenderness along the distal 6 cm of the posterior fibula or tip of the lateral malleolus
- Inability to bear weight for four steps both immediately after the injury and in the emergency department

A foot x-ray series is required if there is midfoot pain and any of the following:

- Bone tenderness at the base of the fifth metatarsal bone
- Bone tenderness at the navicular bone
- Inability to bear weight for four steps both immediately after the injury and in the emergency department

Subtalar joint

The **subtalar joint** is between:

- the large posterior calcaneal facet on the inferior surface of the talus, and
- the corresponding posterior talar facet on the superior surface of the calcaneus.

The articular cavity is enclosed by synovial membrane, which is covered by a fibrous membrane.

The subtalar joint allows gliding and rotation, which are involved in inversion and eversion of the foot. **Lateral**, **medial**, **posterior**, and **interosseous talocalcaneal ligaments** stabilize the joint. The interosseous talocalcaneal ligament lies in the tarsal sinus (Fig. 6.101).

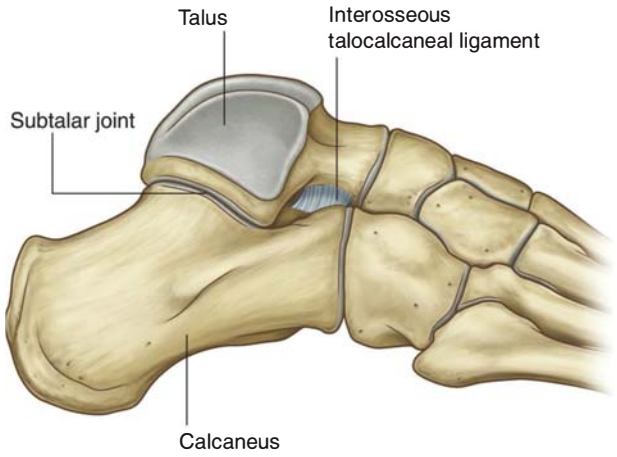


Fig. 6.101 Interosseous talocalcaneal ligament. Lateral view.

Talocalcaneonavicular joint

The **talocalcaneonavicular joint** is a complex joint in which the head of the talus articulates with the calcaneus and plantar calcaneonavicular ligament (spring ligament) below and the navicular in front (Fig. 6.102A).

The talocalcaneonavicular joint allows gliding and rotation movements, which together with similar movements of the subtalar joint are involved with inversion and eversion of the foot. It also participates in pronation and supination.

The parts of the talocalcaneonavicular joint between the talus and calcaneus are:

- the anterior and middle calcaneal facets on the inferior surface of the talar head, and
- the corresponding anterior and middle talar facets on the superior surface and sustentaculum tali, respectively, of the calcaneus (Fig. 6.102B).

The part of the joint between the talus and the plantar calcaneonavicular ligament (spring ligament) is between the ligament and the medial facet on the inferior surface of the talar head.

The joint between the navicular and talus is the largest part of the talocalcaneonavicular joint and is between the ovoid anterior end of the talar head and the corresponding concave posterior surface of the navicular.

Ligaments

The capsule of the talocalcaneonavicular joint, which is a synovial joint, is reinforced:

- posteriorly by the interosseous talocalcaneal ligament,
- superiorly by the **talonavicular ligament**, which passes between the neck of the talus and adjacent regions of the navicular, and
- inferiorly by the plantar calcaneonavicular ligament (spring ligament) (Fig. 6.102C,D).

The lateral part of the talocalcaneonavicular joint is reinforced by the calcaneonavicular part of the **bifurcate**

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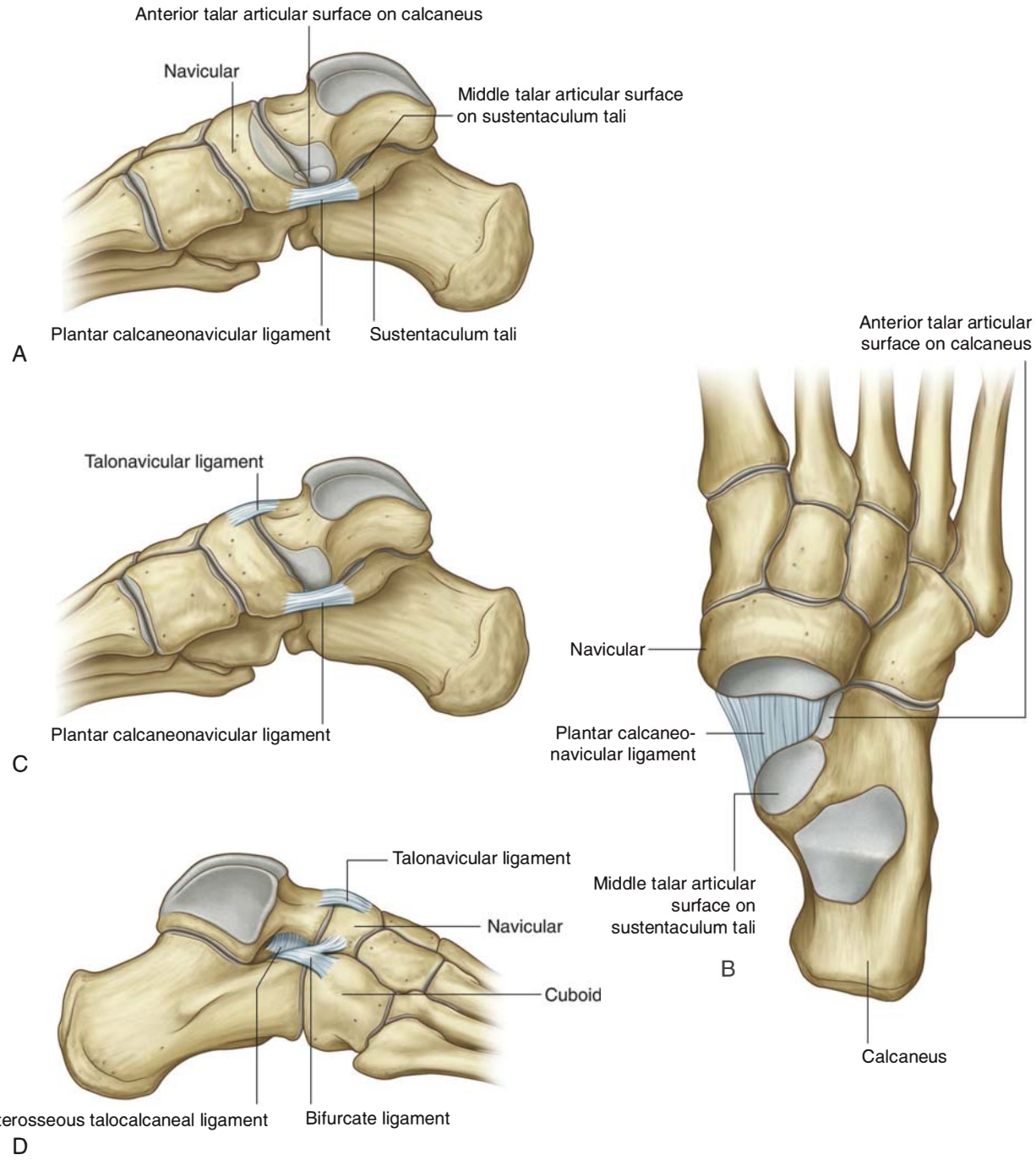


Fig. 6.102 Talocalcaneonavicular joint. **A.** Medial view, right foot. **B.** Superior view, right foot, talus removed. **C.** Ligaments, medial view, right foot. **D.** Ligaments, lateral view, right foot.

ligament, which is a Y-shaped ligament superior to the joint. The base of the bifurcate ligament is attached to the anterior aspect of the superior surface of the calcaneus and its arms are attached to:

- the dorsomedial surface of the cuboid (**calcaneocuboid ligament**), and
- the dorsolateral part of the navicular (**calcaneonavicular ligament**).

The **plantar calcaneonavicular ligament** (spring ligament) is a broad thick ligament that spans the space between the sustentaculum tali behind and the navicular bone in front (Fig. 6.102B,C). It supports the head of the

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talus, takes part in the talocalcaneonavicular joint, and resists depression of the medial arch of the foot.

Calcaneocuboid joint

The **calcaneocuboid joint** is a synovial joint between:

- the facet on the anterior surface of the calcaneus, and
- the corresponding facet on the posterior surface of the cuboid.

The calcaneocuboid joint allows sliding and rotating movements involved with inversion and eversion of the foot, and also contributes to pronation and supination of the forefoot on the hindfoot.

Ligaments

The calcaneocuboid joint is reinforced by the bifurcate ligament (see above) and by the long plantar ligament and the plantar calcaneocuboid ligament (short plantar ligament).

The **plantar calcaneocuboid ligament** (short plantar ligament) is short, wide, and very strong, and connects the calcaneal tubercle to the inferior surface of the cuboid (Fig. 6.103A). It not only supports the calcaneocuboid joint, but also assists the long plantar ligament in resisting depression of the lateral arch of the foot.

The **long plantar ligament** is the longest ligament in the sole of the foot and lies inferior to the plantar calcaneocuboid ligament (Fig. 6.103B):

- Posteriorly, it attaches to the inferior surface of the calcaneus between the tuberosity and the calcaneal tubercle.
- Anteriorly, it attaches to a broad ridge and a tubercle on the inferior surface of the cuboid bone behind the groove for the fibularis longus tendon.

More superficial fibers of the long plantar ligament extend to the bases of the metatarsal bones.

The long plantar ligament supports the calcaneocuboid joint and is the strongest ligament, resisting depression of the lateral arch of the foot.

Tarsometatarsal joints

The **tarsometatarsal joints** between the metatarsal bones and adjacent tarsal bones are plane joints and allow limited sliding movements (Fig. 6.104).

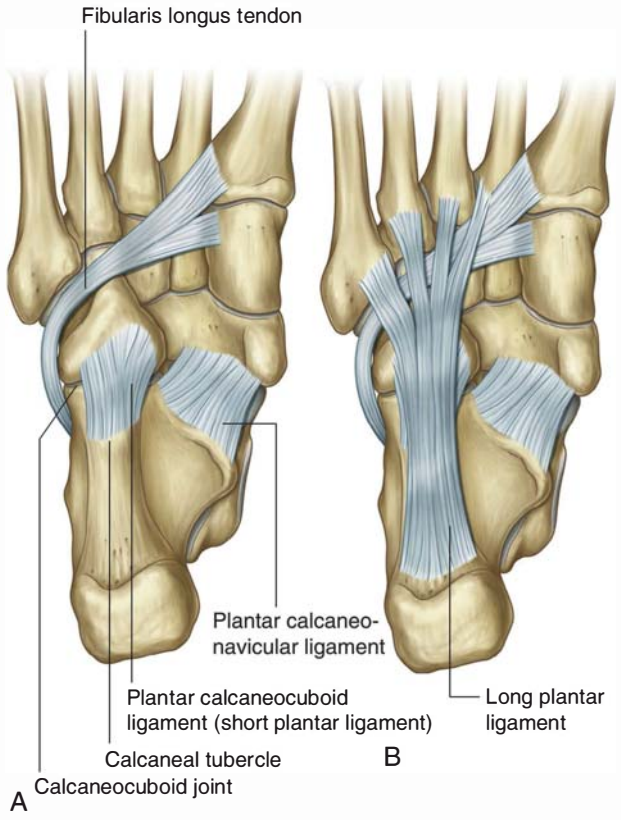


Fig. 6.103 Plantar ligaments. **A.** Plantar calcaneocuboid ligament (short plantar ligament). **B.** Long plantar ligament.

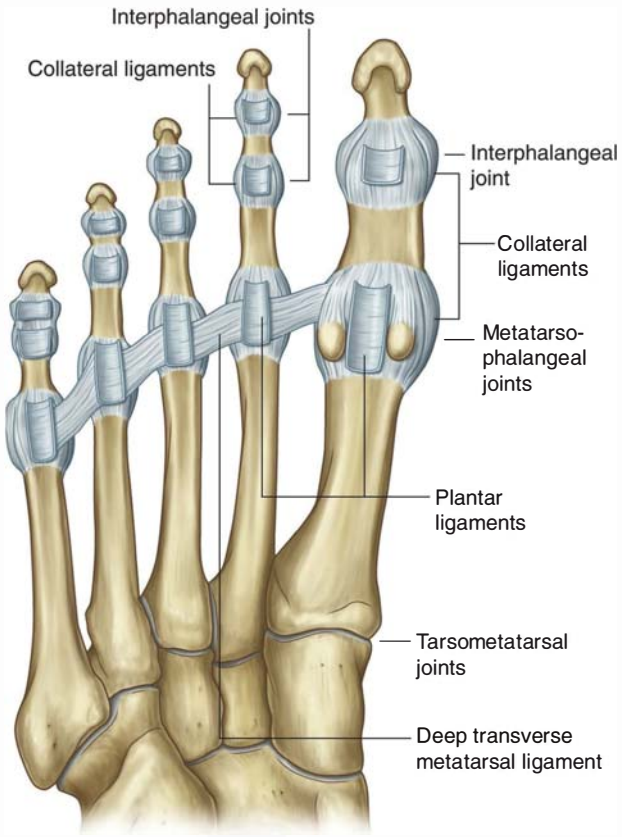


Fig. 6.104 Tarsometatarsal, metatarsophalangeal, and interphalangeal joints, and the deep transverse metatarsal ligaments.

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The range of movement of the tarsometatarsal joint between the metatarsal of the great toe and the medial cuneiform is greater than that of the other tarsometatarsal joints and allows flexion, extension, and rotation. The tarsometatarsal joints, with the transverse tarsal joint, take part in pronation and supination of the foot.

Metatarsophalangeal joints

The metatarsophalangeal joints are ellipsoid synovial joints between the sphere-shaped heads of the metatarsals and the corresponding bases of the proximal phalanges of the digits.

The metatarsophalangeal joints allow extension and flexion, and limited abduction, adduction, rotation, and circumduction.

The joint capsules are reinforced by medial and lateral **collateral ligaments**, and by **plantar ligaments**, which have grooves on their plantar surfaces for the long tendons of the digits (Fig. 6.104).

Deep transverse metatarsal ligaments

Four **deep transverse metatarsal ligaments** link the heads of the metatarsals together and enable the metatarsals to act as a single unified structure (Fig. 6.104). The ligaments blend with the plantar ligaments of the adjacent metatarsophalangeal joints.

The metatarsal of the great toe is oriented in the same plane as the metatarsals of the other toes and is linked to the metatarsal of the second toe by a deep transverse metatarsal ligament. In addition, the joint between the metatarsal of the great toe and medial cuneiform has a limited range of motion. The great toe therefore has a very restricted independent function—unlike the thumb in the hand, where the metacarpal is oriented 90° to the metacarpals of the fingers, there is no deep transverse metacarpal ligament between the metacarpals of the thumb and index finger, and the joint between the metacarpal and carpal bone allows a wide range of motion.

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In the clinic

Bunions

A bunion occurs on the medial aspect of the first metatarsophalangeal joint. This is an extremely important area of the foot because it is crossed by tendons and ligaments, which transmit and distribute the body's weight during movement. It is postulated that abnormal stresses in this region of the joint may produce the bunion deformity.

Clinically, a bunion is a significant protuberance of bone that may include soft tissue around the medial aspect of the first metatarsophalangeal joint. As it progresses, the toe appears to move toward the smaller toes, producing crowding of the digits.

This deformity tends to occur among people who wear high-heeled or pointed shoes, but osteoporosis and a hereditary predisposition are also risk factors.

Typically the patient's symptoms are pain, swelling, and inflammation. The bunion tends to enlarge and may cause problems in obtaining appropriate footwear.

Initial treatment is by adding padding to shoes, changing the type of footwear used, and taking anti-inflammatory drugs. Some patients may need surgery to correct the deformity and realign the toe.

Interphalangeal joints

The interphalangeal joints are hinge joints that allow mainly flexion and extension. They are reinforced by medial and lateral **collateral ligaments** and by **plantar ligaments** (Fig. 6.104).

Tarsal tunnel, retinacula, and arrangement of major structures at the ankle

The tarsal tunnel is formed on the posteromedial side of the ankle by:

- a depression formed by the medial malleolus of the tibia, the medial and posterior surfaces of the talus, the medial surface of the calcaneus, and the inferior surface of the sustentaculum tali of the calcaneus; and
- an overlying flexor retinaculum (Fig. 6.105).

Flexor retinaculum

The flexor retinaculum is a strap-like layer of connective tissue that spans the bony depression formed by the medial malleolus, the medial and posterior surfaces of the talus, the medial surface of the calcaneus, and the inferior surface of the sustentaculum tali (Fig. 6.105). It attaches above to the medial malleolus and below and behind to the inferomedial margin of the calcaneus.

The retinaculum is continuous above with the deep fascia of the leg and below with the deep fascia (plantar aponeurosis) of the foot.

Septa from the flexor retinaculum convert grooves on the bones into tubular connective tissue channels for the tendons of the flexor muscles as they pass into the sole of the foot from the posterior compartment of the leg (Fig. 6.105). Free movement of the tendons in the channels is facilitated by synovial sheaths, which surround the tendons.

Two compartments on the posterior surface of the medial malleolus are for the tendons of the tibialis posterior and flexor digitorum longus muscles. The tendon of the tibialis posterior is medial to the tendon of the flexor digitorum longus.

Immediately lateral to the tendons of the tibialis posterior and flexor digitorum longus, the posterior tibial artery with its associated veins and the tibial nerve pass through the tarsal tunnel into the sole of the foot. The pulse of the posterior tibial artery can be felt through the flexor retinaculum midway between the medial malleolus and the calcaneus.

Lateral to the tibial nerve is the compartment on the posterior surface of the talus and the undersurface of the sustentaculum tali for the tendon of the flexor hallucis longus muscle.

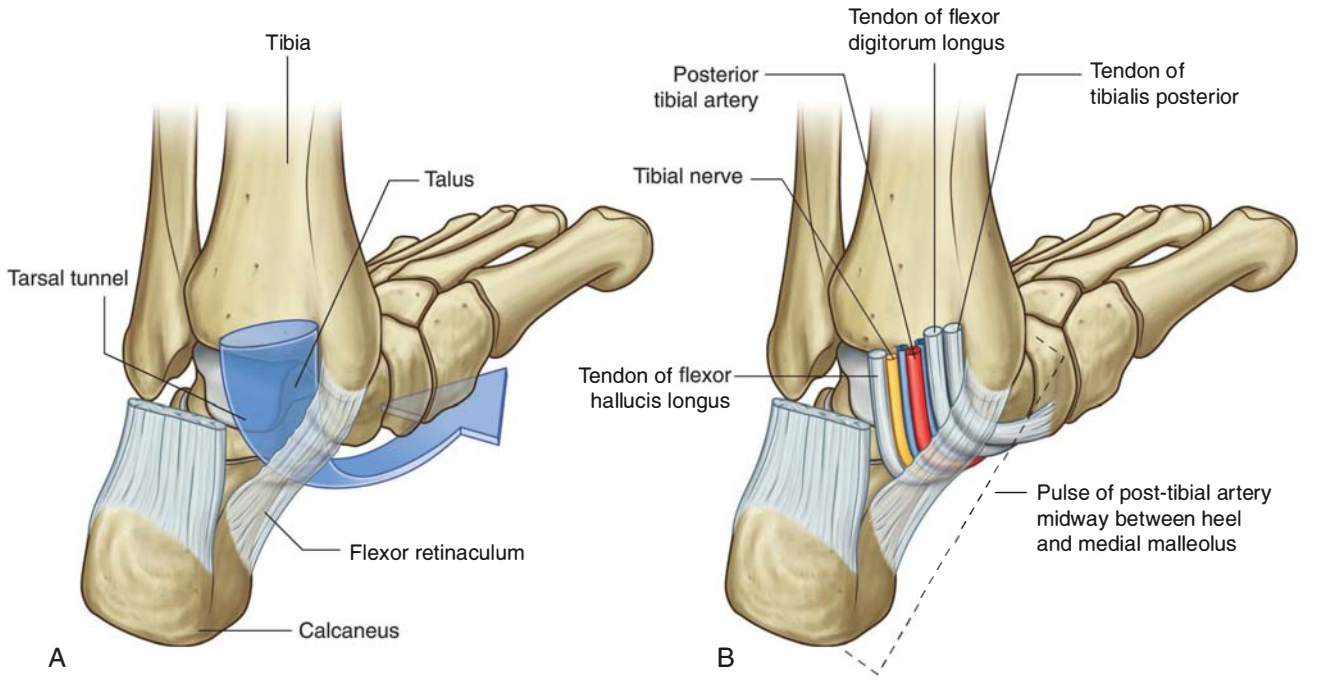


Fig. 6.105 Tarsal tunnel and flexor retinaculum. Posteromedial view. **A.** Bones. **B.** Tarsal tunnel and flexor retinaculum.

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Extensor retinacula

Two extensor retinacula strap the tendons of the extensor muscles to the ankle region and prevent tendon bowing during extension of the foot and toes (Fig. 6.106):

- A **superior extensor retinaculum** is a thickening of deep fascia in the distal leg just superior to the ankle joint and attached to the anterior borders of the fibula and tibia.
- An **inferior retinaculum** is Y-shaped, attached by its base to the lateral side of the upper surface of the calcaneus, and crosses medially over the foot to attach by one of its arms to the medial malleolus, whereas the other arm wraps medially around the foot and attaches to the medial side of the plantar aponeurosis.

The tendons of the extensor digitorum longus and fibularis tertius pass through a compartment on the lateral side of the proximal foot. Medial to these tendons, the dorsalis pedis artery (terminal branch of the anterior tibial artery), the tendon of the extensor hallucis longus muscle, and finally the tendon of the tibialis anterior muscle pass under the extensor retinacula.

Fibular retinacula

Fibular (peroneal) retinacula bind the tendons of the fibularis longus and fibularis brevis muscles to the lateral side of the foot (Fig. 6.107):

- **A superior fibular retinaculum** extends between the lateral malleolus and the calcaneus.
- **An inferior fibular retinaculum** attaches to the lateral surface of the calcaneus around the fibular trochlea and blends above with the fibers of the inferior extensor retinaculum.

At the fibular trochlea, a septum separates the compartment for the tendon of the fibularis brevis muscle above from that for the fibularis longus below.

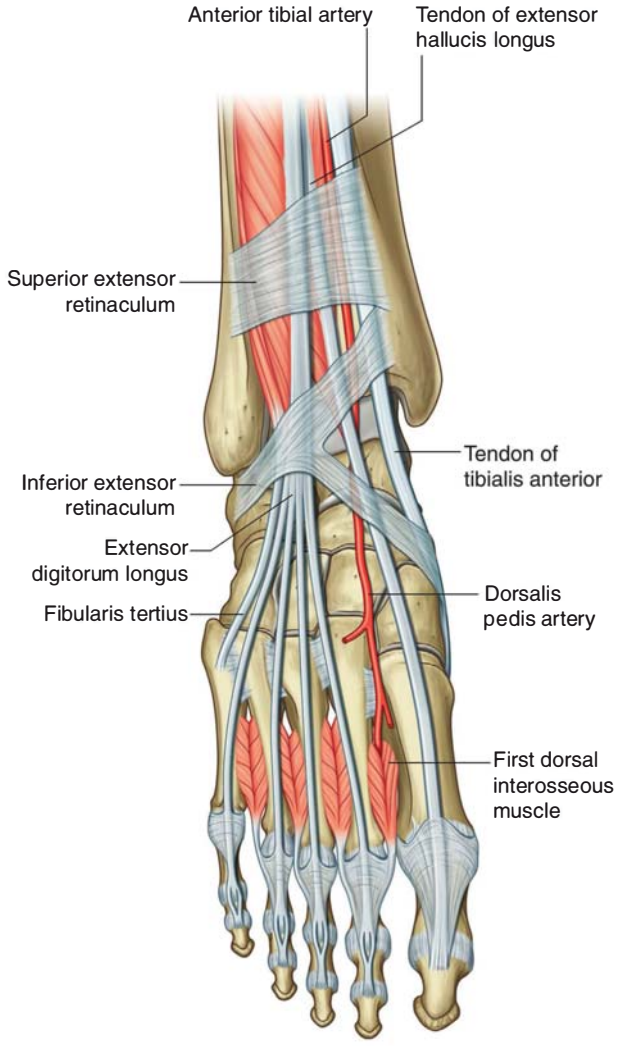


Fig. 6.106 Extensor retinacula.

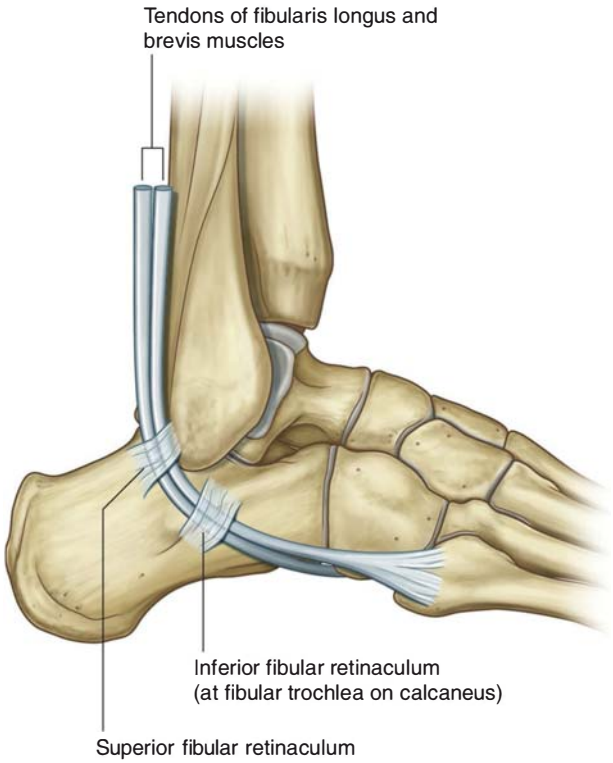


Fig. 6.107 Fibular retinacula. Lateral view, right foot.

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Arches of the foot

The bones of the foot do not lie in a horizontal plane. Instead, they form longitudinal and transverse arches relative to the ground (Fig. 6.108), which absorb and distribute downward forces from the body during standing and moving on different surfaces.

Longitudinal arch

The longitudinal arch of the foot is formed between the posterior end of the calcaneus and the heads of the metatarsals (Fig. 6.108A). It is highest on the medial side, where it forms the medial part of the longitudinal arch, and lowest on the lateral side, where it forms the lateral part.

Transverse arch

The transverse arch of the foot is highest in a coronal plane that cuts through the head of the talus and disappears near the heads of the metatarsals, where these bones are held together by the deep transverse metatarsal ligaments (Fig. 6.108B).

Ligament and muscle support

Ligaments and muscles support the arches of the foot (Fig. 6.109):

- Ligaments that support the arches include the plantar calcaneonavicular (spring ligament), plantar calcaneocuboid (short plantar ligament), and long plantar ligaments, and the plantar aponeurosis.

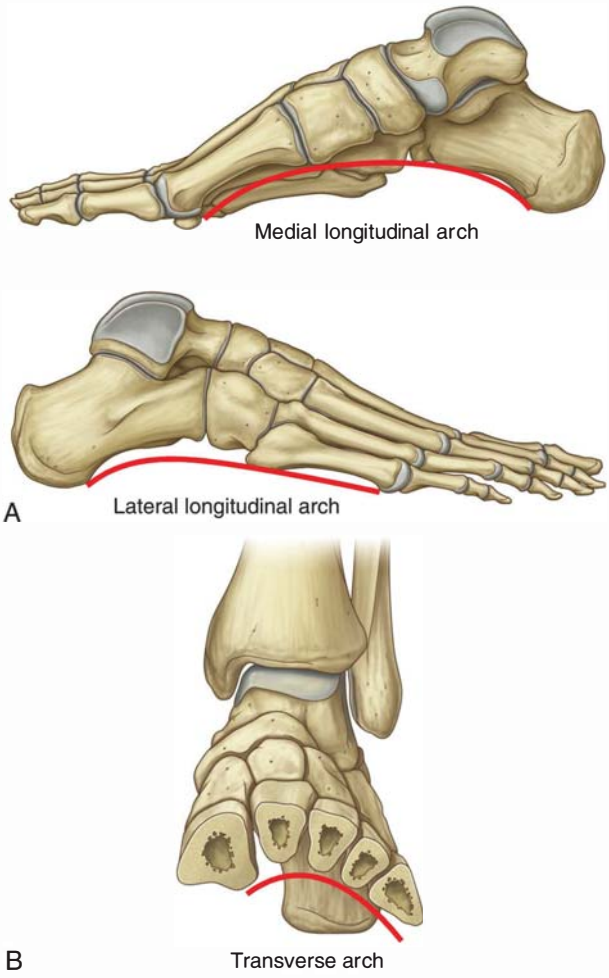


Fig. 6.108 Arches of the foot. **A.** Longitudinal arches, right foot. **B.** Transverse arch, left foot.

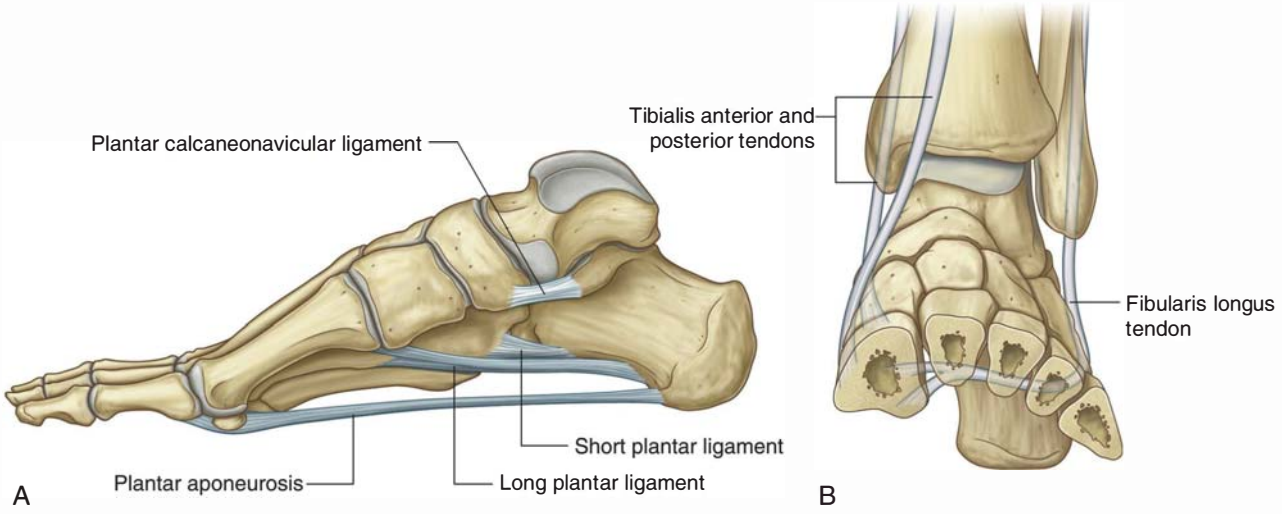


Fig. 6.109 Support for arches of the foot. **A.** Ligaments. Medial view, right foot. **B.** Cross section through the foot to show tendons of muscles supporting the arches.

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- Muscles that provide dynamic support for the arches during walking include the tibialis anterior and posterior and the fibularis longus.

Plantar aponeurosis

The plantar aponeurosis is a thickening of deep fascia in the sole of the foot (Fig. 6.110). It is firmly anchored to the medial process of the calcaneal tuberosity and extends forward as a thick band of longitudinally arranged connective tissue fibers. The fibers diverge as they pass anteriorly and form digital bands, which enter the toes and connect with bones, ligaments, and dermis of the skin.

Distal to the metatarsophalangeal joints, the digital bands of the plantar aponeurosis are interconnected by transverse fibers, which form superficial transverse metatarsal ligaments.

The plantar aponeurosis supports the longitudinal arch of the foot and protects deeper structures in the sole.

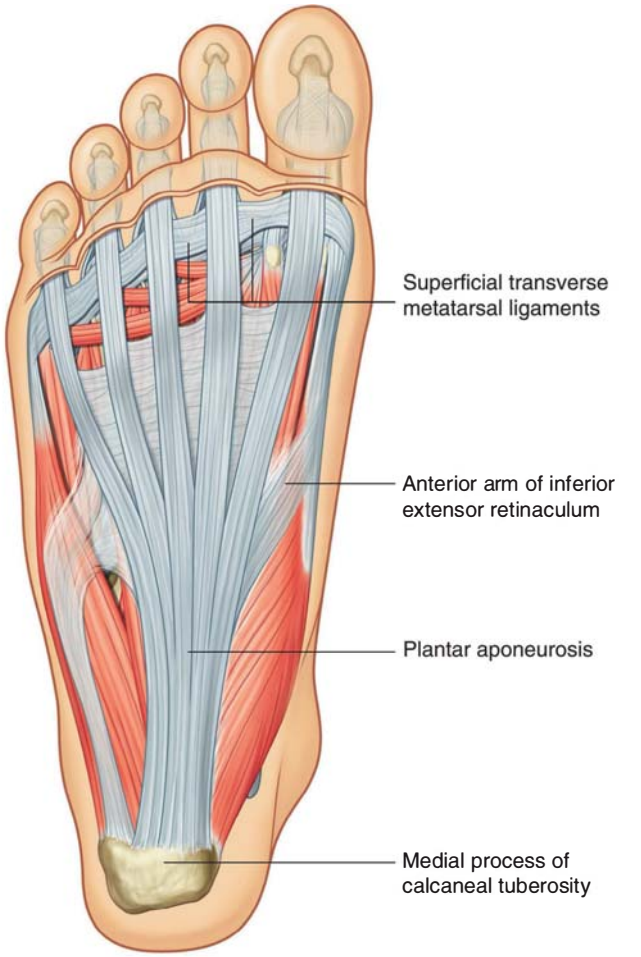


Fig. 6.110 Plantar aponeurosis.

Fibrous sheaths of toes

The tendons of the flexor digitorum longus, flexor digitorum brevis, and flexor hallucis longus muscles enter fibrous digital sheaths or tunnels on the plantar aspect of the digits (Fig. 6.111). These fibrous sheaths begin anterior to the metatarsophalangeal joints and extend to the distal phalanges. They are formed by fibrous arches and cruciate (cross-shaped) ligaments attached posteriorly to the margins of the phalanges and to the plantar ligaments associated with the metatarsophalangeal and interphalangeal joints.

These fibrous tunnels hold the tendons to the bony plane and prevent tendon bowing when the toes are flexed.

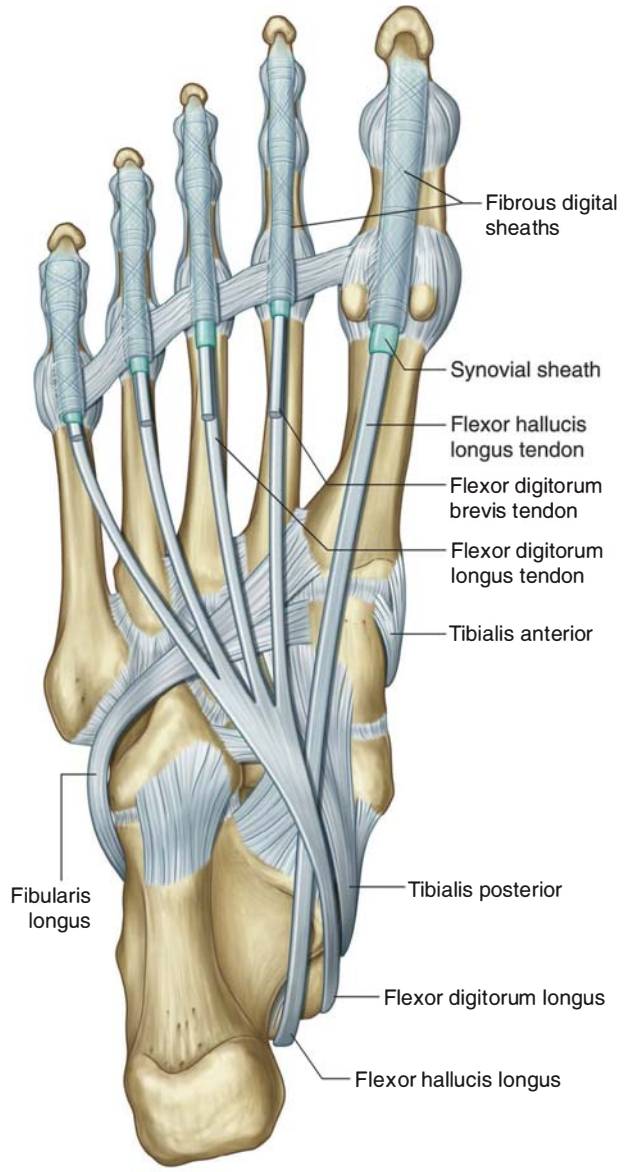


Fig. 6.111 Fibrous digital sheaths.

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Within each tunnel, the tendons are surrounded by a synovial sheath.

Extensor hoods

The tendons of the extensor digitorum longus, extensor digitorum brevis, and extensor hallucis longus pass into the dorsal aspect of the digits and expand over the proximal phalanges to form complex dorsal digital expansions ("extensor hoods") (Fig. 6.112).

Each extensor hood is triangular in shape with the apex attached to the distal phalanx, the central region attached to the middle (toes II to V) or proximal (toe I) phalanx, and each corner of the base wrapped around the sides of the metatarsophalangeal joint. The corners of the hoods attach mainly to the deep transverse metatarsal ligaments.

Many of the intrinsic muscles of the foot insert into the free margin of the hood on each side. The attachment of these muscles into the extensor hoods allows the forces from these muscles to be distributed over the toes to cause flexion of the metatarsophalangeal joints while at the same time extending the interphalangeal joints (Fig. 6.112). The function of these movements in the foot is uncertain, but they may prevent overextension of the metatarsophalangeal joints and flexion of the interphalangeal joints when the heel is elevated off the ground and the toes grip the ground during walking.

Intrinsic muscles

Intrinsic muscles of the foot originate and insert in the foot:

- the extensor digitorum brevis and extensor hallucis brevis on the dorsal aspect of the foot;
- all other intrinsic muscles—the dorsal and plantar interossei, flexor digiti minimi brevis, flexor hallucis brevis, flexor digitorum brevis, quadratus plantae (flexor accessorius), abductor digiti minimi, abductor hallucis, and lumbricals—are on the plantar side of the foot in the sole where they are organized into four layers.

Intrinsic muscles mainly modify the actions of the long tendons and generate fine movements of the toes.

All intrinsic muscles of the foot are innervated by the medial and lateral plantar branches of the tibial nerve except for the extensor digitorum brevis, which is innervated by the deep fibular nerve. The first two dorsal interossei also may receive part of their innervation from the deep fibular nerve.

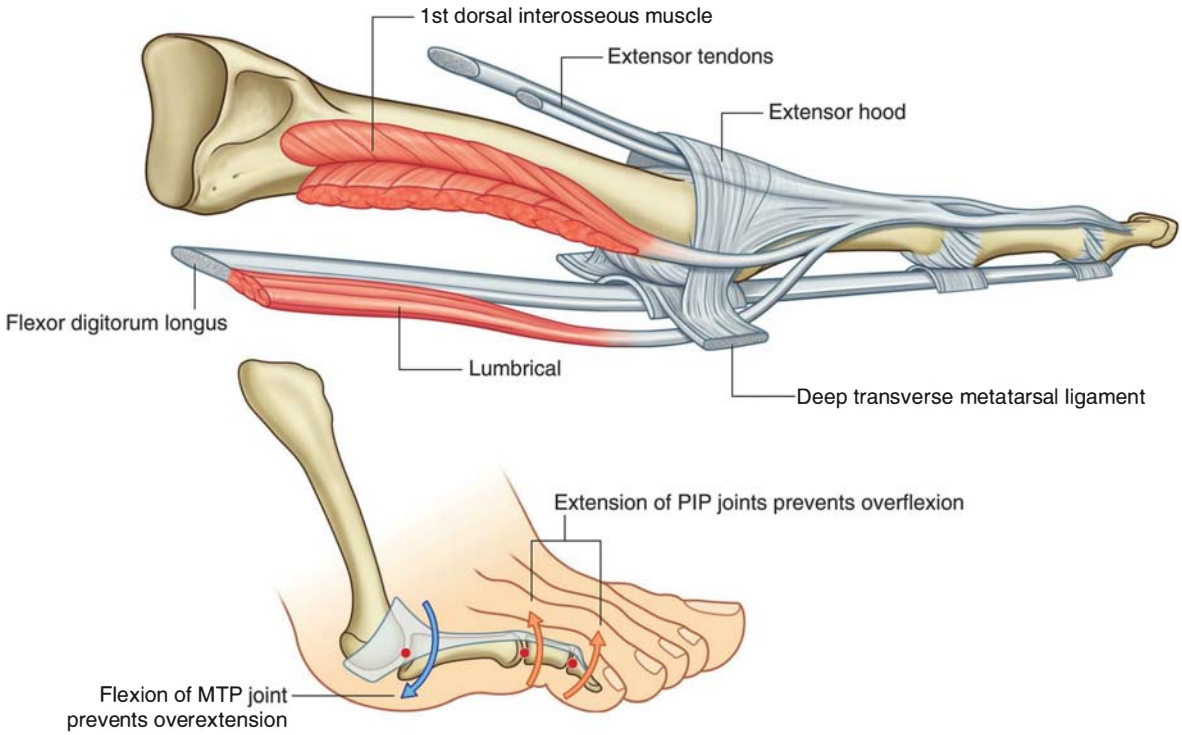


Fig. 6.112 Extensor hoods.

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On the dorsal aspect

Extensor digitorum brevis and extensor hallucis brevis

The **extensor digitorum brevis** is attached to a roughened area on the superolateral surface of the calcaneus lateral to the tarsal sinus (Fig. 6.113 and Table 6.10).

The flat muscle belly passes anteromedially over the foot, deep to the tendons of the extensor digitorum longus, and forms three tendons, which enter digits II, III, and IV. The tendons join the lateral sides of the tendons of the extensor digitorum longus. The extensor digitorum brevis extends the middle three toes through attachments to the long extensor tendons and extensor hoods. It is innervated by the deep fibular nerve.

The **extensor hallucis brevis** originates in conjunction with the extensor digitorum brevis. Its tendon attaches to the base of the proximal phalanx of the great toes. The muscle extends the metatarsophalangeal joint of the great toe and is innervated by the deep fibular nerve.

In the sole

The muscles in the sole of the foot are organized into four layers. From superficial to deep, or plantar to dorsal, these layers are the first, second, third, and fourth layers.

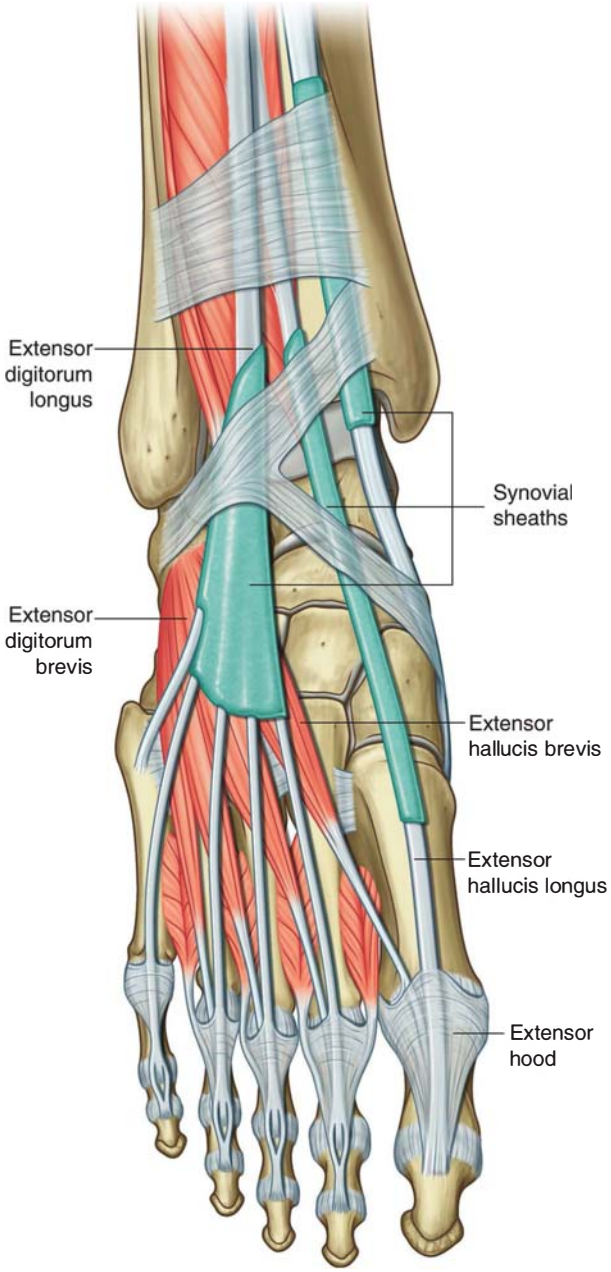


Fig. 6.113 Extensor digitorum brevis muscle.

Table 6.10 Muscles of the dorsal aspect of the foot (spinal segments in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Extensor digitorum brevis	Superolateral surface of the calcaneus	Lateral sides of the tendons of extensor digitorum longus of toes II to IV	Deep fibular nerve (**S1, S2**)	Extension of toes II to IV
Extensor hallucis brevis	Superolateral surface of calcaneus	Base of proximal phalanx of great toe	Deep fibular nerve (**S1, S2**)	Extension of metatarsophalangeal joint of great toe

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First layer

There are three components in the first layer of muscles, which is the most superficial of the four layers and is immediately deep to the plantar aponeurosis (Fig. 6.114 and Table 6.11). From medial to lateral, these muscles are the abductor hallucis, flexor digitorum brevis, and abductor digiti minimi.

Abductor hallucis

The **abductor hallucis** muscle forms the medial margin of the foot and contributes to a soft tissue bulge on the medial side of the sole (Fig. 6.114). It originates from the medial process of the calcaneal tuberosity and adjacent margins of the flexor retinaculum and plantar aponeurosis. It forms a tendon that inserts on the medial side of the base of the proximal phalanx of the great toe and on the medial sesamoid bone associated with the tendon of the flexor hallucis brevis muscle.

The abductor hallucis abducts and flexes the great toe at the metatarsophalangeal joint and is innervated by the medial plantar branch of the tibial nerve.

Flexor digitorum brevis

The **flexor digitorum brevis** muscle lies immediately superior to the plantar aponeurosis and inferior to the tendons of the flexor digitorum longus in the sole of the foot (Fig. 6.114). The flat spindle-shaped muscle belly originates as a tendon from the medial process of the calcaneal tuberosity and from the adjacent plantar aponeurosis.

The muscle fibers of the flexor digitorum brevis converge anteriorly to form four tendons, which each enter one of the lateral four toes. Near the base of the proximal phalanx of the toe, each tendon splits to pass dorsally

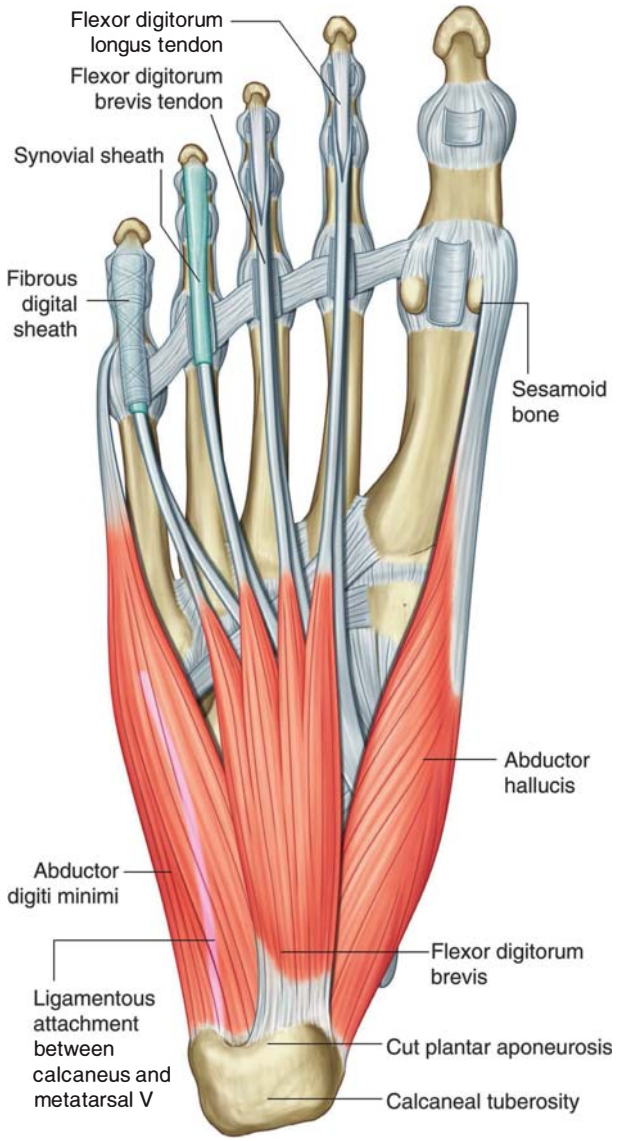


Fig. 6.114 First layer of muscles in the sole of the foot.

Table 6.11 First layer of muscles in the sole of the foot (spinal segments in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Abductor hallucis	Medial process of calcaneal tuberosity	Medial side of base of proximal phalanx of great toe	Medial plantar nerve from the tibial nerve (**S1, S2, S3**)	Abducts and flexes great toe at metatarsophalangeal joint
Flexor digitorum brevis	Medial process of calcaneal tuberosity and plantar aponeurosis	Sides of plantar surface of middle phalanges of lateral four toes	Medial plantar nerve from the tibial nerve (**S1, S2, S3**)	Flexes lateral four toes at proximal interphalangeal joint
Abductor digiti minimi	Lateral and medial processes of calcaneal tuberosity, and band of connective tissue connecting calcaneus with base of metatarsal V	Lateral side of base of proximal phalanx of little toe	Lateral plantar nerve from the tibial nerve (**S1, S2, S3**)	Abducts little toe at the metatarsophalangeal joint

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around each side of the tendon of the flexor digitorum longus and attach to the margins of the middle phalanx.

The flexor digitorum brevis flexes the lateral four toes at the proximal interphalangeal joints and is innervated by the medial plantar branch of the tibial nerve.

Abductor digiti minimi

The **abductor digiti minimi** muscle is on the lateral side of the foot and contributes to the large lateral plantar eminence on the sole (Fig. 6.114). It has a broad base of origin, mainly from the lateral and medial processes of the calcaneal tuberosity and from a fibrous band of connective tissue, which connects the calcaneus with the base of metatarsal V.

The abductor digiti minimi forms a tendon, which travels in a shallow groove on the plantar surface of the base of metatarsal V and continues forward to attach to the lateral side of the base of the proximal phalanx of the little toe.

The abductor digiti minimi abducts the little toe at the metatarsophalangeal joint and is innervated by the lateral plantar branch of the tibial nerve.

Second layer

The second muscle layer in the sole of the foot is associated with the tendons of the flexor digitorum longus muscle, which pass through this layer, and consists of the quadratus plantae and four lumbrical muscles (Fig. 6.115 and Table 6.12).

Quadratus plantae

The **quadratus plantae** muscle is a flat quadrangular muscle with two heads of origin (Fig. 6.115):

- One of the heads originates from the medial surface of the calcaneus inferior to the sustentaculum tali.

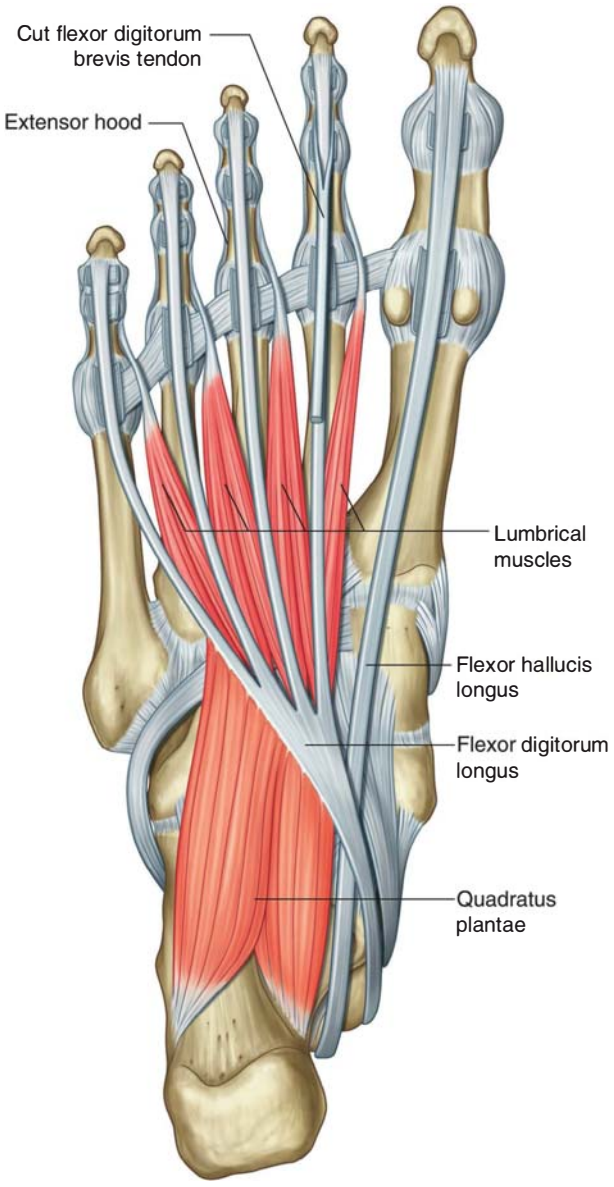


Fig. 6.115 Second layer of muscles in the sole of the foot.

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- The other head originates from the inferior surface of the calcaneus anterior to the lateral process of the calcaneal tuberosity and the attachment of the long plantar ligament.

The quadratus plantae muscle inserts into the lateral side of the tendon of the flexor digitorum longus in the proximal half of the sole of the foot near where the tendon divides.

The quadratus plantae assists the flexor digitorum longus tendon in flexing the toes and may also adjust the "line of pull" of this tendon as it enters the sole of the foot from the medial side. The muscle is innervated by the lateral plantar nerve.

Lumbricals

The lumbrical muscles are four worm-like muscles that originate from the tendons of the flexor digitorum longus and pass dorsally to insert into the free medial margins of the extensor hoods of the four lateral toes (Fig. 6.115).

The first lumbrical originates from the medial side of the tendon of the flexor digitorum longus that is associated with the second toe. The remaining three muscles are bipennate and originate from the sides of adjacent tendons.

The lumbrical muscles act through the extensor hoods to resist excessive extension of the metatarsophalangeal joints and flexion of the interphalangeal joints when the heel leaves the ground during walking.

The first lumbrical is innervated by the medial plantar nerve, while the other three are innervated by the lateral plantar nerve.

Third layer

There are three muscles in the third layer in the sole of the foot (Fig. 6.116 and Table 6.13):

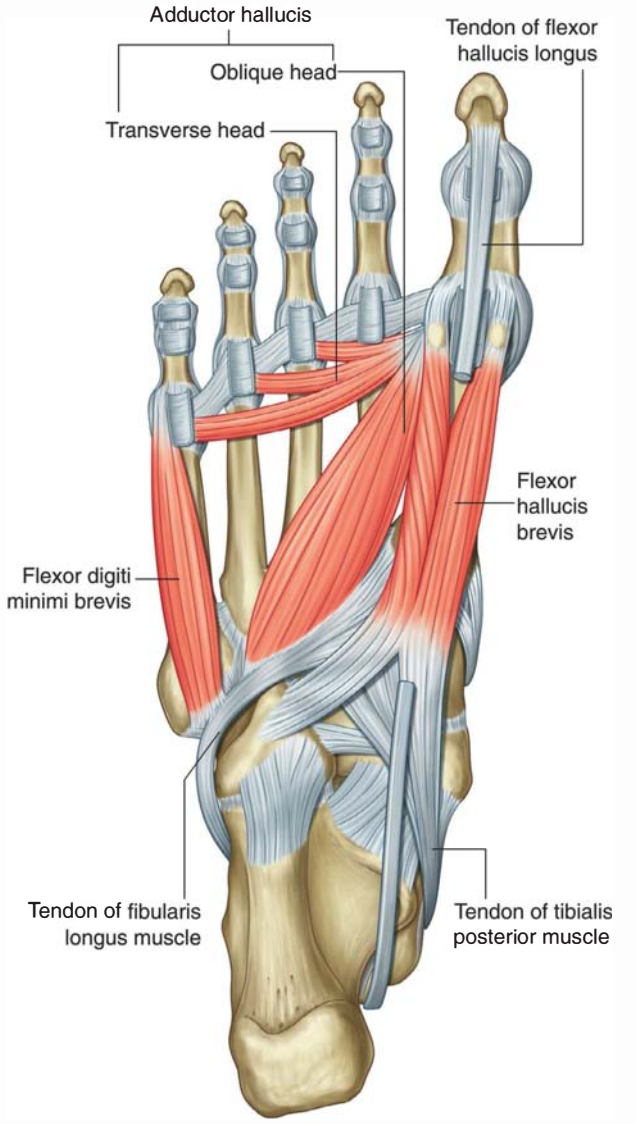


Fig. 6.116 Third layer of muscles in the sole of the foot.

Table 6.12 Second layer of muscles in the sole of the foot (spinal segments in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Quadratus plantae	Medial surface of calcaneus and lateral process of calcaneal tuberosity	Lateral side of tendon of flexor digitorum longus in proximal sole of the foot	Lateral plantar nerve from tibial nerve (**S1, S2, S3**)	Assists flexor digitorum longus tendon in flexing toes II to V
Lumbricals	First lumbrical—medial side of tendon of flexor digitorum longus associated with toe II; second, third, and fourth lumbricals—adjacent surfaces of adjacent tendons of flexor digitorum longus	Medial free margins of extensor hoods of toes II to V	First lumbrical—medial plantar nerve from the tibial nerve; second, third, and fourth lumbricals—lateral plantar nerve from the tibial nerve (**S2, S3**)	Flexion of metatarsophalangeal joint and extension of interphalangeal joints

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- Two (the flexor hallucis brevis and adductor hallucis) are associated with the great toe.
- The third (the flexor digiti minimi brevis) is associated with the little toe.

Flexor hallucis brevis

The **flexor hallucis brevis** muscle has two tendinous heads of origin (Fig. 6.116):

- The **lateral head** originates from the plantar surfaces of the cuboid, behind the groove for the fibularis longus, and adjacent surface of the lateral cuneiform.
- The **medial head** originates from the tendon of the tibialis posterior muscle as it passes into the sole of the foot.

The medial and lateral heads unite and give rise to a muscle belly, which itself is separated into medial and lateral parts adjacent to the plantar surface of metatarsal I. Each part of the muscle gives rise to a tendon that inserts on either the lateral or medial side of the base of the proximal phalanx of the great toe.

A sesamoid bone occurs in each tendon of the flexor hallucis brevis as it crosses the plantar surface of the head of metatarsal I. The tendon of the flexor hallucis longus passes between the sesamoid bones.

The flexor hallucis brevis flexes the metatarsophalangeal joint of the great toe and is innervated by the medial plantar nerve.

Adductor hallucis

The **adductor hallucis** muscle originates by two muscular heads, transverse and oblique, which join near their ends to insert into the lateral side of the base of the proximal phalanx of the great toe (Fig. 6.116):

- The **transverse head** originates from the plantar ligaments associated with the metatarsophalangeal joints of the lateral three toes and from the associated deep transverse metatarsal ligaments—the muscle crosses the sole of the foot transversely from lateral to medial and joins the oblique head near the base of the great toe.
- The **oblique head** is larger than the transverse head and originates from the plantar surfaces of the bases of metatarsals II to IV and from the sheath covering the fibularis longus muscle—this head passes anterolaterally through the sole of the foot and joins the transverse head.

The tendon of insertion of the adductor hallucis attaches to the lateral sesamoid bone associated with the tendon of the flexor hallucis brevis muscle in addition to attaching to the proximal phalanx.

The adductor hallucis adducts the great toe at the metatarsophalangeal joint and is innervated by the lateral plantar nerve.

Flexor digiti minimi brevis

The **flexor digiti minimi brevis** muscle originates from the plantar surface of the base of metatarsal V and adjacent sheath of the fibularis longus tendon (Fig. 6.116). It inserts on the lateral side of the base of the proximal phalanx of the little toe.

The flexor digiti minimi brevis flexes the little toe at the metatarsophalangeal joint and is innervated by the lateral plantar nerve.

Table 6.13 Third layer of muscles in the sole of the foot (spinal segments in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Flexor hallucis brevis	Plantar surface of cuboid and lateral cuneiform; tendon of tibialis posterior	Lateral and medial sides of base of proximal phalanx of the great toe	Medial plantar nerve from tibial nerve (**S1**, S2)	Flexes metatarsophalangeal joint of the great toe
Adductor hallucis	Transverse head—ligaments associated with metatarsophalangeal joints of lateral three toes; oblique head—bases of metatarsals II to IV and from sheath covering fibularis longus	Lateral side of base of proximal phalanx of great toe	Lateral plantar nerve from tibial nerve (**S2, S3**)	Adducts great toe at metatarsophalangeal joint
Flexor digiti minimi brevis	Base of metatarsal V and related sheath of fibularis longus tendon	Lateral side of base of proximal phalanx of little toe	Lateral plantar nerve from tibial nerve (**S2, S3**)	Flexes little toe at metatarsophalangeal joint

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Fourth layer

There are two muscle groups in the deepest muscle layer in the sole of the foot, the dorsal and plantar interossei (Fig. 6.117 and Table 6.14).

Dorsal interossei

The four **dorsal interossei** are the most superior muscles in the sole of the foot and abduct the second to fourth toes relative to the long axis through the second toe (Fig. 6.117). All four muscles are bipennate and originate from the sides of adjacent metatarsals.

The tendons of the dorsal interossei insert into the free margin of the extensor hoods and base of the proximal phalanges of the toes.

The second toe can be abducted to either side of its long axis, so it has two dorsal interossei associated with it, one on each side. The third and fourth toes have a dorsal interosseous muscle on their lateral sides only. The great and little toes have their own abductors (the abductor hallucis and abductor digiti minimi) in the first layer of muscles in the sole of the foot.

In addition to abduction, the dorsal interossei act through the extensor hoods to resist extension of the metatarsophalangeal joints and flexion of the interphalangeal joints.

The dorsal interossei are innervated by the lateral plantar nerve. The first and second dorsal interossei also receive branches on their superior surfaces from the deep fibular nerve.

Plantar interossei

The three plantar interossei adduct the third, fourth, and little toes toward the long axis through the second toe (Fig. 6.117).

Each plantar interosseous muscle originates from the medial side of its associated metatarsal and inserts into

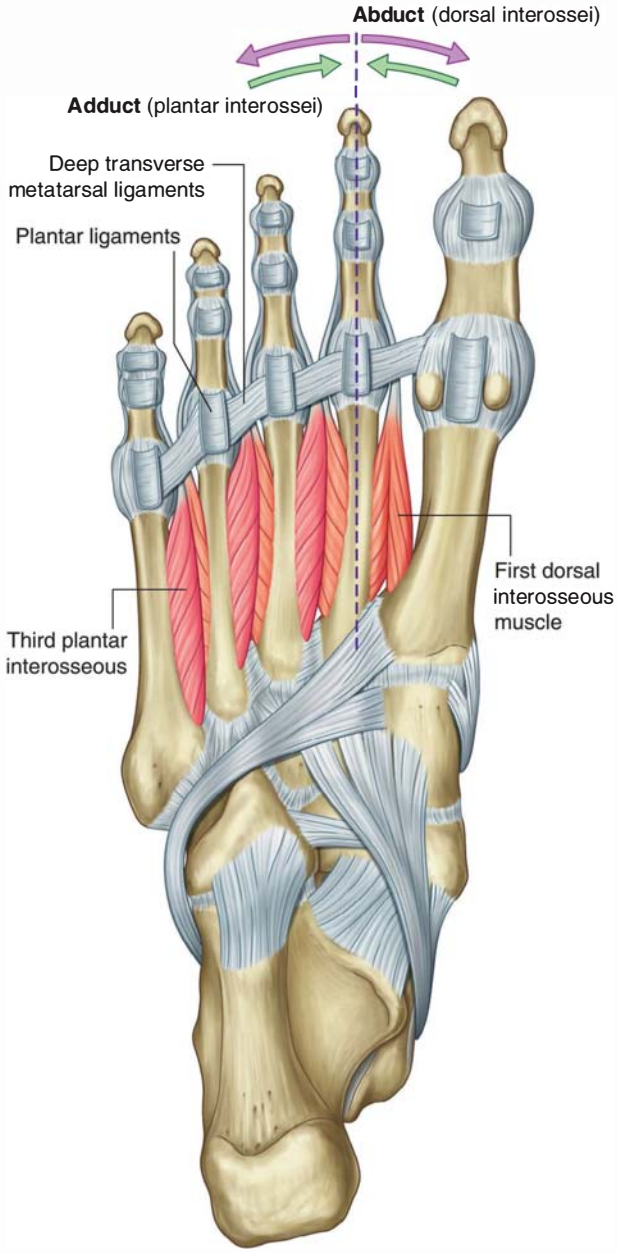


Fig. 6.117 Fourth layer of muscles in the sole of the foot.

Table 6.14 Fourth layer of muscles in the sole of the foot (spinal segments in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Dorsal interossei	Sides of adjacent metatarsals	Extensor hoods and bases of proximal phalanges of toes II to IV	Lateral plantar nerve from tibial nerve; first and second dorsal interossei also innervated by deep fibular nerve (**S2, S3**)	Abduction of toes II to IV at metatarsophalangeal joints; resist extension of metatarsophalangeal joints and flexion of interphalangeal joints
Plantar interossei	Medial sides of metatarsals of toes III to V	Extensor hoods and bases of proximal phalanges of toes III to V	Lateral plantar nerve from tibial nerve (**S2, S3**)	Adduction of toes III to V at metatarsophalangeal joints; resist extension of the metatarsophalangeal joints and flexion of the interphalangeal joints

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the medial free margin of the extensor hood and base of the proximal phalanx.

The great toe has its own adductor (the adductor hallucis) in the third layer of muscles in the sole of the foot and the second toe is adducted back to its longitudinal axis by using one of its dorsal interossei.

In addition to adduction, the plantar interossei act through the extensor hoods to resist extension of the metatarsophalangeal joints and flexion of the interphalangeal joints. All are innervated by the lateral plantar nerve.

Arteries

Blood supply to the foot is by branches of the posterior tibial and dorsalis pedis (dorsal artery of the foot) arteries.

The posterior tibial artery enters the sole and bifurcates into lateral and medial plantar arteries. The lateral plantar artery joins with the terminal end of the dorsalis pedis artery (the deep plantar artery) to form the deep plantar arch. Branches from this arch supply the toes.

The dorsalis pedis artery is the continuation of the anterior tibial artery, passes onto the dorsal aspect of the foot and then inferiorly, as the deep plantar artery, between metatarsals I and II to enter the sole of the foot.

Posterior tibial artery and plantar arch

The posterior tibial artery enters the foot through the tarsal tunnel on the medial side of the ankle and posterior to the medial malleolus. Midway between the medial malleolus and the heel, the pulse of the posterior tibial artery is palpable because here the artery is covered only by a thin layer of retinaculum, by superficial connective tissue, and by skin. Near this location, the posterior tibial artery bifurcates into a small medial plantar artery and a much larger lateral plantar artery.

Lateral plantar artery

The **lateral plantar artery** passes anterolaterally into the sole of the foot, first deep to the proximal end of the abductor hallucis muscle and then between the quadratus plantae and flexor digitorum brevis muscles (Fig. 6.118). It reaches the base of metatarsal V where it lies in the groove between the flexor digitorum brevis and abductor digiti minimi muscles. From here, the lateral plantar artery curves medially to form the **deep plantar arch**, which crosses the deep plane of the sole on the metatarsal bases and the interossei muscles.

Between the bases of metatarsals I and II, the deep plantar arch joins with the terminal branch (deep plantar artery) of the dorsalis pedis artery, which enters the sole from the dorsal side of the foot.

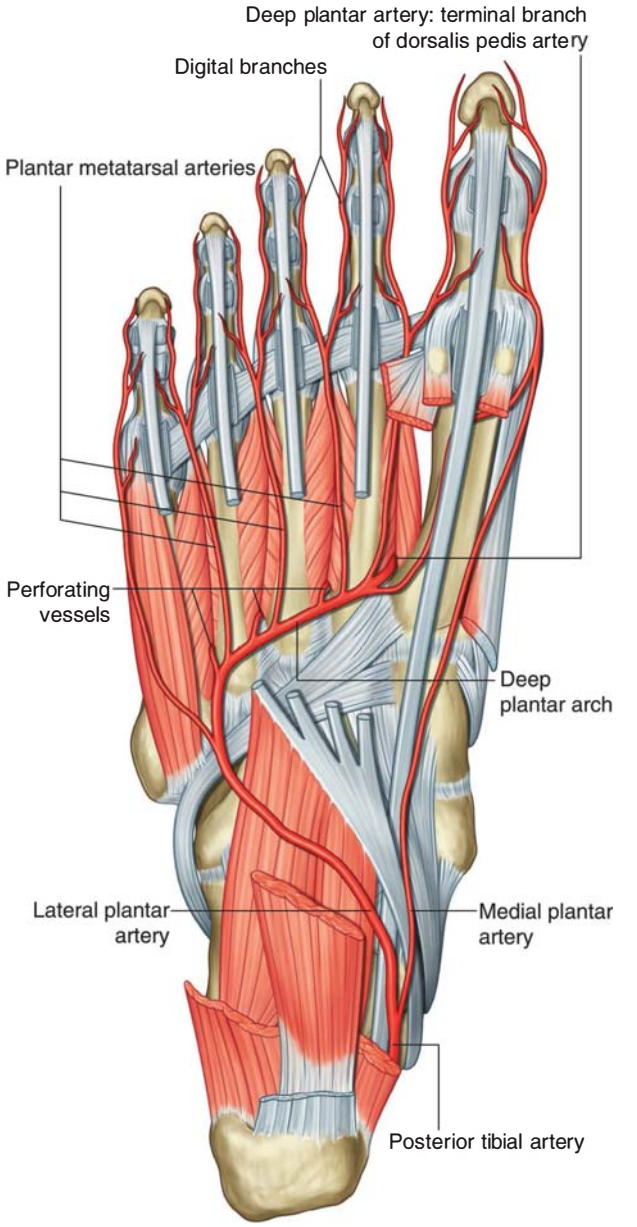


Fig. 6.118 Arteries in the sole of the foot.

Major branches of the deep plantar arch include:

- a digital branch to the lateral side of the little toe;
- four plantar metatarsal arteries, which supply digital branches to adjacent sides of toes I to V and the medial side of the great toe; and
- three perforating arteries, which pass between the bases of metatarsals II to V to anastomose with vessels on the dorsal aspect of the foot.

Medial plantar artery

The **medial plantar artery** passes into the sole of the foot by passing deep to the proximal end of the abductor

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hallucis muscle (Fig. 6.118). It supplies a deep branch to adjacent muscles and then passes forward in the groove between the abductor hallucis and the flexor digitorum brevis muscles. It ends by joining the digital branch of the deep plantar arch, which supplies the medial side of the great toe.

Near the base of metatarsal I, the medial plantar artery gives rise to a superficial branch, which divides into three vessels that pass superficial to the flexor digitorum brevis muscle to join the plantar metatarsal arteries from the deep plantar arch.

Dorsalis pedis artery

The dorsalis pedis artery is the continuation of the anterior tibial artery and begins as the anterior tibial artery crosses the ankle joint (Fig. 6.119). It passes anteriorly over the dorsal aspect of the talus, navicular, and intermediate cuneiform bones, and then passes inferiorly, as the deep plantar artery, between the two heads of the first dorsal interosseous muscle to join the deep plantar arch in the sole of the foot. The pulse of the dorsalis pedis artery on the dorsal surface of the foot can be felt by gently palpating the vessel against the underlying tarsal bones between the tendons of the extensor hallucis longus and the extensor digitorum longus to the second toe.

Branches of the dorsalis pedis artery include lateral and medial tarsal branches, an arcuate artery, and a first dorsal metatarsal artery:

- The **tarsal arteries** pass medially and laterally over the tarsal bones, supplying adjacent structures and anastomosing with a network of vessels formed around the ankle.
- The **arcuate artery** passes laterally over the dorsal aspect of the metatarsals near their bases and gives rise to three **dorsal metatarsal arteries**, which supply **dorsal digital arteries** to adjacent sides of digits II to V, and to a dorsal digital artery that supplies the lateral side of digit V.
- The **first dorsal metatarsal artery** (the last branch of the dorsalis pedis artery before the dorsalis pedis artery continues as the deep plantar artery into the sole of the foot) supplies dorsal digital branches to adjacent sides of the great and second toes.

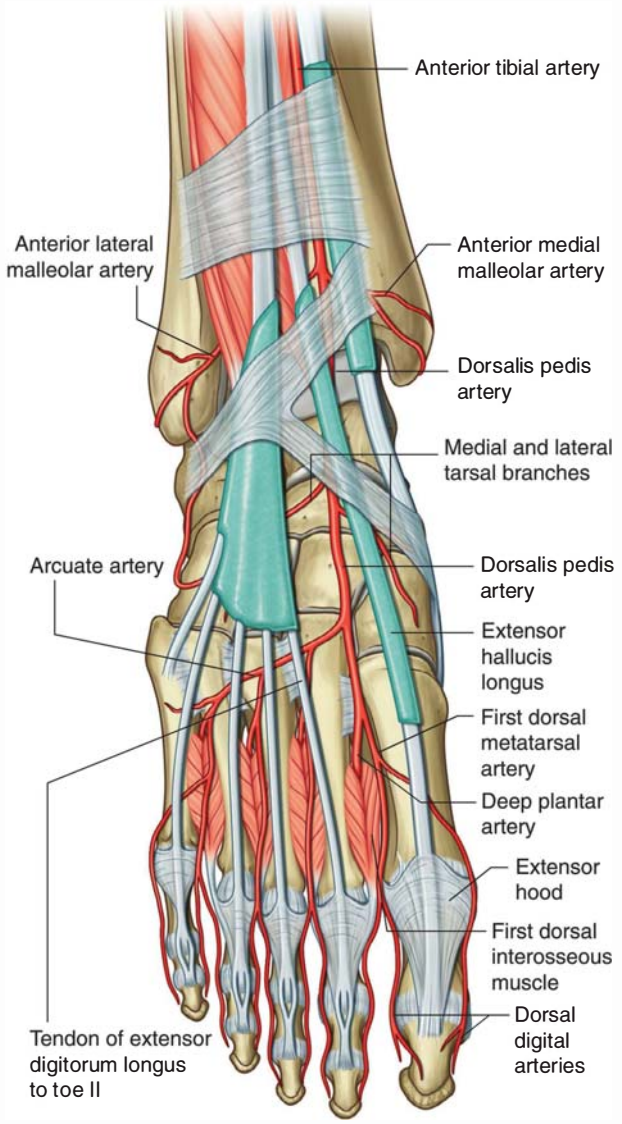


Fig. 6.119 Dorsalis pedis artery.

The dorsal metatarsal arteries connect with perforating branches from the deep plantar arch and similar branches from the plantar metatarsal arteries.

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Veins

There are interconnected networks of deep and superficial veins in the foot. The deep veins follow the arteries. Superficial veins drain into a dorsal venous arch on the dorsal surface of the foot over the metatarsals (Fig. 6.120):

- The **great saphenous vein** originates from the medial side of the arch and passes anterior to the medial malleolus and onto the medial side of the leg.
- The **small saphenous vein** originates from the lateral side of the arch and passes posterior to the lateral malleolus and onto the back of the leg.

Nerves

The foot is supplied by the tibial, deep fibular, superficial fibular, sural, and saphenous nerves:

- All five nerves contribute to cutaneous or general sensory innervation.
- The tibial nerve innervates all intrinsic muscles of the foot except for the extensor digitorum brevis, which is innervated by the deep fibular nerve.
- The deep fibular nerve often also contributes to the innervation of the first and second dorsal interossei.

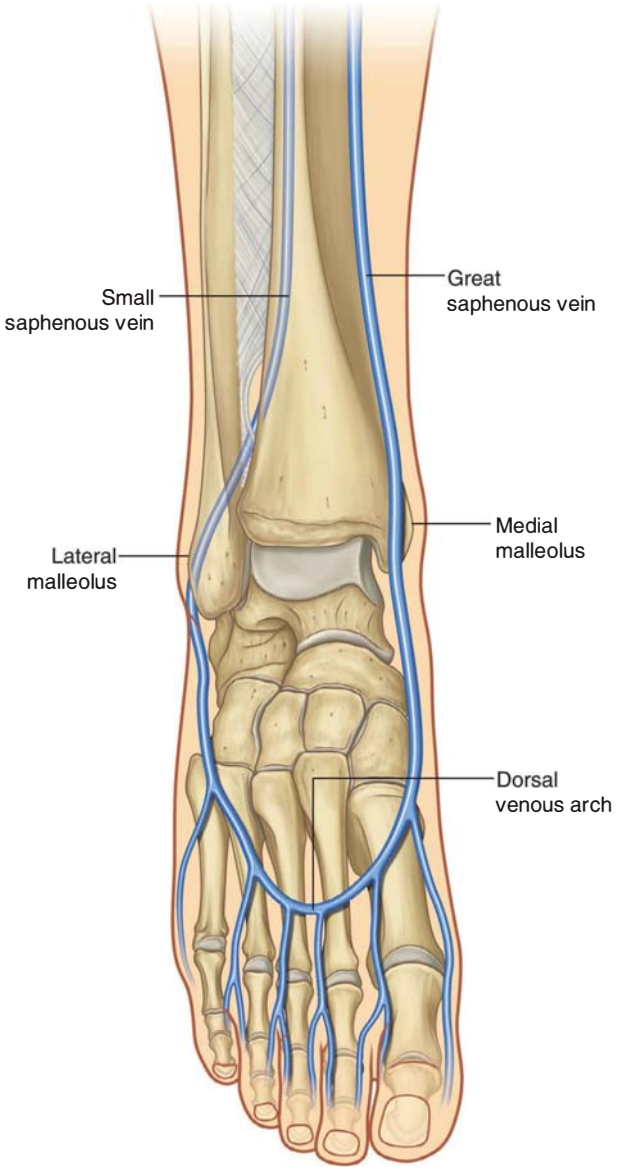


Fig. 6.120 Superficial veins of the foot.

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Tibial nerve

The **tibial nerve** enters the foot through the tarsal tunnel posterior to the medial malleolus. In the tunnel, the nerve is lateral to the posterior tibial artery, and gives origin to **medial calcaneal branches**, which penetrate the flexor retinaculum to supply the heel. Midway between the medial malleolus and the heel, the tibial nerve bifurcates with the posterior tibial artery into:

- a large medial plantar nerve, and
- a smaller lateral plantar nerve (Fig. 6.121).

The medial and lateral plantar nerves lie together between their corresponding arteries.

Medial plantar nerve

The **medial plantar nerve** is the major sensory nerve in the sole of the foot (Fig. 6.121). It innervates skin on most of the anterior two-thirds of the sole and adjacent surfaces of the medial three and one-half toes, which includes the great toe. In addition to this large area of plantar skin, the nerve also innervates four intrinsic muscles—the abductor hallucis, flexor digitorum brevis, flexor hallucis brevis, and first lumbrical.

The medial plantar nerve passes into the sole of the foot deep to the abductor hallucis muscle and forward in the groove between the abductor hallucis and flexor digitorum brevis, supplying branches to both these muscles.

The medial plantar nerve supplies a digital branch (**proper plantar digital nerve**) to the medial side of the great toe and then divides into three nerves (**common plantar digital nerves**) on the plantar surface of the flexor digitorum brevis, which continue forward to supply proper plantar digital branches to adjacent surfaces of toes I to IV. The nerve to the first lumbrical originates from the first common plantar digital nerve.

Lateral plantar nerve

The **lateral plantar nerve** is an important motor nerve in the foot because it innervates all intrinsic muscles in the sole, except for the muscles supplied by the medial plantar nerve (the abductor hallucis, flexor digitorum brevis, flexor hallucis brevis, and first lumbrical) (Fig. 6.121). It also innervates a strip of skin on the lateral side of the anterior two-thirds of the sole and the adjacent plantar surfaces of the lateral one and one-half digits.

The lateral plantar nerve enters the sole of the foot by passing deep to the proximal attachment of the abductor hallucis muscle. It continues laterally and anteriorly across the sole between the flexor digitorum brevis and quadratus plantae muscles, supplying branches to both these muscles,

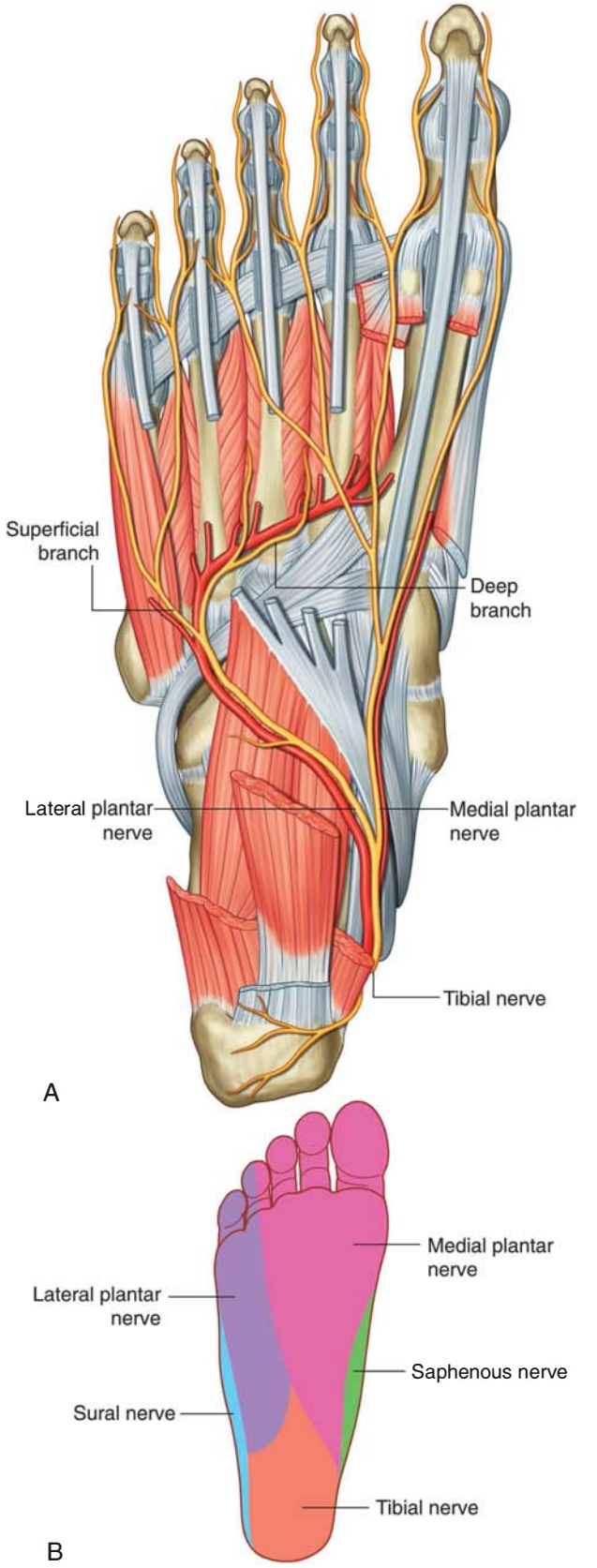


Fig. 6.121 Lateral and medial plantar nerves. **A.** Sole of the right foot. **B.** Cutaneous distribution.

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and then divides near the head of metatarsal V into deep and superficial branches.

The **superficial branch** of the lateral plantar nerve gives rise to a **proper plantar digital nerve**, which supplies skin on the lateral side of the little toe, and to a **common plantar digital nerve**, which divides to supply proper plantar digital nerves to skin on the adjacent sides of toes IV and V.

The proper plantar digital nerve to the lateral side of the little toe also innervates the flexor digiti minimi brevis and the dorsal and plantar interossei muscles between metatarsals IV and V.

The **deep branch** of the lateral plantar nerve is motor and accompanies the lateral plantar artery deep to the long flexor tendons and the adductor hallucis muscle. It supplies branches to the second to fourth lumbrical muscles, the adductor hallucis muscle, and all interossei except those between metatarsals IV and V, which are innervated by the superficial branch.

In the clinic

Morton's neuroma

A Morton's neuroma is an enlarged common plantar nerve, usually in the third interspace between the third and fourth toes. In this region of the foot the lateral plantar nerve often unites with the medial plantar nerve. As the two nerves join, the resulting nerve is typically larger in diameter than those of the other toes. Also, it is in a relatively subcutaneous position, just above the fat pad of the foot close to the artery and the vein. Above the nerve is the deep transverse metatarsal ligament, which is a broad strong structure holding the metatarsals together. Typically, as the patient enters the "push-off" phase of walking the interdigital nerve is sandwiched between the ground and the deep transverse metatarsal ligament. The forces tend to compress the common plantar nerve, which can be irritated, in which case there is usually some associated inflammatory change and thickening.

Typically, patients experience pain in the third interspace, which may be sharp or dull and is usually worsened by wearing shoes and walking.

Treatment may include injection of anti-inflammatory drugs, or it may be necessary to surgically remove the lesion.

Deep fibular nerve

The **deep fibular nerve** innervates the extensor digitorum brevis, contributes to the innervation of the first two dorsal interossei muscles, and supplies general sensory branches to the skin on the adjacent dorsal sides of the first and second toes and to the web space between them (Fig. 6.122).

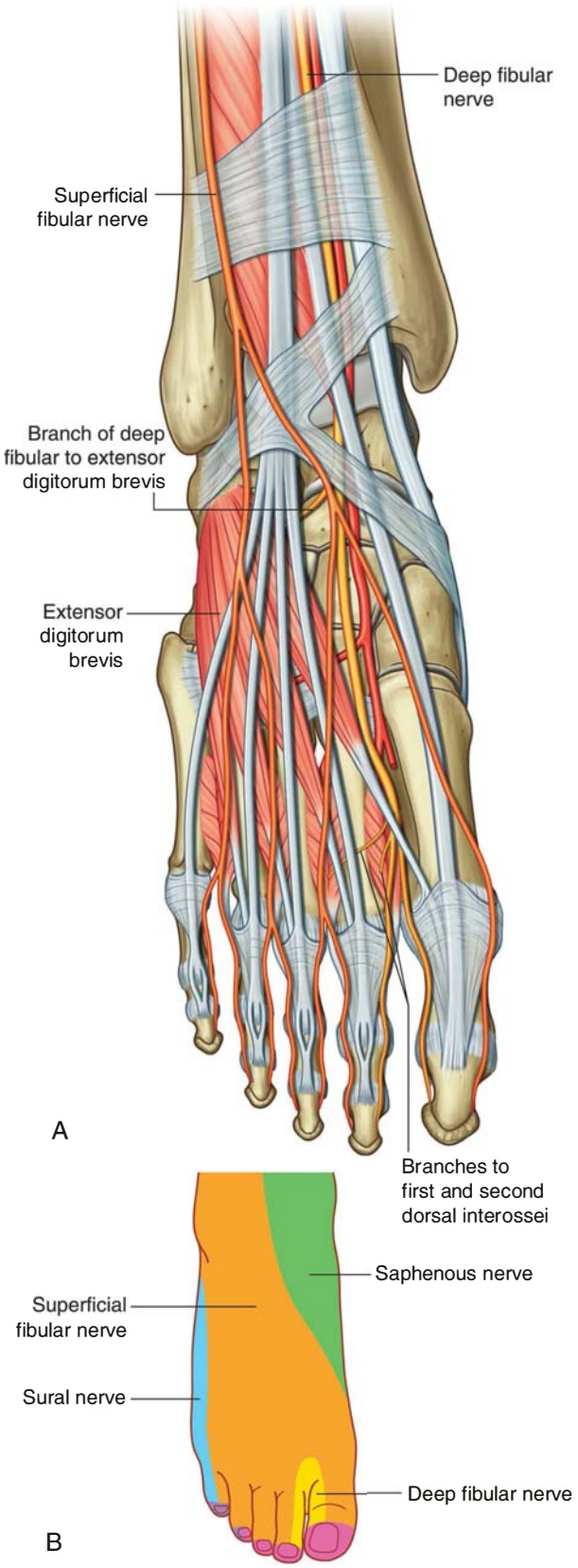


Fig. 6.122 **A.** Terminal branches of superficial and deep fibular nerves in the foot. **B.** Cutaneous distribution.

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The deep fibular nerve enters the dorsal aspect of the foot on the lateral side of the dorsalis pedis artery, and is parallel with and lateral to the tendon of the extensor hallucis longus muscle. Just distal to the ankle joint, the nerve gives origin to a lateral branch, which innervates the extensor digitorum brevis from its deep surface.

The deep fibular nerve continues forward on the dorsal surface of the foot, penetrates deep fascia between metatarsals I and II near the metatarsophalangeal joints, and then divides into two **dorsal digital nerves**, which supply skin over adjacent surfaces of toes I and II down to the beginning of the nail beds.

Small motor branches, which contribute to the supply of the first two dorsal interossei muscles, originate from the deep fibular nerve before it penetrates deep fascia.

Superficial fibular nerve

The **superficial fibular nerve** is sensory to most skin on the dorsal aspect of the foot and toes except for skin on adjacent sides of toes I and II (which is innervated by the deep fibular nerve) and skin on the lateral side of the foot and little toe (which is innervated by the sural nerve; Fig. 6.122).

The superficial fibular nerve penetrates deep fascia on the anterolateral side of the lower leg and enters the dorsal aspect of the foot in superficial fascia. It gives rise to cutaneous branches and **dorsal digital nerves** along its course.

Sural nerve

The sural nerve is a cutaneous branch of the tibial nerve that originates high in the leg. It enters the foot in superficial fascia posterior to the lateral malleolus close to the short saphenous vein. Terminal branches innervate skin on the lateral side of the foot and dorsolateral surface of the little toe (Fig. 6.122B).

Saphenous nerve

The saphenous nerve is a cutaneous branch of the femoral nerve that originates in the thigh. Terminal branches enter the foot in superficial fascia on the medial side of the ankle and supply skin on the medial side of the proximal foot (Fig. 6.122B).

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Surface anatomy

Lower limb surface anatomy

Tendons, muscles, and bony landmarks in the lower limb are used to locate major arteries, veins, and nerves.

Because vessels are large, they can be used as entry points to the vascular system. In addition, vessels in the lower limb are farthest from the heart and the most inferior in the body. Therefore, the nature of peripheral pulses in the lower limb can give important information about the status of the circulatory system in general.

Sensation and muscle action in the lower limb are tested to assess lumbar and sacral regions of the spinal cord.

Avoiding the sciatic nerve

The sciatic nerve innervates muscles in the posterior compartment of the thigh, muscles in the leg and foot, and an appreciable area of skin. It enters the lower limb in the gluteal region (Fig. 6.123) and passes inferiorly midway between two major palpable bony landmarks, the greater trochanter and the ischial tuberosity. The greater trochanter can be easily felt as a hard bony protuberance about one hand's width inferior to the midpoint of the iliac crest. The ischial tuberosity is palpable just above the gluteal fold.

The gluteal region can be divided into quadrants by two lines positioned using palpable bony landmarks.

- One line descends vertically from the highest point of the iliac crest.
- The other line passes horizontally through the first line midway between the highest point of the iliac crest and the horizontal plane through the ischial tuberosity.

The sciatic nerve curves through the upper lateral corner of the lower medial quadrant and descends along the lateral margin of the lower medial quadrant. Injections can be carried out in the anterior corner of the upper lateral quadrant to avoid injury to the sciatic nerve and major vessels in the region (Fig. 6.123B).

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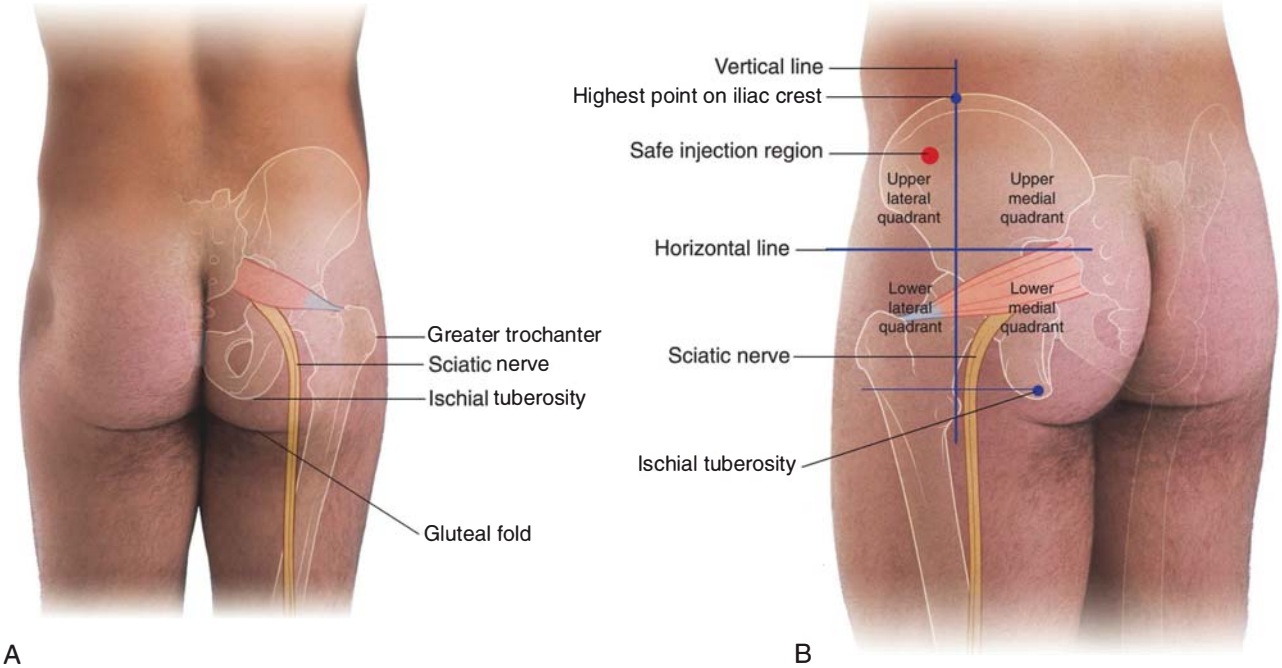


Fig. 6.123 Avoiding the sciatic nerve. **A.** Posterior view of the gluteal region of a man with the position of the sciatic nerve indicated. **B.** Posterolateral view of the left gluteal region with gluteal quadrants and the position of the sciatic nerve indicated.

Finding the femoral artery in the femoral triangle

The femoral artery passes into the femoral triangle (Fig. 6.124) of the lower limb from the abdomen.

The femoral triangle is the depression formed in the anterior thigh between the medial margin of the adductor longus muscle, the medial margin of the sartorius muscle, and the inguinal ligament.

The tendon of the adductor longus muscle can be palpated as a cord-like structure that attaches to bone immediately inferior to the pubic tubercle.

The sartorius muscle originates from the anterior superior iliac spine and crosses anteriorly over the thigh to attach to the medial aspect of the tibia below the knee joint.

The inguinal ligament attaches to the anterior superior iliac spine laterally and the pubic tubercle medially.

The femoral artery descends into the thigh from the abdomen by passing under the inguinal ligament and into the femoral triangle. In the femoral triangle, its pulse is easily felt just inferior to the inguinal ligament midway between the pubic symphysis and the anterior superior iliac spine. Medial to the artery is the femoral vein and medial to the vein is the femoral canal, which contains lymphatics and lies immediately lateral to the pubic tubercle. The femoral nerve lies lateral to the femoral artery.

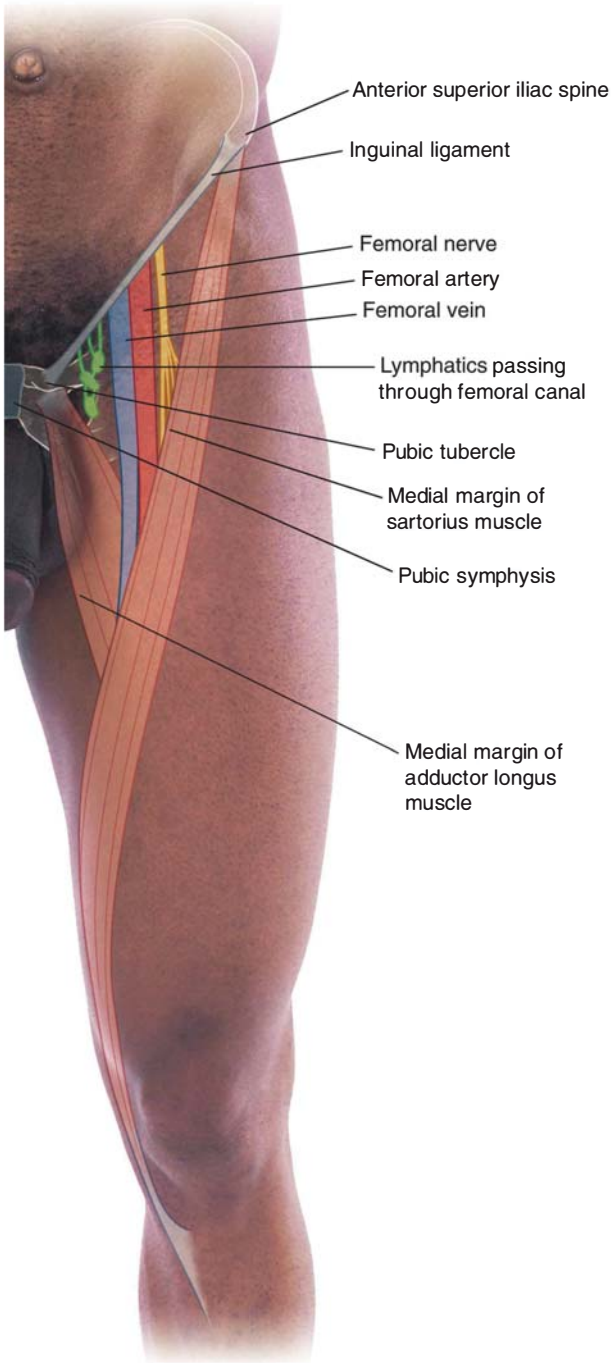


Fig. 6.124 Position of the femoral artery in the femoral triangle. Anterior thigh.

Identifying structures around the knee

The patella is a prominent palpable feature at the knee. The quadriceps femoris tendon attaches superiorly to it and the patellar ligament connects the inferior surface of the patella to the tibial tuberosity (Fig. 6.125). The patellar ligament and the tibial tuberosity are easily palpable. A tap on the patellar ligament (tendon) tests reflex activity mainly at spinal cord levels L3 and L4.

The head of the fibula is palpable as a protuberance on the lateral surface of the knee just inferior to the lateral condyle of the tibia. It can also be located by following the tendon of the biceps femoris inferiorly.

The common fibular nerve passes around the lateral surface of the neck of the fibula just inferior to the head and can often be felt as a cord-like structure in this position.

Another structure that can usually be located on the lateral side of the knee is the iliotibial tract. This flat tendinous structure, which attaches to the lateral tibial condyle, is most prominent when the knee is fully extended. In this position, the anterior edge of the tract raises a sharp vertical fold of skin posterior to the lateral edge of the patella.

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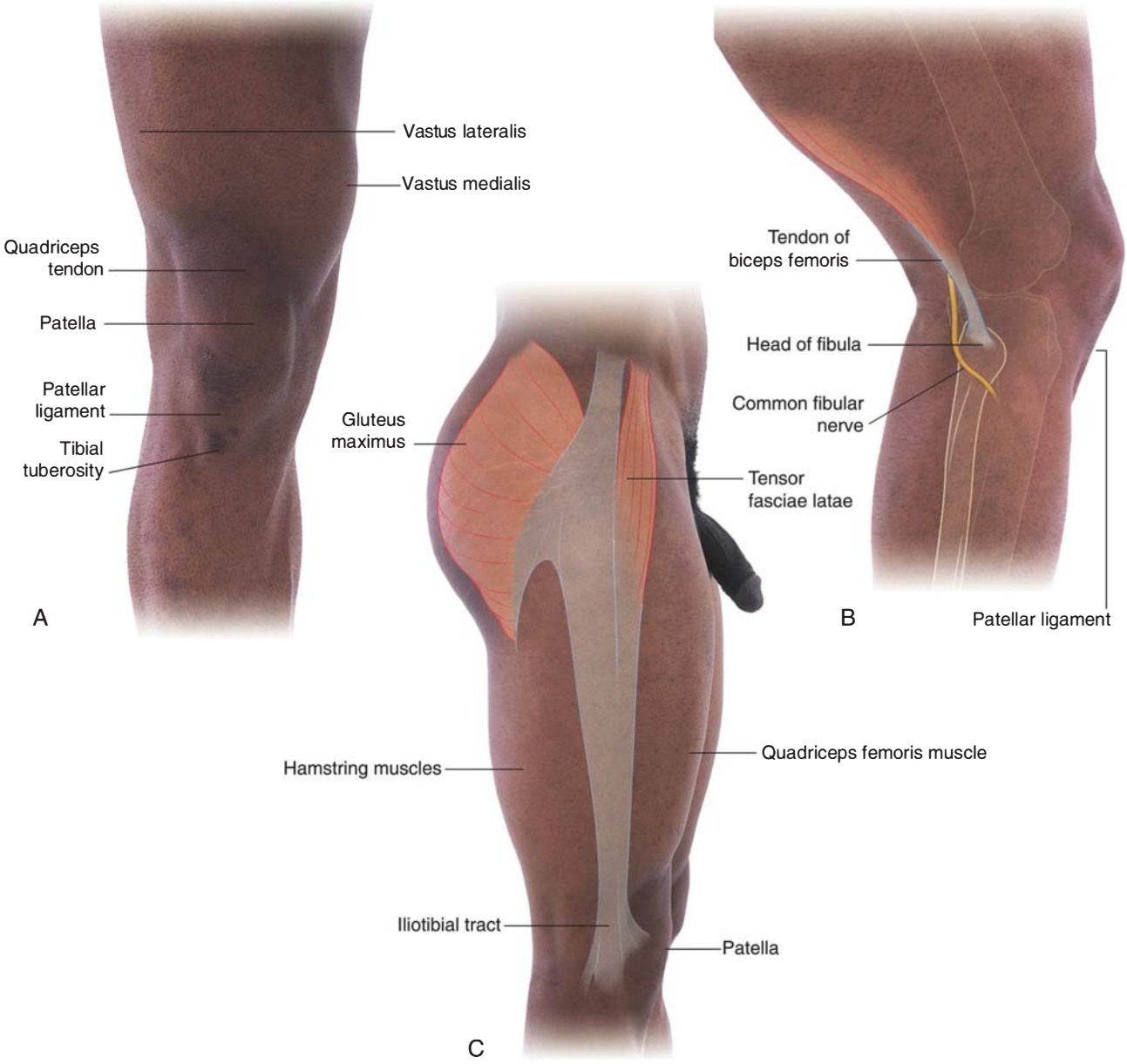


Fig. 6.125 Identifying structures around the knee. **A.** Anterior view of the right knee. **B.** Lateral view of the partially flexed right knee. **C.** Lateral view of the extended right knee, thigh, and gluteal region.

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Visualizing the contents of the popliteal fossa

The popliteal fossa is a diamond-shaped depression formed between the hamstrings and gastrocnemius muscle posterior to the knee. The inferior margins of the diamond are formed by the medial and lateral heads of the gastrocnemius muscle. The superior margins are formed laterally by the biceps femoris muscle and medially by the semimembranosus and semitendinosus muscles. The tendons of the biceps femoris muscle and the semitendinosus muscle are palpable and often visible.

The head of the fibula is palpable on the lateral side of the knee and can be used as a landmark for identifying the biceps femoris tendon and the common fibular nerve, which curves laterally out of the popliteal fossa and crosses the neck of the fibula just inferior to the head.

The popliteal fossa contains the popliteal artery, the popliteal vein, the tibial nerve, and the common fibular nerve (Fig. 6.126). The popliteal artery is the deepest of the structures in the fossa and descends through the region from the upper medial side. As a consequence of its position, the popliteal artery pulse is difficult to find, but usually can be detected on deep palpation just medial to the midline of the fossa.

The small saphenous vein penetrates deep fascia in the upper part of the posterior leg and joins the popliteal vein.

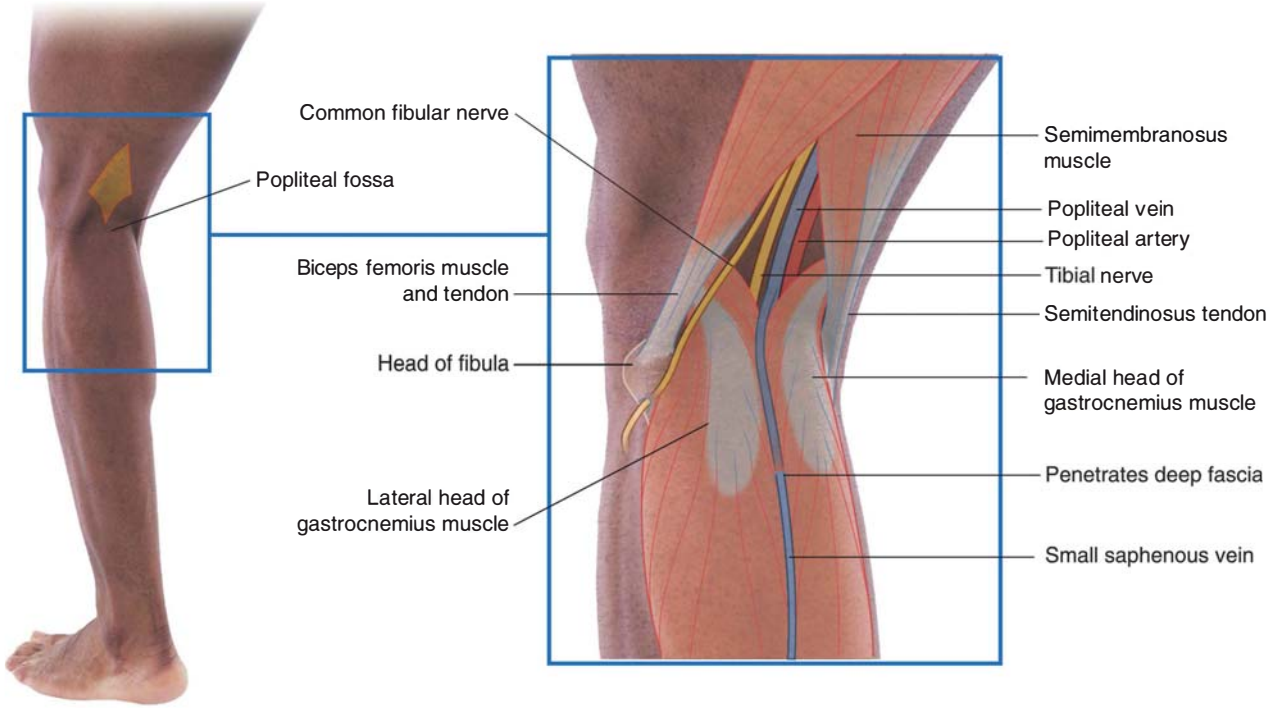


Fig. 6.126 Visualizing the contents of the popliteal fossa. Posterior view of the left knee.

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Finding the tarsal tunnel—the gateway to the foot

The tarsal tunnel (Fig. 6.127) is formed on the medial side of the foot in the groove between the medial malleolus and the heel (calcaneal tuberosity) and by the overlying flexor retinaculum.

The posterior tibial artery and tibial nerve enter the foot through the tarsal tunnel. The tendons of the tibialis posterior, flexor digitorum longus, and flexor hallucis longus also pass through the tarsal tunnel in compartments formed by septa of the flexor retinaculum.

The order of structures passing through the tunnel from the anteromedial to posterolateral are the tendon of the tibialis posterior, the tendon of the flexor digitorum longus, the posterior tibial artery and associated veins, the tibial nerve, and the tendon of the flexor hallucis longus ("Tom, Dick, and a very nervous Harry").

The tibial artery is palpable just posteroinferior to the medial malleolus on the anterior face of the visible groove between the heel and medial malleolus.

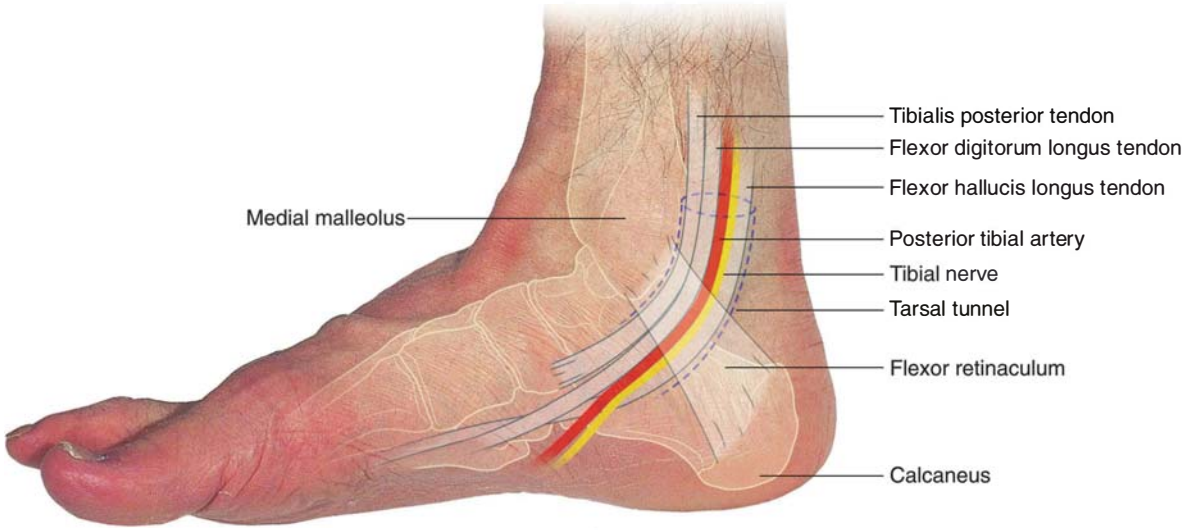


Fig. 6.127 Finding the tarsal tunnel—the gateway to the foot.

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Identifying tendons around the ankle and in the foot

Numerous tendons can be identified around the ankle and in the foot (Fig. 6.128) and can be used as useful landmarks for locating vessels or testing spinal reflexes.

The tibialis anterior tendon is visible on the medial side of the ankle anterior to the medial malleolus.

The calcaneal tendon is the largest tendon entering the foot and is prominent on the posterior aspect of the foot as it descends from the leg to the heel. A tap with a tendon hammer on this tendon tests reflex activity of spinal cord levels S1 and S2.

When the foot is everted, the tendons of the fibularis longus and fibularis brevis raise a linear fold of skin, which descends from the lower leg to the posterior edge of the lateral malleolus.

The tendon of the fibularis brevis is often evident on the lateral surface of the foot descending obliquely to the base of metatarsal V. The tendons of the fibularis tertius,



Fig. 6.128 Identifying tendons around the ankle and in the foot. **A.** Medial side of the right foot. **B.** Posterior aspect of the right foot. **C.** Lateral side of the right foot. **D.** Dorsal aspect of the right foot.

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extensor digitorum longus, and extensor hallucis longus are visible on the dorsal aspect of the foot from lateral to medial.

Finding the dorsalis pedis artery

The nature of the dorsalis pedis pulse (Fig. 6.129) is important for assessing peripheral circulation because the dorsalis pedis artery is the farthest palpable vessel from the heart. Also, it is the lowest palpable artery in the body when a person is standing.

The dorsalis pedis artery passes onto the dorsal aspect of the foot and anteriorly over the tarsal bones where it lies between and is parallel to the tendon of the extensor hallucis longus and the tendon of the extensor digitorum longus to the second toe. It is palpable in this position. The terminal branch of the dorsalis pedis artery passes into the plantar surface of the foot between the two heads of the first dorsal interosseous muscle.

Approximating the position of the plantar arterial arch

The blood supply of the foot is provided by branches of the posterior tibial and dorsalis pedis arteries.

The posterior tibial artery enters the plantar surface of the foot through the tarsal tunnel and divides into a lateral and a medial plantar artery.

The lateral plantar artery curves laterally across the posterior half of the sole and then curves medially as the plantar arch (Fig. 6.130) through the anterior sole. Between the bases of metatarsals I and II, the plantar arch joins the terminal branch (deep plantar artery) of the dorsalis pedis artery. Most of the foot is supplied by the plantar arch.

The medial plantar artery passes anteriorly through the sole, connects with branches of the plantar arch, and supplies the medial side of the great toe.

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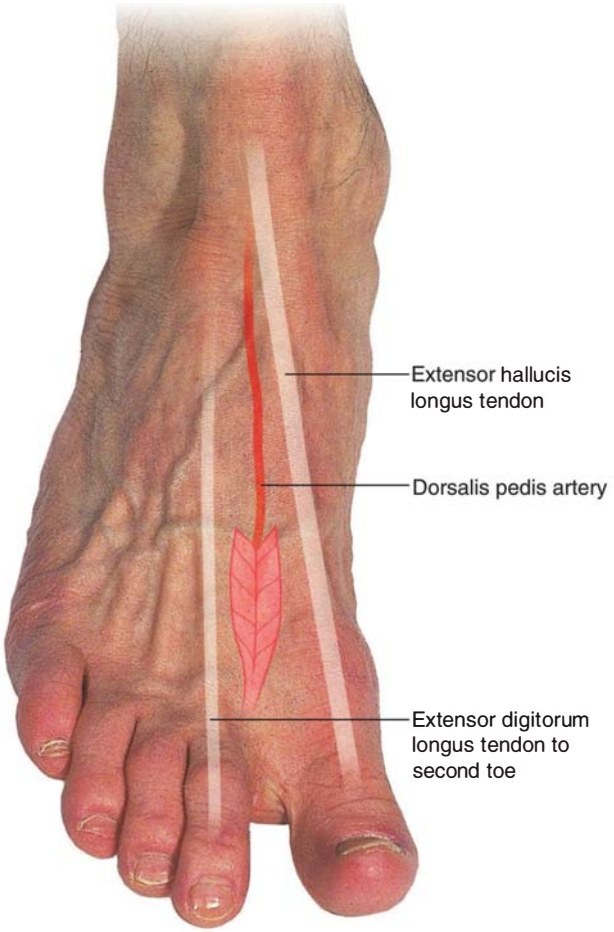


Fig. 6.129 Finding the dorsalis pedis artery.

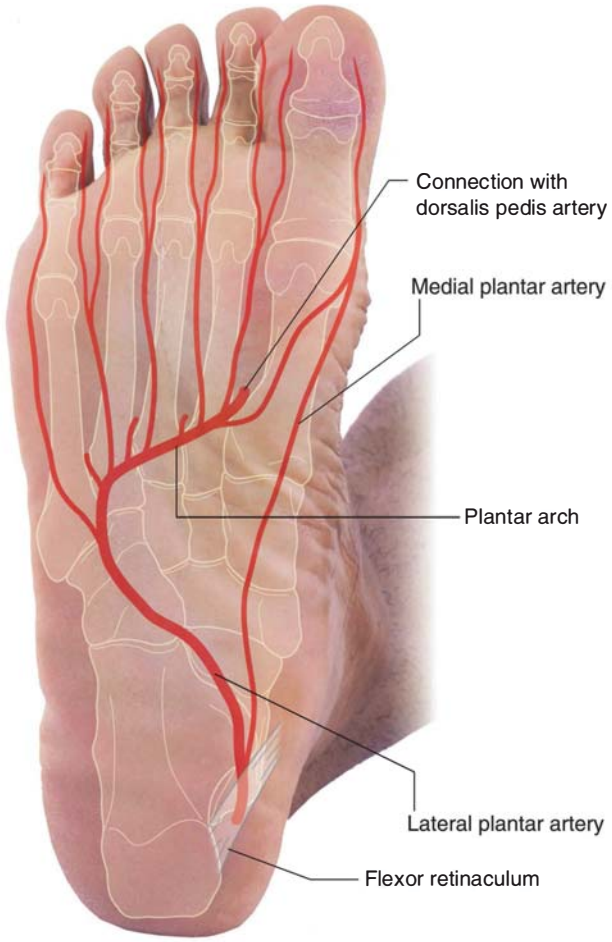


Fig. 6.130 Position of the plantar arch.

Major superficial veins

Superficial veins in the lower limb often become enlarged. Also, because the veins are long, they can be removed and used elsewhere in the body as vascular grafts.

Superficial veins (Fig. 6.131) in the lower limb begin as a dorsal venous arch in the foot. The medial side of the arch curves superiorly anterior to the medial malleolus and passes up the leg and thigh as the great saphenous vein. This vein passes through an aperture in the fascia lata (saphenous ring) to join with the femoral vein in the femoral triangle.

The lateral side of the dorsal venous arch in the foot passes posterior to the lateral malleolus and up the posterior surface of the leg as the small saphenous vein. This vessel passes through the deep fascia in the upper one-third of the leg and connects with the popliteal vein in the popliteal fossa posterior to the knee.



Fig. 6.131 Major superficial veins. **A.** Dorsal aspect of the right foot. **B.** Anterior view of right lower limb. **C.** Posterior aspect of the left thigh, leg, and foot.

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Pulse points

Peripheral pulses can be felt at four locations in the lower limb (Fig. 6.132):

- **femoral pulse** in the femoral triangle—femoral artery inferior to the inguinal ligament and midway between the anterior superior iliac spine and the pubic symphysis;
- **popliteal pulse** in the popliteal fossa—popliteal artery deep in the popliteal fossa near the midline;
- **posterior tibial pulse** in the tarsal tunnel—posterior tibial artery posteroinferior to the medial malleolus in the groove between the medial malleolus and the heel (calcaneal tuberosity);
- **dorsalis pedis pulse** on the dorsal aspect of the foot—dorsalis pedis artery as it passes distally over the tarsal bones between the tendon of the extensor hallucis longus and the tendon of the extensor digitorum longus to the second toe.



Fig. 6.132 Where to feel peripheral arterial pulses in the lower limb.

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Clinical cases

Case 1

VARICOSE VEINS

A young woman came to a vascular surgeon with a series of large dilated tortuous veins in her right leg. The rest of her leg was otherwise unremarkable.

A diagnosis of varicose veins was made and the surgeon needs to determine the site of valvular incompetence.

There are typical points where incompetent valves occur between the superficial and the deep veins. In these regions the varicosities tend to become marked. The typical sites are:

- *at the saphena varix—the saphenofemoral junction where the femoral vein is joined by the great saphenous vein;*
- *in the midthigh perforating vein between the great saphenous vein and the femoral vein;*
- *in the calf the three sites where perforators occur, 5, 10, and 15 cm above the medial malleolus between the great saphenous vein and the deep veins of the calf; and*
- *at the junction of the small saphenous vein and the popliteal vein.*

The surgeon asked the patient to lie supine on the bed and elevated the leg. A tourniquet was placed around the upper thigh below the saphenofemoral junction and the patient was asked to stand up. No veins were demonstrated filling on the medial aspect of the thigh and lower limb.

The effect of the tourniquet is to compress the great saphenous vein while permitting blood to flow in the deep venous system of the femoral vein and the deep femoral vein.

Because there was no filling of the medial varicose veins below the level of the tourniquet, the surgeon assumed that the valve at the saphenofemoral junction was incompetent and would require surgical treatment.

However, during the tourniquet maneuver the surgeon also noted some veins around the posterior and posterolateral aspect of the calf.

A similar technique was performed by application of a tourniquet just below the level of the knee joint while the leg was elevated. The patient stood up and no veins were demonstrated filling in the posterior and posterolateral aspect of the calf. These findings suggested to the surgeon that there was also incompetence of the valve for the small saphenous system where it anastomoses with the popliteal vein.

Surgery was planned.

A small transverse incision was made below the level of the inguinal ligament where the great saphenous vein passes through the saphenous ring in the deep fascia. This can be easily palpated as a small circular defect in the fascia. The saphenofemoral junction was identified and the great saphenous vein was ligated, at its anastomosis with the femoral vein. The great saphenous vein was stripped using special surgical techniques and removed.

The patient was placed prone for the second part of the operation.

A small incision was made transversely at the level of the skin crease in the popliteal fossa. However, the surgeon had difficulty identifying the junction between the small saphenous vein and the popliteal vein. After considerable time the surgeon located what he thought was the small saphenous vein and the structure was ligated and the wound closed.

The following day the patient was sent home, but returned to the clinic after 2 weeks complaining of problems walking. On examination there was absence of dorsiflexion, a sensory disturbance over the lateral aspect

(continues)

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Case 1 *(continued)*

of the leg and foot, and obvious wasting of the fibular muscles. As the patient walked, the foot was dragged between steps. A clinical diagnosis of footdrop was made and a common fibular nerve injury was diagnosed. The injury occurred at the time of surgery.

Within the popliteal fossa are the popliteal artery, popliteal vein, and sciatic nerve (and its divisions). The popliteal artery is the deepest structure. The popliteal vein is superficial to the artery and the sciatic nerve is superficial to the vein (Fig. 6.133). Importantly, the sciatic nerve divides at the apex of the popliteal fossa. The tibial nerve continues into the lower popliteal fossa. The common fibular nerve passes laterally adjacent to the biceps femoris muscle to become superficial and wrap around the fibula neck.

It was concluded that the surgeon had accidentally ligated the common fibular nerve rather than the small saphenous vein, thus producing this patient's symptoms.

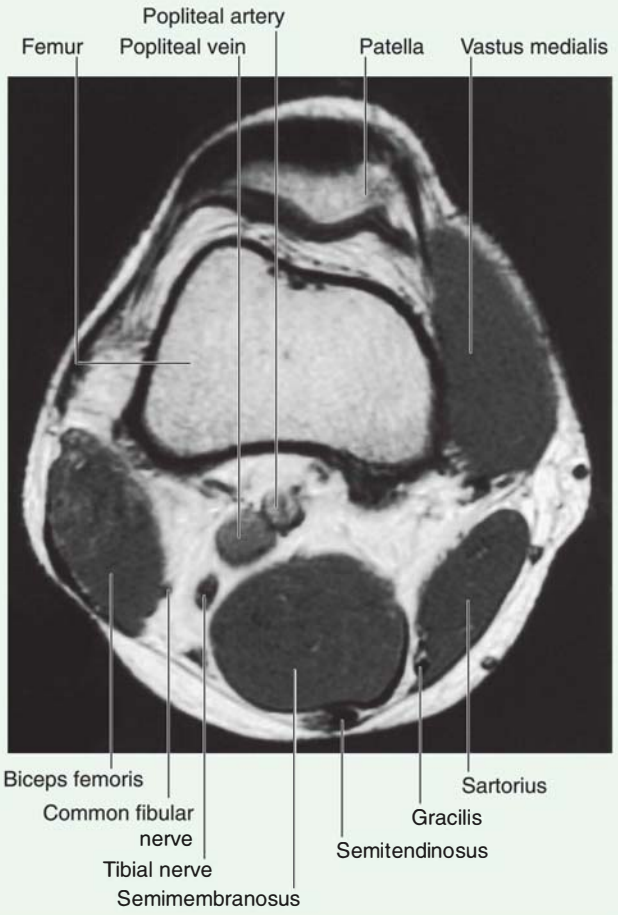


Fig. 6.133 Popliteal fossa showing position of the popliteal artery and vein and sciatic nerve. T1-weighted magnetic resonance image in the axial plane.

Case 2

KNEE JOINT INJURY

A young man was enjoying a long weekend skiing at a European ski resort. While racing a friend he caught an inner edge of his right ski. He lost his balance and fell. During his tumble he heard an audible "click." After recovering from his spill, he developed tremendous pain in his right knee. He was unable to carry on skiing for that day, and by the time he returned to his chalet, his knee was significantly swollen. He went immediately to see an orthopedic surgeon.

The orthopedic surgeon carefully reviewed the mechanism of injury.

The man was skiing down the slope with both skis in parallel. The ankles were held rigid in the boots and the knees were slightly flexed. A momentary loss of concentration led to the skier catching the inner edge of his right ski. This effect was to force the boot and calf into external rotation. Furthermore, the knee was forced into a valgus position (bowed laterally away from the midline) and the skier tumbled. Both skis were detached from the boots as the bindings released them.

A series of structures within the knee joint were damaged sequentially.

(continues)

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Case 2 *(continued)*

As the knee went into external rotation and valgus, the anterior cruciate ligament became taut, acting as a fulcrum. The tibial collateral ligament was stressed and the lateral compartment of the knee compressed. As the force increased, the tibial collateral ligament was torn (Fig. 6.134A,B), as was the medial meniscus (Fig. 6.135C). Finally, the anterior cruciate ligament, which was taut, gave way (Fig. 6.135A,B).

The joint became swollen some hours afterward.

Disruption of the anterior cruciate ligament characteristically produces marked joint swelling. The ligament is extrasynovial and intracapsular and has a rich blood supply. As the ligament was torn it ruptured into the joint. Blood from the tear irritates the synovial membrane and also enters the joint. These factors produce gradual swelling of the joint over the ensuing hours with significant fluid accumulation in the joint cavity.

The patient had a surgical reconstruction of the anterior cruciate ligament.

It is difficult to find a man-made substance that can act in the same way as the anterior cruciate ligament and demonstrate the same physical properties. Surgeons have devised ingenious ways of reconstructing the anterior cruciate ligament. Two of the commonest methods use the patellar ligament (tendon) and hamstrings to reconstruct the ligament.

The patient had further surgical procedures.

The tibial collateral ligament was explored and resutured. Using arthroscopic techniques, the tear in the medial meniscus was débrided to prevent further complications.

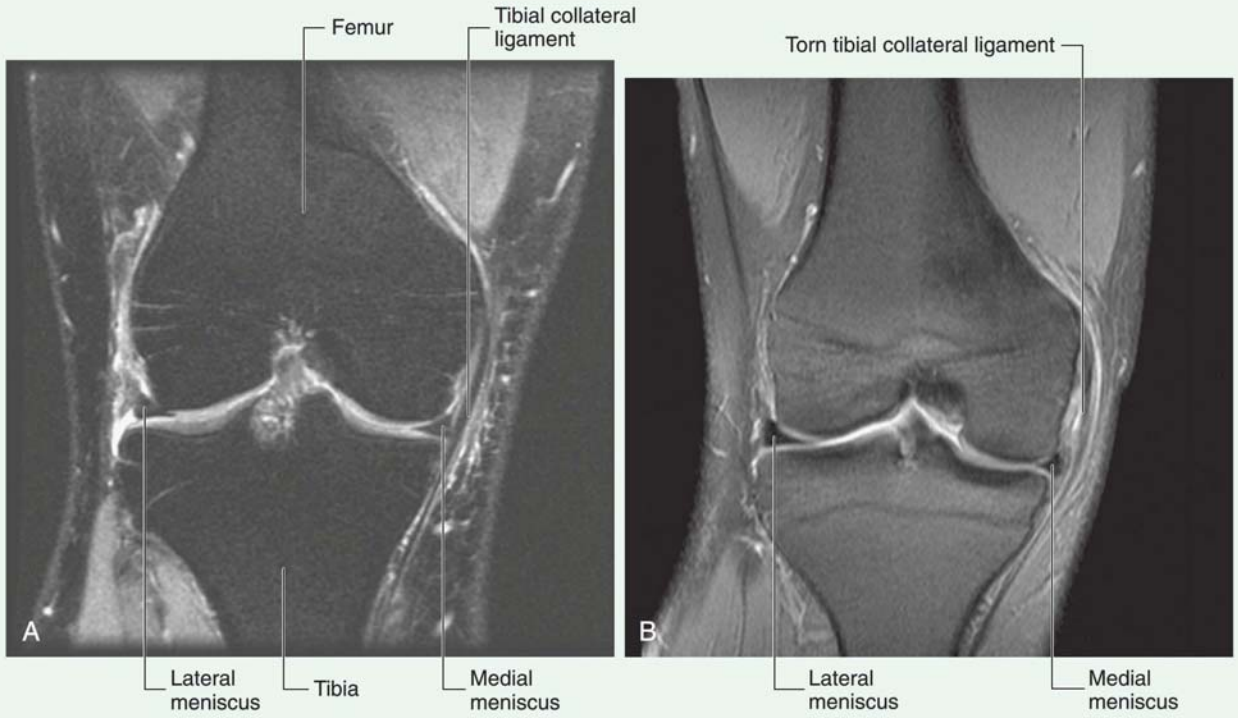


Fig. 6.134 A. Normal knee joint showing the tibial collateral ligament and the medial and lateral menisci. Proton density (PD)-weighted magnetic resonance image in the coronal plane. **B.** Knee joint showing a torn tibial collateral ligament. PD-weighted magnetic resonance image in the coronal plane.

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Case 2 *(continued)*

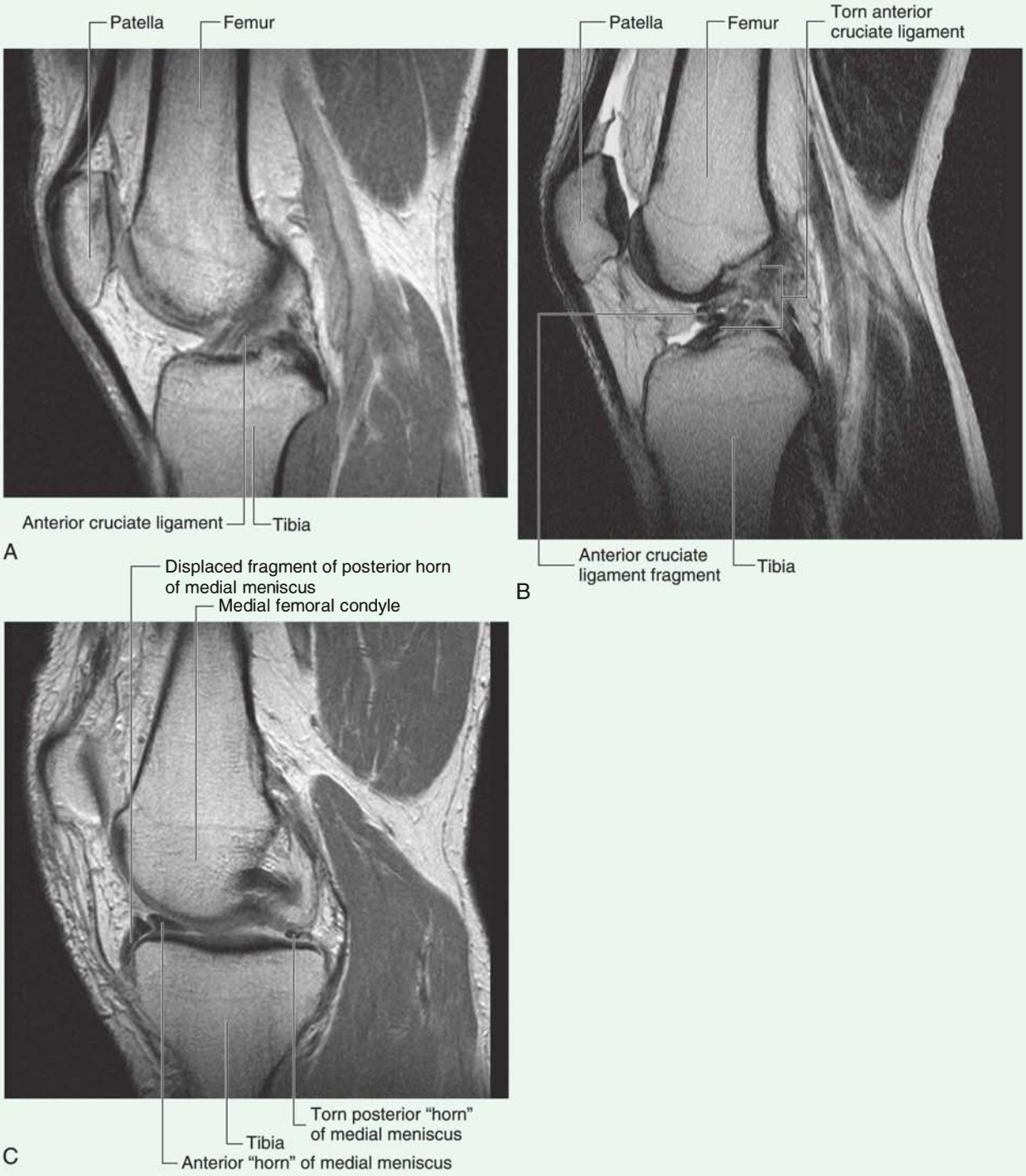


Fig. 6.135 A. Knee joint showing an intact anterior cruciate ligament. T2-weighted magnetic resonance image in the sagittal plane. **B.** Knee joint showing a torn anterior cruciate ligament. T2-weighted magnetic resonance image in the sagittal plane. **C.** Knee joint showing a torn medial meniscus (the broken off portion of the posterior horn has moved into the anterior aspect of the joint giving the impression of a 'double meniscus' in this location). Proton density-weighted magnetic resonance image in the sagittal plane.

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Case 3

FRACTURE OF NECK OF FEMUR

A 72-year-old woman was admitted to the emergency room after falling at home. She complained of a severe pain in her right hip and had noticeable bruising on the right side of the face.

On admission it was noted that the patient's right leg was shorter than her left leg and externally rotated.

An initial series of investigations was carried out, including a plain radiograph of the pelvis.

The plain radiograph of the pelvis demonstrated a displaced fracture through the right neck of the femur.

The apparent shortening and external rotation of the leg on clinical examination were accounted for by spasm of the muscles connecting the pelvis to the trochanters and proximal femur. Of the muscles that surround the hip joint the largest group is the adductor group (adductor longus, brevis, and magnus) and the psoas major. The psoas major inserts onto the lesser trochanter and its action is to externally rotate and flex the hip. The fulcrum of action of the psoas major is the femoral head in the acetabulum. However, when the femoral neck is detached its overriding action pulls the femur proximally and into external rotation. The external rotation is exacerbated by the spasm in the adductor muscles.

Extensive medical testing was necessary before surgery. It is important to remember that elderly patients may have a number of coexisting diseases.

The patient then underwent a hemiarthroplasty.

Hemiarthroplasty is a surgical procedure in which the femoral head is removed from the acetabulum. The femoral neck is trimmed close to the trochanters and the medullary cavity of the femoral shaft is reamed. A metal hip prosthesis is inserted into the medullary cavity of the femur and the head of the prosthesis is placed into the acetabulum, in which it articulates. Importantly, the acetabulum is not replaced in straightforward cases, though a prosthetic acetabulum may be inserted if clinically appropriate.

An arthroplasty was the only procedure that could be performed.

The blood supply to the femoral head is from three sources—the artery within the ligament of the head of the femur, vessels in the medullary cavity, and vessels deep to the synovium running in the retinacula of the fibrous capsule of the hip joint. With increasing age, the medullary cavity undergoes fatty replacement of the normal red marrow, thus attenuating the medullary blood supply. The artery within the ligament of the head of the femur also becomes attenuated and this is often associated with atherosclerotic arterial disease.

Unfortunately for this patient, the sole blood supply to the head of the femur was via the vessels in the retinacula fibers, which were transected at the time of the fracture. If the patient had an intertrochanteric fracture instead, the vessels of the retinacula fibers would not have been damaged and a different approach to surgical fixation could be undertaken without the need for a hemiarthroplasty.

The patient has osteoporosis.

Osteoporosis is a common condition affecting older people, but is significantly more frequent in postmenopausal women. Many fractures of the femoral neck in elderly patients occur because the strength of the bone is significantly reduced when it is osteoporotic. Other common sites for osteoporotic fractures include the distal radius and the thoracic and lumbar vertebrae.

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Case 4

DEEP VEIN THROMBOSIS

A 28-year-old woman was seen by her family practitioner for a routine pregnancy checkup at 36 weeks' gestational age. Neither the patient nor the family physician had any concerns about the pregnancy. However, the patient did complain of unilateral swelling of her left leg, which had gradually increased over the previous 2 days. Furthermore, the evening before her visit she developed some sharp chest pain, which was exacerbated by deep breaths.

The family physician ordered a duplex ultrasound scan of the left leg venous system.

Ultrasound scanning of the vascular tree can demonstrate flow and occlusion of arteries and veins.

The probe was placed over the left femoral vein and no flow was demonstrated. Furthermore, the vein could not be compressed, and no alteration of flow with breathing could be demonstrated. Some flow was demonstrated in the deep femoral vein and in the great saphenous vein. No flow was demonstrated throughout the length of the left femoral vein, the popliteal vein, and the tibial veins.

The technician scanned the opposite side, where excellent flow was demonstrated within the right femoral venous system. In addition, when the calf was gently massaged an augmentation to the flow was noted. It was possible to demonstrate alteration of flow with respiratory excursion and venous compression was satisfactory. A diagnosis of extensive left-sided deep vein thrombosis was made.

Certain patients are more prone to deep vein thrombosis.

Three major factors predispose a patient to thrombosis:

- *reduced or stagnant blood flow in the veins—significant stasis of blood (which may be due to lack of movement), reduced muscular calf pump effect, and obstruction to flow may occur;*
- *injury to the vein wall—venous trauma may damage the vessel walls, promoting thrombus formation;*
- *hypercoagulability of the blood—hypercoagulable states are often associated with abnormal levels of certain clotting factors, such as antithrombin III, protein C, and protein S.*

In this patient, compression of the left external iliac veins by the gravid uterus was the likely cause of stasis, which led to the deep vein thrombosis.

The chest pain was due to pulmonary emboli.

Small emboli were thrown off from the leading edge of the thrombus through the heart to lodge in the lungs. Small emboli cause typical pleuritic chest pain, which is exacerbated by breathing. In isolation these small pulmonary emboli may affect respiratory function, but they may be the precursor to a large and potentially fatal pulmonary embolus (Fig. 6.136).

Anticoagulation was instituted and the patient had an uneventful delivery.

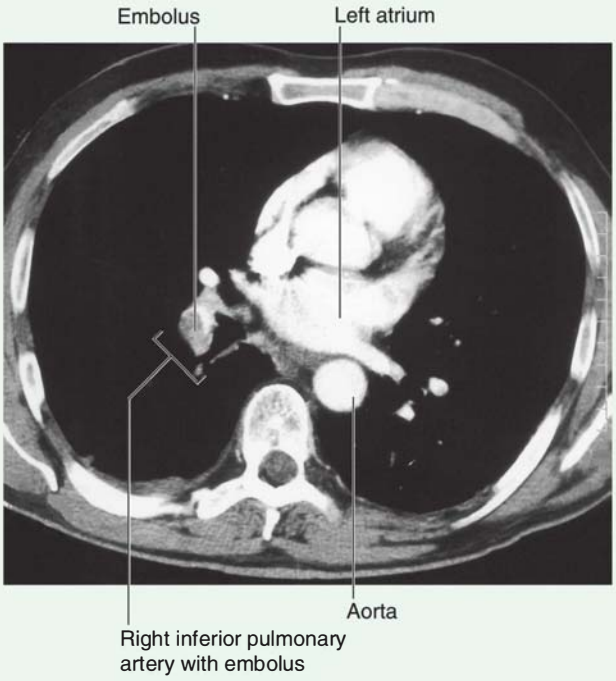


Fig. 6.136 Pulmonary embolus. Axial computed tomogram.

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Case 5

RUPTURED CALCANEAL TENDON

A 45-year-old man had recently taken up squash. During a game he attempted a forehand shot and noticed severe sudden pain in his heel. He thought his opponent had hit him with his racket. When he turned, though, he realized his opponent was too far away to have hit him.

Within minutes there was marked swelling of the ankle. The patient was unable to plantarflex his foot and had to stop the game. Afterward an appreciable subcutaneous hematoma developed in the ankle.

The diagnostic possibilities include a bone or soft tissue injury.

A bony injury was excluded because there was no bone tenderness.

The patient had a significant soft tissue injury. On examination he had significant swelling of the ankle with a subcutaneous hematoma. He was unable to stand on tiptoe on the right leg, and in the prone position a palpable defect was demonstrated within the calcaneal tendon.

A diagnosis of calcaneal tendon rupture was made.

This patient has a typical history of ruptured calcaneal tendon and the clinical findings support this. Magnetic resonance imaging was carried out and confirmed the diagnosis (Fig. 6.137).

The patient underwent an operative repair.

The tendon healed well, though the patient has not gone back to playing squash.



Fig. 6.137 Ankle showing a ruptured calcaneal tendon. T2-weighted magnetic resonance image in the sagittal plane.

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Case 6

POPLITEAL ARTERY ANEURYSM

A 67-year-old man was noted to have a mass at the back of his knee. The mass measured approximately 4 cm in transverse diameter. The patient was otherwise fit and well and had no other history of note.

The mass was arising from one of the structures in the popliteal fossa.

Within the popliteal fossa there is a neurovascular bundle that contains the sciatic nerve (and its two divisions), the popliteal artery and the popliteal vein. There are also numerous small bursae associated with the posterior aspect of the knee joint and the muscles and tendons in this region. It is possible that this mass is arising from the posterior structures of the knee joint, which include synovial outpouchings, the menisci, and the muscles and tendons of this region.

The commonest masses demonstrated within the popliteal fossa are a popliteal cyst, a popliteal aneurysm, and an arterial adventitial cyst.

Further clinical examination revealed that this mass was pulsatile and demonstrated a bruit (an audible rumbling made by turbulent blood flow) on auscultation.

A diagnosis of popliteal artery aneurysm was made.

A popliteal artery aneurysm is an abnormal dilatation of the popliteal artery. It is unusual for it to be greater than 5 cm because symptoms usually develop before it reaches this size.

Unlike aneurysms elsewhere in the body, the natural history of popliteal aneurysms is to embolize, with the mural thrombus producing ischemia distal to the lesion, rather than rupture. It is mandatory to examine the rest of the arterial tree in patients with a popliteal aneurysm because aneurysms may be bilateral and there is an association with abdominal aortic aneurysms.

The other diagnostic possibilities include a popliteal cyst and an adventitial cyst.

A popliteal cyst (Baker's cyst) is a synovial outpouching that arises from the posteromedial aspect of the knee joint. The synovial membrane of the knee joint outpouches between the medial head of the gastrocnemius and the semimembranosus tendon to lie medially within the popliteal fossa. Occasionally it tracks inferiorly to lie in and around the tendons that form the pes anserinus (sartorius, gracilis, and semitendinosus).

An arterial adventitial cyst is an uncommon cystic structure that arises from the wall of the artery.

An ultrasound investigation was carried out.

Using real-time ultrasound the dimensions of the popliteal aneurysm were characterized and the flow in the vessels was demonstrated. Furthermore, a popliteal cyst and adventitial cyst were completely excluded.

The patient underwent a surgical excision and graft interposition and has made an uneventful recovery.

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Case 7

ANTERIOR TALOFIBULAR LIGAMENT TEAR

A young long-distance runner came to her physician with acute swelling around the lateral aspect of her ankle. This injury occurred directly after accidentally running into a pothole in the pavement.

A fractured ankle was suspected.

Plain anteroposterior and lateral radiographs of the ankle revealed no evidence of any bone injury to account for the patient's soft tissue swelling.

The patient was given a pair of crutches and analgesics and told to rest. A diagnosis of a simple sprain was made.

Over the ensuing weeks the swelling and edema within the soft tissue of the ankle decreased and the patient began to run, but noted that the ankle kept "giving way." She went to an orthopedic surgeon for further assessment.

There was a positive anterior drawer test of the ankle joint.

At this stage it is important to review the mechanism of injury. Typically when running on a hard surface the final phase of push-off involves supination of the foot. If the foot is caught in a pothole or divot, this supinating maneuver continues and inverts the ankle joint in plantarflexion. This puts significant strain on the lateral ligament complex and, given the appropriate circumstances, disruption (in order) of the ligament structures occurs from anterior to posterior. First, the anterior talofibular ligament is disrupted, followed by the calcaneofibular ligament, and then the posterior talofibular ligament. As each of these ligaments is disrupted, the severity of the soft tissue injury is significantly enhanced and the chance of permanent ankle instability is increased.

On examination any positive anterior drawer test of the ankle (4–5 mm compared to the opposite side) suggests an injury to the anterior talofibular ligament.

The anterior talofibular ligament can be assessed by placing the foot in marked plantarflexion. If there is over 10° of difference between the affected foot and nonaffected foot, an anterior talofibular ligament disruption is suspected.

It is extremely rare for all three ligaments to be disrupted, and if so there are usually other significant ankle injuries.

Magnetic resonance imaging (MRI) was carried out to assess ligament damage.

MRI is excellent for demonstrating the medial and lateral ligament complexes of the ankle as well as the soft tissues that support the bones of the posterior foot.

Unfortunately for this patient there was a tear of the anterior talofibular ligament (Fig. 6.138), which had to be repaired surgically.

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Case 7 *(continued)*

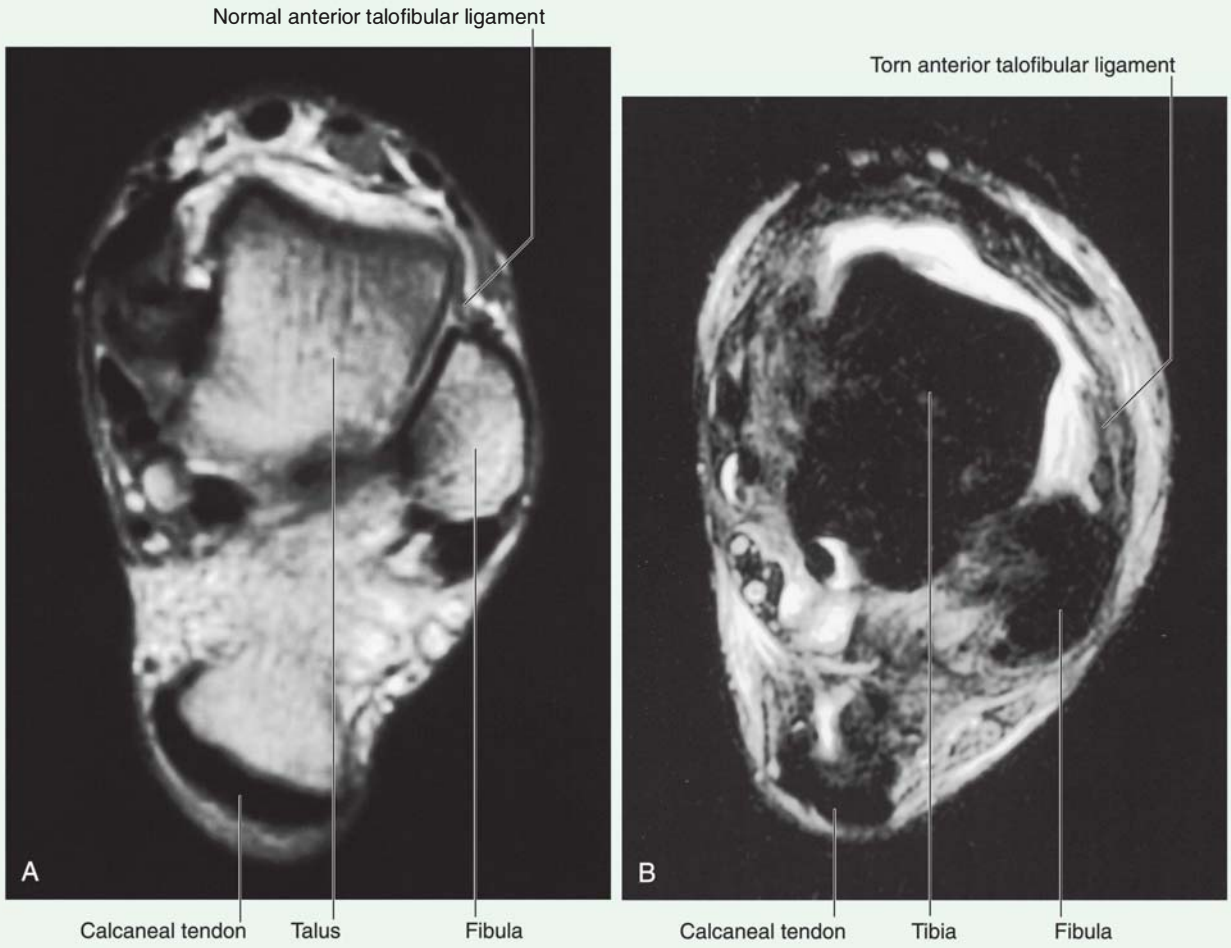


Fig. 6.138 A. Normal ankle joint showing an intact anterior talofibular ligament. T1-weighted magnetic resonance image in the axial plane. **B.** Ankle joint showing a torn anterior talofibular ligament. T2-weighted magnetic resonance image in the axial plane.

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7

Upper Limb

ADDITIONAL LEARNING RESOURCES for Chapter 7, *Upper Limb*, on STUDENT CONSULT (www.studentconsult.com):

- Image Library—illustrations of upper limb anatomy, Chapter 7
- Self-Assessment—National Board style multiple-choice questions, Chapter 7
- Short Questions—these are questions requiring short responses, Chapter 7
- Interactive Surface Anatomy—interactive surface animations, Chapter 7
- PT Case Studies, Chapter 7
  Upper quarter screening examination
  Thoracic outlet syndrome
  Rotator cuff tendinitis
  Adhesive capsulitis
  Subacromial impingement
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  Pancoast tumor
  Clavicular fracture
  Extensor tendon injury of the hand
  Hand infection
  High median nerve palsy
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  Brachial plexus nerve block
  Complication of a fractured first rib
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Conceptual overview

GENERAL DESCRIPTION

The upper limb is associated with the lateral aspect of the lower portion of the neck and with the thoracic wall. It is suspended from the trunk by muscles and a small skeletal articulation between the clavicle and the sternum—the sternoclavicular joint. Based on the position of its major joints and component bones, the upper limb is divided into shoulder, arm, forearm, and hand (Fig. 7.1A).

The shoulder is the area of upper limb attachment to the trunk (Fig. 7.1B).

The arm is the part of the upper limb between the shoulder and the elbow joint; the forearm is between the elbow joint and the wrist joint; and the hand is distal to the wrist joint.

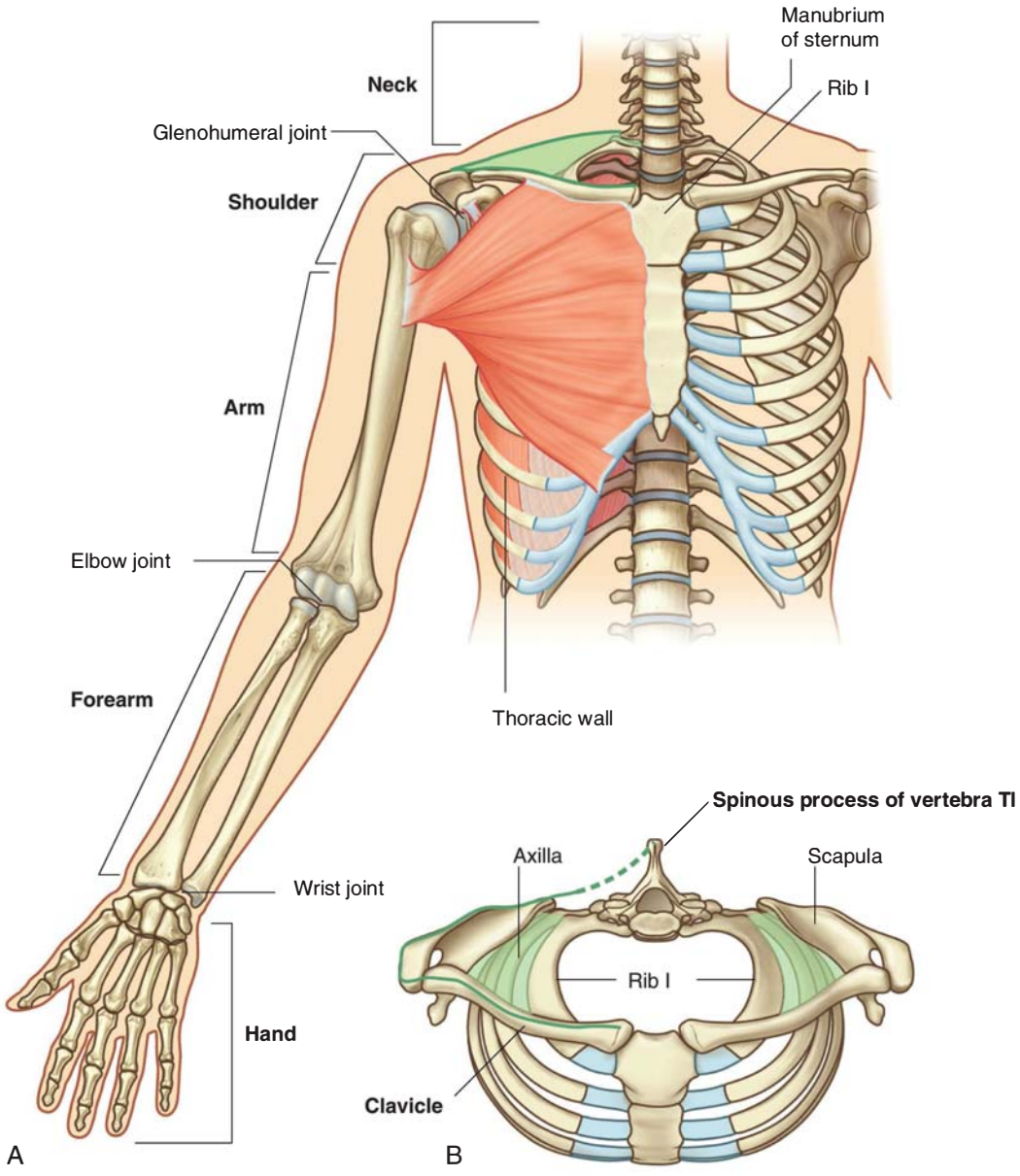


Fig. 7.1 Upper limb. **A.** Anterior view of the upper limb. **B.** Superior view of the shoulder.

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The axilla, cubital fossa, and carpal tunnel are significant areas of transition between the different parts of the limb (Fig. 7.2). Important structures pass through, or are related to, each of these areas.

The axilla is an irregularly shaped pyramidal area formed by muscles and bones of the shoulder and the lateral surface of the thoracic wall. The apex or inlet opens directly into the lower portion of the neck. The skin of the armpit forms the floor. All major structures that pass between the neck and arm pass through the axilla.

The cubital fossa is a triangularly shaped depression formed by muscles anterior to the elbow joint. The major artery, the brachial artery, passing from the arm to the forearm passes through this fossa, as does one of the major nerves of the upper limb, the median nerve.

The carpal tunnel is the gateway to the palm of the hand. Its posterior, lateral, and medial walls form an arch, which is made up of small carpal bones in the proximal region of the hand. A thick band of connective tissue, the flexor retinaculum, spans the distance between each side of the arch and forms the anterior wall of the tunnel. The median nerve and all the long flexor tendons passing from the forearm to the digits of the hand pass through the carpal tunnel.

FUNCTIONS

Positioning the hand

Unlike the lower limb, which is used for support, stability, and locomotion, the upper limb is highly mobile for positioning the hand in space.

The shoulder is suspended from the trunk predominantly by muscles and can therefore be moved relative to the body. Sliding (protraction and retraction) and rotating the scapula on the thoracic wall changes the position of the **glenohumeral joint** (**shoulder joint**) and extends the reach of the hand (Fig. 7.3). The glenohumeral joint allows the arm to move around three axes with a wide range of motion. Movements of the arm at this joint are

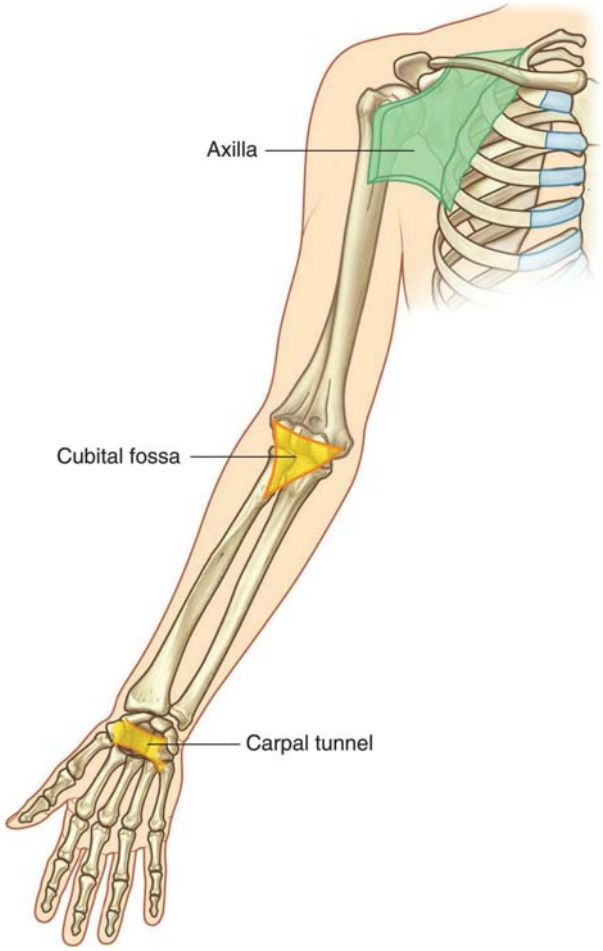


Fig. 7.2 Areas of transition in the upper limb.

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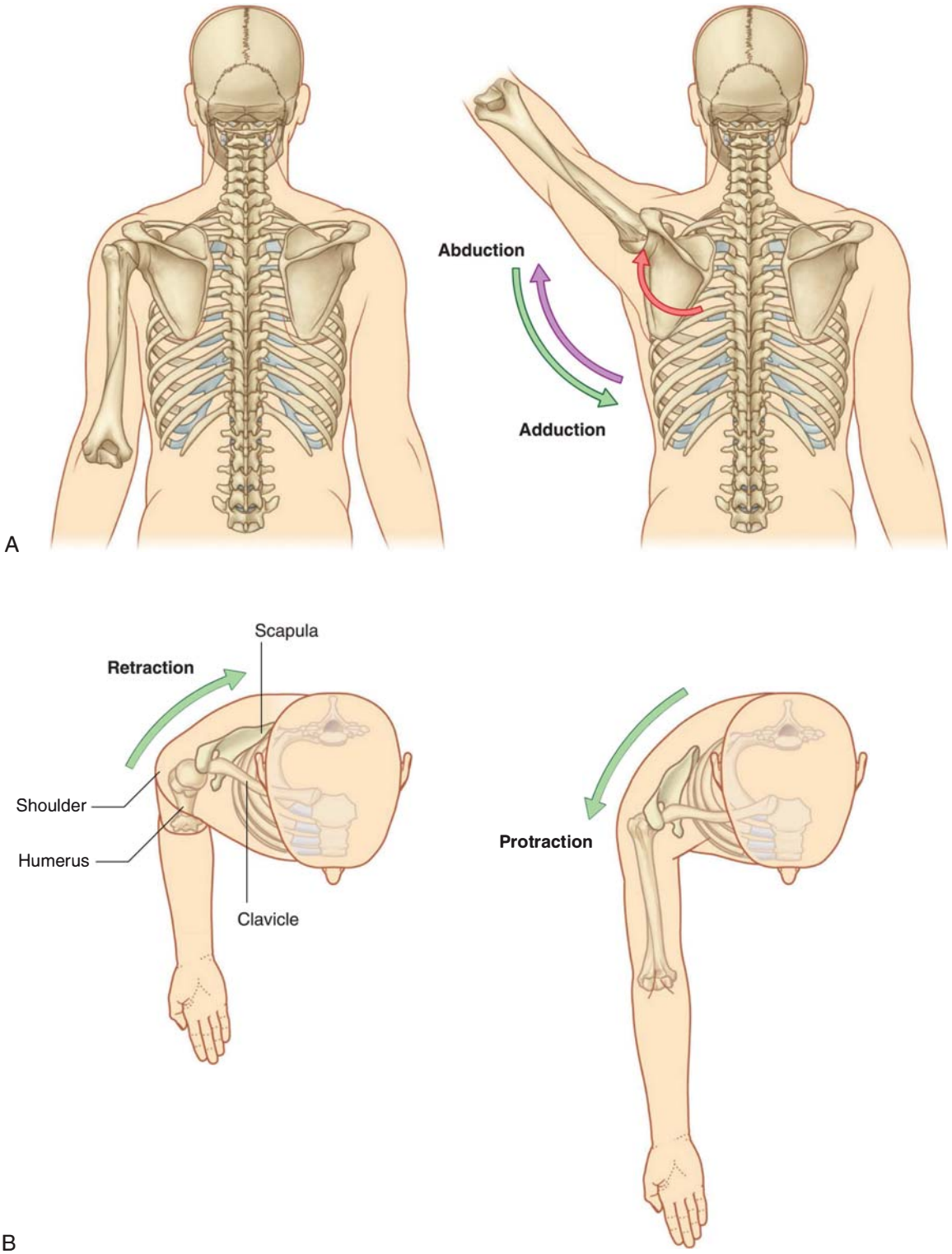


Fig. 7.3 Movements of the scapula. **A.** Rotation. **B.** Protraction and retraction.

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flexion, extension, abduction, adduction, medial rotation (internal rotation), lateral rotation (external rotation), and circumduction (Fig. 7.4).

The major movements at the **elbow joint** are flexion and extension of the forearm (Fig. 7.5A). At the other end of the forearm, the distal end of the lateral bone, the radius, can be flipped over the adjacent head of the medial bone, the ulna. Because the hand is articulated with the radius, it can be efficiently moved from a palm-anterior position to a palm-posterior position simply by crossing the distal end

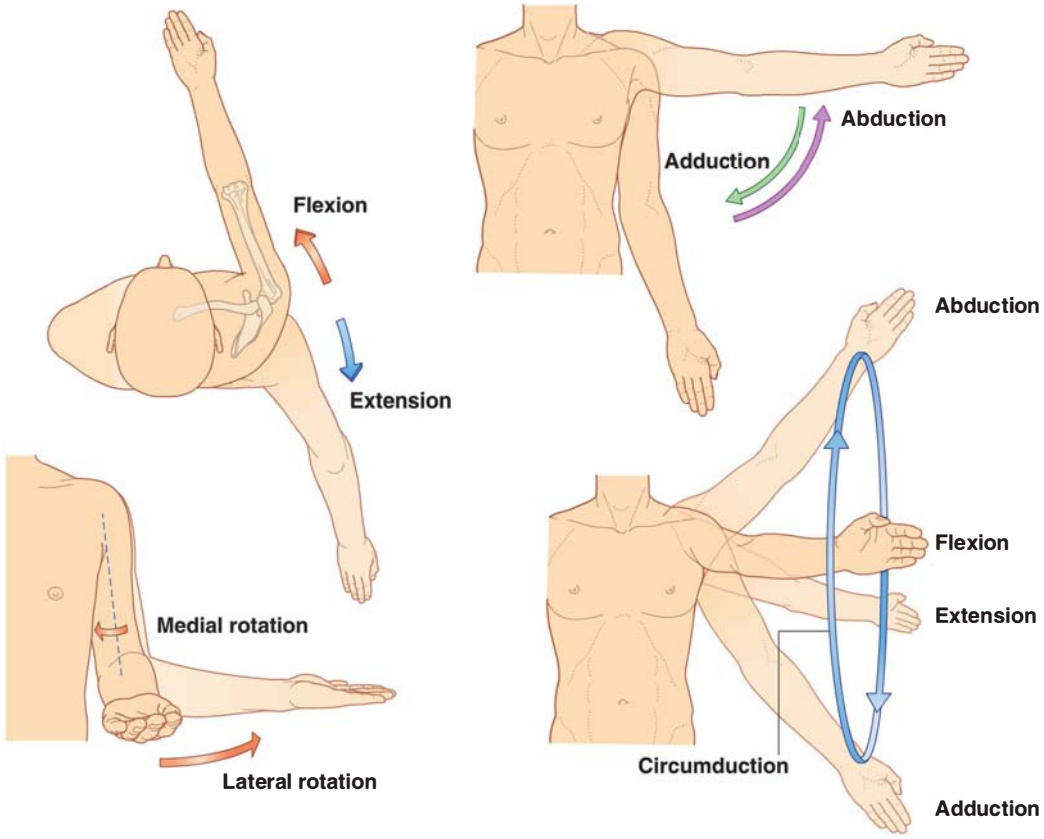


Fig. 7.4 Movements of the arm at the glenohumeral joint.

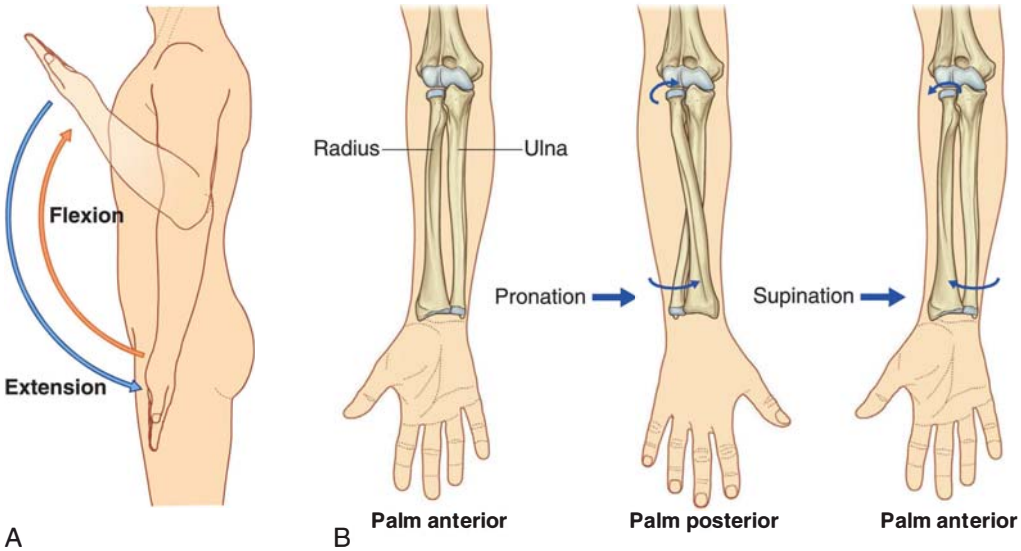


Fig. 7.5 Movements of the forearm. **A.** Flexion and extension at the elbow joint. **B.** Pronation and supination.

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of the radius over the ulna (Fig. 7.5B). This movement, termed pronation, occurs solely in the forearm. Supination returns the hand to the anatomical position.

At the **wrist joint**, the hand can be abducted, adducted, flexed, extended, and circumducted (Fig. 7.6). These movements, combined with those of the shoulder, arm, and forearm, enable the hand to be placed in a wide range of positions relative to the body.

The hand as a mechanical tool

One of the major functions of the hand is to grip and manipulate objects. Gripping objects generally involves flexing the fingers against the thumb. Depending on the type of grip, muscles in the hand act to:

- modify the actions of long tendons that emerge from the forearm and insert into the digits of the hand, and
- produce combinations of joint movements within each digit that cannot be generated by the long flexor and extensor tendons alone coming from the forearm.

The hand as a sensory tool

The hand is used to discriminate between objects on the basis of touch. The pads on the palmar aspect of the fingers

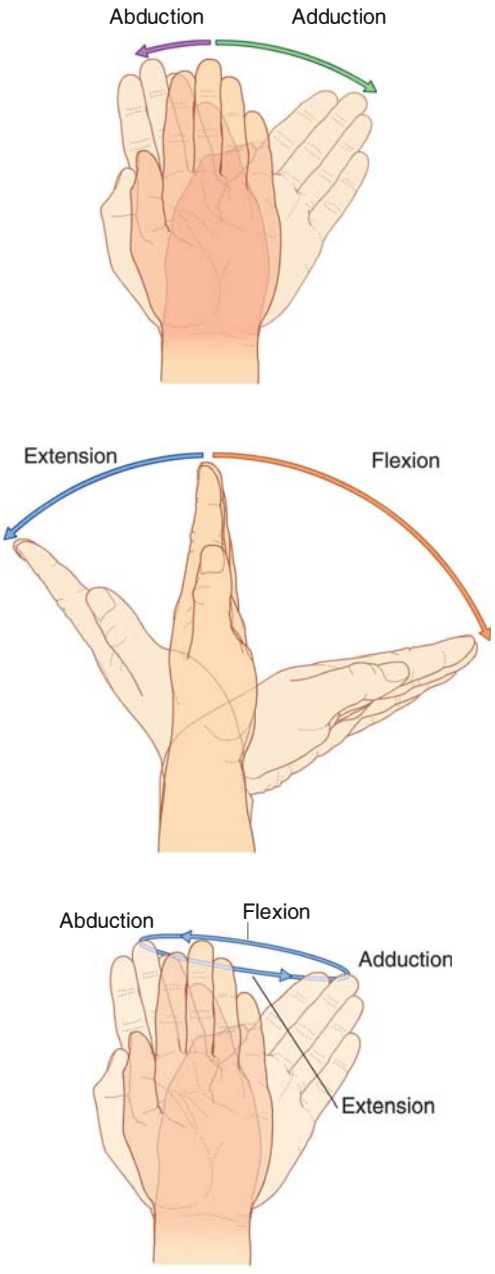


Fig. 7.6 Movements of the hand at the wrist joint.

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contain a high density of somatic sensory receptors. Also, the sensory cortex of the brain devoted to interpreting information from the hand, particularly from the thumb, is disproportionately large relative to that for many other regions of skin.

COMPONENT PARTS

Bones and joints

The bones of the shoulder consist of the scapula, clavicle, and proximal end of the humerus (Fig. 7.7).

The clavicle articulates medially with the manubrium of the sternum and laterally with the acromion of the scapula, which arches over the joint between the glenoid cavity of the scapula and the head of the humerus (the glenohumeral joint).

The humerus is the bone of the arm (Fig. 7.7). The distal end of the humerus articulates with the bones of the forearm at the elbow joint, which is a hinge joint that allows flexion and extension of the forearm.

The forearm contains two bones:

- The lateral bone is the radius.
- The medial bone is the ulna (Fig. 7.7).

At the elbow joint, the proximal ends of the radius and ulna articulate with each other as well as with the humerus.

In addition to flexing and extending the forearm, the elbow joint allows the radius to spin on the humerus while sliding against the head of the ulna during pronation and supination of the hand.

The distal portions of the radius and the ulna also articulate with each other. This joint allows the end of the

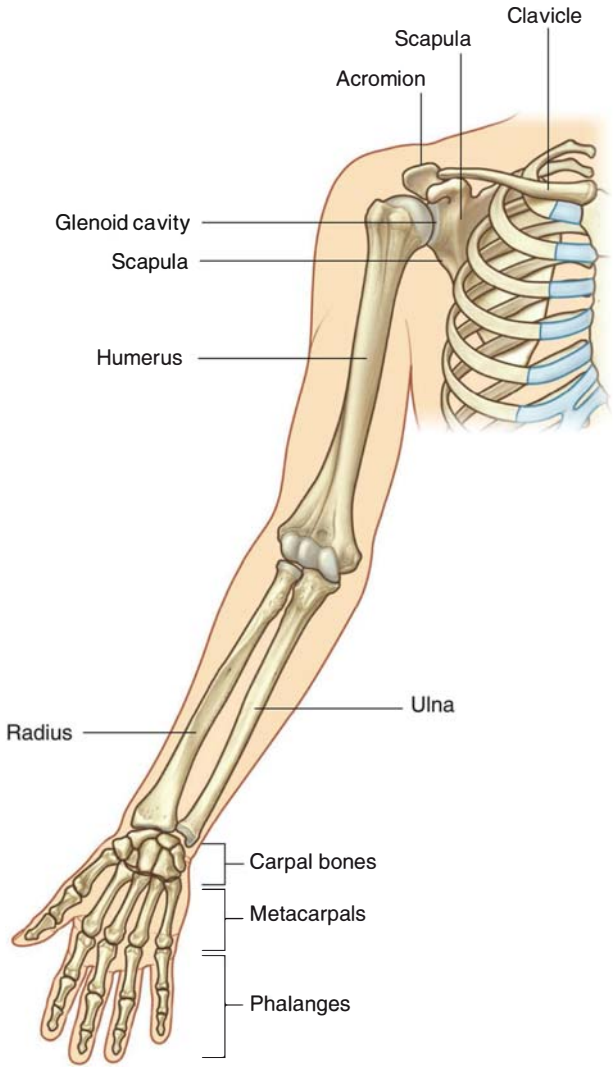


Fig. 7.7 Bones of the upper limb.

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radius to flip from the lateral side to the medial side of the ulna during pronation of the hand.

The wrist joint is formed between the radius and carpal bones of the hand and between an articular disc, distal to the ulna, and carpal bones.

The bones of the hand consist of the carpal bones, the metacarpals, and the phalanges (Fig. 7.7).

The five digits in the hand are the thumb and the index, middle, ring, and little fingers.

Joints between the eight small carpal bones allow only limited amounts of movement; as a result, the bones work together as a unit.

The five metacarpals, one for each digit, are the primary skeletal foundation of the palm (Fig. 7.7).

The joint between the metacarpal of the thumb (metacarpal I) and one of the carpal bones allows greater mobility than the limited sliding movement that occurs at the carpometacarpal joints of the fingers.

Distally, the heads of metacarpals II to V (i.e., except that of the thumb) are interconnected by strong ligaments. Lack of this ligamentous connection between the metacarpal bones of the thumb and index finger together with the biaxial **saddle joint** between the metacarpal bone of the thumb and the carpus provide the thumb with greater freedom of movement than the other digits of the hand.

The bones of the digits are the phalanges (Fig. 7.7). The thumb has two phalanges, while each of the other digits has three.

The metacarpophalangeal joints are biaxial **condylar joints** (**ellipsoid joints**) that allow abduction, adduction, flexion, extension, and circumduction (Fig. 7.8). Abduction and adduction of the fingers is defined in reference to an axis passing through the center of the middle finger in the anatomical position. The middle finger can therefore abduct both medially and laterally and adduct back to the central axis from either side. The interphalangeal joints are primarily **hinge joints** that allow only flexion and extension.

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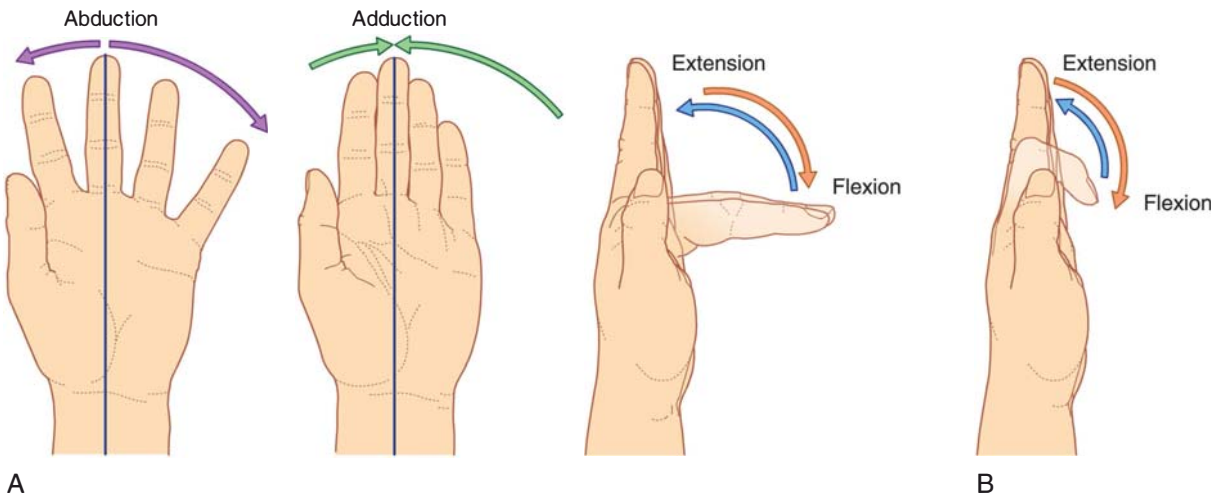


Fig. 7.8 Movements of the metacarpophalangeal (**A**) and interphalangeal (**B**) joints.

Muscles

Some muscles of the shoulder, such as the trapezius, levator scapulae, and rhomboids, connect the scapula and clavicle to the trunk. Other muscles connect the clavicle, scapula, and body wall to the proximal end of the humerus. These include the pectoralis major, pectoralis minor, latissimus dorsi, teres major, and deltoid (Fig. 7.9A,B). The most important of these muscles are the four rotator cuff muscles—the subscapularis, supraspinatus, infraspinatus, and teres minor muscles—which connect the scapula to the humerus and provide support for the glenohumeral joint (Fig. 7.9C).

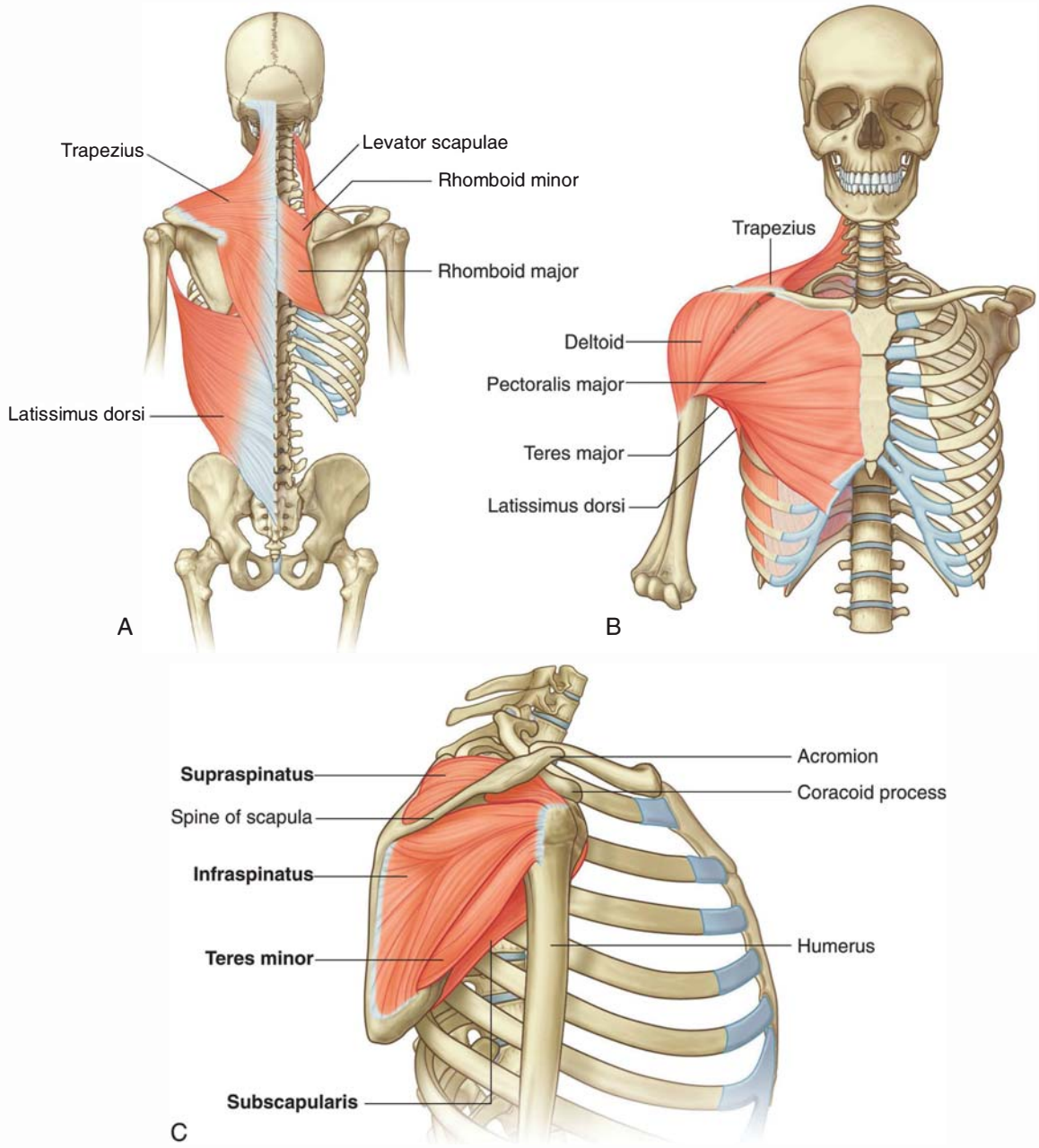


Fig. 7.9 Muscles of the shoulder. **A.** Posterior shoulder. **B.** Anterior shoulder. **C.** Rotator cuff muscles.

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Muscles in the arm and forearm are separated into anterior (flexor) and posterior (extensor) compartments by layers of fascia, bones, and ligaments (Fig. 7.10).

The anterior compartment of the arm lies anteriorly in position and is separated from muscles of the posterior compartment by the humerus and by medial and lateral intermuscular septa. These intermuscular septa are continuous with the deep fascia enclosing the arm and attach to the sides of the humerus.

In the forearm, the anterior and posterior compartments are separated by a lateral intermuscular septum, the radius, the ulna, and an interosseous membrane, which joins adjacent sides of the radius and ulna (Fig. 7.10).

Muscles in the arm act mainly to move the forearm at the elbow joint, while those in the forearm function predominantly to move the hand at the wrist joint and the fingers and thumb.

Muscles found entirely in the hand, the intrinsic muscles, generate delicate movements of the digits of the hand and modify the forces produced by tendons coming into the fingers and thumb from the forearm. Included among the intrinsic muscles of the hand are three small thenar muscles, which form a soft tissue mound, called the **thenar eminence**, over the palmar aspect of metacarpal I. The thenar muscles allow the thumb to move freely relative to the other fingers.

RELATIONSHIP TO OTHER REGIONS

Neck

The upper limb is directly related to the neck. Lying on each side of the **superior thoracic aperture** at the base of the neck is an **axillary inlet,** which is formed by:

- the lateral margin of rib I,
- the posterior surface of the clavicle,
- the superior margin of the scapula, and

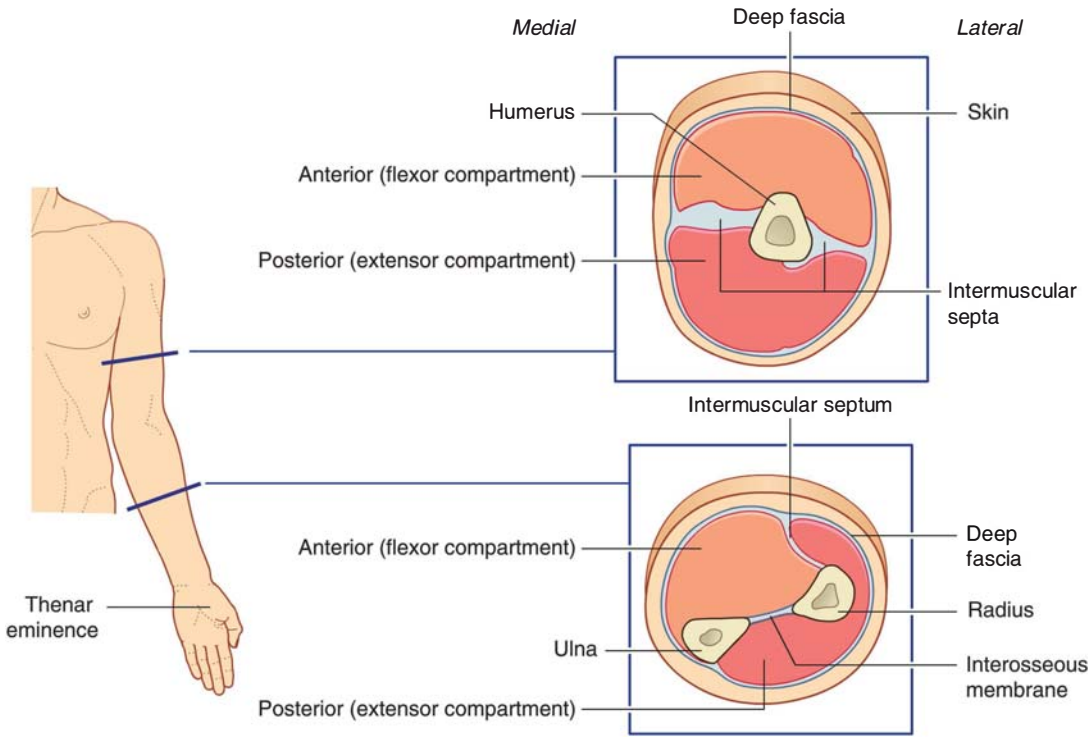


Fig. 7.10 Muscle components in the arm and forearm.

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- the medial surface of the coracoid process of the scapula (Fig. 7.11).

The major artery and vein of the upper limb pass between the thorax and the limb by passing over rib I and through the axillary inlet. Nerves, predominantly derived from the cervical portion of the spinal cord, also pass through the axillary inlet and the axilla to supply the upper limb.

Back and thoracic wall

Muscles that attach the bones of the shoulder to the trunk are associated with the back and the thoracic wall and

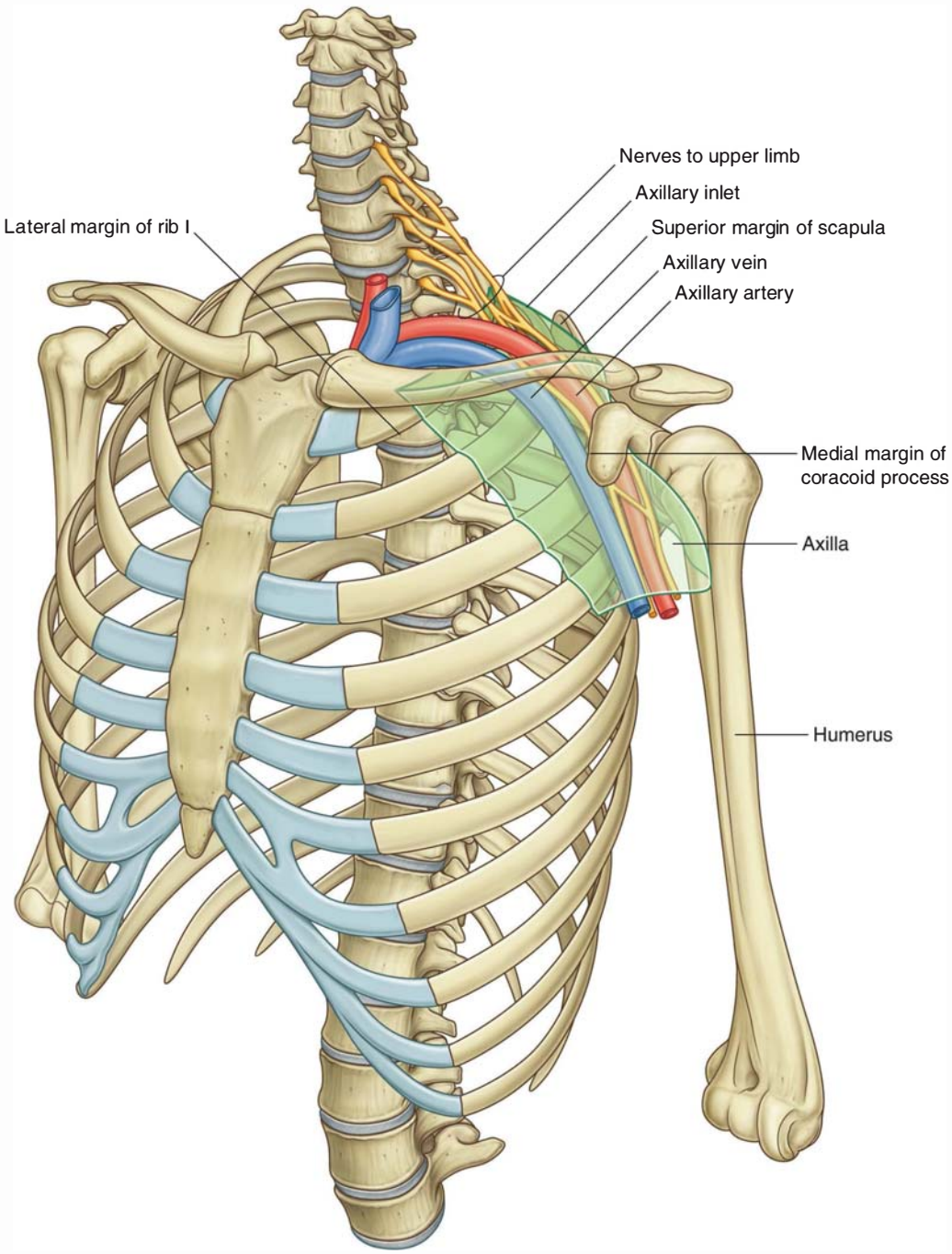


Fig. 7.11 Relationship of the upper limb to the neck.

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include the trapezius, levator scapulae, rhomboid major, rhomboid minor, and latissimus dorsi (Fig. 7.12).

The breast on the anterior thoracic wall has a number of significant relationships with the axilla and upper limb. It overlies the pectoralis major muscle, which forms most of the anterior wall of the axilla and attaches the humerus to the chest wall (Fig. 7.13). Often, part of the breast known as the axillary process extends around the lateral margin of the pectoralis major into the axilla.

Lymphatic drainage from lateral and superior parts of the breast is predominantly into lymph nodes in the axilla. Several arteries and veins that supply or drain the gland also originate from, or drain into, major axillary vessels.

KEY POINTS

Innervation by cervical and upper thoracic nerves

Innervation of the upper limb is by the brachial plexus, which is formed by the anterior rami of cervical spinal

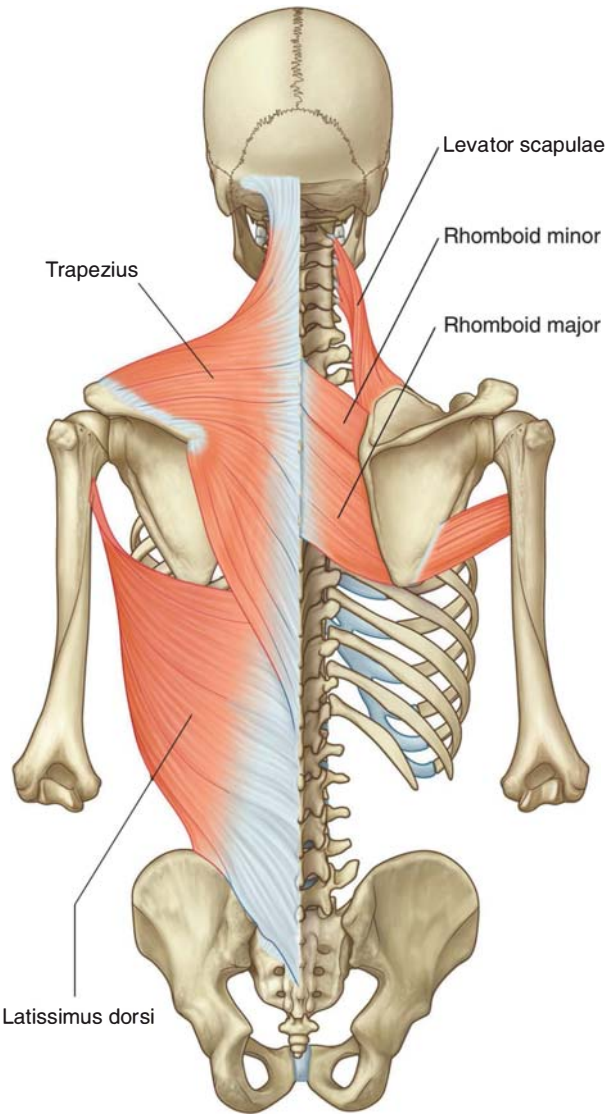


Fig. 7.12 Muscles of the back and thoracic wall.

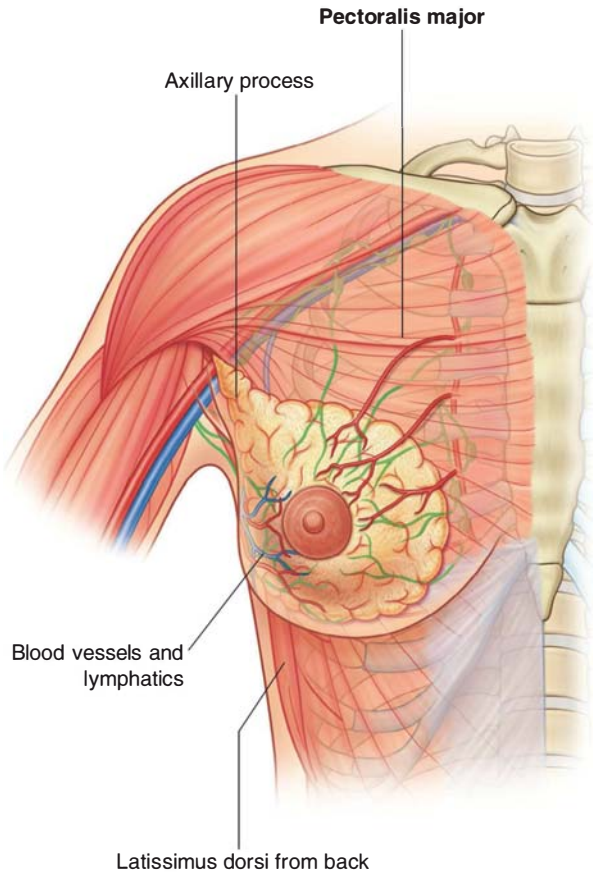


Fig. 7.13 Breast.

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nerves C5 to C8, and T1 (Fig. 7.14). This plexus is initially formed in the neck and then continues through the axillary inlet into the axilla. Major nerves that ultimately innervate the arm, forearm, and hand originate from the brachial plexus in the axilla.

As a consequence of this innervation pattern, clinical testing of lower cervical and T1 nerves is carried out by examining dermatomes, myotomes, and tendon reflexes in the upper limb. Another consequence is that the clinical signs of problems related to lower cervical nerves—pain; pins-and-needles sensations, or paresthesia; and muscle twitching—appear in the upper limb.

Dermatomes of the upper limb (Fig. 7.15A) are often tested for sensation. Areas where overlap of dermatomes is minimal include the:

- upper lateral region of the arm for spinal cord level C5,
- palmar pad of the thumb for spinal cord level C6,
- pad of the index finger for spinal cord level C7,
- pad of the little finger for spinal cord level C8, and
- skin on the medial aspect of the elbow for spinal cord level T1.

Selected joint movements are used to test myotomes (Fig. 7.15B):

- Abduction of the arm at the glenohumeral joint is controlled predominantly by C5.
- Flexion of the forearm at the elbow joint is controlled primarily by C6.
- Extension of the forearm at the elbow joint is controlled mainly by C7.
- Flexion of the fingers is controlled mainly by C8.
- Abduction and adduction of the index, middle, and ring fingers is controlled predominantly by T1.

In an unconscious patient, both somatic sensory and motor functions of spinal cord levels can be tested using tendon reflexes:

- A tap on the tendon of the biceps in the cubital fossa tests mainly for spinal cord level C6.
- A tap on the tendon of the triceps posterior to the elbow tests mainly for C7.

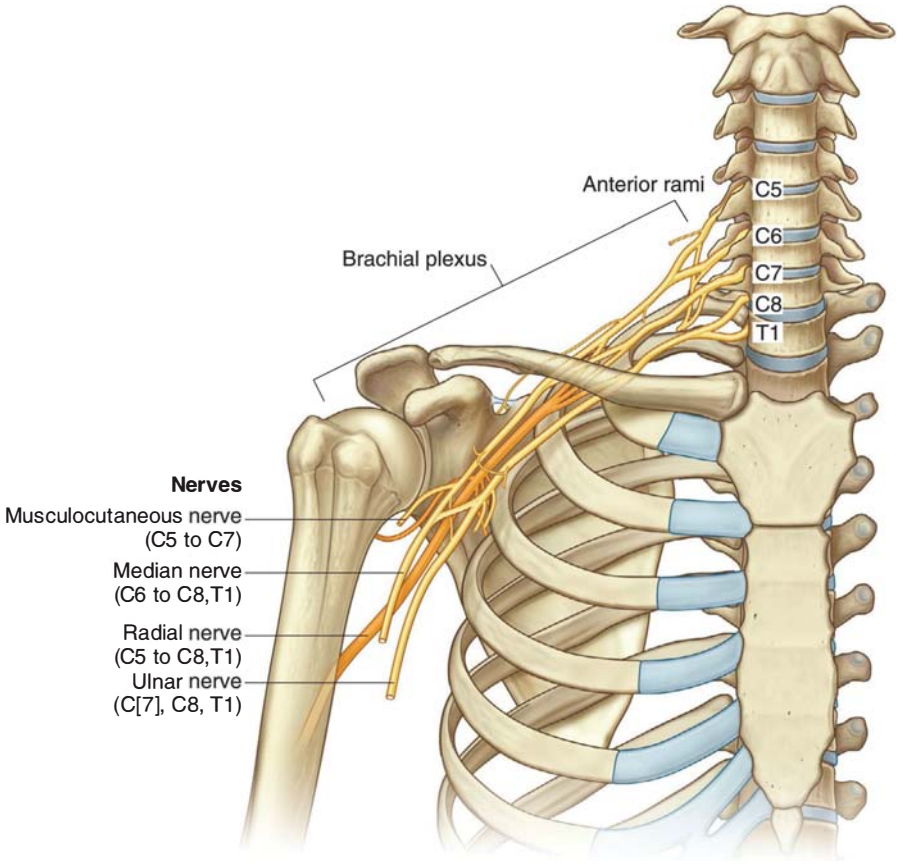


Fig. 7.14 Innervation of the upper limb.

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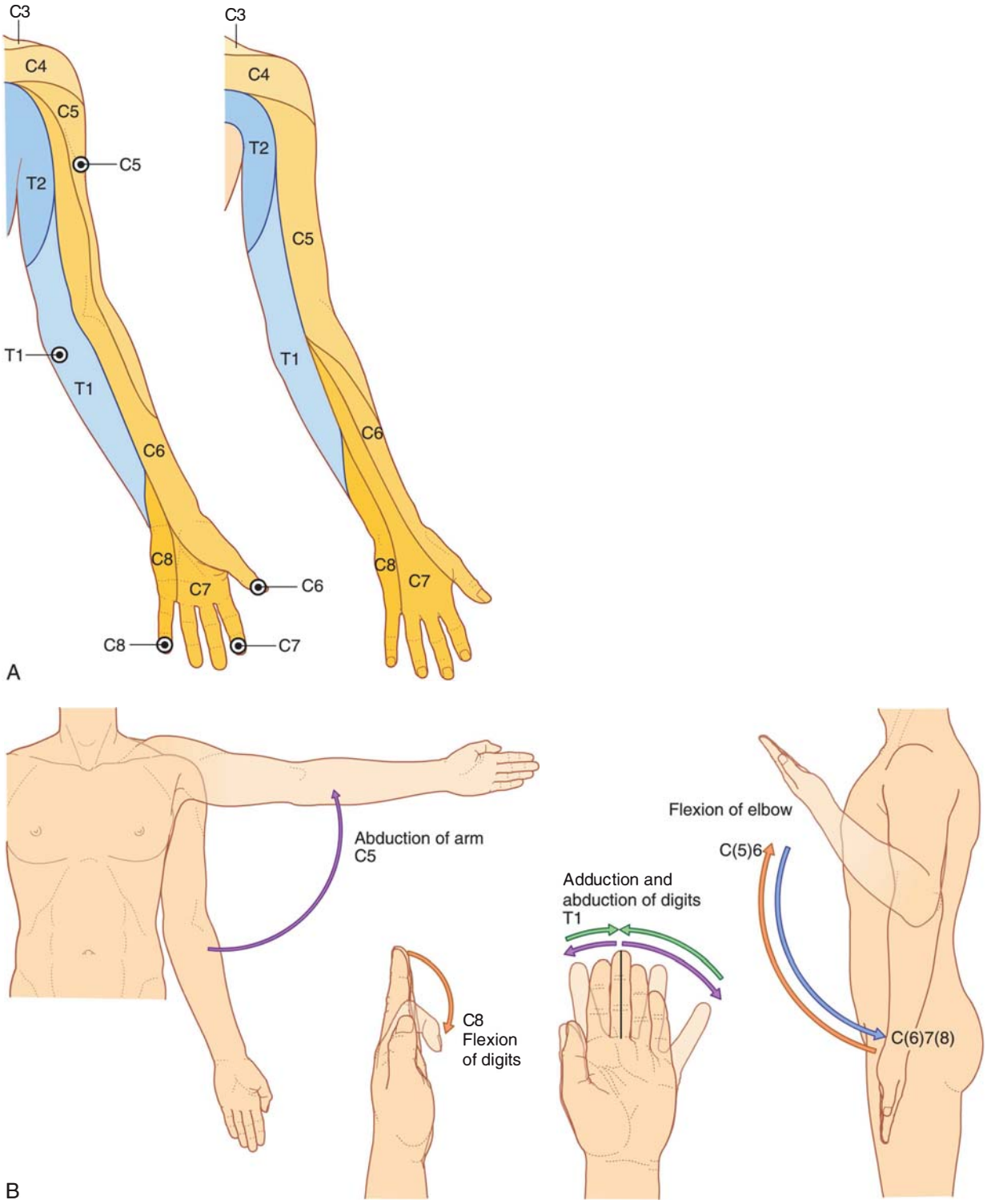


Fig. 7.15 Dermatomes and myotomes in the upper limb. **A.** Dermatomes. **B.** Movements produced by myotomes.

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The major spinal cord level associated with innervation of the diaphragm, C4, is immediately above the spinal cord levels associated with the upper limb.

Evaluation of dermatomes and myotomes in the upper limb can provide important information about potential breathing problems that might develop as complications of damage to the spinal cord in regions just below the C4 spinal level.

Each of the major muscle compartments in the arm and forearm and each of the intrinsic muscles of the hand is innervated predominantly by one of the major nerves that originate from the brachial plexus in the axilla (Fig. 7.16A):

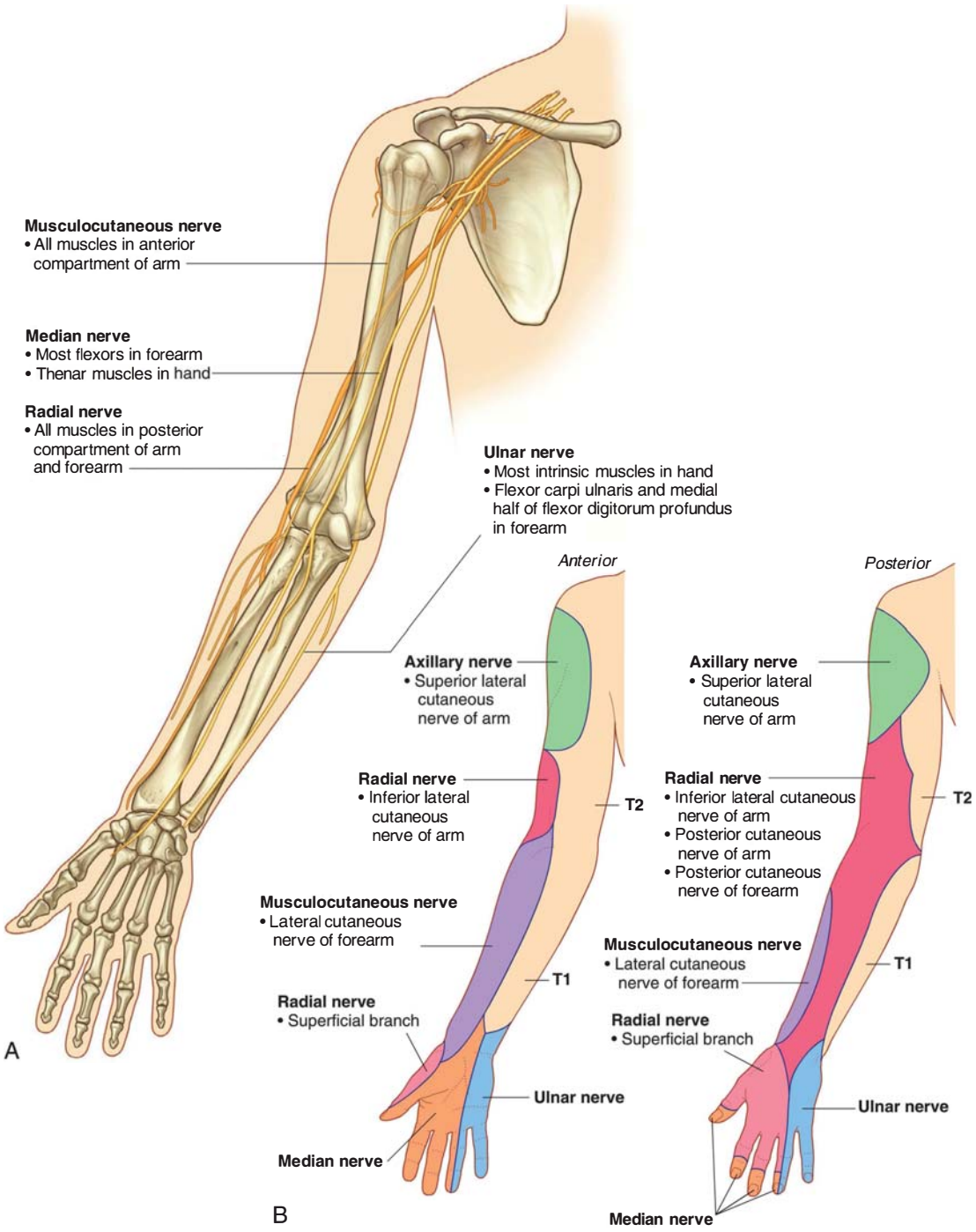


Fig. 7.16 Nerves of upper limb. **A.** Major nerves in the arm and forearm. **B.** Anterior and posterior areas of skin innervated by major peripheral nerves in the arm and forearm.

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- All muscles in the anterior compartment of the arm are innervated by the musculocutaneous nerve.
- The median nerve innervates the muscles in the anterior compartment of the forearm, with two exceptions—one flexor of the wrist (the flexor carpi ulnaris muscle) and part of one flexor of the fingers (the medial half of the flexor digitorum profundus muscle) are innervated by the ulnar nerve.
- Most intrinsic muscles of the hand are innervated by the ulnar nerve, except for the thenar muscles and two lateral lumbrical muscles, which are innervated by the median nerve.
- All muscles in the posterior compartments of the arm and forearm are innervated by the radial nerve.

In addition to innervating major muscle groups, each of the major peripheral nerves originating from the brachial plexus carries somatic sensory information from patches of skin quite different from dermatomes (Fig. 7.16B). Sensation in these areas can be used to test for peripheral nerve lesions:

- The musculocutaneous nerve innervates skin on the anterolateral side of the forearm.
- The median nerve innervates the palmar surface of the lateral three and one-half digits, and the ulnar nerve innervates the medial one and one-half digits.
- The radial nerve supplies skin on the posterior surface of the forearm and the dorsolateral surface of the hand.

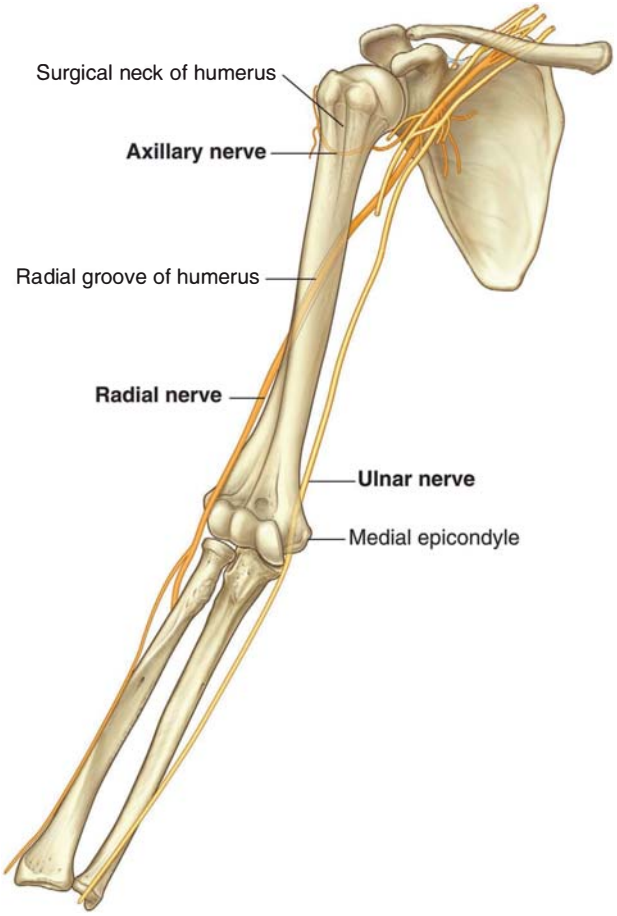


Fig. 7.17 Nerves related to the humerus.

Nerves related to bone

Three important nerves are directly related to parts of the humerus (Fig. 7.17):

- The axillary nerve, which supplies the deltoid muscle, a major abductor of the humerus at the glenohumeral joint, passes around the posterior aspect of the upper part of the humerus (the surgical neck).
- The radial nerve, which supplies all of the extensor muscles of the upper limb, passes diagonally around the posterior surface of the middle of the humerus in the radial groove.
- The ulnar nerve, which is ultimately destined for the hand, passes posteriorly to a bony protrusion, the medial epicondyle, on the medial side of the distal end of the humerus.

Fractures of the humerus in any one of these three regions can endanger the related nerve.

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Superficial veins

Large veins embedded in the superficial fascia of the upper limb are often used to access a patient's vascular system and to withdraw blood. The most significant of these veins are the cephalic, basilic, and median cubital veins (Fig. 7.18).

The **cephalic** and **basilic veins** originate from the **dorsal venous network** on the back of the hand.

The cephalic vein originates over the anatomical snuffbox at the base of the thumb, passes laterally around the distal forearm to reach the anterolateral surface of the limb, and then continues proximally. It crosses the elbow, then passes up the arm into a triangular depression—the **clavipectoral triangle** (**deltopectoral triangle**)—between the pectoralis major muscle, deltoid muscle, and clavicle. In this depression, the vein passes into the axilla by penetrating deep fascia just inferior to the clavicle.

The basilic vein originates from the medial side of the dorsal venous network of the hand and passes proximally up the posteromedial surface of the forearm. It passes onto the anterior surface of the limb just inferior to the elbow and then continues proximally to penetrate deep fascia about midway up the arm.

At the elbow, the cephalic and basilic veins are connected by the **median cubital vein**, which crosses the roof of the cubital fossa.

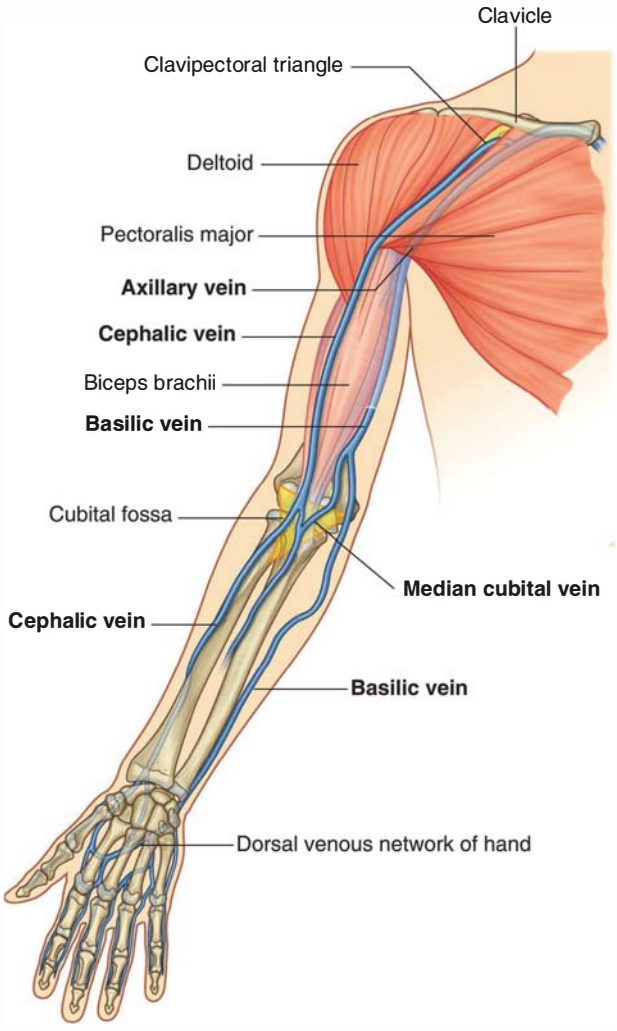


Fig. 7.18 Veins in the superficial fascia of upper limb. The area of the cubital fossa is shown in yellow.

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Orientation of the thumb

The thumb is positioned at right angles to the orientation of the index, middle, ring, and little fingers (Fig. 7.19). As a result, movements of the thumb occur at right angles to those of the other digits. For example, flexion brings the thumb across the palm, whereas abduction moves it away from the fingers at right angles to the palm.

Importantly, with the thumb positioned at right angles to the palm, only a slight rotation of metacarpal I on the wrist brings the pad of the thumb into a position directly facing the pads of the other fingers. This opposition of the thumb is essential for normal hand function.

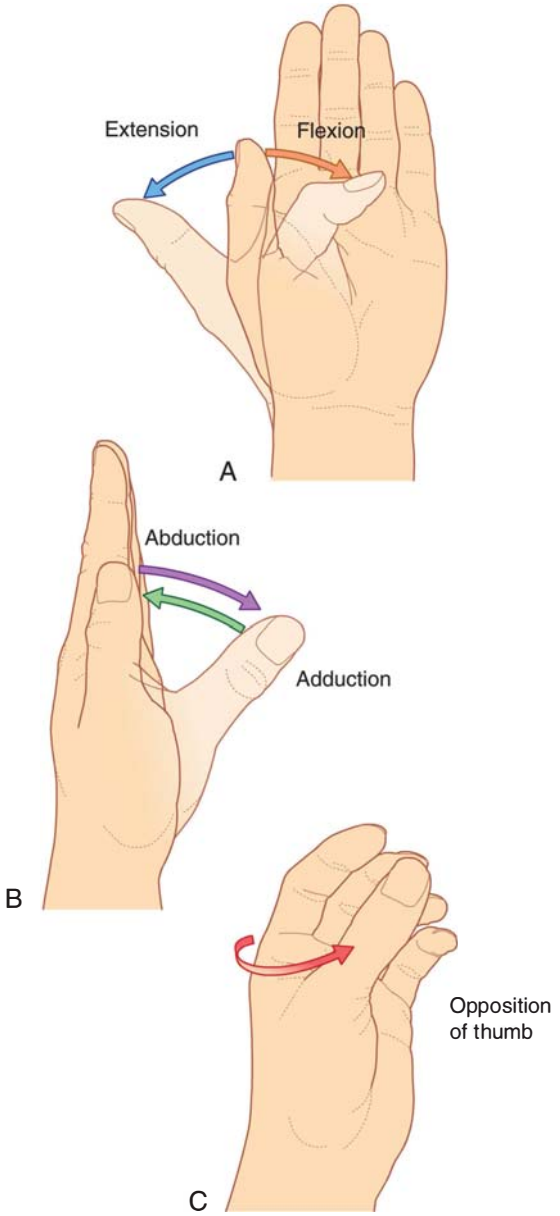


Fig. 7.19 A to **C.** Movements of the thumb.

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Regional anatomy

SHOULDER

The shoulder is the region of upper limb attachment to the trunk.

The bone framework of the shoulder consists of:

- the clavicle and scapula, which form the **pectoral girdle** (**shoulder girdle**), and
- the proximal end of the humerus.

The superficial muscles of the shoulder consist of the trapezius and deltoid muscles, which together form the smooth muscular contour over the lateral part of the shoulder. These muscles connect the scapula and clavicle to the trunk and to the arm, respectively.

Bones

Clavicle

The clavicle is the only bony attachment between the trunk and the upper limb. It is palpable along its entire length and has a gentle S-shaped contour, with the forward-facing convex part medial and the forward-facing concave part lateral. The acromial (lateral) end of the clavicle is flat, whereas the sternal (medial) end is more robust and somewhat quadrangular in shape (Fig. 7.20).

The acromial end of the clavicle has a small oval facet on its surface for articulation with a similar facet on the medial surface of the acromion of the scapula.

The sternal end has a much larger facet for articulation mainly with the manubrium of the sternum, and to a lesser extent, with the first costal cartilage.

The inferior surface of the lateral third of the clavicle possesses a distinct tuberosity consisting of a tubercle (the **conoid tubercle**) and lateral roughening (the **trapezoid line**), for attachment of the important coracoclavicular ligament.

In addition, the surfaces and margins of the clavicle are roughened by the attachment of muscles that connect the clavicle to the thorax, neck, and upper limb. The superior surface is smoother than the inferior surface.

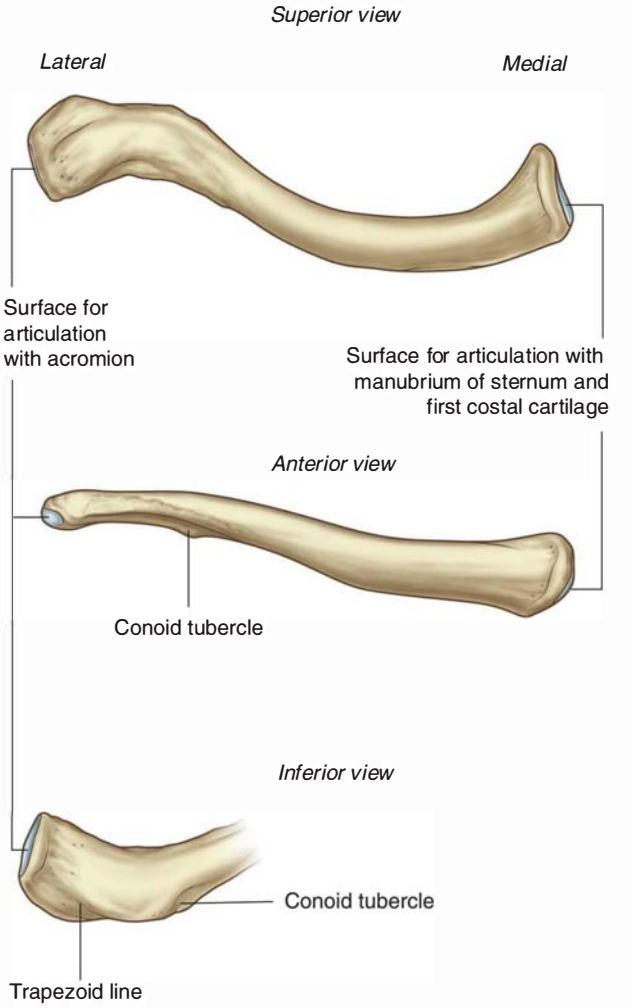


Fig. 7.20 Right clavicle.

Scapula

The scapula is a large, flat triangular bone with:

- three angles (lateral, superior, and inferior),
- three borders (superior, lateral, and medial),
- two surfaces (costal and posterior), and
- three processes (acromion, spine, and coracoid process) (Fig. 7.21).

The **lateral angle** of the scapula is marked by a shallow, somewhat comma-shaped **glenoid cavity**, which articulates with the head of the humerus to form the glenohumeral joint (Fig. 7.21B,C).

A large triangular-shaped roughening (the **infraglenoid tubercle**) inferior to the glenoid cavity is the site of attachment for the long head of the triceps brachii muscle.

A less distinct **supraglenoid tubercle** is located superior to the glenoid cavity and is the site of attachment for the long head of the biceps brachii muscle.

A prominent **spine** subdivides the **posterior surface** of the scapula into a small, superior **supraspinous**

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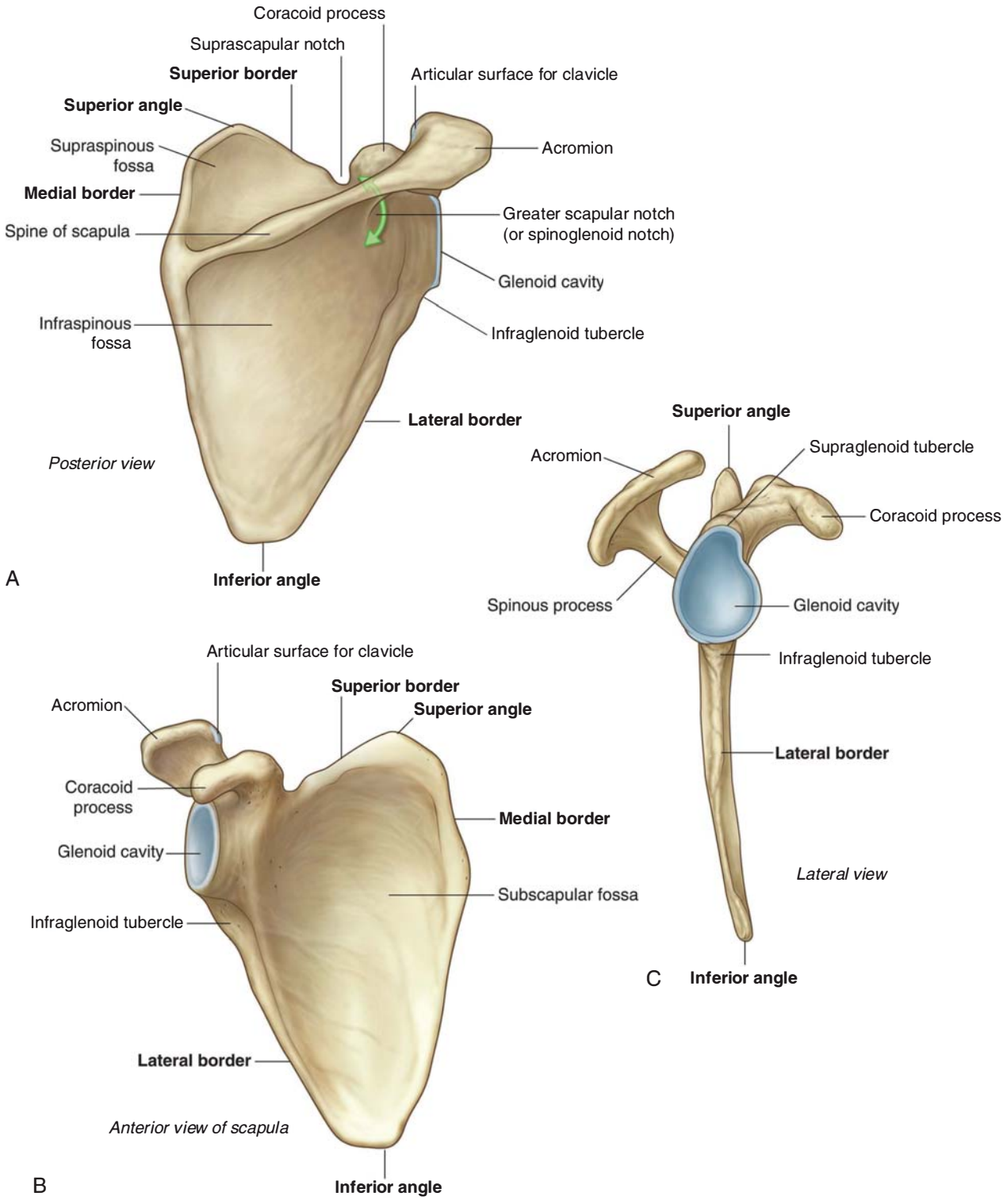


Fig. 7.21 Scapula. **A.** Posterior view of right scapula. **B.** Anterior view of costal surface. **C.** Lateral view.

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fossa and a much larger, inferior **infraspinous fossa** (Fig. 7.21A).

The **acromion**, which is an anterolateral projection of the spine, arches over the glenohumeral joint and articulates, via a small oval facet on its distal end, with the clavicle.

The region between the lateral angle of the scapula and the attachment of the spine to the posterior surface of the scapula is the **greater scapular notch** (**spinoglenoid notch**).

Unlike the posterior surface, the **costal surface** of the scapula is unremarkable, being characterized by a shallow concave **subscapular fossa** over much of its extent (Fig. 7.21B). The costal surface and margins provide for muscle attachment, and the costal surface, together with its related muscle (**subscapularis**), moves freely over the underlying thoracic wall.

The lateral border of the scapula is strong and thick for muscle attachment, whereas the medial border and much of the superior border is thin and sharp.

The superior border is marked on its lateral end by:

- the **coracoid process**, a hook-like structure that projects anterolaterally and is positioned directly inferior to the lateral part of the clavicle; and
- the small but distinct **suprascapular notch**, which lies immediately medial to the root of the coracoid process.

The spine and acromion can be readily palpated on a patient, as can the tip of the coracoid process, the inferior angle, and much of the medial border of the scapula.

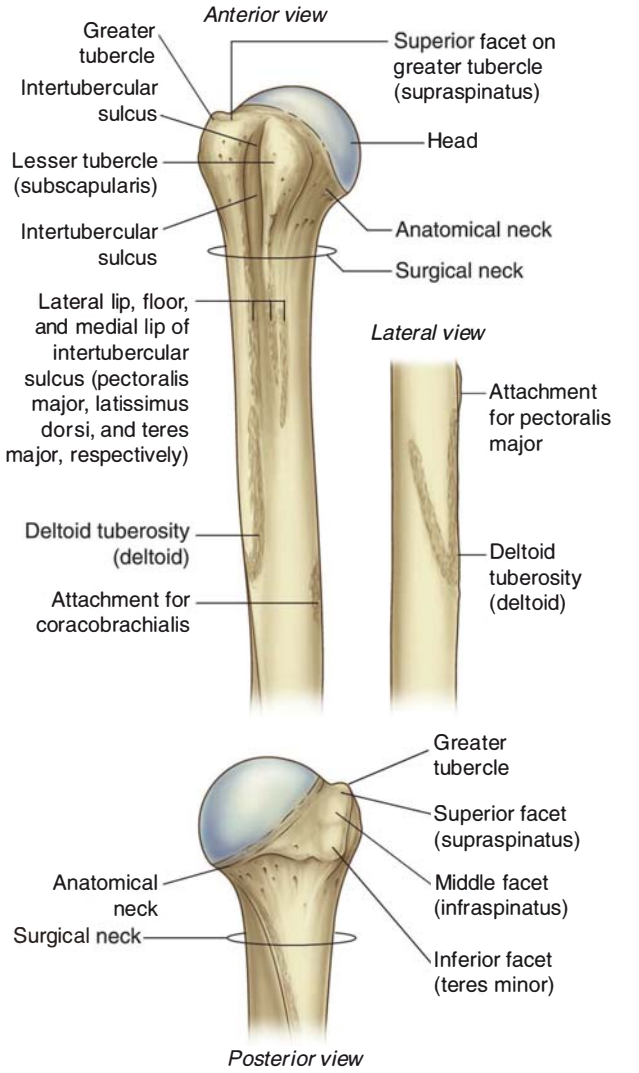


Fig. 7.22 Proximal end of right humerus.

Proximal humerus

The proximal end of the humerus consists of the head, the anatomical neck, the greater and lesser tubercles, the surgical neck, and the superior half of the shaft of the humerus (Fig. 7.22).

The **head** is half-spherical in shape and projects medially and somewhat superiorly to articulate with the much smaller glenoid cavity of the scapula.

The **anatomical neck** is very short and is formed by a narrow constriction immediately distal to the head. It lies between the head and the greater and lesser tubercles laterally, and between the head and the shaft more medially.

Greater and lesser tubercles

The **greater** and **lesser tubercles** are prominent landmarks on the proximal end of the humerus and serve as attachment sites for the four rotator cuff muscles of the glenohumeral joint.

The greater tubercle is lateral in position. Its superior surface and posterior surface are marked by three large smooth facets for muscle tendon attachments:

- The superior facet is for attachment of the supraspinatus muscle.
- The middle facet is for attachment of the infraspinatus.
- The inferior facet is for attachment of the teres minor.

The lesser tubercle is anterior in position and its surface is marked by a large smooth impression for attachment of the subscapularis muscle.

A deep **intertubercular sulcus** (**bicipital groove**) separates the lesser and greater tubercles and continues inferiorly onto the proximal shaft of the humerus (Fig. 7.22). The tendon of the long head of the biceps brachii passes through this sulcus.

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Roughenings on the lateral and medial lips and on the floor of the intertubercular sulcus mark sites for the attachment of the pectoralis major, teres major, and latissimus dorsi muscles, respectively.

The lateral lip of the intertubercular sulcus is continuous inferiorly with a large V-shaped **deltoid tuberosity** on the lateral surface of the humerus midway along its length (Fig. 7.22), which is where the deltoid muscle inserts onto the humerus.

In approximately the same position, but on the medial surface of the bone, there is a thin vertical roughening for attachment of the coracobrachialis muscle.

Surgical neck

One of the most important features of the proximal end of the humerus is the **surgical neck** (Fig. 7.22). This region is oriented in the horizontal plane between the expanded proximal part of the humerus (head, anatomical neck, and tubercles) and the narrower shaft. The axillary nerve and the posterior circumflex humeral artery, which pass into the deltoid region from the axilla, do so immediately posterior to the surgical neck. Because the surgical neck is weaker than more proximal regions of the bone, it is one of the sites where the humerus commonly fractures. The associated nerve (axillary) and artery (posterior circumflex humeral) can be damaged by fractures in this region.

In the clinic

Fracture of the proximal humerus

It is extremely rare for fractures to occur across the anatomical neck of the humerus because the obliquity of such a fracture would have to traverse the thickest region of bone. Typically fractures occur around the surgical neck of the humerus. Although the axillary nerve and posterior circumflex humeral artery may be damaged with this type of fracture, this rarely happens. It is important that the axillary nerve is tested before relocation to be sure that the injury has not damaged the nerve and that the treatment itself does not cause a neurological deficit.

Joints

The three joints in the shoulder complex are the sternoclavicular, acromioclavicular, and glenohumeral joints.

The sternoclavicular joint and the acromioclavicular joint link the two bones of the pectoral girdle to each other and to the trunk. The combined movements at these two joints enable the scapula to be positioned over a wide range on the thoracic wall, substantially increasing "reach" by the upper limb.

The glenohumeral joint (shoulder joint) is the articulation between the humerus of the arm and the scapula.

Sternoclavicular joint

The sternoclavicular joint occurs between the proximal end of the clavicle and the **clavicular notch** of the **manubrium of the sternum** together with a small part of the first costal cartilage (Fig. 7.23). It is synovial and saddle shaped. The articular cavity is completely separated into two compartments by an articular disc. The sternoclavicular joint allows movement of the clavicle, predominantly in the anteroposterior and vertical planes, although some rotation also occurs.

The sternoclavicular joint is surrounded by a joint capsule and is reinforced by four ligaments:

- The **anterior** and **posterior sternoclavicular ligaments** are anterior and posterior, respectively, to the joint.

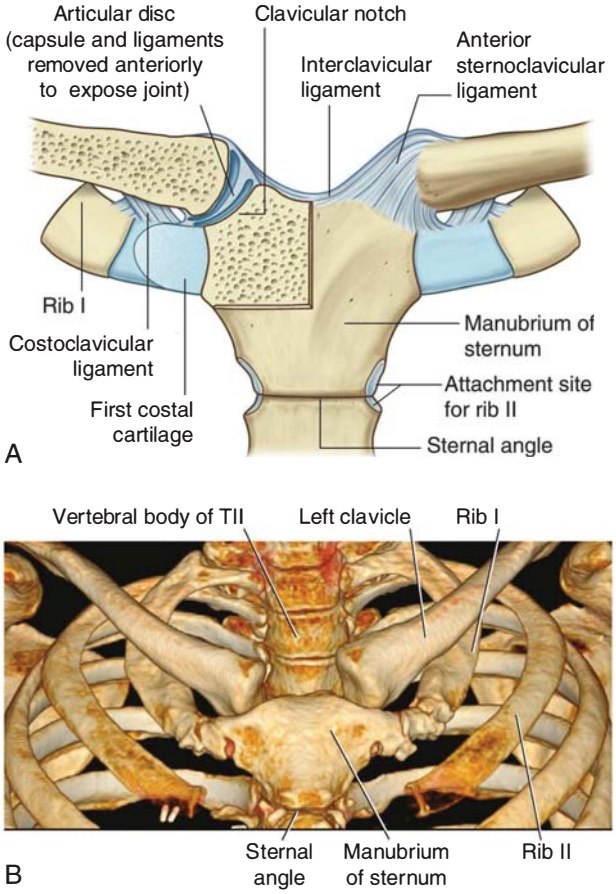


Fig. 7.23 Sternoclavicular joint. **A.** Bones and ligaments. **B.** Volume-rendered reconstruction using multidetector computed tomography.

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- An **interclavicular ligament** links the ends of the two clavicles to each other and to the superior surface of the manubrium of the sternum.
- The **costoclavicular ligament** is positioned laterally to the joint and links the proximal end of the clavicle to the first rib and related costal cartilage.

Acromioclavicular joint

The acromioclavicular joint is a small synovial joint between an oval facet on the medial surface of the acromion and a similar facet on the acromial end of the clavicle (Fig. 7.24, also see Fig. 7.31). It allows movement in the anteroposterior and vertical planes together with some axial rotation.

The acromioclavicular joint is surrounded by a joint capsule and is reinforced by:

- a small **acromioclavicular ligament** superior to the joint and passing between adjacent regions of the clavicle and acromion, and
- a much larger **coracoclavicular ligament**, which is not directly related to the joint, but is an important strong accessory ligament, providing much of the weight-bearing support for the upper limb on the clavicle and maintaining the position of the clavicle on the acromion—it spans the distance between the coracoid process of the scapula and the inferior surface of the acromial end of the clavicle and comprises an anterior **trapezoid ligament** (which attaches to the trapezoid line on the clavicle) and a posterior **conoid ligament** (which attaches to the related conoid tubercle).

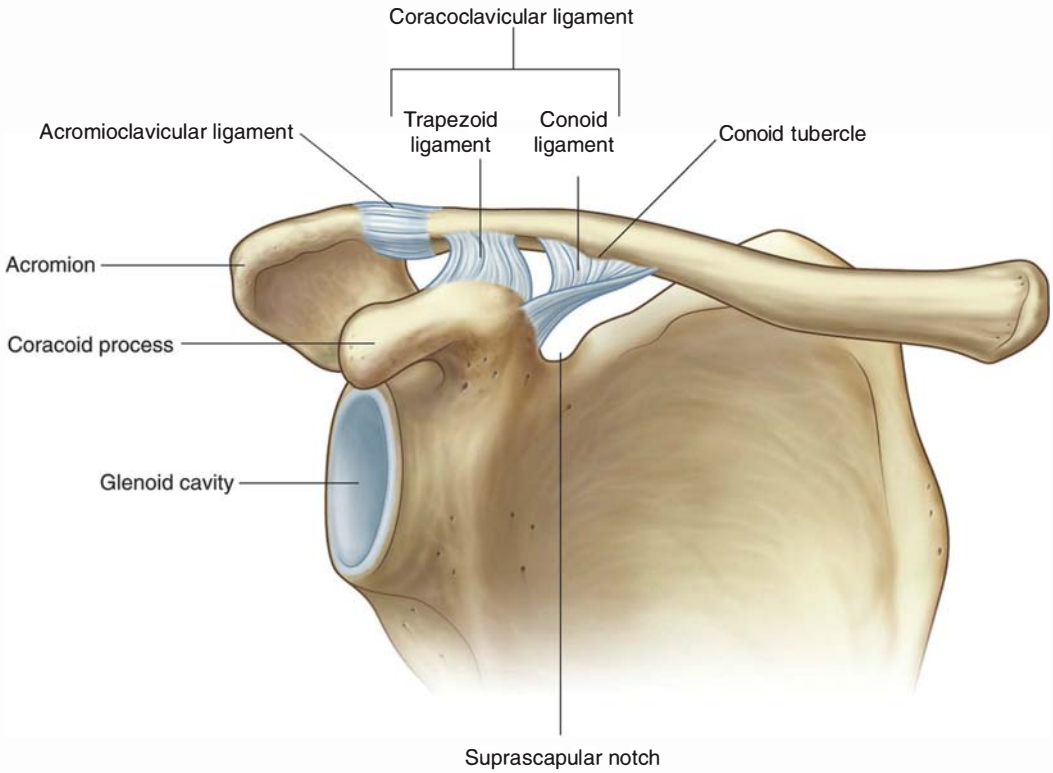


Fig. 7.24 Right acromioclavicular joint.

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Glenohumeral joint

The **glenohumeral joint** is a synovial ball and socket articulation between the head of the humerus and the glenoid cavity of the scapula (Fig. 7.25). It is multiaxial with a wide range of movements provided at the cost of skeletal stability. Joint stability is provided, instead, by the rotator cuff muscles, the long head of the biceps brachii muscle, related bony processes, and extracapsular ligaments. Movements at the joint include flexion, extension, abduction, adduction, medial rotation, lateral rotation, and circumduction.

The articular surfaces of the glenohumeral joint are the large spherical head of the humerus and the small glenoid cavity of the scapula (Fig. 7.25). Each of the surfaces is covered by hyaline cartilage.

The glenoid cavity is deepened and expanded peripherally by a fibrocartilaginous collar (the **glenoid labrum**), which attaches to the margin of the fossa. Superiorly, this labrum is continuous with the tendon of the long head of the biceps brachii muscle, which attaches to the supraglenoid tubercle and passes through the articular cavity superior to the head of the humerus.

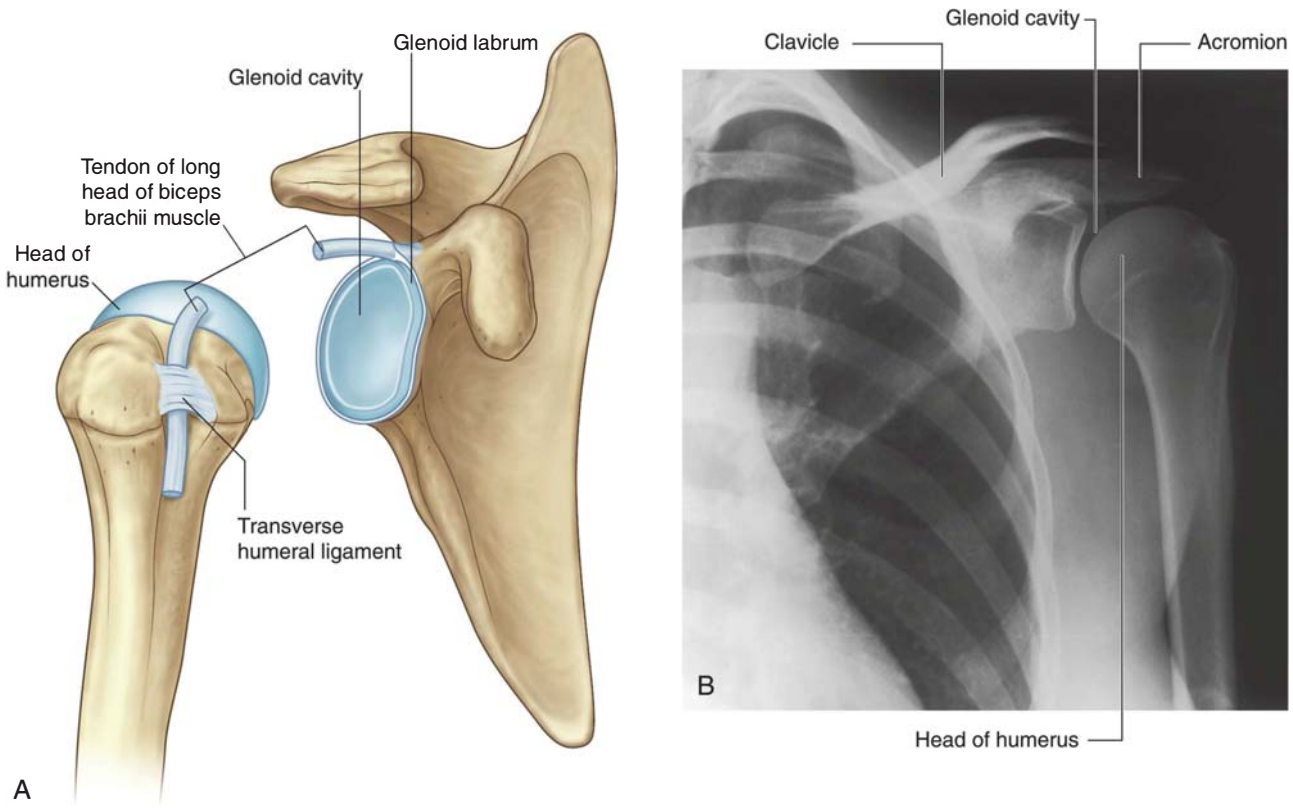


Fig. 7.25 Glenohumeral joint. **A.** Articular surfaces of right glenohumeral joint. **B.** Radiograph of a normal glenohumeral joint.

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The synovial membrane attaches to the margins of the articular surfaces and lines the fibrous membrane of the joint capsule (Fig. 7.26). The synovial membrane is loose inferiorly. This redundant region of synovial membrane and related fibrous membrane accommodates abduction of the arm.

The synovial membrane protrudes through apertures in the fibrous membrane to form bursae, which lie between the tendons of surrounding muscles and the fibrous membrane. The most consistent of these is the **subtendinous bursa of the subscapularis**, which lies between the subscapularis muscle and the fibrous membrane. The synovial membrane also folds around the tendon of the long head of the biceps brachii muscle in the joint and extends along the tendon as it passes into the intertubercular sulcus. All these synovial structures reduce friction between the tendons and adjacent joint capsule and bone.

In addition to bursae that communicate with the articular cavity through apertures in the fibrous membrane, other bursae are associated with the joint but are not connected to it. These occur:

- between the acromion (or deltoid muscle) and supraspinatus muscle (or joint capsule) (the **subacromial** or **subdeltoid bursa**),
- between the acromion and skin,
- between the coracoid process and the joint capsule, and
- in relationship to tendons of muscles around the joint (coracobrachialis, teres major, long head of triceps brachii, and latissimus dorsi muscles).

The fibrous membrane of the joint capsule attaches to the margin of the glenoid cavity, outside the attachment of the glenoid labrum and the long head of the biceps brachii muscle, and to the anatomical neck of the humerus (Fig. 7.27).

On the humerus, the medial attachment occurs more inferiorly than the neck and extends onto the shaft. In this region, the fibrous membrane is also loose or folded in the anatomical position. This redundant area of the fibrous membrane accommodates abduction of the arm.

Openings in the fibrous membrane provide continuity of the articular cavity with bursae that occur between the joint capsule and surrounding muscles and around the tendon of the long head of the biceps brachii muscle in the intertubercular sulcus.

The fibrous membrane of the joint capsule is thickened:

- anterosuperiorly in three locations to form **superior**, **middle**, and **inferior glenohumeral ligaments**, which pass from the superomedial margin of the glenoid cavity to the lesser tubercle and inferiorly related anatomical neck of the humerus (Fig. 7.27);

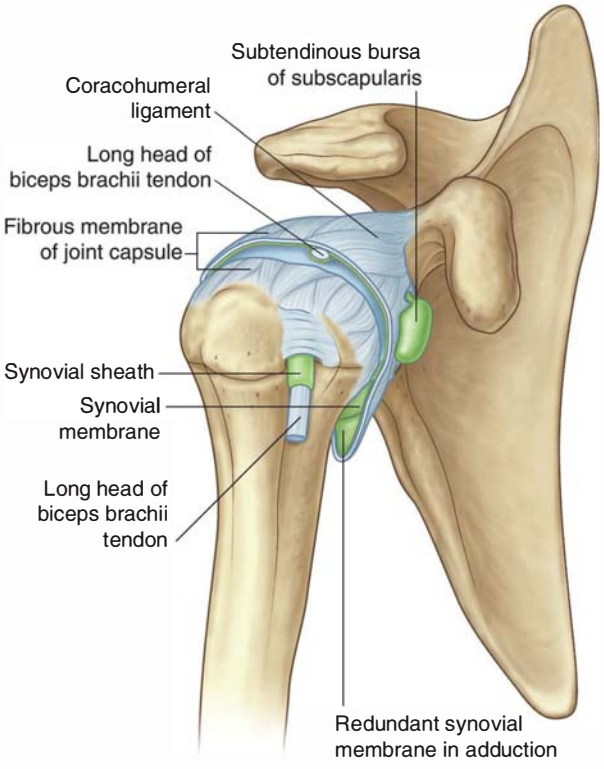


Fig. 7.26 Synovial membrane and joint capsule of right glenohumeral joint.

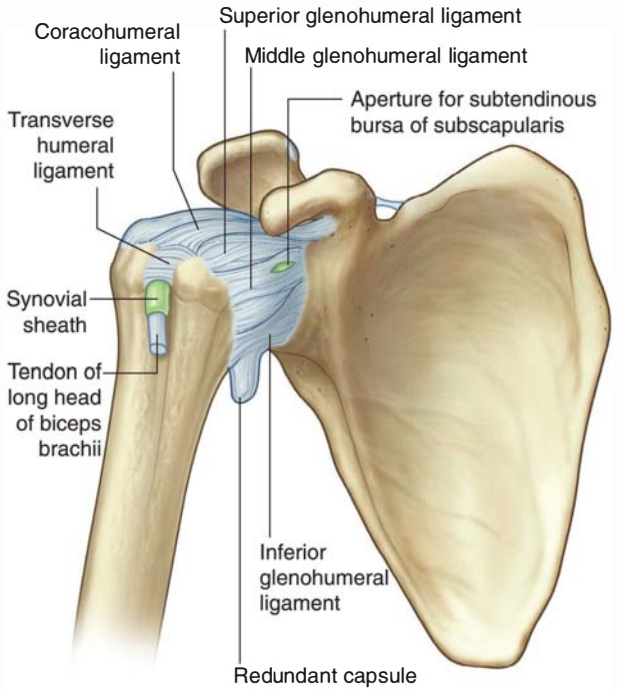


Fig. 7.27 Capsule of right glenohumeral joint.

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- superiorly between the base of the coracoid process and the greater tubercle of the humerus (the **coracohumeral ligament**); and
- between the greater and lesser tubercles of the humerus (**transverse humeral ligament**)—this holds the tendon of the long head of the biceps brachii muscle in the intertubercular sulcus (Fig. 7.27).

Joint stability is provided by surrounding muscle tendons and a skeletal arch formed superiorly by the coracoid process and acromion and the coraco-acromial ligament (Fig. 7.28).

Tendons of the rotator cuff muscles (the supraspinatus, infraspinatus, teres minor, and subscapularis muscles) blend with the joint capsule and form a musculotendinous

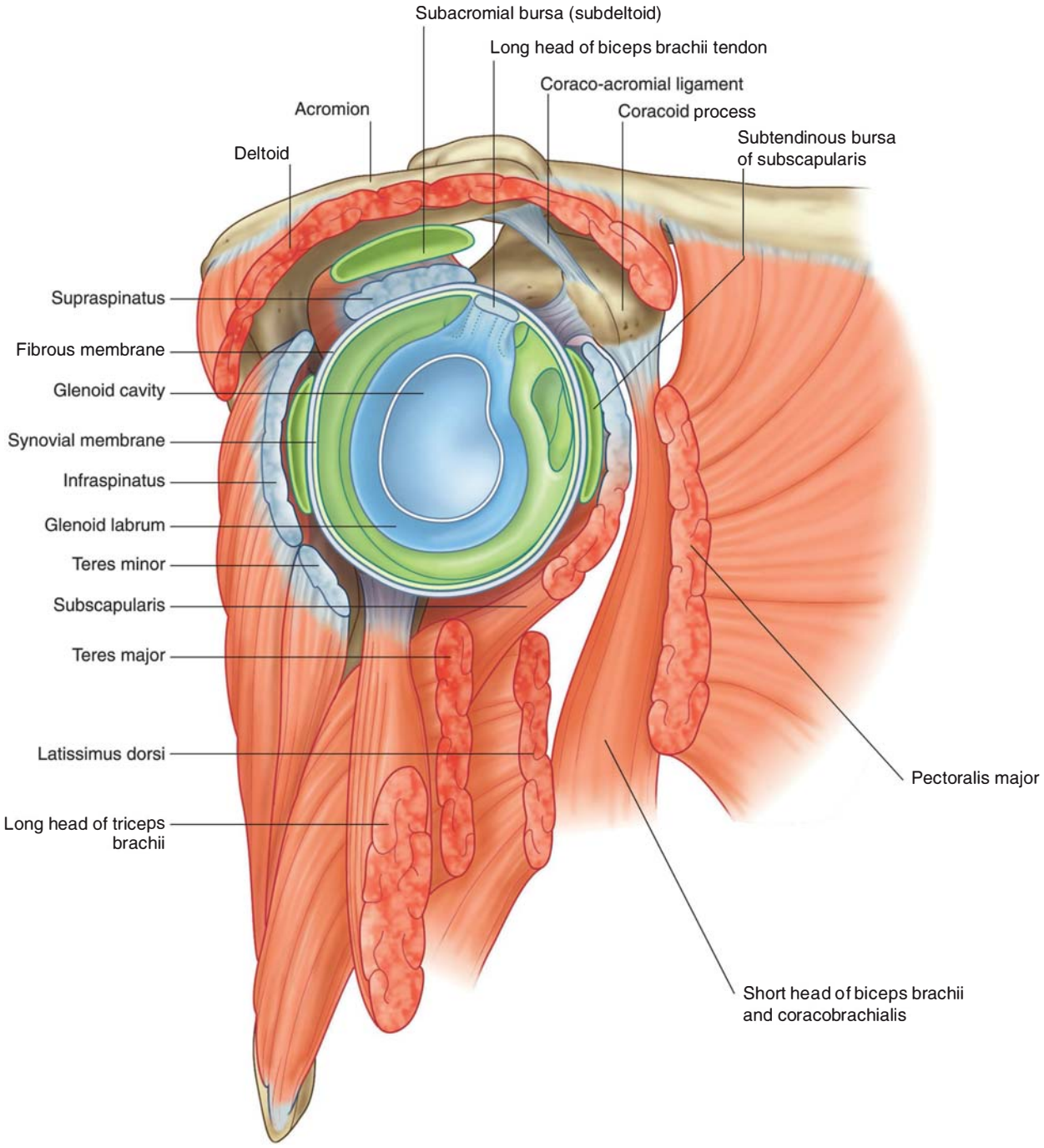


Fig. 7.28 Lateral view of right glenohumeral joint and surrounding muscles with proximal end of humerus removed.

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collar that surrounds the posterior, superior, and anterior aspects of the glenohumeral joint (Figs. 7.28 and 7.29). This cuff of muscles stabilizes and holds the head of the humerus in the glenoid cavity of the scapula without compromising the arm's flexibility and range of motion. The tendon of the long head of the biceps brachii muscle passes superiorly through the joint and restricts upward movement of the humeral head on the glenoid cavity.

Vascular supply to the glenohumeral joint is predominantly through branches of the anterior and posterior circumflex humeral and suprascapular arteries.

The glenohumeral joint is innervated by branches from the posterior cord of the brachial plexus, and from the suprascapular, axillary, and lateral pectoral nerves.

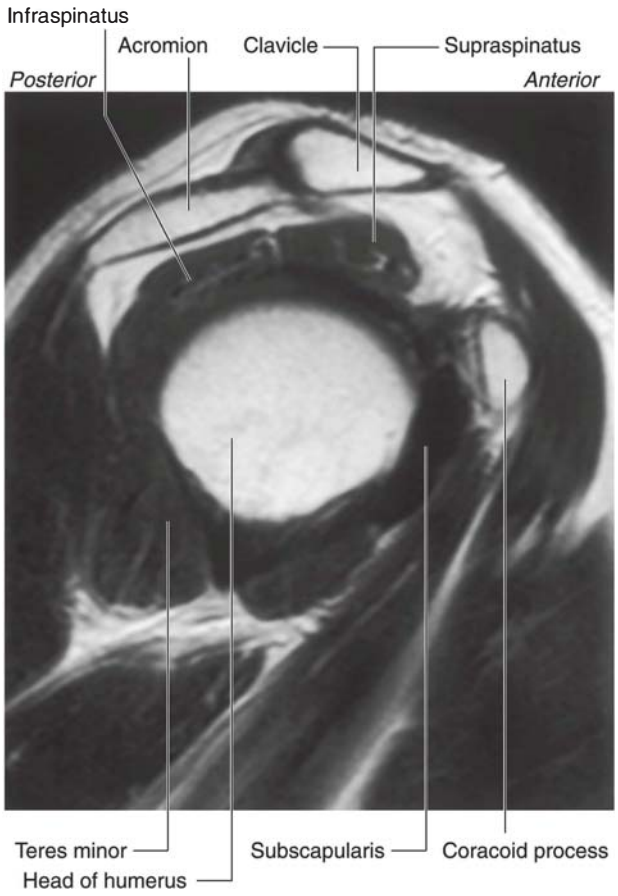


Fig. 7.29 Magnetic resonance image (T1-weighted) of a normal glenohumeral joint in the sagittal plane.

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In the clinic

Fractures of the clavicle and dislocations of the acromioclavicular and sternoclavicular joints

The clavicle provides osseous continuity between the upper limb and thorax. Given its relative size and the potential forces that it transmits from the upper limb to the trunk, it is not surprising that it is often fractured. The typical site of fracture is the middle third (Fig. 7.30). The medial and lateral thirds are rarely fractured.

The acromial end of the clavicle tends to dislocate at the acromioclavicular joint with trauma (Fig. 7.31). The outer third of the clavicle is joined to the scapula by the conoid and trapezoid ligaments of the coracoclavicular ligament.

A minor injury tends to tear the fibrous joint capsule and ligaments of the acromioclavicular joint, resulting in acromioclavicular separation on a plain radiograph. More severe trauma will disrupt the conoid and trapezoid ligaments of the coracoclavicular ligament, which results in elevation and upward subluxation of the clavicle.

The typical injury at the medial end of the clavicle is an anterior or posterior dislocation of the sternoclavicular joint. Importantly, a posterior dislocation of the clavicle may impinge on the great vessels in the root of the neck and compress or disrupt them.

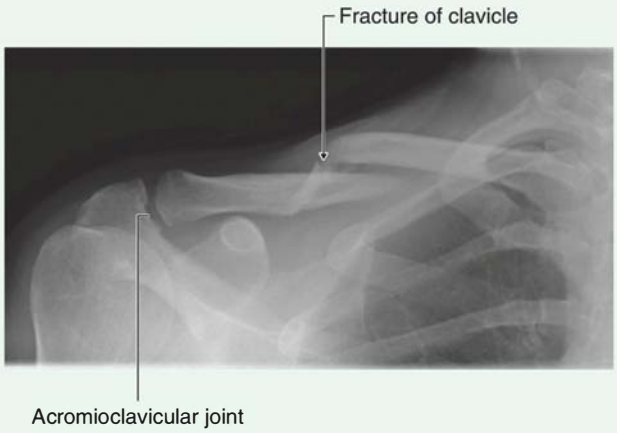


Fig. 7.30 There is an oblique fracture of the middle third of the right clavicle.

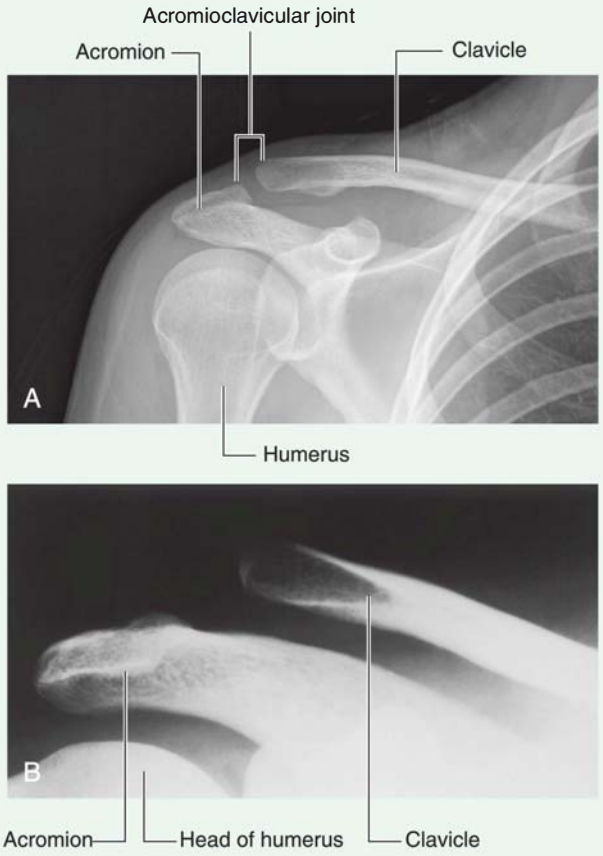


Fig. 7.31 Radiographs of acromioclavicular joints. **A.** Normal right acromioclavicular joint. **B.** Dislocated right acromioclavicular joint.

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In the clinic

Dislocations of the glenohumeral joint

The glenohumeral joint is extremely mobile, providing a wide range of movement at the expense of stability. The relatively small bony glenoid cavity, supplemented by the less robust fibrocartilaginous glenoid labrum and the ligamentous support, make it susceptible to dislocation.

Anterior dislocation (Fig. 7.32) occurs most frequently and is usually associated with an isolated traumatic incident (clinically, all anterior dislocations are anteroinferior). In some cases, the anteroinferior glenoid labrum is torn with or without a small bony fragment. Once the joint capsule and cartilage are disrupted, the joint is susceptible to further (recurrent) dislocations. When an anteroinferior dislocation occurs, the axillary nerve may be injured by direct compression of the humeral head on the nerve inferiorly as it passes through the quadrangular space. Furthermore, the "lengthening" effect of the humerus may stretch the radial nerve, which is tightly bound within the radial groove, and produce a radial nerve paralysis. Occasionally, an anteroinferior dislocation is associated with a fracture, which may require surgical reduction.

Posterior dislocation is extremely rare; when seen, the clinician should focus on its cause, the most common being extremely vigorous muscle contractions, which may be associated with an epileptic seizure caused by electrocution.

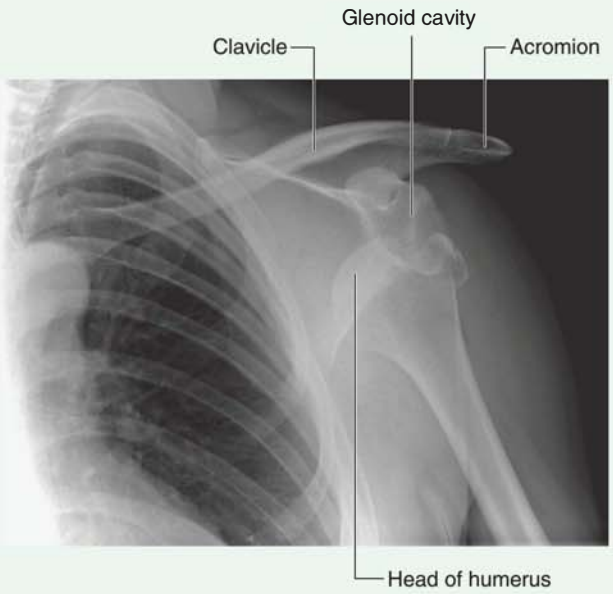


Fig. 7.32 Radiograph showing an anteroinferior dislocation of the shoulder joint.

In the clinic

Rotator cuff disorders

The two main disorders of the rotator cuff are impingement and tendinopathy. The muscle most commonly involved is supraspinatus as it passes beneath the acromion and the acromioclavicular ligament. This space, beneath which the supraspinatus tendon passes, is of fixed dimensions. Swelling of the supraspinatus muscle, excessive fluid within the subacromial/subdeltoid bursa, or subacromial bony spurs may produce significant impingement when the arm is abducted.

The blood supply to the supraspinatus tendon is relatively poor. Repeated trauma, in certain circumstances, makes the tendon susceptible to degenerative change, which may result in calcium deposition, producing extreme pain.

When the supraspinatus tendon has undergone significant degenerative change, it is more susceptible to trauma, and partial- or full-thickness tears may develop (Fig. 7.33). These tears are most common in older patients and may result in considerable difficulty in carrying out normal activities of daily living such as combing hair. However, complete tears may be entirely unsymptomatic.

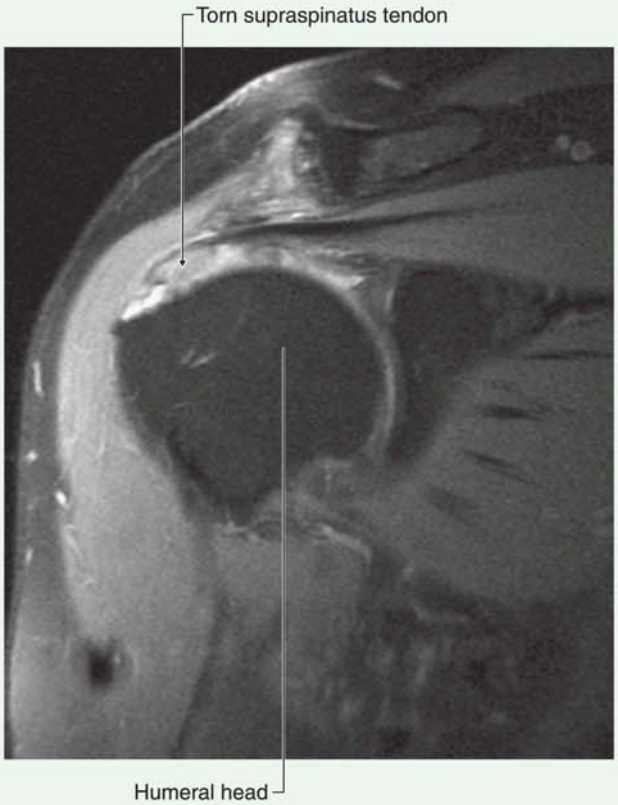


Fig. 7.33 Magnetic resonance image of a full-thickness tear of the supraspinatus tendon as it inserts onto the greater tubercle of the humerus.

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In the clinic

Inflammation of the subacromial (subdeltoid) bursa

Between the supraspinatus and deltoid muscles laterally and the acromion medially, there is a bursa referred to clinically as the subacromial or subdeltoid bursa. In patients who have injured the shoulder or who have supraspinatus tendinopathy, this bursa may become inflamed, making movements of the glenohumeral joint painful. These inflammatory changes may be treated by injection of a corticosteroid and local anesthetic agent (Fig. 7.34).

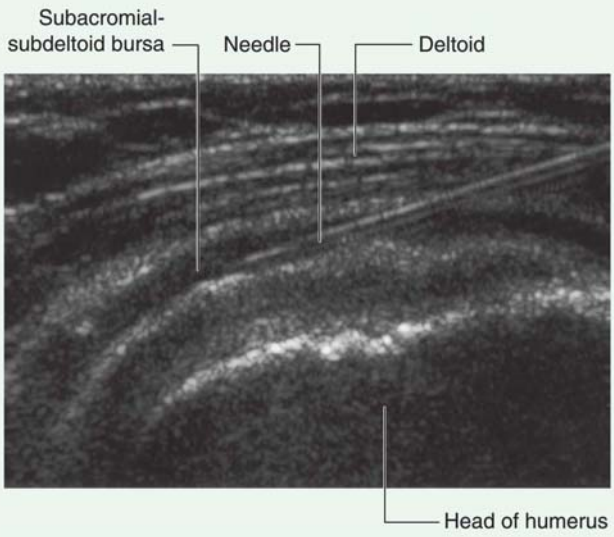


Fig. 7.34 Ultrasound of shoulder showing needle placement into the subdeltoid/subacromial bursa.

Muscles

The two most superficial muscles of the shoulder are the trapezius and deltoid muscles (Fig. 7.35 and Table 7.1). Together, they provide the characteristic contour of the shoulder:

- The trapezius attaches the scapula and clavicle to the trunk.
- The deltoid attaches the scapula and clavicle to the humerus.

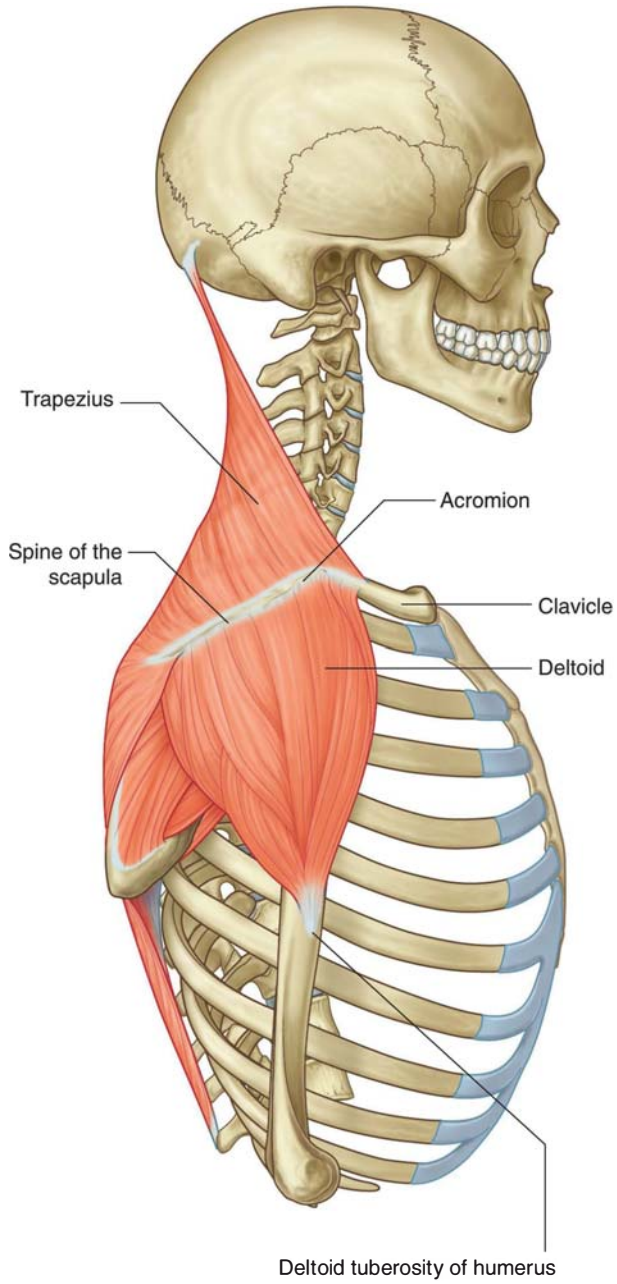


Fig. 7.35 Lateral view of trapezius and deltoid muscles.

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Table 7.1 Muscles of the shoulder (spinal segments in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Trapezius	Superior nuchal line, external occipital protuberance, medial margin of the ligamentum nuchae, spinous processes of CVII to TXII and the related supraspinous ligaments	Superior edge of the crest of the spine of the scapula, acromion, posterior border of lateral one-third of clavicle	Motor spinal part of accessory nerve (XI). Sensory (proprioception) anterior rami of C3 and C4	Powerful elevator of the scapula; rotates the scapula during abduction of humerus above horizontal; middle fibers retract scapula; lower fibers depress scapula
Deltoid	Inferior edge of the crest of the spine of the scapula, lateral margin of the acromion, anterior border of lateral one-third of clavicle	Deltoid tuberosity of humerus	Axillary nerve (**C5,** C6)	Major abductor of arm (abducts arm beyond initial 15° done by supraspinatus); clavicular fibers assist in flexing the arm; posterior fibers assist in extending the arm
Levator scapulae	Transverse processes of CI and CII vertebrae and posterior tubercles of transverse processes of CIII and CIV vertebrae	Posterior surface of medial border of scapula from superior angle to root of spine of the scapula	Branches directly from anterior rami of **C3** and **C4** spinal nerves and by branches (**C5**) from the dorsal scapular nerve	Elevates the scapula
Rhomboid minor	Lower end of ligamentum nuchae and spinous processes of CVII and TI vertebrae	Posterior surface of medial border of scapula at the root of the spine of the scapula	Dorsal scapular nerve (C4, C5)	Elevates and retracts the scapula
Rhomboid major	Spinous processes of TII–TV vertebrae and intervening supraspinous ligaments	Posterior surface of medial border of scapula from the root of the spine of the scapula to the inferior angle	Dorsal scapular nerve **(C4, C5)**	Elevates and retracts the scapula

Both the trapezius and deltoid are attached to opposing surfaces and margins of the spine of the scapula, acromion, and clavicle. The scapula, acromion, and clavicle can be palpated between the attachments of the trapezius and deltoid.

Deep to the trapezius the scapula is attached to the vertebral column by three muscles—the levator scapulae, rhomboid minor, and rhomboid major. These three muscles work with the trapezius (and with muscles found anteriorly) to position the scapula on the trunk.

Trapezius

The **trapezius** muscle has an extensive origin from the axial skeleton, which includes sites on the skull and the vertebrae, from CI to TXII (Fig. 7.36). From CI to CVII, the muscle attaches to the vertebrae through the ligamentum nuchae. The muscle inserts onto the skeletal framework of the shoulder along the inner margins of a continuous U-shaped line of attachment oriented in the horizontal plane, with the bottom of the U directed laterally. Together, the left and right trapezius muscles form a diamond or trapezoid shape, from which the name is derived.

The trapezius muscle is a powerful elevator of the shoulder and also rotates the scapula to extend the reach superiorly.

Innervation of the trapezius muscle is by the accessory nerve [XI] and the anterior rami of cervical nerves C 3 and C4 (Fig. 7.36). These nerves pass vertically along the deep surface of the muscle. The accessory nerve can be evaluated by testing the function of the trapezius muscle. This is most easily done by asking patients to shrug their shoulders against resistance.

Deltoid

The **deltoid** muscle is large and triangular in shape, with its base attached to the scapula and clavicle and its apex attached to the humerus (Fig. 7.36). It originates along a continuous U-shaped line of attachment to the clavicle and the scapula, mirroring the adjacent insertion sites of the trapezius muscle. It inserts into the deltoid tuberosity on the lateral surface of the shaft of the humerus.

The major function of the deltoid muscle is abduction of the arm beyond the initial 15° accomplished by the supraspinatus muscle.

The deltoid muscle is innervated by the axillary nerve, which is a branch of the posterior cord of the brachial plexus. The axillary nerve and associated blood vessels (the posterior circumflex humeral artery and vein) enter the deltoid by passing posteriorly around the surgical neck of the humerus.

Levator scapulae

The levator scapulae originates from the transverse processes of CI to CIV vertebrae (Fig. 7.36). It descends

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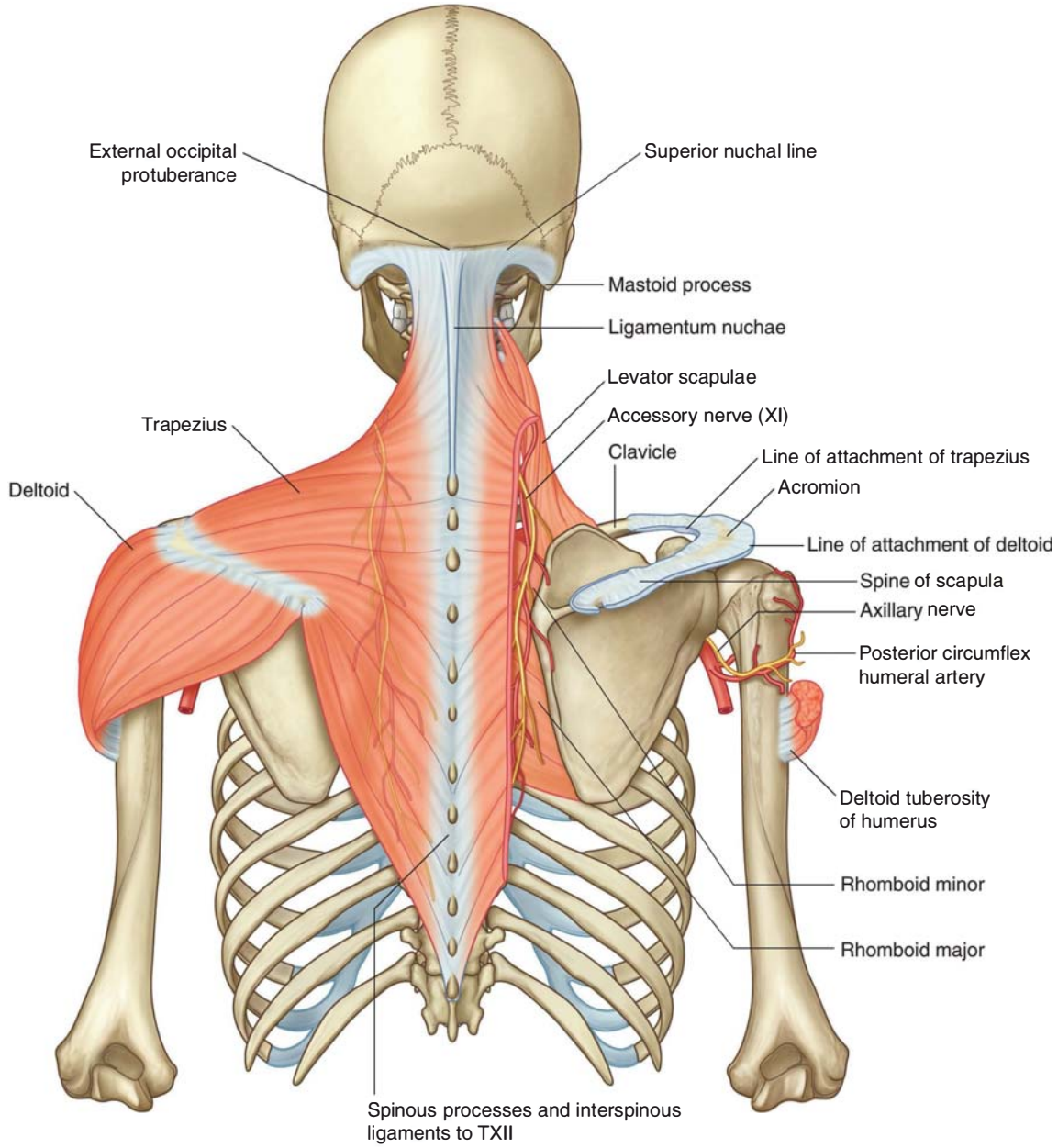


Fig. 7.36 Attachments and neurovascular supply of the trapezius and deltoid muscles.

laterally to attach to the posterior surface of the medial border of the scapula from the superior angle to the smooth triangular area of bone at the root of the spine.

The levator scapulae muscle is innervated by the dorsal scapular nerve and directly from C3 and C4 spinal nerves.

The levator scapulae elevates the scapula.

Rhomboid minor and major

The rhomboid minor and major muscles attach medially to the vertebral column and descend laterally to attach to the medial border of the scapula inferior to the levator scapulae muscle (Fig. 7.36).

The rhomboid minor originates from the lower end of the ligamentum nuchae and the spines of CVII and TI vertebrae. It inserts laterally into the smooth triangular area of bone at the root of the spine of the scapula on the posterior surface.

The rhomboid major originates from the spines of vertebrae TII to TV and from the intervening supraspinous ligaments. It descends laterally to insert along the posterior surface of the medial border of the scapula from the insertion of the rhomboid minor to the inferior angle.

The rhomboid muscles are innervated by the dorsal scapular nerve, which is a branch of the brachial plexus.

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The rhomboid minor and major retract and elevate the scapula.

POSTERIOR SCAPULAR REGION

The posterior scapular region occupies the posterior aspect of the scapula and is located deep to the trapezius and deltoid muscles (Fig. 7.37 and Table 7.2). It contains four muscles, which pass between the scapula and proximal end of the humerus: the supraspinatus, infraspinatus, teres minor, and teres major muscles.

The posterior scapular region also contains part of one additional muscle, the long head of the triceps brachii, which passes between the scapula and the proximal end of the forearm. This muscle, along with other muscles of the region and the humerus, participates in forming a number of spaces through which nerves and vessels enter and leave the region.

The supraspinatus, infraspinatus, and teres minor muscles are components of the rotator cuff, which stabilizes the glenohumeral joint.

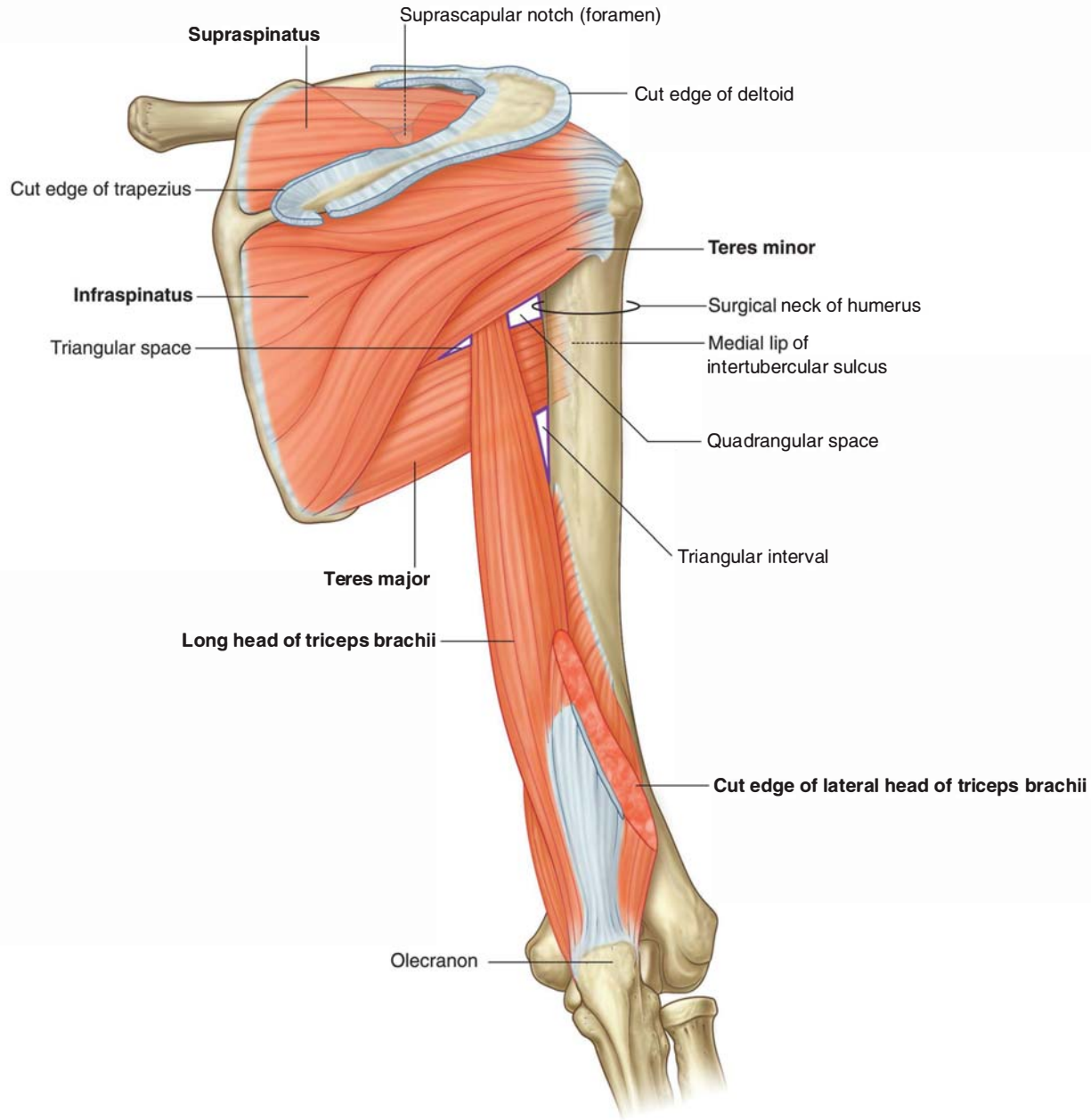


Fig. 7.37 Right posterior scapular region.

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Table 7.2 Muscles of the posterior scapular region (spinal segments in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Supraspinatus	Medial two-thirds of the supraspinous fossa of the scapula and the deep fascia that covers the muscle	Most superior facet on the greater tubercle of the humerus	Suprascapular nerve (**C5**, C6)	**Rotator cuff muscle**; initiation of abduction of arm to 15° at glenohumeral joint
Infraspinatus	Medial two-thirds of the infraspinous fossa of the scapula and the deep fascia that covers the muscle	Middle facet on posterior surface of the greater tubercle of the humerus	Suprascapular nerve (**C5**, C6)	**Rotator cuff muscle**; lateral rotation of arm at the glenohumeral joint
Teres minor	Upper two-thirds of a flattened strip of bone on the posterior surface of the scapula immediately adjacent to the lateral border of the scapula	Inferior facet on the posterior surface of the greater tubercle of the humerus	Axillary nerve (**C5**, C6)	**Rotator cuff muscle**; lateral rotation of arm at the glenohumeral joint
Teres major	Elongate oval area on the posterior surface of the inferior angle of the scapula	Medial lip of the intertubercular sulcus on the anterior surface of the humerus	Inferior subscapular nerve (**C5**, **C6**, **C7**)	Medial rotation and extension of the arm at the glenohumeral joint
Long head of triceps brachii	Infraglenoid tubercle on scapula	Common tendon of insertion with medial and lateral heads on the olecranon process of ulna	Radial nerve (C6, **C7**, C8)	Extension of the forearm at the elbow joint; accessory adductor and extensor of the arm at the glenohumeral joint

Muscles

Supraspinatus and infraspinatus

The **supraspinatus** and **infraspinatus** muscles originate from two large fossae, one above and one below the spine, on the posterior surface of the scapula (Fig. 7.37). They form tendons that insert on the greater tubercle of the humerus.

- The tendon of the supraspinatus passes under the acromion, where it is separated from the bone by a subacromial bursa, passes over the glenohumeral joint, and inserts on the superior facet of the greater tubercle.
- The tendon of the infraspinatus passes posteriorly to the glenohumeral joint and inserts on the middle facet of the greater tubercle.

The supraspinatus initiates abduction of the arm. The infraspinatus laterally rotates the humerus.

Teres minor and teres major

The **teres minor** muscle is a cord-like muscle that originates from a flattened area of the scapula immediately adjacent to its lateral border below the infraglenoid tubercle (Fig. 7.37). Its tendon inserts on the inferior facet of the greater tubercle of the humerus. The teres minor laterally rotates the humerus and is a component of the rotator cuff.

The **teres major** muscle originates from a large oval region on the posterior surface of the inferior angle of the scapula (Fig. 7.37). This broad cord-like muscle passes superiorly and laterally and ends as a flat tendon that attaches to the medial lip of the intertubercular sulcus on the anterior surface of the humerus. The teres major medially rotates and extends the humerus.

Long head of triceps brachii

The **long head of the triceps brachii** muscle originates from the infraglenoid tubercle and passes somewhat vertically down the arm to insert, with the medial and lateral heads of this muscle, on the olecranon of the ulna (Fig. 7.37).

The triceps brachii is the primary extensor of the forearm at the elbow joint. Because the long head crosses the glenohumeral joint, it can also extend and adduct the humerus.

The importance of the triceps brachii in the posterior scapular region is that its vertical course between the teres minor and teres major, together with these muscles and the humerus, forms spaces through which nerves and vessels pass between regions.

Gateways to the posterior scapular region

Suprascapular foramen

The suprascapular foramen is the route through which structures pass between the base of the neck and the posterior scapular region (Fig. 7.37). It is formed by the suprascapular notch of the scapula and the superior transverse scapular (suprascapular) ligament, which converts the notch into a foramen.

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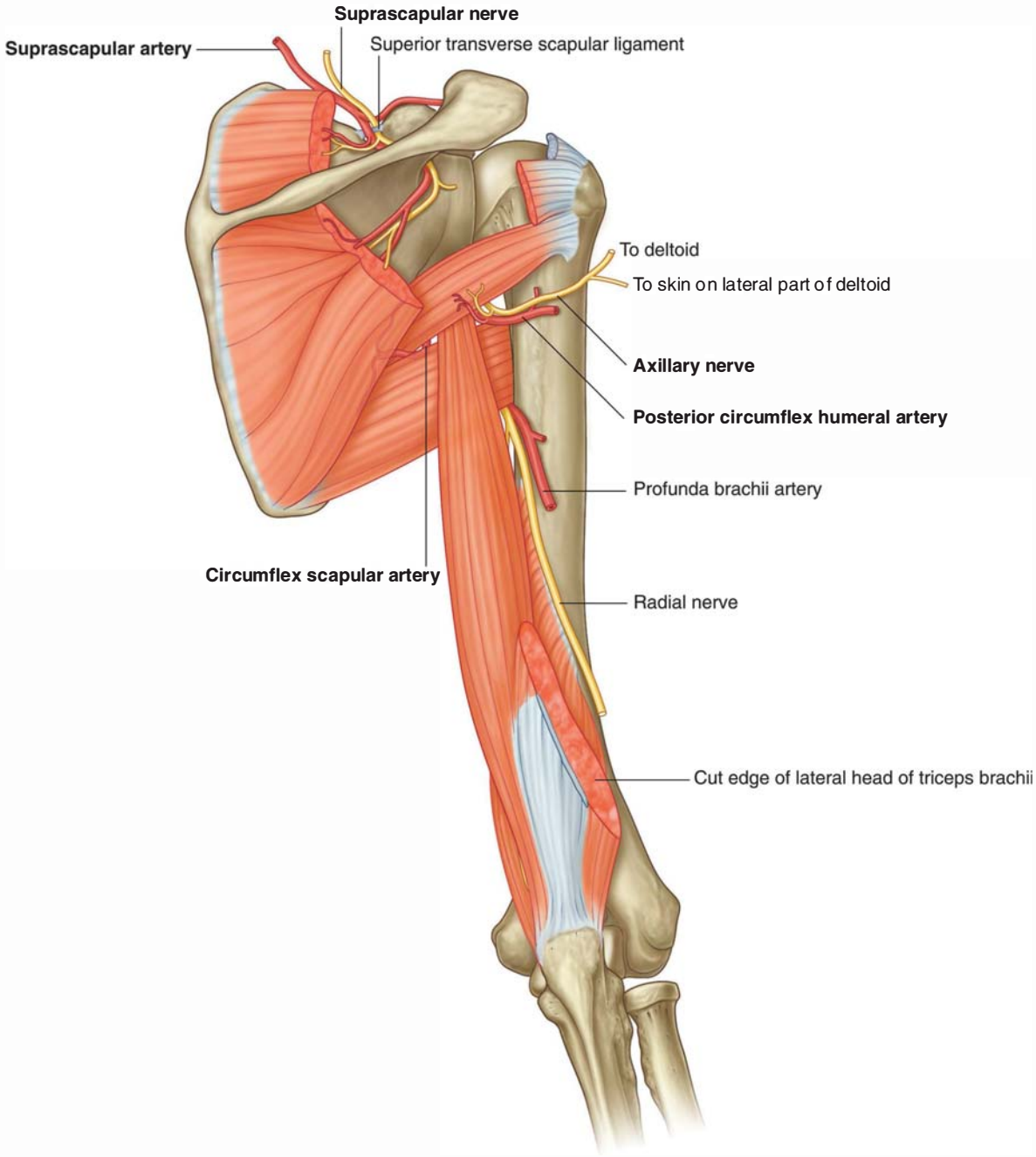
The suprascapular nerve passes through the suprascapular foramen; the suprascapular artery and the suprascapular vein follow the same course as the nerve, but normally pass immediately superior to the superior transverse scapular ligament and not through the foramen (Fig. 7.38).

Quadrangular space (from posterior)

The quadrangular space provides a passageway for nerves and vessels passing between more anterior regions (the axilla) and the posterior scapular region (Fig. 7.37). In the posterior scapular region, its boundaries are formed by:

- the inferior margin of the teres minor,
- the surgical neck of the humerus,
- the superior margin of the teres major, and
- the lateral margin of the long head of the triceps brachii.

The axillary nerve and the posterior circumflex humeral artery and vein pass through this space (Fig. 7.38).



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Fig. 7.38 Arteries and nerves associated with gateways in the posterior scapular region.

Triangular space

The triangular space is an area of communication between the axilla and the posterior scapular region (Fig. 7.37). When viewed from the posterior scapular region, the triangular space is formed by:

- the medial margin of the long head of the triceps brachii,
- the superior margin of the teres major, and
- the inferior margin of the teres minor.

The circumflex scapular artery and vein pass through this gap (Fig. 7.38).

Triangular interval

The triangular interval is formed by:

- the lateral margin of the long head of the triceps brachii,
- the shaft of the humerus, and
- the inferior margin of the teres major (Fig. 7.37).

Because this space is below the inferior margin of the teres major, which defines the inferior boundary of the axilla, the triangular interval serves as a passageway between the anterior and posterior compartments of the arm and between the posterior compartment of the arm and the axilla. The radial nerve, the **profunda brachii artery** (**deep artery of arm**), and associated veins pass through it (Fig. 7.38).

Nerves

The two major nerves of the posterior scapular region are the suprascapular and axillary nerves, both of which originate from the brachial plexus in the axilla (Fig. 7.38).

Suprascapular nerve

The **suprascapular nerve** originates in the base of the neck from the superior trunk of the brachial plexus. It passes posterolaterally from its origin, through the suprascapular foramen to reach the posterior scapular region, where it lies in the plane between bone and muscle (Fig. 7.38).

It innervates the supraspinatus muscle and then passes through the greater scapular (spinoglenoid) notch, between the root of the spine of the scapula and the glenoid cavity, to terminate in and innervate the infraspinatus muscle.

Generally, the suprascapular nerve has no cutaneous branches.

Axillary nerve

The **axillary nerve** originates from the posterior cord of the brachial plexus. It exits the axilla by passing through the quadrangular space in the posterior wall of the axilla, and enters the posterior scapular region (Fig. 7.38). Together with the posterior circumflex humeral artery and vein, it is directly related to the posterior surface of the surgical neck of the humerus.

The axillary nerve innervates the deltoid and teres minor muscles. In addition, it has a cutaneous branch, the superior lateral cutaneous nerve of the arm, which carries general sensation from the skin over the inferior part of the deltoid muscle.

Arteries and veins

Three major arteries are found in the posterior scapular region: the suprascapular, posterior circumflex humeral, and circumflex scapular arteries. These arteries contribute

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to an interconnected vascular network around the scapula (Fig. 7.39).

In the clinic

Quadrangular space syndrome

Hypertrophy of the quadrangular space muscles or fibrosis of the muscle edges may impinge on the axillary nerve. Uncommonly, this produces weakness of the deltoid muscle. Typically it produces atrophy of the teres minor muscle, which may affect the control that the rotator cuff muscles exert on shoulder movement.

Suprascapular artery

The **suprascapular artery** originates in the base of the neck as a branch of the thyrocervical trunk, which, in turn, is a major branch of the subclavian artery (Figs. 7.38 and 7.39). The vessel may also originate directly from the third part of the subclavian artery.

The suprascapular artery normally enters the posterior scapular region superior to the suprascapular foramen, whereas the nerve passes through the foramen. In the posterior scapular region, the vessel runs with the suprascapular nerve.

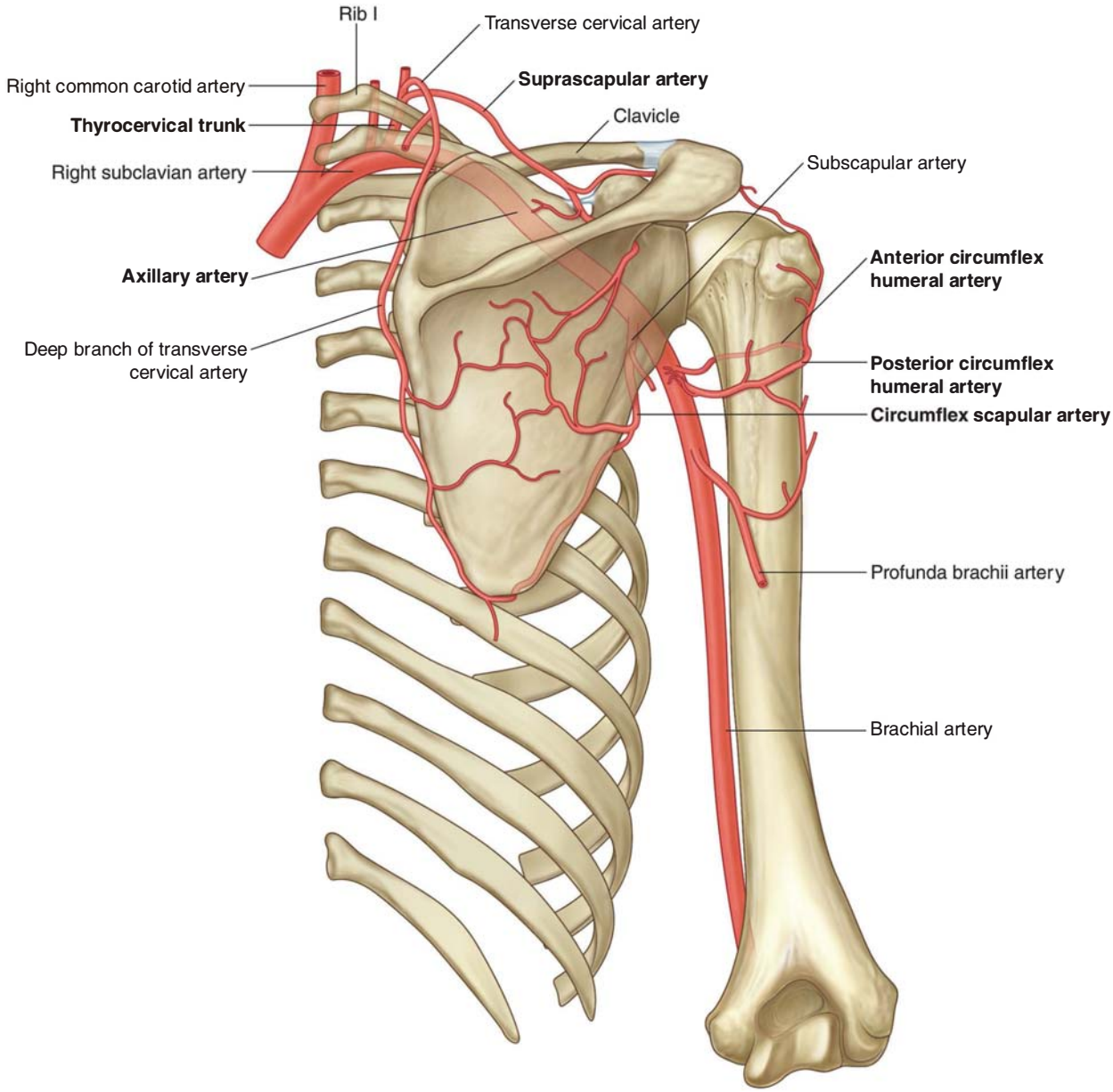


Fig. 7.39 Arterial anastomoses around the shoulder.

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In addition to supplying the supraspinatus and infraspinatus muscles, the suprascapular artery contributes branches to numerous structures along its course.

Posterior circumflex humeral artery

The **posterior circumflex humeral artery** originates from the third part of the axillary artery in the axilla (Fig. 7.39).

The posterior circumflex humeral artery and axillary nerve leave the axilla through the quadrangular space in the posterior wall and enter the posterior scapular region. The vessel supplies the related muscles and the glenohumeral joint.

Circumflex scapular artery

The **circumflex scapular artery** is a branch of the subscapular artery that also originates from the third part of the axillary artery in the axilla (Fig. 7.39). The circumflex scapular artery leaves the axilla through the triangular space and enters the posterior scapular region, passes through the origin of the teres minor muscle, and forms anastomotic connections with other arteries in the region.

Veins

Veins in the posterior scapular region generally follow the arteries and connect with vessels in the neck, back, arm, and axilla.

AXILLA

The axilla is the gateway to the upper limb, providing an area of transition between the neck and the arm (Fig. 7.40A). Formed by the clavicle, the scapula, the upper

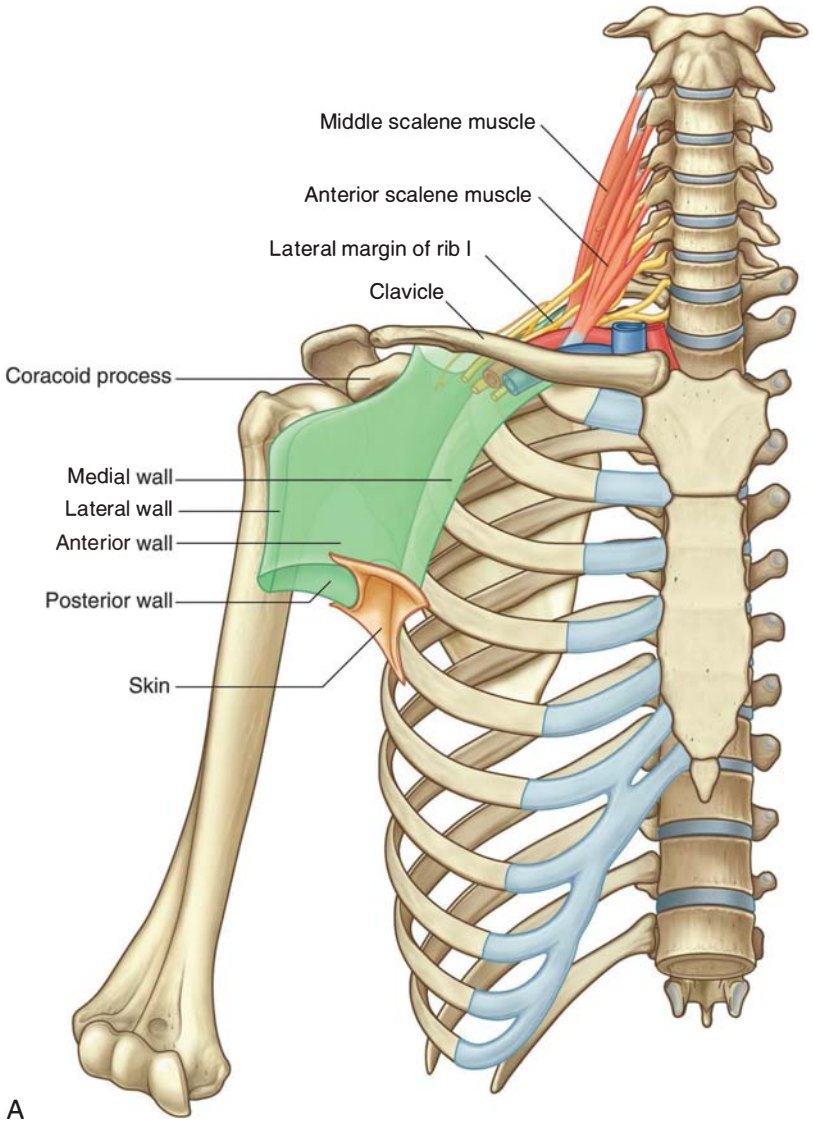


Fig. 7.40 Axilla. **A.** Walls and transition between neck and arm. *Continued*

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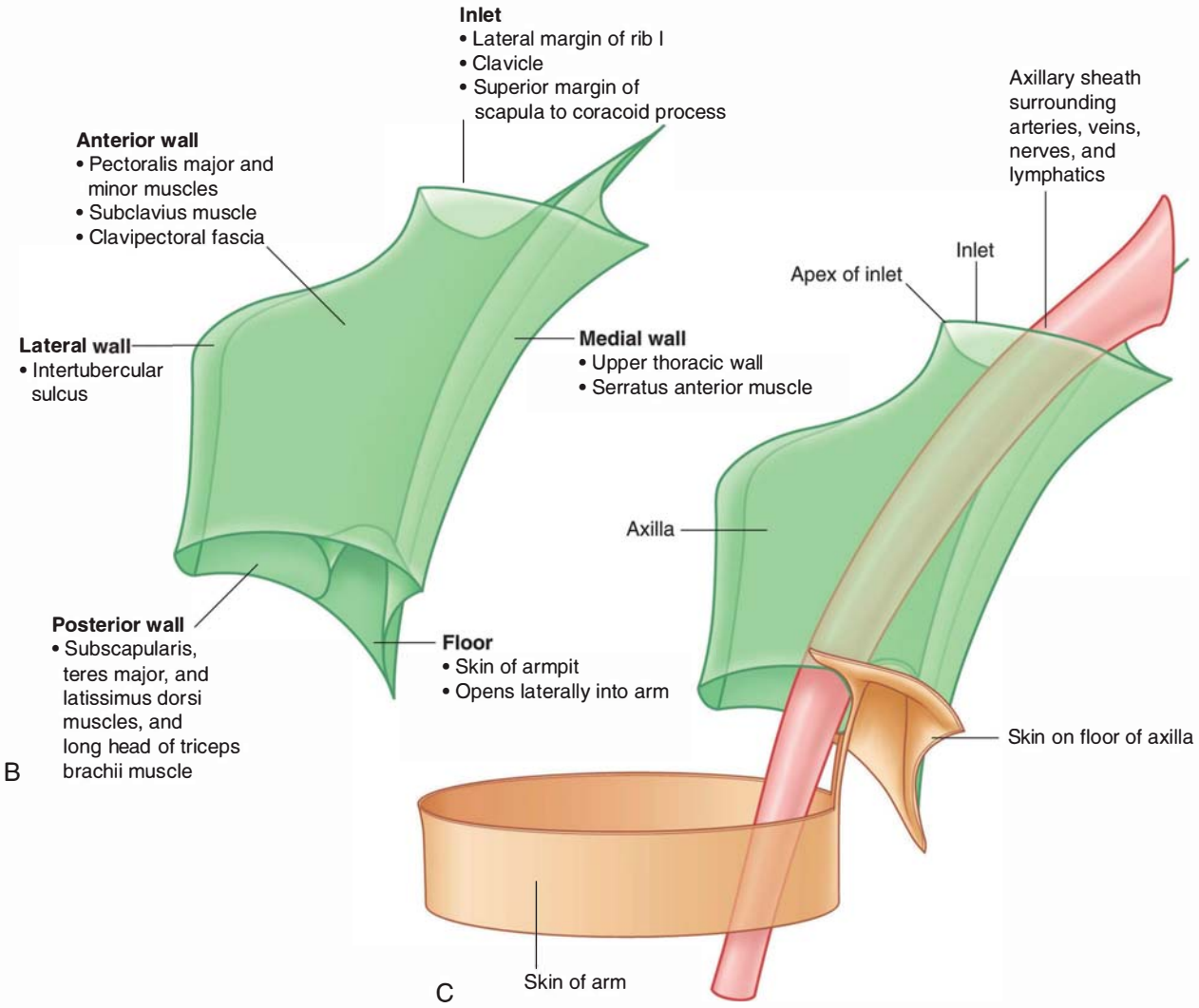


Fig. 7.40, cont'd Axilla. **B.** Boundaries. **C.** Continuity with the arm.

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thoracic wall, the humerus, and related muscles, the axilla is an irregularly shaped pyramidal space with:

- four sides,
- an inlet, and
- a floor (base) (Fig. 7.40A,B).

The axillary inlet is continuous superiorly with the neck, and the lateral part of the floor opens into the arm.

All major structures passing into and out of the upper limb pass through the axilla (Fig. 7.40C). Apertures formed between muscles in the anterior and posterior walls enable structures to pass between the axilla and immediately adjacent regions (the posterior scapular, pectoral, and deltoid regions).

Axillary inlet

The axillary inlet is oriented in the horizontal plane and is somewhat triangular in shape, with its apex directed laterally (Fig. 7.40A,B). The margins of the inlet are completely formed by bone:

- The medial margin is the lateral border of rib I.
- The anterior margin is the posterior surface of the clavicle.
- The posterior margin is the superior border of the scapula up to the coracoid process.

The apex of the triangularly shaped axillary inlet is lateral in position and is formed by the medial aspect of the coracoid process.

Major vessels and nerves pass between the neck and the axilla by crossing over the lateral border of rib I and through the axillary inlet (Fig. 7.40A).

The subclavian artery, the major blood vessel supplying the upper limb, becomes the axillary artery as it crosses the lateral margin of rib I and enters the axilla. Similarly, the axillary vein becomes the subclavian vein as it passes over the lateral margin of rib I and leaves the axilla to enter the neck.

At the axillary inlet, the axillary vein is anterior to the axillary artery, which, in turn, is anterior to the trunks of the brachial plexus.

The inferior trunk (lower trunk) of the brachial plexus lies directly on rib I in the neck, as does the subclavian artery and vein. As they pass over rib I, the vein and artery are separated by the insertion of the anterior scalene muscle (Fig. 7.40A).

Anterior wall

The anterior wall of the axilla is formed by the lateral part of the pectoralis major muscle, the underlying pectoralis minor and subclavius muscles, and the clavipectoral fascia (Table 7.3).

Table 7.3 Muscles of the anterior wall of the axilla (spinal segments in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Pectoralis major	Clavicular head—anterior surface of medial half of clavicle; sternocostal head—anterior surface of sternum; first seven costal cartilages; sternal end of sixth rib; aponeurosis of external oblique	Lateral lip of intertubercular sulcus of humerus	Medial and lateral pectoral nerves; clavicular head (**C5**, C6); sternocostal head (C6, **C7**, C8, T1)	Flexion, adduction, and medial rotation of arm at glenohumeral joint; clavicular head—flexion of extended arm; sternocostal head—extension of flexed arm
Subclavius	First rib at junction between rib and costal cartilage	Groove on inferior surface of middle one-third of clavicle	Nerve to subclavius (**C5**, **C6**)	Pulls tip of shoulder down; pulls clavicle medially to stabilize sternoclavicular joint
Pectoralis minor	Anterior surfaces and superior borders of ribs III to V; and from deep fascia overlying the related intercostal spaces	Coracoid process of scapula (medial border and upper surface)	Medial pectoral nerve (C5, C6, **C7**, **C8**, T1)	Pulls tip of shoulder down; protracts scapula

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Pectoralis major

The **pectoralis major** muscle is the largest and most superficial muscle of the anterior wall (Fig. 7.41). Its inferior margin underlies the anterior axillary fold, which marks the anteroinferior border of the axilla. The muscle has two heads:

- The clavicular head originates from the medial half of the clavicle.
- The sternocostal head originates from the medial part of the anterior thoracic wall—often, fibers from this head continue inferiorly and medially to attach to the anterior abdominal wall, forming an additional abdominal part of the muscle.

The muscle inserts into the lateral lip of the intertubercular sulcus of the humerus. The parts of the muscle that have a superior origin on the trunk insert lower and more anteriorly on the lateral lip of the intertubercular sulcus than the parts of the muscle that originate inferiorly.

Acting together, the two heads of the pectoralis major flex, adduct, and medially rotate the arm at the glenohumeral joint. The clavicular head flexes the arm from an extended position, whereas the sternocostal head extends the arm from a flexed position, particularly against resistance.

The pectoralis major is innervated by the lateral and medial pectoral nerves, which originate from the brachial plexus in the axilla.

Subclavius

The **subclavius** muscle is a small muscle that lies deep to the pectoralis major muscle and passes between the clavicle and rib I (Fig. 7.42). It originates medially, as a tendon, from rib I at the junction between the rib and its costal cartilage. It passes laterally and superiorly to insert via a muscular attachment into an elongate shallow groove on the inferior surface of the middle third of the clavicle.

The function of the subclavius is not entirely clear, but it may act to pull the shoulder down by depressing the

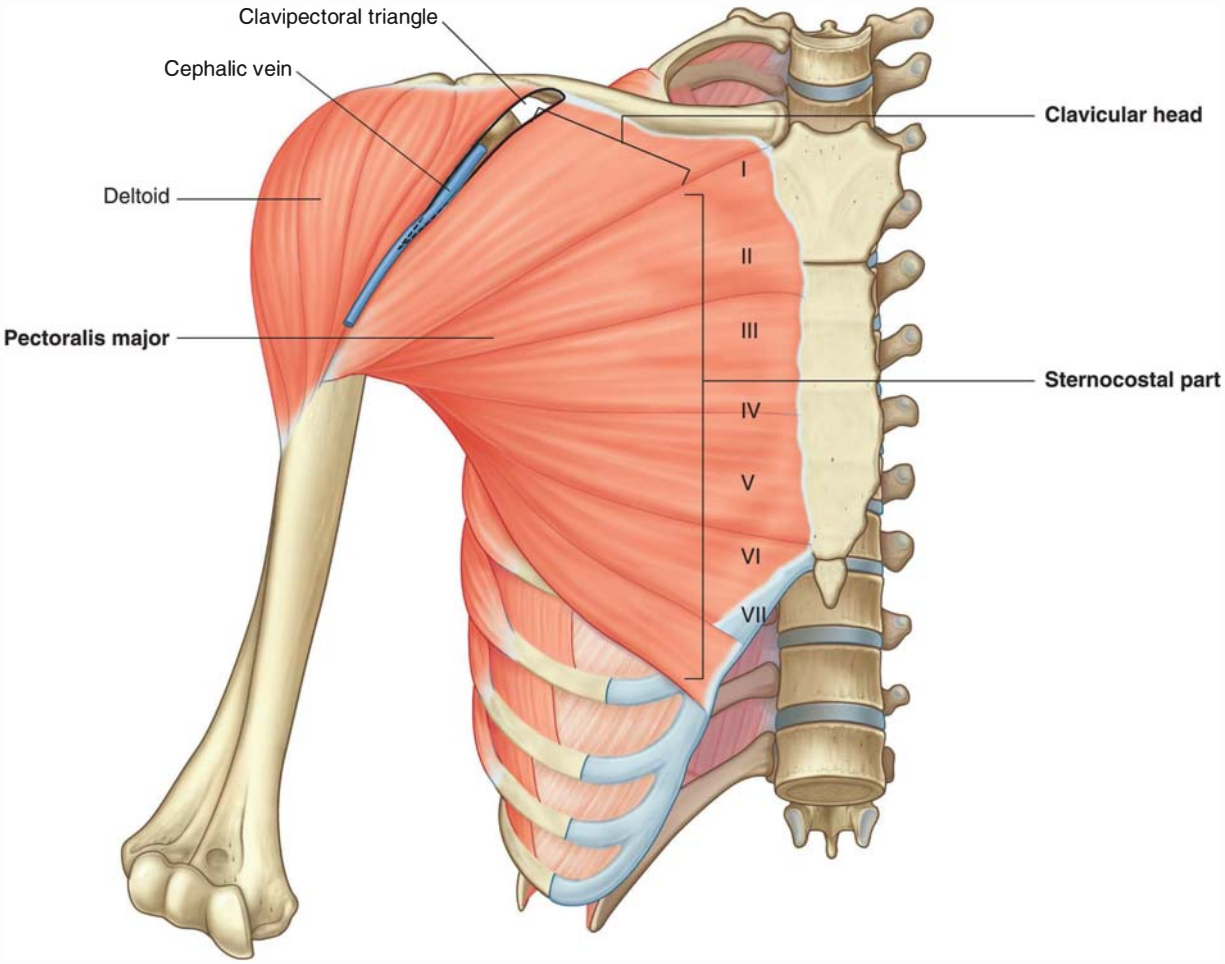


Fig. 7.41 Pectoralis major muscle.

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clavicle and may also stabilize the sternoclavicular joint by pulling the clavicle medially.

The subclavius muscle is innervated by a small branch from the superior trunk of the brachial plexus.

Pectoralis minor

The **pectoralis minor** muscle is a small triangular-shaped muscle that lies deep to the pectoralis major muscle and passes from the thoracic wall to the coracoid process of the scapula (Fig. 7.42). It originates as three muscular slips from the anterior surfaces and upper margins of ribs III to V and from the fascia overlying muscles of the related intercostal spaces. The muscle fibers pass superiorly and laterally to insert into the medial and upper aspects of the coracoid process.

The pectoralis minor muscle protracts the scapula (by pulling the scapula anteriorly on the thoracic wall) and depresses the lateral angle of the scapula.

The pectoralis minor is innervated by the medial pectoral nerve, which originates from the brachial plexus in the axilla.

Clavipectoral fascia

The clavipectoral fascia is a thick sheet of connective tissue that connects the clavicle to the floor of the axilla (Fig. 7.42). It encloses the subclavius and pectoralis minor muscles and spans the gap between them.

Structures travel between the axilla and the anterior wall of the axilla by passing through the clavipectoral

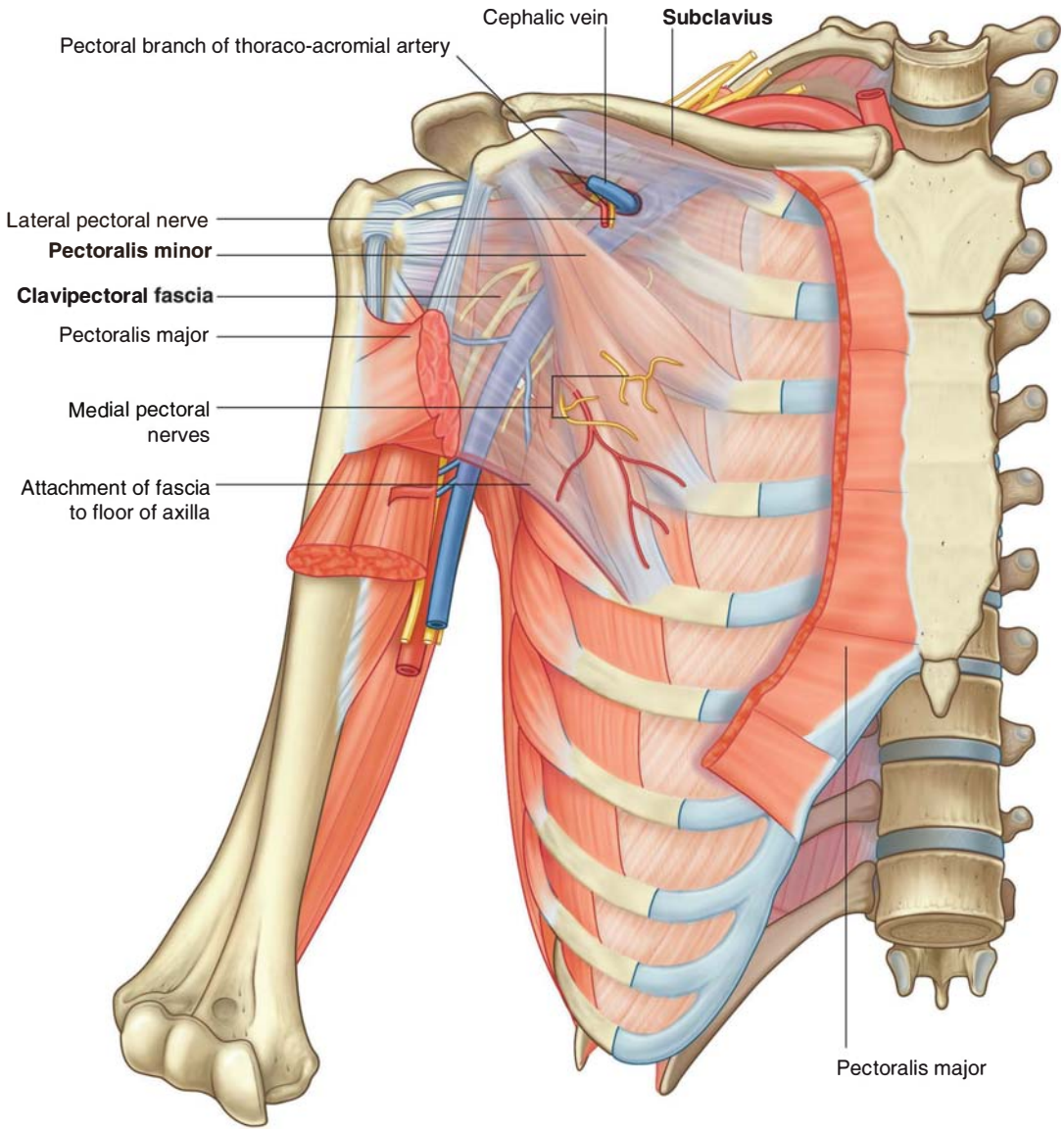


Fig. 7.42 Pectoralis minor and subclavius muscles and clavipectoral fascia.

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fascia either between the pectoralis minor and subclavius muscles or inferior to the pectoralis minor muscle.

Important structures that pass between the subclavius and pectoralis minor muscles include the cephalic vein, the thoraco-acromial artery, and the lateral pectoral nerve.

The lateral thoracic artery leaves the axilla by passing through the fascia inferior to the pectoralis minor muscle.

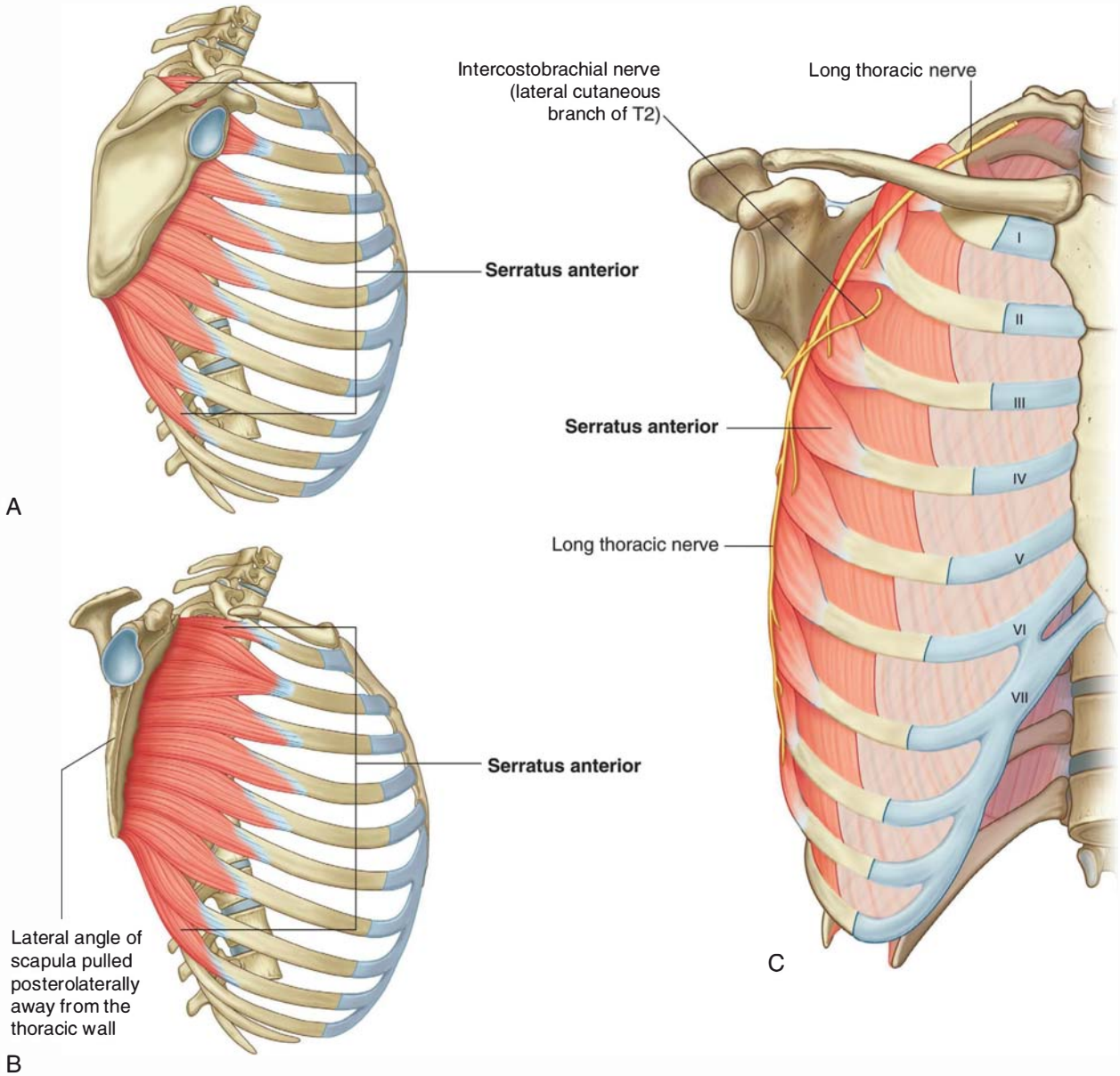
The medial pectoral nerve leaves the axilla by penetrating directly through the pectoralis minor muscle to supply this muscle and to reach the pectoralis major muscle. Occasionally, branches of the medial pectoral nerve pass around the lower margin of the pectoralis minor to reach and innervate the overlying pectoralis major muscle.

Medial wall

The medial wall of the axilla consists of the upper thoracic wall (the ribs and related intercostal tissues) and the serratus anterior muscle (Fig. 7.43 and Table 7.4, and see Fig. 7.40).

Serratus anterior

The **serratus anterior** muscle originates as a number of muscular slips from the lateral surfaces of ribs I to IX and the intervening deep fascia overlying the related intercostal spaces (Fig. 7.43). The muscle forms a flattened sheet, which passes posteriorly around the thoracic wall to insert



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Fig. 7.43 Medial wall of the axilla. **A.** Lateral view. **B.** Lateral view with lateral angle of scapula retracted posteriorly. **C.** Anterior view.

primarily on the costal surface of the medial border of the scapula.

The serratus anterior pulls the scapula forward over the thoracic wall and facilitates scapular rotation. It also keeps the costal surface of the scapula closely opposed to the thoracic wall.

The serratus anterior is innervated by the long thoracic nerve, which is derived from the roots of the brachial plexus, passes through the axilla along the medial wall, and passes vertically down the serratus anterior muscle on its external surface, just deep to skin and superficial fascia.

Intercostobrachial nerve

The only major structure that passes directly through the medial wall and into the axilla is the intercostobrachial nerve (Fig. 7.43). This nerve is the lateral cutaneous branch of the second intercostal nerve (anterior ramus of T2). It communicates with a branch of the brachial plexus (the medial cutaneous nerve of the arm) in the axilla and supplies skin on the upper posteromedial side of the arm, which is part of the T2 dermatome.

In the clinic

"Winging" of the scapula

Because the long thoracic nerve passes down the lateral thoracic wall on the external surface of the serratus anterior muscle, just deep to skin and subcutaneous fascia, it is vulnerable to damage. Loss of function of this muscle causes the medial border, and particularly the inferior angle, of the scapula to elevate away from the thoracic wall, resulting in characteristic "winging" of the scapula, on pushing forward with the arm. Furthermore, normal elevation at the arm is no longer possible.

Lateral wall

The lateral wall of the axilla is narrow and formed entirely by the intertubercular sulcus of the humerus (Fig. 7.44). The pectoralis major muscle of the anterior wall attaches to the lateral lip of the intertubercular sulcus. The latissimus dorsi and teres major muscles of the posterior wall

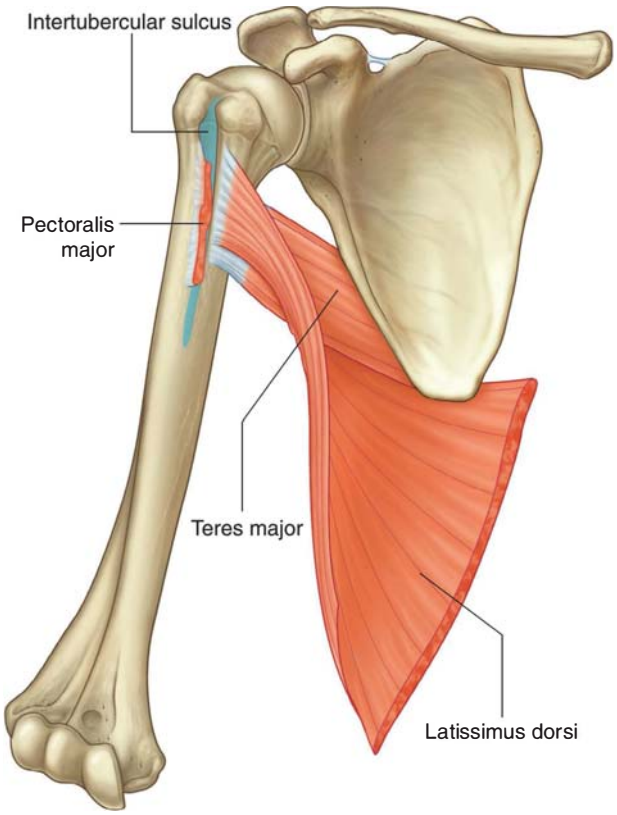


Fig. 7.44 Lateral wall of the axilla.

Table 7.4 Muscle of the medial wall of the axilla (spinal segment in bold is the major segment innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Serratus anterior	Lateral surfaces of upper 8–9 ribs and deep fascia overlying the related intercostal spaces	Costal surface of medial border of scapula	Long thoracic nerve (**C5**, C6, C7)	Protraction and rotation of the scapula; keeps medial border and inferior angle of scapula opposed to thoracic wall

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attach to the floor and medial lip of the intertubercular sulcus, respectively (Table 7.5).

Posterior wall

The posterior wall of the axilla is complex (Fig. 7.45 and see Fig. 7.50). Its bone framework is formed by the costal surface of the scapula. Muscles of the wall are:

- the subscapularis muscle (associated with the costal surface of the scapula),
- the distal parts of the latissimus dorsi and teres major muscles (which pass into the wall from the back and posterior scapular region), and
- the proximal part of the long head of the triceps brachii muscle (which passes vertically down the wall and into the arm).

Gaps between the muscles of the posterior wall form apertures through which structures pass between the axilla, posterior scapular region, and posterior compartment of the arm.

Table 7.5 Muscles of the lateral and posterior wall of the axilla (spinal segments in bold are the major segments innervating the muscle; spinal segments in parentheses do not consistently innervate the muscle)

Muscle	Origin	Insertion	Innervation	Function
Subscapularis	Medial two-thirds of subscapular fossa	Lesser tubercle of humerus	Upper and lower subscapular nerves (C5, **C6**, (C7))	**Rotator cuff muscle**; medial rotation of the arm at the glenohumeral joint
Teres major	Elongate oval area on the posterior surface of the inferior angle of the scapula	Medial lip of the intertubercular sulcus on the anterior surface of the humerus	Lower subscapular nerve (**C5, C6, C7**)	Medial rotation and extension of the arm at the glenohumeral joint
Latissimus dorsi	Spinous processes of lower six thoracic vertebrae and related interspinous ligaments; via the thoracolumbar fascia to the spinous processes of the lumbar vertebrae, related interspinous ligaments, and iliac crest; lower 3–4 ribs	Floor of intertubercular sulcus	Thoracodorsal nerve (C6, **C7**, C8)	Adduction, medial rotation, and extension of the arm at the glenohumeral joint
Long head of triceps brachii	Infraglenoid tubercle on scapula	Common tendon of insertion with medial and lateral heads on the olecranon process of ulna	Radial nerve (C6, **C7**, C8)	Extension of the forearm at the elbow joint; accessory adductor and extensor of the arm at the glenohumeral joint

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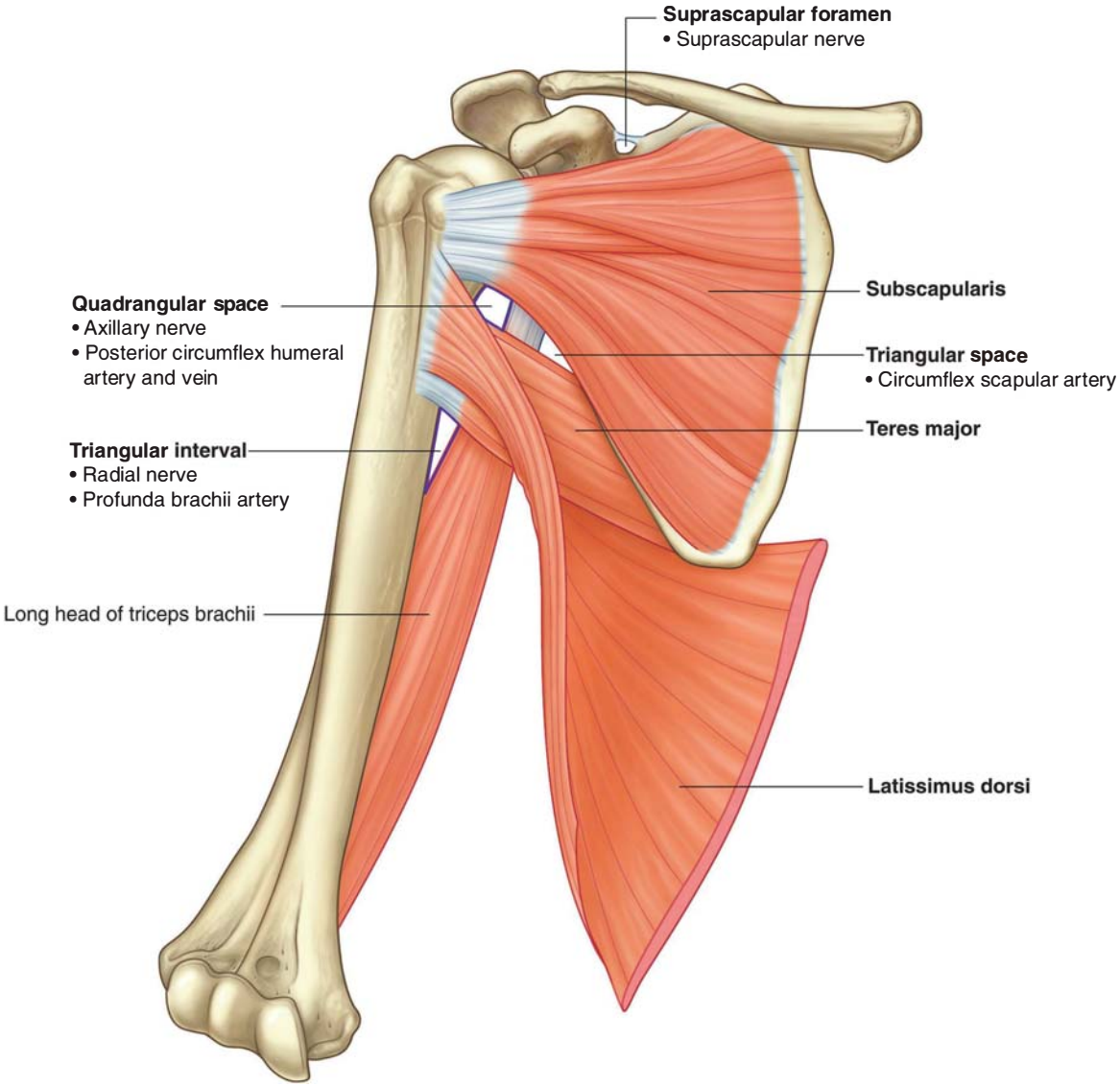


Fig. 7.45 Posterior wall of the axilla.

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Subscapularis

The **subscapularis** muscle forms the largest component of the posterior wall of the axilla. It originates from, and fills, the subscapular fossa and inserts on the lesser tubercle of the humerus (Figs. 7.45 and 7.46). The tendon crosses immediately anterior to the joint capsule of the glenohumeral joint.

Together with three muscles of the posterior scapular region (the supraspinatus, infraspinatus, and teres minor muscles), the subscapularis is a member of the rotator cuff muscle group, which stabilize the glenohumeral joint.

The subscapularis is innervated by branches of the brachial plexus (the **superior** and **inferior subscapular nerves**), which originate in the axilla.

Teres major and latissimus dorsi

The inferolateral aspect of the posterior wall of the axilla is formed by the terminal part of the **teres major** muscle and the tendon of the **latissimus dorsi** muscle (Fig. 7.45). These two structures lie under the posterior axillary fold, which marks the posteroinferior border of the axilla.

The flat tendon of the latissimus dorsi muscle curves around the inferior margin of the teres major muscle on the posterior wall to insert into the floor of the intertubercular sulcus of the humerus, anterior to and slightly above the most distal attachment of the teres major muscle to the medial lip of the intertubercular sulcus. As a consequence, the inferior margin of the teres major muscle defines the inferior limit of the axilla laterally.

The axillary artery becomes the brachial artery of the arm as it crosses the inferior margin of the teres major muscle.

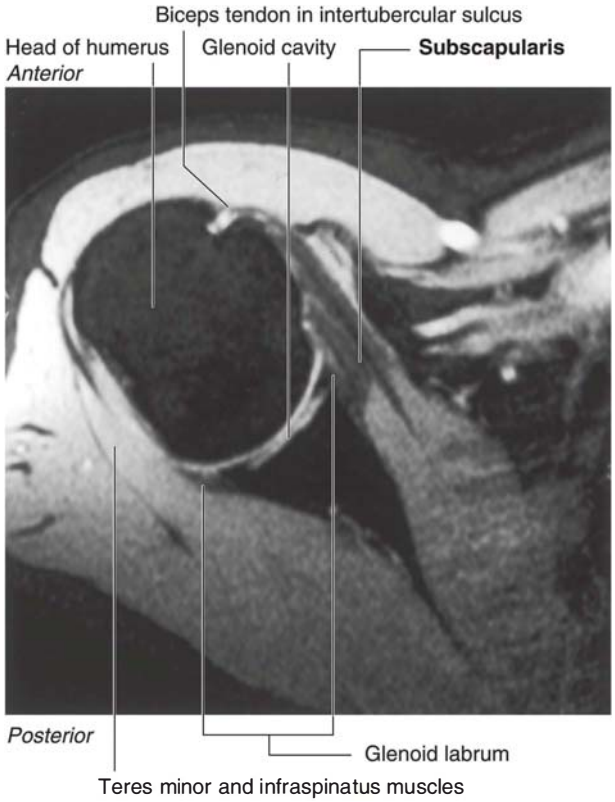


Fig. 7.46 Magnetic resonance image of the glenohumeral joint in the transverse or horizontal plane.

Long head of the triceps brachii

The **long head of the triceps brachii** muscle passes vertically through the posterior wall of the axilla, and, together with surrounding muscles and adjacent bones, results in the formation of three apertures through which major structures pass through the posterior wall:

- the quadrangular space,
- the triangular space, and
- the triangular interval (Fig. 7.45).

Gateways in the posterior wall

(See also "Gateways to the posterior scapular region," pp. 717–721, and Figs. 7.37 and 7.38)

Quadrangular space

The quadrangular space provides a passageway for nerves and vessels passing between the axilla and the more posterior scapular and deltoid regions (Fig. 7.45). When viewed from anteriorly, its boundaries are formed by:

- the inferior margin of the subscapularis muscle,
- the surgical neck of the humerus,
- the superior margin of the teres major muscle, and
- the lateral margin of the long head of the triceps brachii muscle.

Passing through the quadrangular space are the axillary nerve and the posterior circumflex humeral artery and vein.

Triangular space

The **triangular space** is an area of communication between the axilla and the posterior scapular region (Fig. 7.45). When viewed from anteriorly, it is formed by:

- the medial margin of the long head of the triceps brachii muscle,
- the superior margin of the teres major muscle, and
- the inferior margin of the subscapularis muscle.

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The circumflex scapular artery and vein pass into this space.

Triangular interval

This triangular interval is formed by:

- the lateral margin of the long head of the triceps brachii muscle,
- the shaft of the humerus, and
- the inferior margin of the teres major muscle (Fig. 7.45).

The radial nerve passes out of the axilla traveling through this interval to reach the posterior compartment of the arm.

Floor

The floor of the axilla is formed by fascia and a dome of skin that spans the distance between the inferior margins of the walls (Fig. 7.47 and see Fig. 7.40B). It is supported by the clavipectoral fascia. On a patient, the anterior axillary fold is more superior in position than is the posterior axillary fold.

Inferiorly, structures pass into and out of the axilla immediately lateral to the floor where the anterior and posterior walls of the axilla converge and where the axilla is continuous with the anterior compartment of the arm.

Contents of the axilla

Passing through the axilla are the major vessels, nerves, and lymphatics of the upper limb. The space also contains the proximal parts of two muscles of the arm, the axillary process of the breast, and collections of lymph nodes, which drain the upper limb, chest wall, and breast.

The proximal parts of the biceps brachii and coracobrachialis muscles pass through the axilla (Table 7.6).

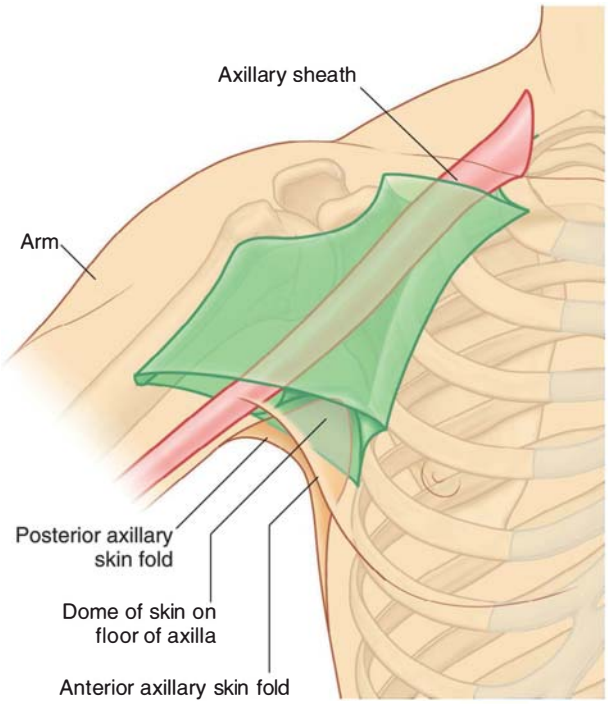


Fig. 7.47 Floor of the axilla.

Table 7.6 Muscles having parts that pass through the axilla (spinal segments in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Biceps brachii	Long head—supraglenoid tubercle of scapula; short head—apex of coracoid process	Tuberosity of radius	Musculocutaneous nerve (**C5, C6**)	Powerful flexor of the forearm at the elbow joint and supinator of the forearm; accessory flexor of the arm at the glenohumeral joint
Coracobrachialis	Apex of coracoid process	Linear roughening on midshaft of humerus on medial side	Musculocutaneous nerve (**C5, C6, C7**)	Flexor of the arm at the glenohumeral joint; adducts arm

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Biceps brachii

The **biceps brachii** muscle originates as two heads (Fig. 7.48):

- The short head originates from the apex of the coracoid process of the scapula and passes vertically through the axilla and into the arm where it joins the long head.
- The long head originates as a tendon from the supraglenoid tubercle of the scapula, passes over the head of the humerus deep to the joint capsule of the glenohumeral joint, and enters the intertubercular sulcus where it is held in position by a ligament, the transverse humeral ligament, which spans the distance between the greater and lesser tubercles; the tendon passes through the axilla in the intertubercular sulcus and forms a muscle belly in the proximal part of the arm.

The long and short heads of the muscle join in distal regions of the arm and primarily insert as a single tendon into the radial tuberosity in the forearm.

The biceps brachii muscle is primarily a powerful flexor of the forearm at the elbow joint and a powerful supinator in the forearm. Because both heads originate from the scapula, the muscle also acts as an accessory flexor of the arm at the glenohumeral joint. In addition, the long head prevents superior movement of the humerus on the glenoid cavity.

The biceps brachii muscle is innervated by the musculocutaneous nerve.

Coracobrachialis

The **coracobrachialis** muscle, together with the short head of the biceps brachii muscle, originates from the apex of the coracoid process (Fig. 7.48). It passes vertically through the axilla to insert on a small linear roughening on the medial aspect of the humerus, approximately midshaft.

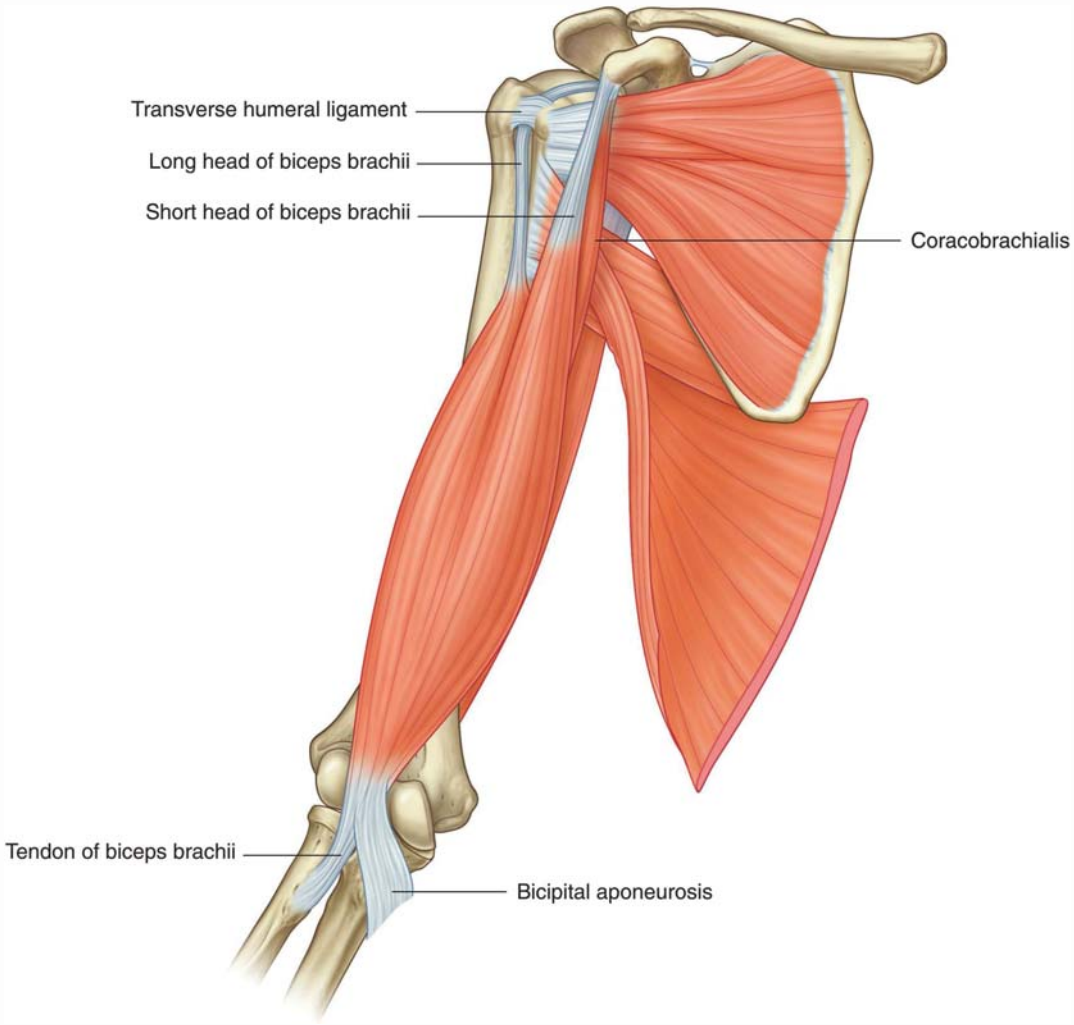


Fig. 7.48 Contents of the axilla: muscles.

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The coracobrachialis muscle flexes the arm at the glenohumeral joint.

In the axilla, the medial surface of the coracobrachialis muscle is pierced by the musculocutaneous nerve, which innervates and then passes through the muscle to enter the arm.

Axillary artery

The axillary artery supplies the walls of the axilla and related regions, and continues as the major blood supply to the more distal parts of the upper limb (Fig. 7.49).

The subclavian artery in the neck becomes the axillary artery at the lateral margin of rib I and passes through the axilla, becoming the brachial artery at the inferior margin of the teres major muscle.

The axillary artery is separated into three parts by the pectoralis minor muscle, which crosses anteriorly to the vessel (Fig. 7.49):

- The first part is proximal to the pectoralis minor.
- The second part is posterior to the pectoralis minor.
- The third part is distal to the pectoralis minor.

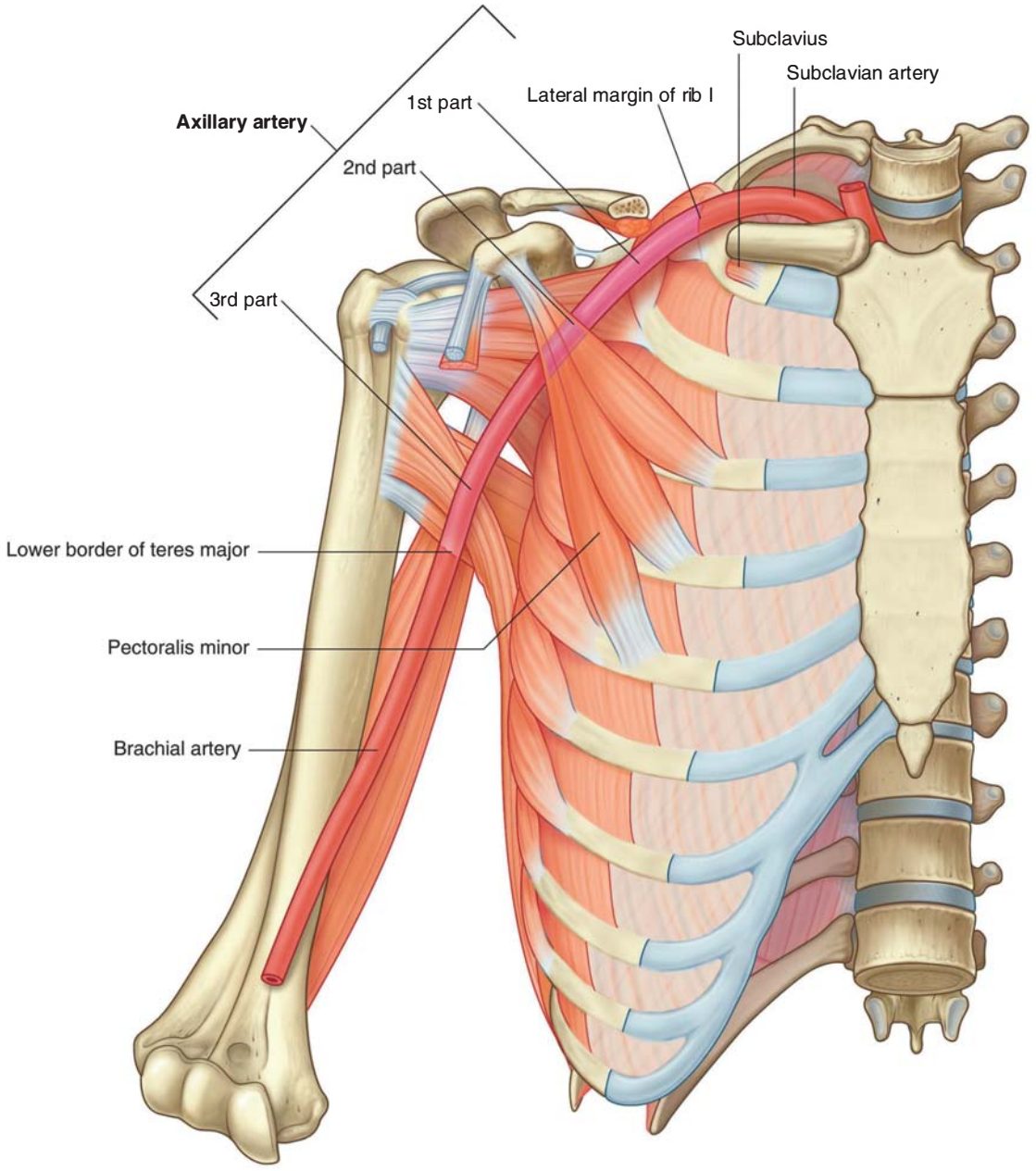


Fig. 7.49 Contents of the axilla: the axillary artery.

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Generally, six branches arise from the axillary artery:

- One branch, the **superior thoracic artery**, originates from the first part.
- Two branches, the **thoraco-acromial artery** and the **lateral thoracic artery**, originate from the second part.
- Three branches, the **subscapular artery**, the **anterior circumflex humeral artery**, and the **posterior circumflex humeral artery**, originate from the third part (Fig. 7.50).

Superior thoracic artery

The superior thoracic artery is small and originates from the anterior surface of the first part of the axillary artery (Fig. 7.50). It supplies upper regions of the medial and anterior axillary walls.

Thoraco-acromial artery

The thoraco-acromial artery is short and originates from the anterior surface of the second part of the axillary

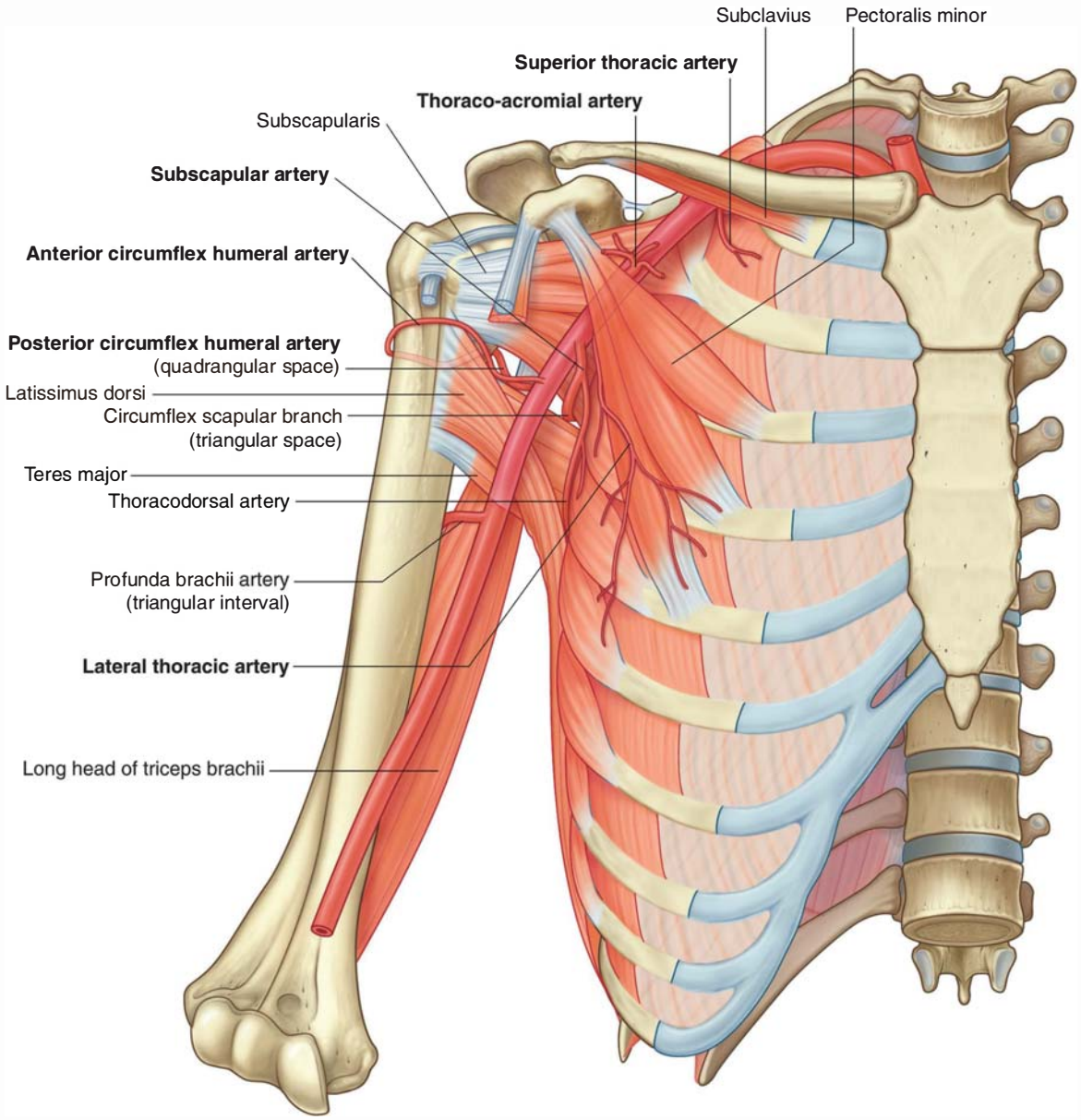


Fig. 7.50 Branches of the axillary artery.

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artery just posterior to the medial (superior) margin of the pectoralis minor muscle (Fig. 7.50). It curves around the superior margin of the muscle, penetrates the clavipectoral fascia, and immediately divides into four branches—the pectoral, deltoid, clavicular, and acromial branches, which supply the anterior axillary wall and related regions.

Additionally, the pectoral branch contributes vascular supply to the breast, and the deltoid branch passes into the clavipectoral triangle where it accompanies the cephalic vein and supplies adjacent structures (see Fig. 7.41).

Lateral thoracic artery

The lateral thoracic artery arises from the anterior surface of the second part of the axillary artery posterior to the lateral (inferior) margin of the pectoralis minor (Fig. 7.50). It follows the margin of the muscle to the thoracic wall and supplies the medial and anterior walls of the axilla. In women, branches emerge from around the inferior margin of the pectoralis major muscle and contribute to the vascular supply of the breast.

Subscapular artery

The subscapular artery is the largest branch of the axillary artery and is the major blood supply to the posterior wall of the axilla (Fig. 7.50). It also contributes to the blood supply of the posterior scapular region.

The subscapular artery originates from the posterior surface of the third part of the axillary artery, follows the inferior margin of the subscapularis muscle for a short distance, and then divides into its two terminal branches, the **circumflex scapular artery** and the **thoracodorsal artery**.

- The circumflex scapular artery passes through the triangular space between the subscapularis, teres major, and long head of the triceps muscle. Posteriorly, it passes inferior to, or pierces, the origin of the teres minor muscle to enter the infraspinous fossa. It anastomoses with the suprascapular artery and the **deep branch** (**dorsal scapular artery**) of the transverse cervical artery, thereby contributing to an anastomotic network of vessels around the scapula.
- The thoracodorsal artery approximately follows the lateral border of the scapula to the inferior angle. It contributes to the vascular supply of the posterior and medial walls of the axilla.

Anterior circumflex humeral artery

The **anterior circumflex humeral artery** is small compared to the posterior circumflex humeral artery, and originates from the lateral side of the third part of the axillary artery (Fig. 7.50). It passes anterior to the surgical neck of the humerus and anastomoses with the posterior circumflex humeral artery.

This anterior circumflex humeral artery supplies branches to surrounding tissues, which include the glenohumeral joint and the head of the humerus.

Posterior circumflex humeral artery

The **posterior circumflex humeral artery** originates from the lateral surface of the third part of the axillary artery immediately posterior to the origin of the anterior circumflex humeral artery (Fig. 7.50). With the axillary nerve, it leaves the axilla by passing through the quadrangular space between the teres major, teres minor, and long head of the triceps brachii muscle and the surgical neck of the humerus.

The posterior circumflex humeral artery curves around the surgical neck of the humerus and supplies the surrounding muscles and the glenohumeral joint. It anastomoses with the anterior circumflex humeral artery and with branches from the profunda brachii, suprascapular, and thoraco-acromial arteries.

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Axillary vein

The axillary vein begins at the lower margin of the teres major muscle and is the continuation of the basilic vein (Fig. 7.51), which is a superficial vein that drains the posteromedial surface of the hand and forearm and penetrates the deep fascia in the middle of the arm.

The axillary vein passes through the axilla medial and anterior to the axillary artery and becomes the subclavian vein as the vessel crosses the lateral border of rib I at the axillary inlet. Tributaries of the axillary vein generally follow the branches of the axillary artery. Other tributaries include brachial veins that follow the brachial artery, and the cephalic vein.

The cephalic vein is a superficial vein that drains the lateral and posterior parts of the hand, the forearm, and the arm. In the area of the shoulder, it passes into an inverted triangular cleft (the clavipectoral triangle) between the deltoid muscle, pectoralis major muscle, and clavicle. In the superior part of the clavipectoral triangle, the cephalic vein passes deep to the clavicular head of the pectoralis major muscle and pierces the clavipectoral fascia to join the axillary vein. Many patients who are critically unwell have lost blood or fluid, which requires replacement. Access to a peripheral vein is necessary to replace the fluid. The typical sites for venous access are the cephalic vein in the hand or veins that lie within the superficial tissues of the cubital fossa.

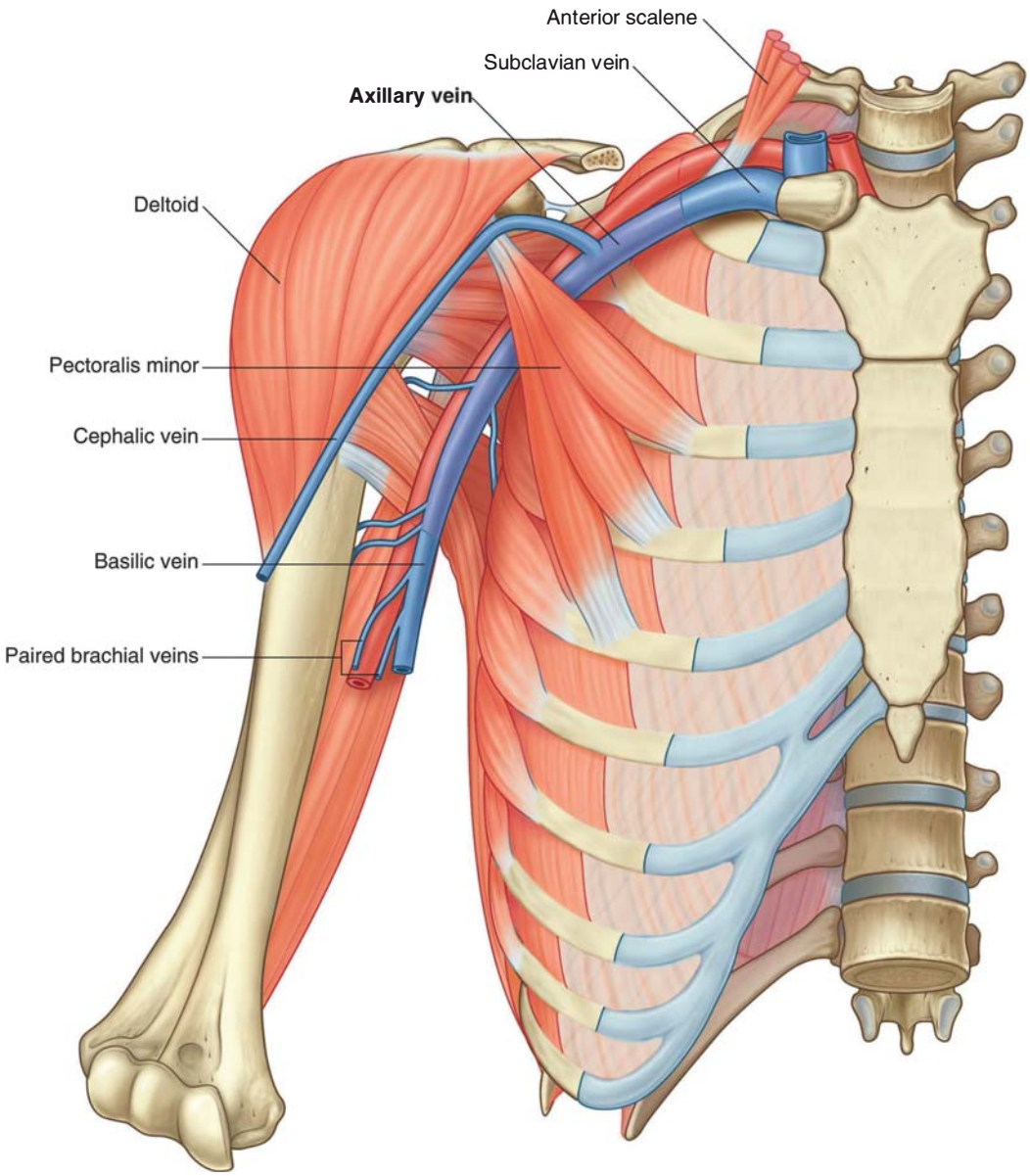


Fig. 7.51 Axillary vein.

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In the clinic

Imaging the blood supply to the upper limb

When there is clinical evidence of vascular compromise to the upper limb, or vessels are needed to form an arteriovenous fistula (which is necessary for renal dialysis), imaging is required to assess the vessels.

Ultrasound is a useful tool for carrying out a noninvasive assessment of the vessels of the upper limb from the third part of the subclavian artery to the deep and superficial palmar arteries. Blood flow can be quantified and anatomical variants can be noted.

Angiography is carried out in certain cases. The femoral artery is punctured below the inguinal ligament and a long catheter is placed through the iliac arteries and around the arch of the aorta to enter either the left subclavian artery or the brachiocephalic trunk and then the right subclavian artery. Radiopaque contrast agents are injected into the vessel and radiographs are obtained as the contrast agents pass first through the arteries, then the capillaries, and finally the veins.

In the clinic

Trauma to the arteries of the upper limb

The arterial supply to the upper limb is particularly susceptible to trauma in places where it is relatively fixed or in a subcutaneous position.

Fracture of rib I

As the subclavian artery passes out of the neck and into the axilla, it is fixed in position by the surrounding muscles to the superior surface of rib I. A rapid deceleration injury involving upper thoracic trauma may cause a first rib fracture, which may significantly compromise the distal part of the subclavian artery or the first part of the axillary artery. Fortunately, there are anastomotic connections between branches of the subclavian artery and the axillary artery, which form a network around the scapula and proximal end of the humerus; therefore, even with complete vessel transection, the arm is rarely rendered completely ischemic (ischemia is poor blood supply to an organ or a limb).

Anterior dislocation of the humeral head

Anterior dislocation of the humeral head may compress the axillary artery, resulting in vessel occlusion. This is unlikely to render the upper limb completely ischemic, but it may be necessary to surgically reconstruct the axillary artery to obtain pain-free function. Importantly, the axillary artery is intimately related to the brachial plexus, which may be damaged at the time of anterior dislocation.

In the clinic

Subclavian/axillary venous access

There are a number of routes through which central venous access may be obtained. The "subclavian route" and the jugular routes are commonly used by clinicians. The subclavian route is a misnomer that remains the preferred term in clinical practice. In fact, most clinicians enter the first part of the axillary vein.

There are a number of patients that undergo catheterization of the subclavian vein/axillary vein. Entering the subclavian vein/axillary vein is a relatively straightforward technique. The clavicle is identified and a sharp needle is placed in the infraclavicular region, aiming superomedially. When venous blood is aspirated, access has been obtained. This route is popular for long-term venous access, such as Hickman lines, and for shorter-term access where multiple-lumen catheters are inserted (e.g., intensive care unit).

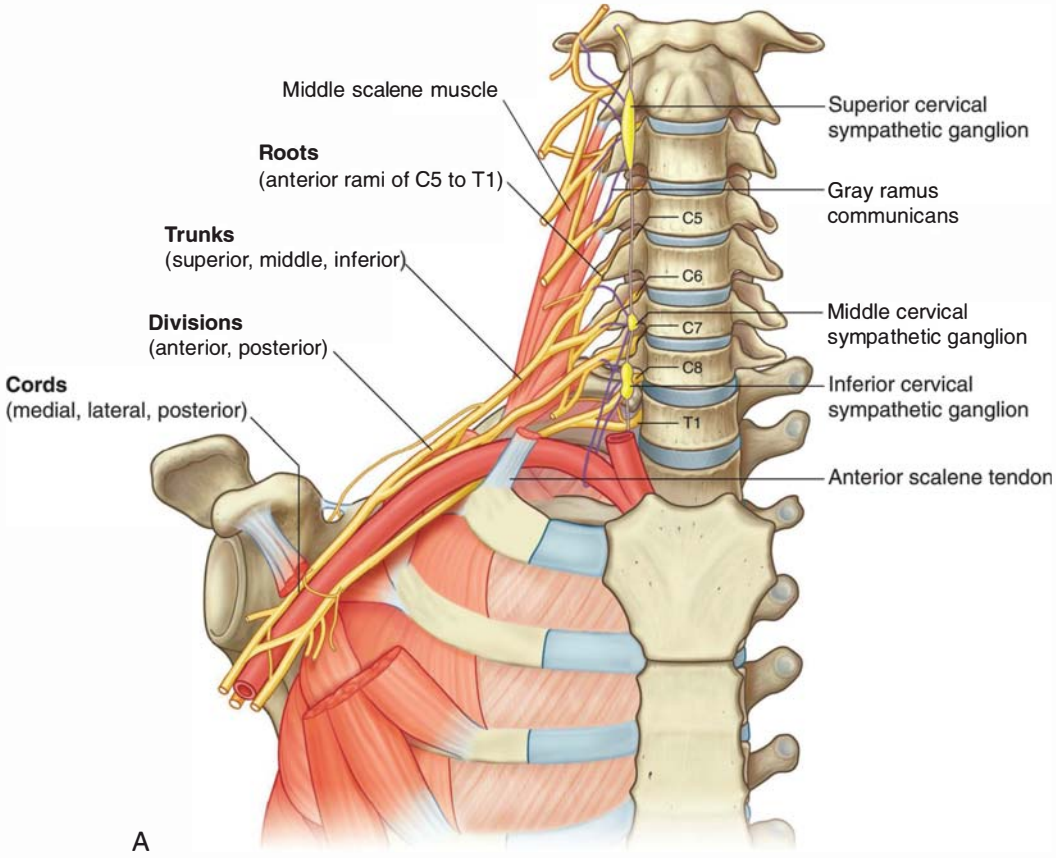
The subclavian vein/axillary vein is also the preferred site for insertion of pacemaker wires. There is, however, a preferred point of entry into the vein to prevent complications. The vein should be punctured in the midclavicular line or lateral to this line. The reason for this puncture site is the course of the vein and its relationship to other structures. The vein passes anterior to the artery, superior to the first rib, and inferior to the clavicle as it courses toward the thoracic inlet. Beneath the clavicle is situated the subclavius muscle. Should the puncture of the vein enter where the subclavius muscle is related to the axillary vein, the catheter or wire may become kinked at this point. Moreover, the constant contraction and relaxation of this muscle will induce fatigue in the line and wire, which may ultimately lead to fracture. A fractured pacemaker wire or a rupture in a chemotherapy catheter can have severe consequences for the patient.

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Brachial plexus

The brachial plexus is a somatic plexus formed by the **anterior rami** of C5 to C8, and most of the anterior ramus of T1 (Fig. 7.52). The plexus originates in the neck, passes laterally and inferiorly over rib I, and enters the axilla.

The parts of the brachial plexus, from medial to lateral, are roots, trunks, divisions, and cords. All major nerves that innervate the upper limb originate from the brachial plexus, mostly from the cords. Proximal parts of the brachial plexus are posterior to the subclavian artery in the neck, while more distal regions of the plexus surround the axillary artery.



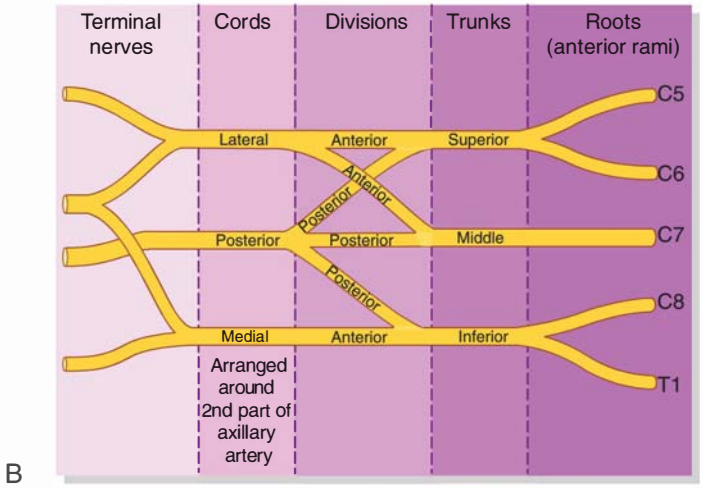


Fig. 7.52 Brachial plexus. **A.** Major components in the neck and axilla. **B.** Schematic showing parts of the brachial plexus.

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Roots

The roots of the brachial plexus are the anterior rami of C5 to C8, and most of T1. Close to their origin, the roots receive **gray rami communicantes** from the sympathetic trunk (Fig. 7.52). These carry postganglionic sympathetic fibers onto the roots for distribution to the periphery. The roots and trunks enter the **posterior triangle** of the neck by passing between the anterior scalene and middle scalene muscles and lie superior and posterior to the subclavian artery.

Trunks

The three trunks of the brachial plexus originate from the roots, pass laterally over rib I, and enter the axilla (Fig. 7.52):

- The superior trunk is formed by the union of C5 and C6 roots.
- The middle trunk is a continuation of the C7 root.
- The inferior trunk is formed by the union of the C8 and T1 roots.

The inferior trunk lies on rib I posterior to the subclavian artery; the middle and superior trunks are more superior in position.

Divisions

Each of the three trunks of the brachial plexus divides into an **anterior** and a **posterior division** (Fig. 7.52):

- The three anterior divisions form parts of the brachial plexus that ultimately give rise to peripheral nerves associated with the anterior compartments of the arm and forearm.
- The three posterior divisions combine to form parts of the brachial plexus that give rise to nerves associated with the posterior compartments.

No peripheral nerves originate directly from the divisions of the brachial plexus.

Cords

The three cords of the brachial plexus originate from the divisions and are related to the second part of the axillary artery (Fig. 7.52):

- The **lateral cord** results from the union of the anterior divisions of the upper and middle trunks and therefore has contributions from C5 to C7—it is positioned lateral to the second part of the axillary artery.
- The **medial cord** is medial to the second part of the axillary artery and is the continuation of the anterior division of the inferior trunk—it contains contributions from C8 and T1.
- The **posterior cord** occurs posterior to the second part of the axillary artery and originates as the union of all three posterior divisions—it contains contributions from all roots of the brachial plexus (C5 to T1).

Most of the major peripheral nerves of the upper limb originate from the cords of the brachial plexus. Generally, nerves associated with the anterior compartments of the upper limb arise from the medial and lateral cords and nerves associated with the posterior compartments originate from the posterior cord.

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Branches (Table 7.7)

Branches of the roots

In addition to small segmental branches from C5 to C8 to muscles of the neck and a contribution of C5 to the phrenic nerve, the roots of the brachial plexus give rise to the dorsal scapular and long thoracic nerves (Fig. 7.53).

The **dorsal scapular nerve**:

- originates from the C5 root of the brachial plexus,
- passes posteriorly, often piercing the middle scalene muscle in the neck, to reach and travel along the medial border of the scapula (Fig. 7.54), and
- innervates the rhomboid major and minor muscles from their deep surfaces.

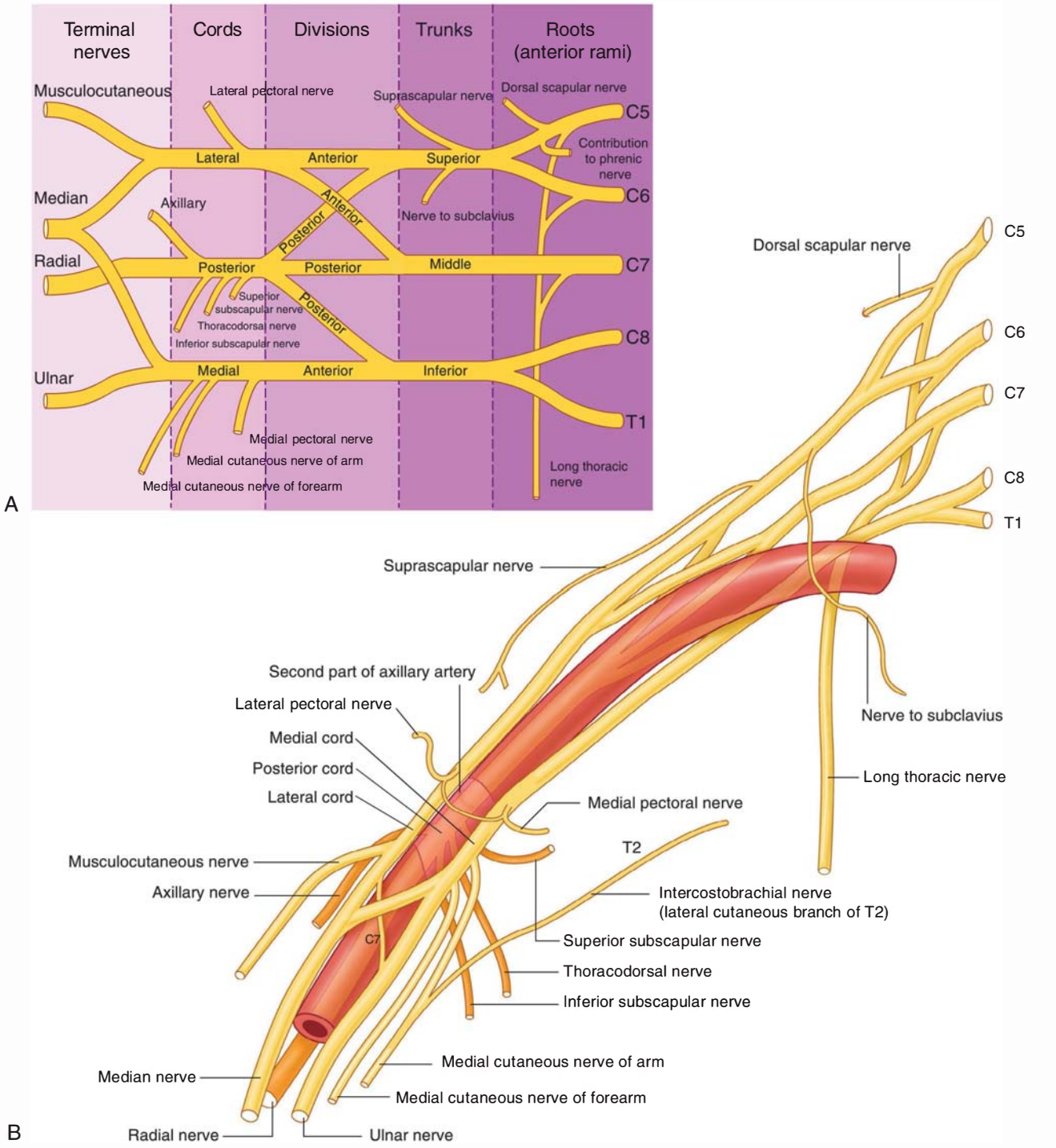


Fig. 7.53 Brachial plexus. **A.** Schematic showing branches of the brachial plexus. **B.** Relationships to the axillary artery.

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The **long thoracic nerve**:

- originates from the anterior rami of C5 to C7,
- passes vertically down the neck, through the axillary inlet, and down the medial wall of the axilla to supply the serratus anterior muscle (Fig. 7.54), and
- lies on the superficial aspect of the serratus anterior muscle.

Branches of the trunks

The only branches from the trunks of the brachial plexus are two nerves that originate from the superior trunk (upper trunk): the suprascapular nerve and the nerve to the subclavius muscle (Fig. 7.53).

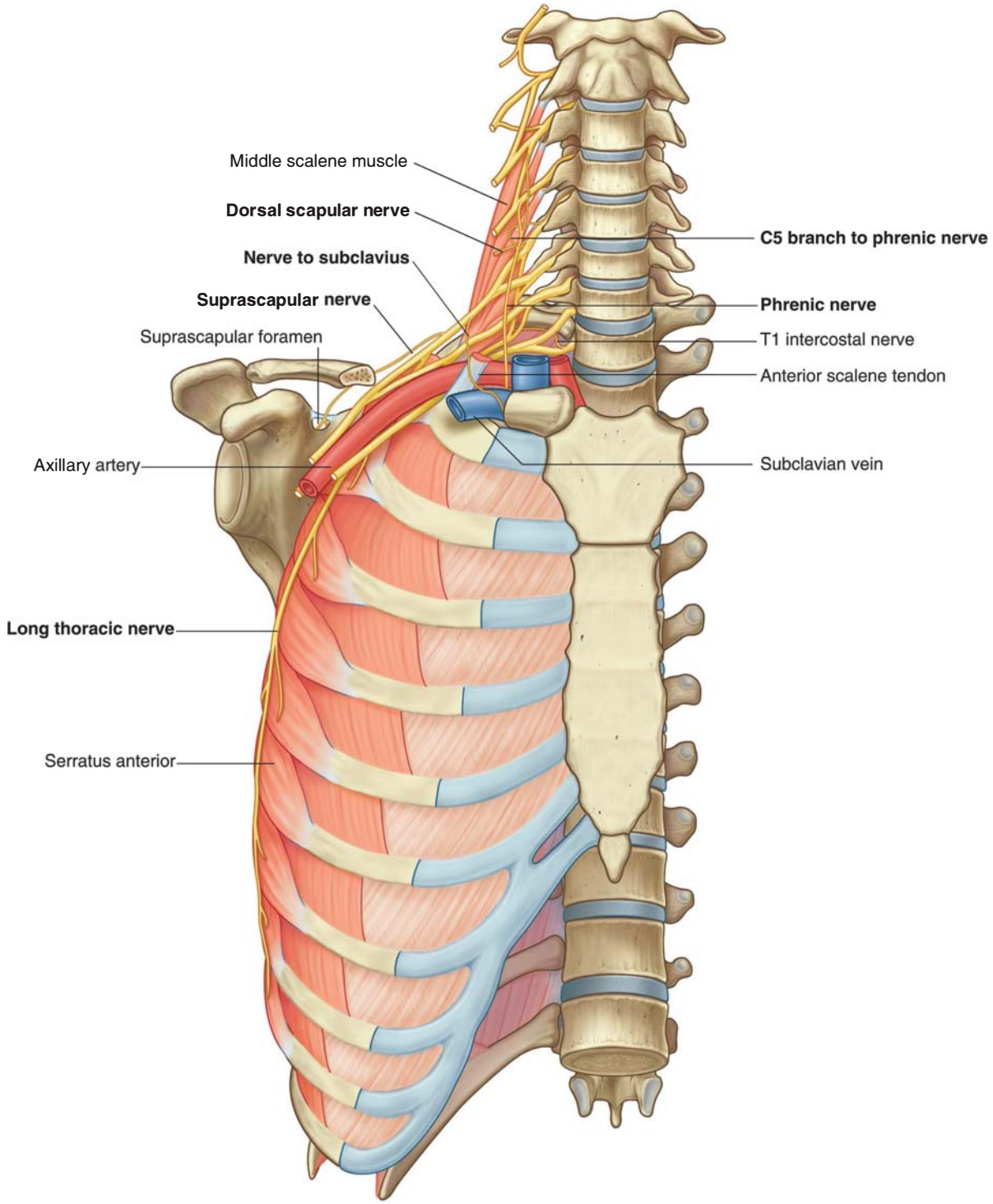


Fig. 7.54 Branches of the roots and trunks of the brachial plexus.

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The **suprascapular nerve** (C5 and C6):

- originates from the superior trunk of the brachial plexus,
- passes laterally through the posterior triangle of the neck (Fig. 7.54) and through the suprascapular foramen to enter the posterior scapular region,
- innervates the supraspinatus and infraspinatus muscles, and
- is accompanied in the lateral parts of the neck and in the posterior scapular region by the suprascapular artery.

The **nerve to the subclavius muscle** (C5 and C6) is a small nerve that:

- originates from the superior trunk of the brachial plexus,
- passes anteroinferiorly over the subclavian artery and vein, and
- innervates the subclavius muscle.

Branches of the lateral cord

Three nerves originate entirely or partly from the lateral cord (Fig. 7.53).

- The **lateral pectoral nerve** is the most proximal of the branches from the lateral cord. It passes anteriorly, together with the thoraco-acromial artery, to penetrate the clavipectoral fascia that spans the gap between the subclavius and pectoralis minor muscles (Fig. 7.55), and innervates the pectoralis major muscle.
- The **musculocutaneous nerve** is a large terminal branch of the lateral cord. It passes laterally to penetrate the coracobrachialis muscle and pass between the biceps brachii and brachialis muscles in the arm, and innervates all three flexor muscles in the anterior compartment of the arm, terminating as the **lateral cutaneous nerve of the forearm**.
- The **lateral root of the median nerve** is the largest terminal branch of the lateral cord and passes medially to join a similar branch from the medial cord to form the median nerve (Fig. 7.55).

Branches of the medial cord

The medial cord has five branches (Fig. 7.55).

- The **medial pectoral nerve** is the most proximal branch. It receives a communicating branch from the lateral pectoral nerve and then passes anteriorly between the axillary artery and axillary vein. Branches of the nerve penetrate and supply the pectoralis minor muscle. Some of these branches pass through the muscle to reach and supply the pectoralis major muscle. Other branches occasionally pass around the inferior or lateral margin of the pectoralis minor muscle to reach the pectoralis major muscle.
- The **medial cutaneous nerve of the arm** (**medial brachial cutaneous nerve**) passes through the axilla and into the arm where it penetrates deep fascia and supplies skin over the medial side of the distal third of the arm. In the axilla, the nerve communicates with the **intercostobrachial nerve** of T2. Fibers of the medial cutaneous nerve of the arm innervate the upper part of the medial surface of the arm and floor of the axilla.
- The **medial cutaneous nerve of the forearm (medial antebrachial cutaneous nerve)** originates just distal to the origin of the medial cutaneous nerve of the arm. It passes out of the axilla and into the arm where it gives off a branch to the skin over the biceps brachii muscle, and then continues down the arm to penetrate the deep fascia with the basilic vein, continuing inferiorly to supply the skin over the anterior surface of the forearm. It innervates skin over the medial surface of the forearm down to the wrist.
- The **medial root of the median nerve** passes laterally to join with a similar root from the lateral cord to form the median nerve anterior to the third part of the axillary artery.
- The **ulnar nerve** is a large terminal branch of the medial cord (Fig. 7.55). However, near its origin, it often receives a communicating branch from the lateral root of the median nerve originating from the lateral cord and carrying fibers from C7. The ulnar nerve passes through the arm and forearm into the hand where it innervates all intrinsic muscles of the hand (except for the three thenar muscles and the two lateral lumbrical muscles). On passing through the forearm, branches of the ulnar nerve innervate the flexor carpi ulnaris muscle and the medial half of the flexor digitorum profundus muscle. The ulnar nerve innervates skin over the palmar surface of the little finger, medial half of the ring finger, and associated palm and wrist, and the skin over the dorsal surface of the medial part of the hand.

Median nerve. The median nerve is formed anterior to the third part of the axillary artery by the union of lateral and medial roots originating from the lateral and medial cords of the brachial plexus (Fig. 7.55). It passes into the arm anterior to the brachial artery and through the arm into the forearm, where branches innervate most of the muscles in the anterior compartment of the forearm (except for the flexor carpi ulnaris muscle and the medial half of the flexor digitorum profundus muscle, which are innervated by the ulnar nerve).

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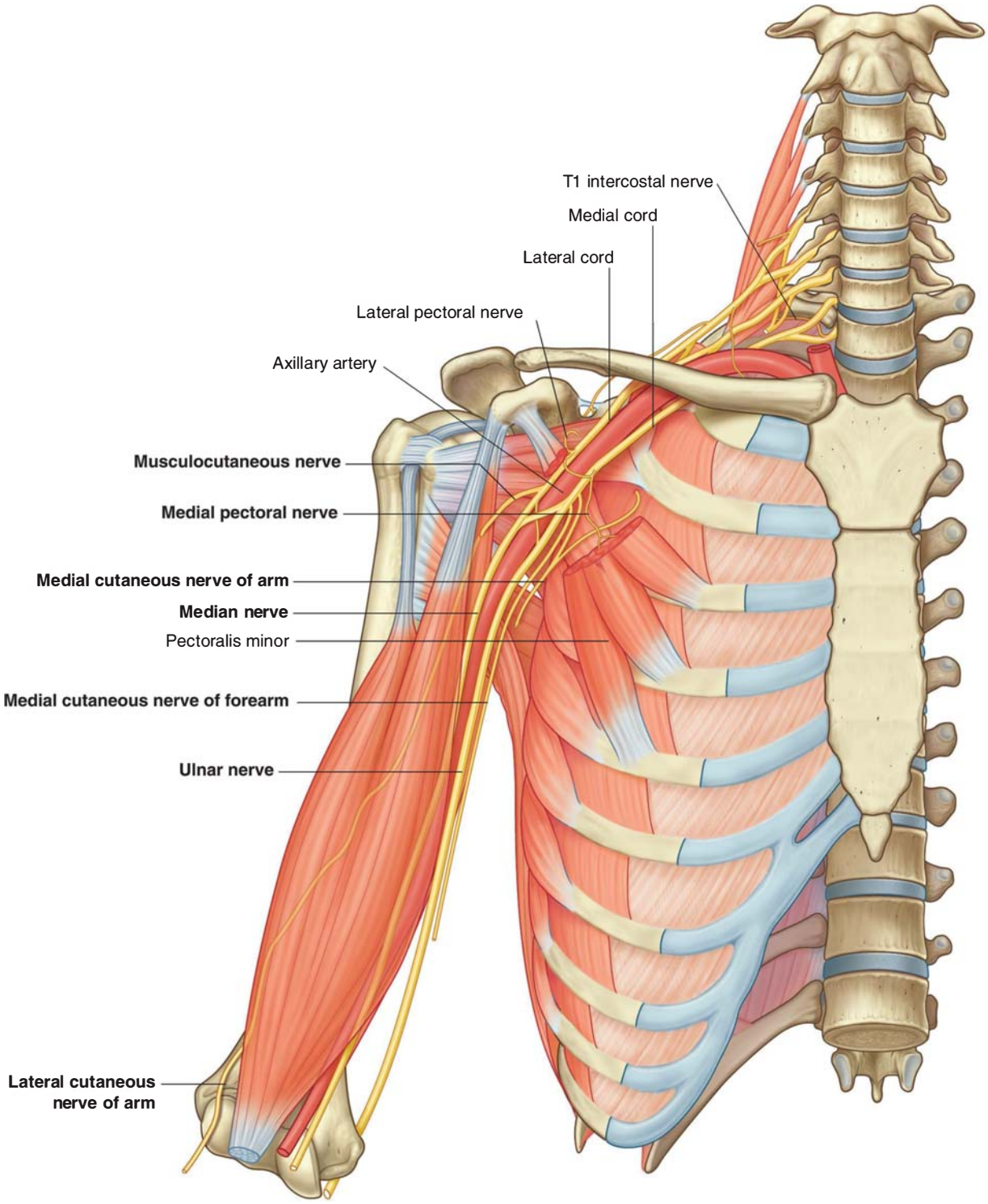


Fig. 7.55 Branches of the lateral and medial cords of the brachial plexus.

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Table 7.7 Branches of brachial plexus (parentheses indicate that a spinal segment is a minor component of the nerve or is inconsistently present in the nerve)

Branch		
Dorsal scapular Origin: C5 root Spinal segment: C5		Function: motor Rhomboid major, rhomboid minor
Long thoracic Origin: C5 to C7 roots Spinal segments: C5 to C7		Function: motor Serratus anterior
Suprascapular Origin: Superior trunk Spinal segments: C5, C6		Function: motor Supraspinatus, infraspinatus
Nerve to subclavius Origin: Superior trunk Spinal segments: C5, C6		Function: motor Subclavius
Lateral pectoral Origin: Lateral cord Spinal segments: C5 to C7		Function: motor Pectoralis major
Musculocutaneous Origin: Lateral cord Spinal segments: C5 to C7		Function: motor All muscles in the anterior compartment of the arm Function: sensory Skin on lateral side of forearm
Medial pectoral Origin: Medial cord Spinal segments: C8, T1 (also receives contributions from spinal segments C5 to C7 through a communication with the lateral pectoral nerve)		Function: motor Pectoralis major, pectoralis minor
Medial cutaneous of arm Origin: Medial cord Spinal segments: C8, T1		Function: sensory Skin on medial side of distal one-third of arm

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Table 7.7 Branches of brachial plexus (parentheses indicate that a spinal segment is a minor component of the nerve or is inconsistently present in the nerve)—cont'd

Branch		
Medial cutaneous of forearm Origin: Medial cord Spinal segments: C8, T1		Function: sensory Skin on medial side of forearm
Median Origin: Medial and lateral cords Spinal segments: (C5), C6 to T1		Function: motor All muscles in the anterior compartment of the forearm (except flexor carpi ulnaris and medial half of flexor digitorum profundus), three thenar muscles of the thumb and two lateral lumbrical muscles Function: sensory Skin over the palmar surface of the lateral three and one-half digits and over the lateral side of the palm and middle of the wrist
Ulnar Origin: Medial cord Spinal segments: (C7), C8, T1		Function: motor All intrinsic muscles of the hand (except three thenar muscles and two lateral lumbricals); also flexor carpi ulnaris and the medial half of flexor digitorum profundus in the forearm Function: sensory Skin over the palmar surface of the medial one and one-half digits and associated palm and wrist, and skin over the dorsal surface of the medial one and one-half digits
Superior subscapular Origin: Posterior cord Spinal segments: C5, C6		Function: motor Subscapularis
Thoracodorsal Origin: Posterior cord Spinal segments: C6 to C8		Function: motor Latissimus dorsi
Inferior subscapular Origin: Posterior cord Spinal segments: C5, C6		Function: motor Subscapularis, teres major
Axillary Origin: Posterior cord Spinal segments: C5, C6		Function: motor Deltoid, teres minor Function: sensory Skin over upper lateral part of arm
Radial Origin: Posterior cord Spinal segments: C5 to C8, (T1)		Function: motor All muscles in the posterior compartments of arm and forearm Function: sensory Skin on the posterior aspects of the arm and forearm, the lower lateral surface of the arm, and the dorsal lateral surface of the hand

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The median nerve continues into the hand to innervate:

- the three thenar muscles associated with the thumb,
- the two lateral lumbrical muscles associated with movement of the index and middle fingers, and
- the skin over the palmar surface of the lateral three and one-half digits and over the lateral side of the palm and middle of the wrist.

The musculocutaneous nerve, the lateral root of the median nerve, the median nerve, the medial root of the median nerve, and the ulnar nerve form an M over the third part of the axillary artery (Fig. 7.55). This feature, together with penetration of the coracobrachialis muscle by the musculocutaneous nerve, can be used to identify components of the brachial plexus in the axilla.

Branches of the posterior cord

Five nerves originate from the posterior cord of the brachial plexus:

- the superior subscapular nerve,
- the thoracodorsal nerve,
- the inferior subscapular nerve,
- the axillary nerve, and
- the radial nerve (Fig. 7.53).

All these nerves except the radial nerve innervate muscles associated with the posterior wall of the axilla; the radial nerve passes into the arm and forearm.

The superior subscapular, thoracodorsal, and inferior subscapular nerves originate sequentially from the posterior cord and pass directly into muscles associated with the posterior axillary wall (Fig. 7.56). The **superior subscapular nerve** is short and passes into and supplies the subscapularis muscle. The **thoracodorsal nerve** is the longest of these three nerves and passes vertically along the posterior axillary wall. It penetrates and innervates the latissimus dorsi muscle. The **inferior subscapular nerve** also passes inferiorly along the posterior axillary wall and innervates the subscapularis and teres major muscles.

The **axillary nerve** originates from the posterior cord and passes inferiorly and laterally along the posterior wall to exit the axilla through the quadrangular space (Fig. 7.56). It passes posteriorly around the surgical neck of the humerus and innervates both the deltoid and teres minor muscles. A **superior lateral cutaneous nerve of the arm** originates from the axillary nerve after passing through the quadrangular space and loops around the posterior margin of the deltoid muscle to innervate skin in that region. The axillary nerve is accompanied by the posterior circumflex humeral artery.

The **radial nerve** is the largest terminal branch of the posterior cord (Fig. 7.56). It passes out of the axilla and into the posterior compartment of the arm by passing through the triangular interval between the inferior border of the teres major muscle, the long head of the triceps brachii muscle, and the shaft of the humerus. It is accompanied through the triangular interval by the profunda brachii artery, which originates from the brachial artery in the anterior compartment of the arm. The radial nerve and its branches innervate:

- all muscles in the posterior compartments of the arm and forearm, and
- the skin on the posterior aspect of the arm and forearm, the lower lateral surface of the arm, and the dorsal lateral surface of the hand.

The **posterior cutaneous nerve of the arm** (**posterior brachial cutaneous nerve**) originates from the radial nerve in the axilla and innervates skin on the posterior surface of the arm.

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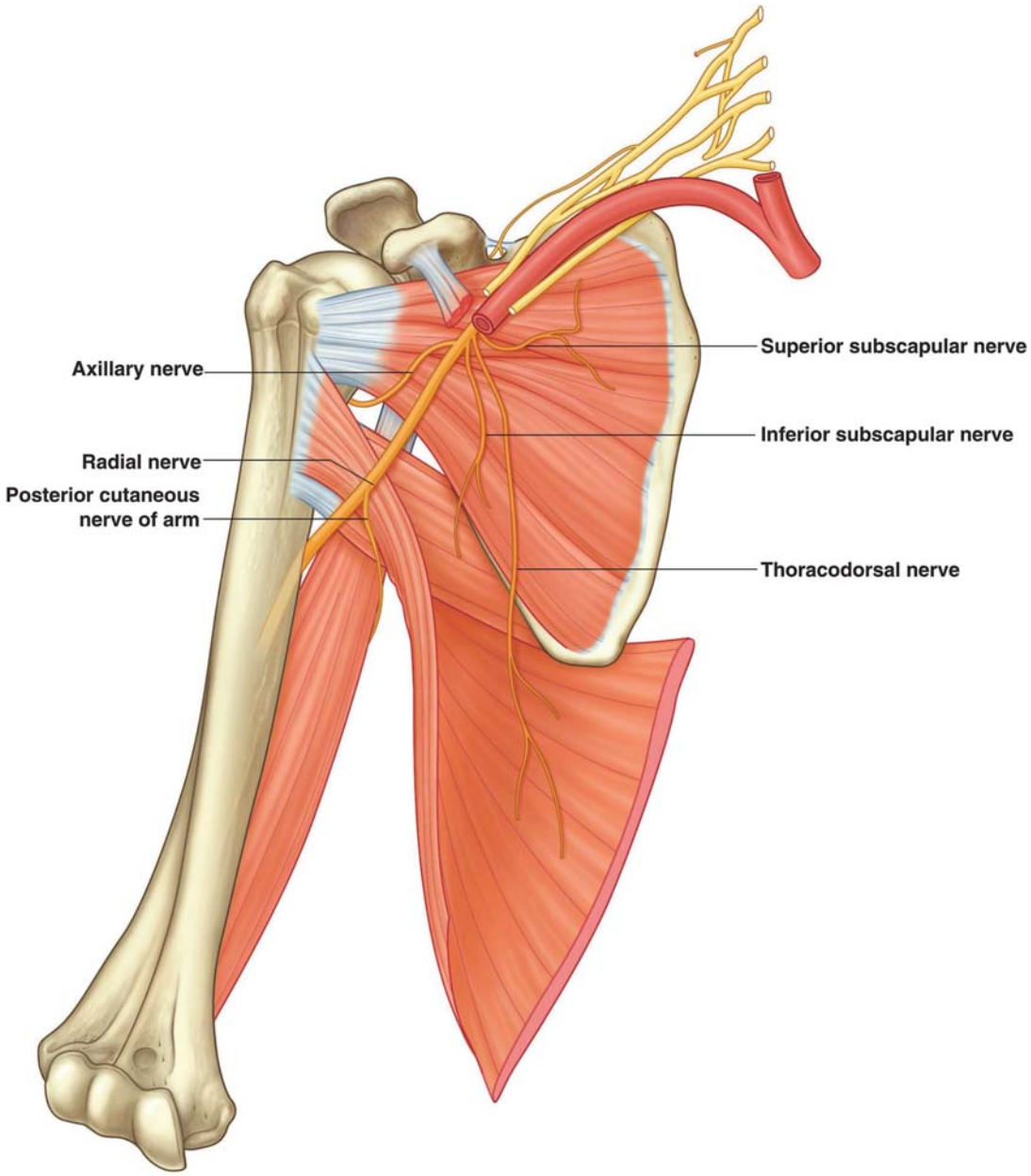


Fig. 7.56 Branches of the posterior cord of the brachial plexus.

In the clinic

Injuries to the brachial plexus

The brachial plexus is an extremely complex structure. When damaged, it requires meticulous clinical history taking and examination. Assessment of the individual nerve functions can be obtained by nerve conduction studies and electromyography, which assess the latency of muscle contraction when the nerve is artificially stimulated.

Brachial plexus injuries are usually the result of blunt trauma producing nerve avulsions and disruption. These injuries are usually devastating for the function of the upper limb and require many months of dedicated rehabilitation for even a small amount of function to return.

Spinal cord injuries in the cervical region and direct pulling injuries tend to affect the roots of the brachial plexus. Severe trauma to the first rib usually affects the trunks. The divisions and cords of the brachial plexus can be injured by dislocation of the glenohumeral joint.

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Lymphatics

All lymphatics from the upper limb drain into lymph nodes in the axilla (Fig. 7.57).

In addition, axillary nodes receive drainage from an extensive area on the adjacent trunk, which includes regions of the upper back and shoulder, the lower neck, the chest, and the upper anterolateral abdominal wall. Axillary nodes also receive drainage from approximately 75% of the mammary gland.

The 20–30 axillary nodes are generally divided into five groups on the basis of location.

- **Humeral (lateral) nodes** posteromedial to the axillary vein receive most of the lymphatic drainage from the upper limb.
- **Pectoral (anterior) nodes** occur along the inferior margin of the pectoralis minor muscle along the course of the lateral thoracic vessels and receive drainage from the abdominal wall, the chest, and the mammary gland.

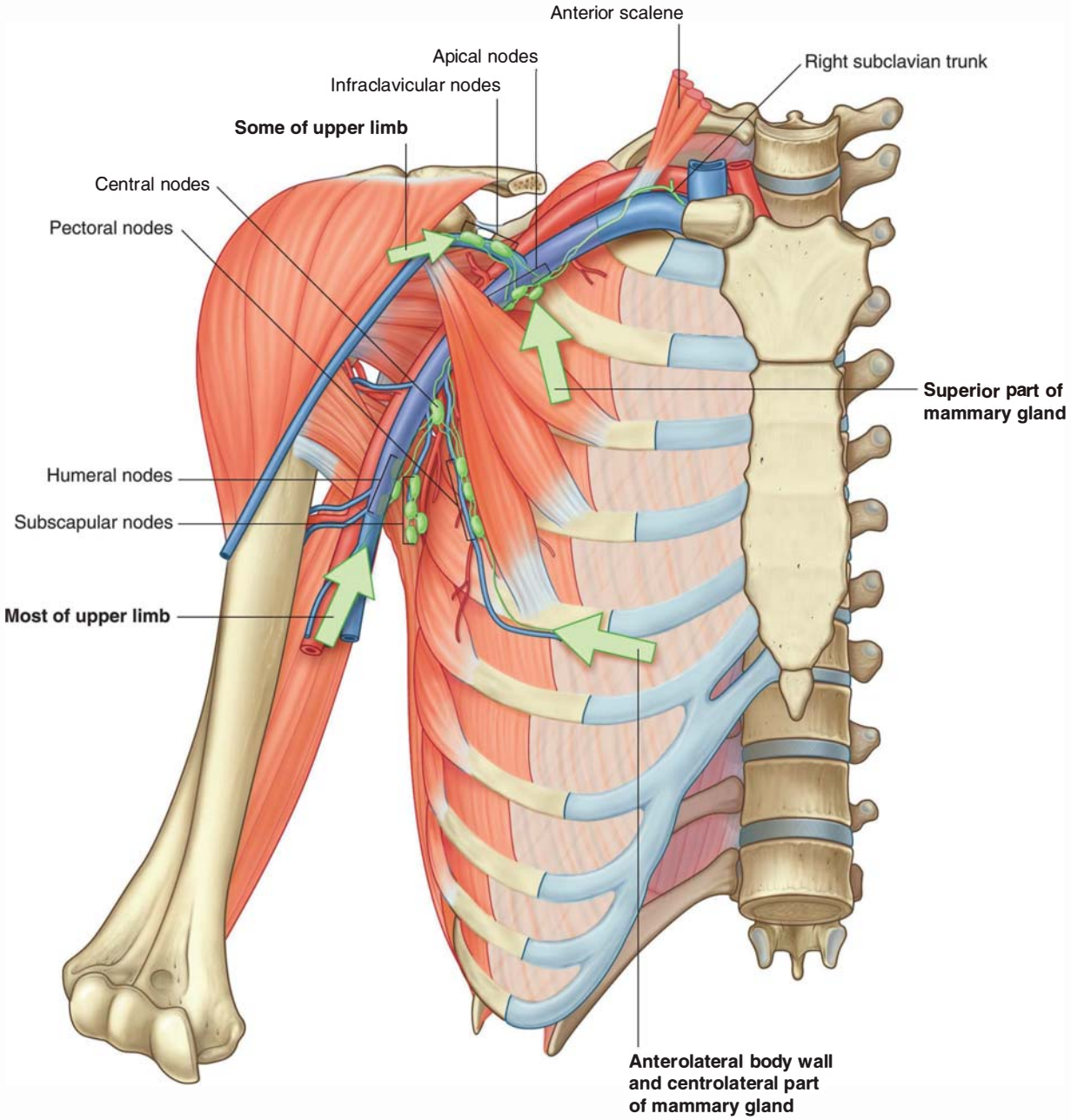


Fig. 7.57 Lymph nodes and vessels in the axilla.

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- **Subscapular (posterior) nodes** on the posterior axillary wall in association with the subscapular vessels drain the posterior axillary wall and receive lymphatics from the back, the shoulder, and the neck.
- **Central nodes** are embedded in axillary fat and receive tributaries from humeral, subscapular, and pectoral groups of nodes.
- **Apical nodes** are the most superior group of nodes in the axilla and drain all other groups of nodes in the region. In addition, they receive lymphatic vessels that accompany the cephalic vein as well as vessels that drain the superior region of the mammary gland.

Efferent vessels from the apical group converge to form the subclavian trunk, which usually joins the venous system at the junction between the right subclavian vein and the right internal jugular vein in the neck. On the left, the subclavian trunk usually joins the thoracic duct in the base of the neck.

In the clinic

Breast cancer

Lymphatic drainage from the lateral part of the breast passes through nodes in the axilla. Significant disruption to the normal lymphatic drainage of the upper limb may occur if a mastectomy or a surgical axillary nodal clearance has been carried out for breast cancer. Furthermore, some patients have radiotherapy to the axilla to prevent the spread of metastatic disease, but a side effect of this is the destruction of the tiny lymphatics as well as the cancer cells.

If the lymphatic drainage of the upper limb is damaged, the arm may swell and pitting edema (lymphedema) may develop.

Axillary process of the mammary gland

Although the mammary gland is in superficial fascia overlying the thoracic wall, its superolateral region extends along the inferior margin of the pectoralis major muscle toward the axilla. In some cases, this may pass around the margin of the muscle to penetrate deep fascia and enter the axilla (Fig. 7.58). This axillary process rarely reaches as high as the apex of the axilla.

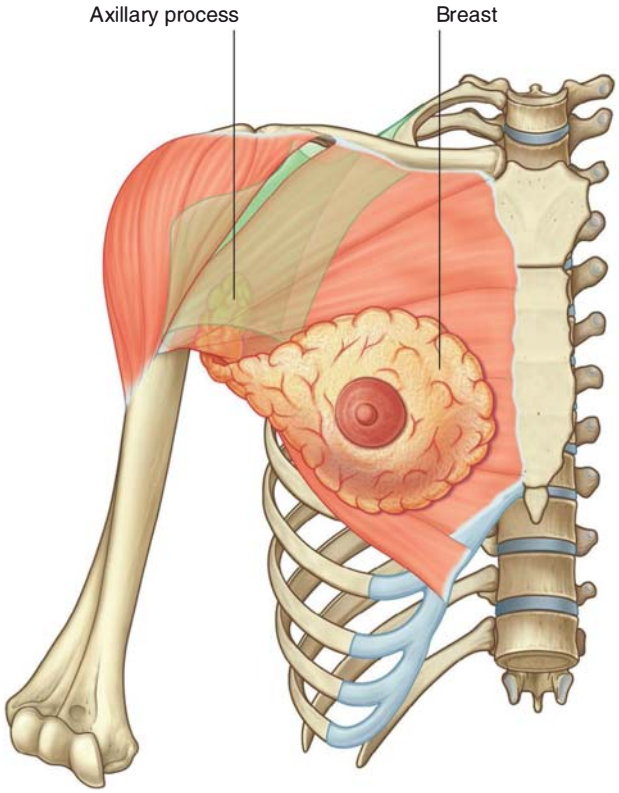


Fig. 7.58 Axillary process of the breast.

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ARM

The arm is the region of the upper limb between the shoulder and the elbow (Fig. 7.59). The superior aspect of the arm communicates medially with the axilla. Inferiorly, a number of important structures pass between the arm and the forearm through the cubital fossa, which is positioned anterior to the elbow joint.

The arm is divided into two compartments by medial and lateral intermuscular septa, which pass from each side of the humerus to the outer sleeve of deep fascia that surrounds the limb (Fig. 7.59).

The anterior compartment of the arm contains muscles that predominantly flex the elbow joint; the posterior compartment contains muscles that extend the joint. Major nerves and vessels supply and pass through each compartment.

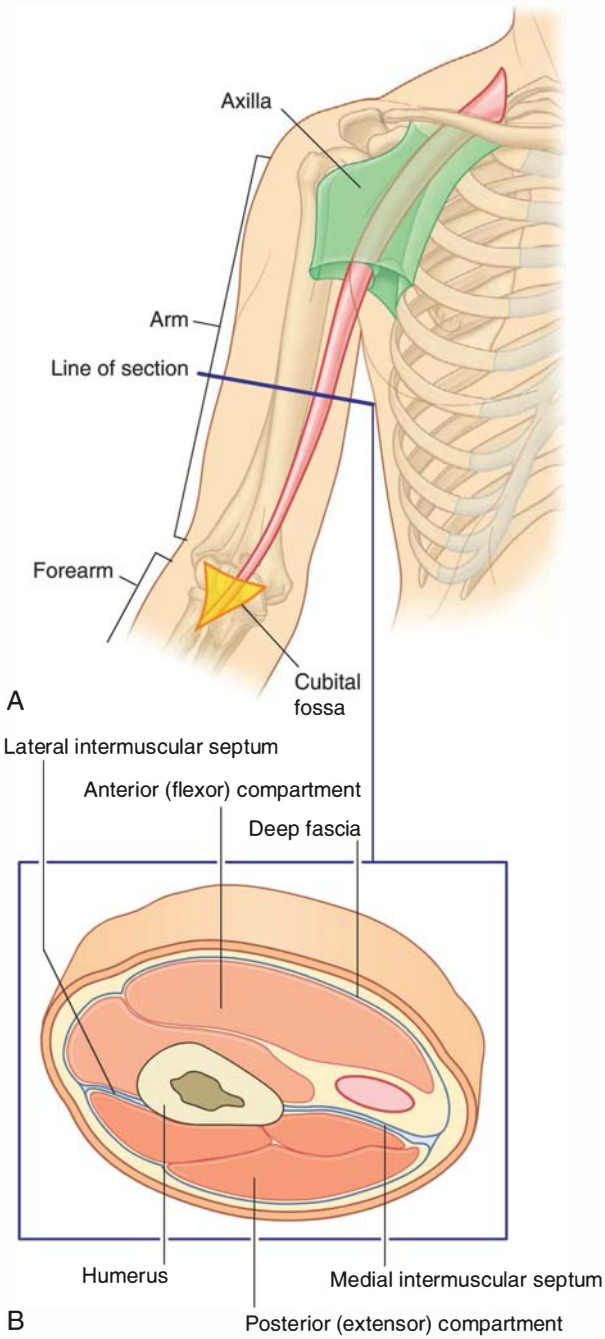


Fig. 7.59 Arm. **A.** Proximal and distal relationships. **B.** Transverse section through the middle of the arm.

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Bones

The skeletal support for the arm is the humerus (Fig. 7.60). Most of the large muscles of the arm insert into the proximal ends of the two bones of the forearm, the radius and the ulna, and flex and extend the forearm at the elbow joint. In addition, the muscles predominantly situated in the forearm that move the hand originate at the distal end of the humerus.

Shaft and distal end of the humerus

In cross section, the shaft of the humerus is somewhat triangular with:

- **anterior, lateral**, and **medial borders**, and
- **anterolateral, anteromedial**, and **posterior surfaces** (Fig. 7.60).

The posterior surface of the humerus is marked on its superior aspect by a linear roughening for the attachment of the lateral head of the triceps brachii muscle, beginning just inferior to the surgical neck and passing diagonally across the bone to the **deltoid tuberosity**.

The middle part of the posterior surface and adjacent part of the anterolateral surface are marked by the shallow **radial groove**, which passes diagonally down the bone and parallel to the sloping posterior margin of the deltoid tuberosity. The radial nerve and the profunda brachii artery lie in this groove.

Approximately in the middle of the shaft, the medial border is marked by thin elongate roughening for the attachment of the coracobrachialis muscle.

Intermuscular septa, which separate the anterior compartment from the posterior compartment, attach to the medial and lateral borders (Fig. 7.61).

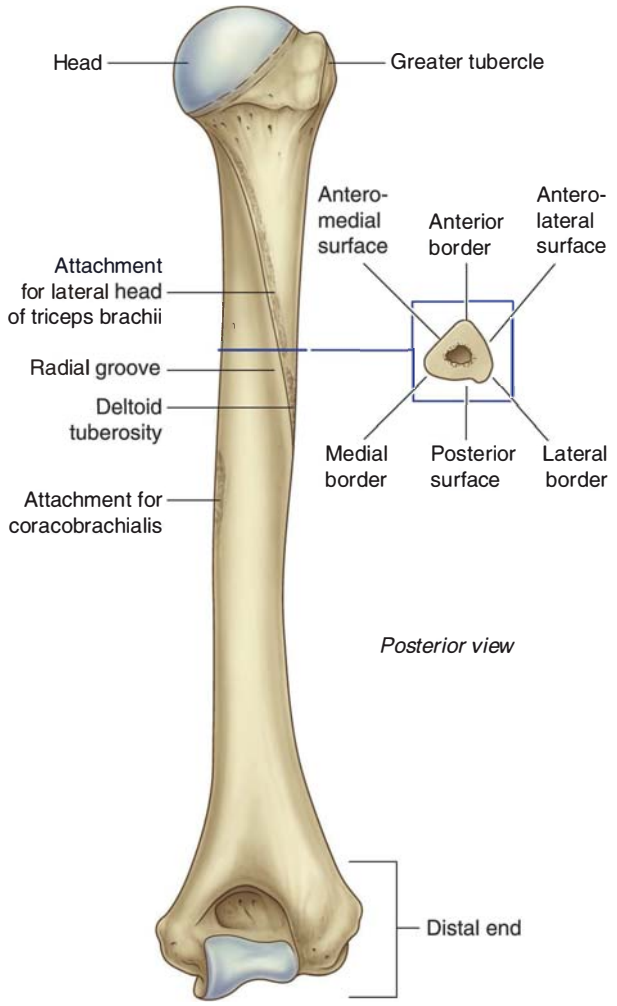


Fig. 7.60 Humerus. Posterior view.

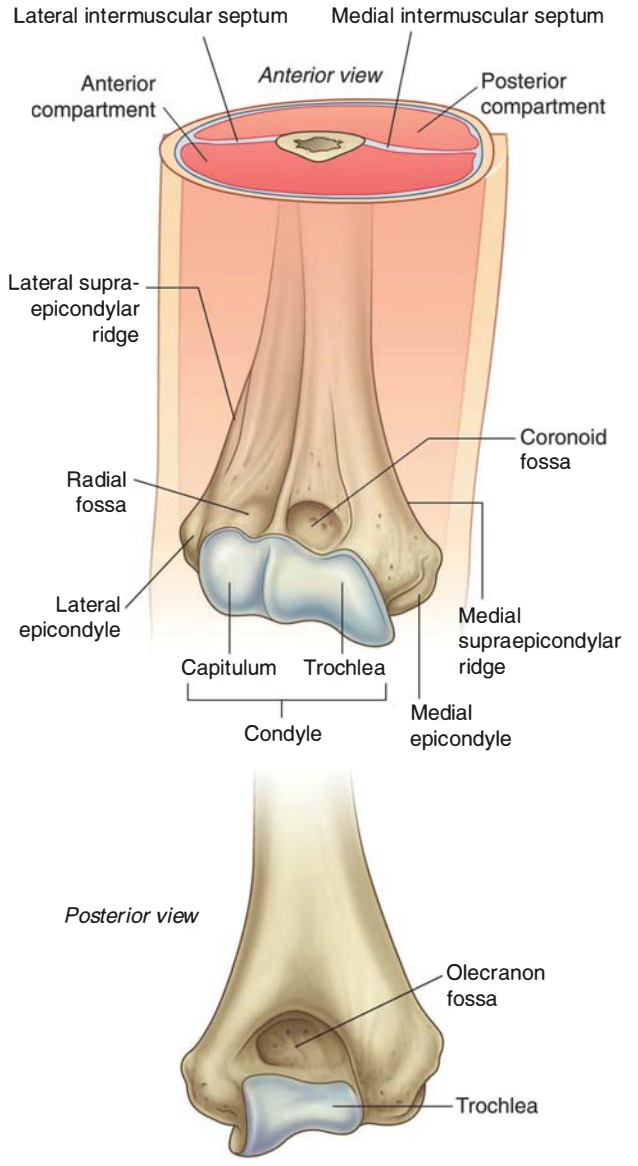


Fig. 7.61 Distal end of the humerus.

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Distally, the bone becomes flattened, and these borders expand as the **lateral supraepicondylar ridge** (**lateral supracondylar ridge**) and the **medial supraepicondylar ridge** (**medial supracondylar ridge**). The lateral supraepicondylar ridge is more pronounced than the medial ridge and is roughened for the attachment of muscles found in the posterior compartment of the forearm.

The distal end of the humerus, which is flattened in the anteroposterior plane, bears a condyle, two epicondyles, and three fossae, as follows (Fig. 7.61).

The condyle

The two articular parts of the condyle, the **capitulum** and the **trochlea**, articulate with the two bones of the forearm.

The **capitulum** articulates with the radius of the forearm. Lateral in position and hemispherical in shape, it projects anteriorly and somewhat inferiorly and is not visible when the humerus is viewed from the posterior aspect.

The **trochlea** articulates with the ulna of the forearm. It is pulley shaped and lies medial to the capitulum. Its medial edge is more pronounced than its lateral edge and, unlike the capitulum, it extends onto the posterior surface of the bone.

The two epicondyles

The two epicondyles lie adjacent, and somewhat superior, to the trochlea and capitulum (Fig. 7.61).

The **medial epicondyle**, a large bony protuberance, is the major palpable landmark on the medial side of the elbow, and projects medially from the distal end of the humerus. On its surface, it bears a large oval impression for the attachment of muscles in the anterior compartment of the forearm. The ulnar nerve passes from the arm into the forearm around the posterior surface of the medial epicondyle and can be palpated against the bone in this location.

The **lateral epicondyle** is much less pronounced than the medial epicondyle. It is lateral to the capitulum and has a large irregular impression for the attachment of muscles in the posterior compartment of the forearm.

The three fossae

Three fossae occur superior to the trochlea and capitulum on the distal end of the humerus (Fig. 7.61).

The **radial fossa** is the least distinct of the fossae and occurs immediately superior to the capitulum on the anterior surface of the humerus.

The **coronoid fossa** is adjacent to the radial fossa and is superior to the trochlea.

The largest of the fossae, the **olecranon fossa**, occurs immediately superior to the trochlea on the posterior surface of the distal end of the humerus.

These three fossae accommodate projections from the bones in the forearm during movements of the elbow joint.

Proximal end of the radius

The proximal end of the radius consists of a head, a neck, and the radial tuberosity (Fig. 7.62A,B).

The **head** of the radius is a thick disc-shaped structure oriented in the horizontal plane. The circular superior surface is concave for articulation with the capitulum of the humerus. The thick margin of the disc is broad medially where it articulates with the radial notch on the proximal end of the ulna.

The **neck** of the radius is a short and narrow cylinder of bone between the expanded head and the radial tuberosity on the shaft.

The **radial tuberosity** is a large blunt projection on the medial surface of the radius immediately inferior to the neck. Much of its surface is roughened for the attachment of the biceps brachii tendon. The oblique line of the radius continues diagonally across the shaft of the bone from the inferior margin of the radial tuberosity.

Proximal end of the ulna

The proximal end of the ulna is much larger than the proximal end of the radius and consists of the olecranon, the coronoid process, the trochlear notch, the radial notch, and the tuberosity of the ulna (Fig. 7.63A,B).

The **olecranon** is a large projection of bone that extends proximally from the ulna. Its anterolateral surface is articular and contributes to the formation of the trochlear notch, which articulates with the trochlea of the humerus. The superior surface is marked by a large roughened impression for the attachment of the triceps brachii muscle. The posterior surface is smooth, shaped somewhat triangularly, and can be palpated as the "tip of the elbow."

The **coronoid process** projects anteriorly from the proximal end of the ulna (Fig. 7.63). Its superolateral surface is articular and participates, with the olecranon, in forming the **trochlear notch**. The lateral surface is marked by the **radial notch** for articulation with the head of the radius.

Just inferior to the radial notch is a fossa that allows the radial tuberosity to change position during pronation and supination. The posterior margin of this fossa is broadened to form the **supinator crest**. The anterior surface of the coronoid process is triangular, with the apex directed distally, and has a number of roughenings for muscle attachment. The largest of these roughenings, the **tuberosity of the ulna**, is at the apex of the anterior surface and is the attachment site for the brachialis muscle.

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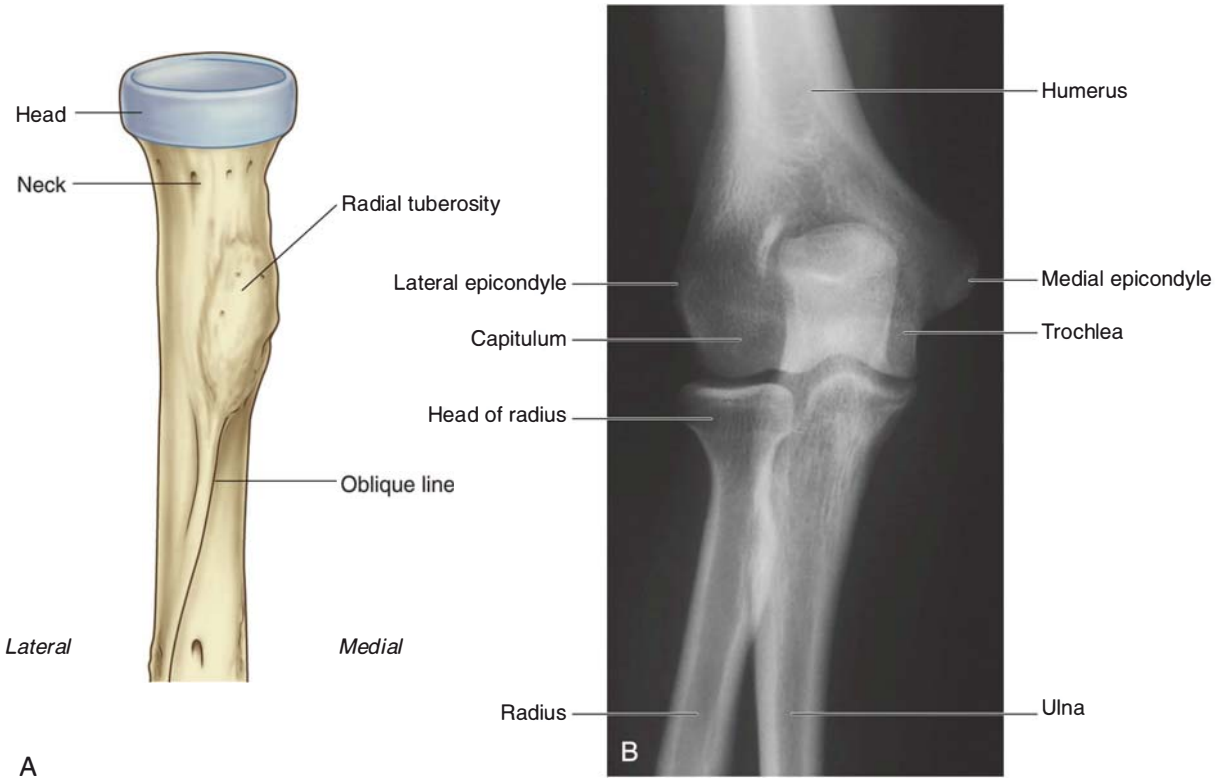


Fig. 7.62 A. Anterior view of the proximal end of the radius. **B.** Radiograph of the elbow joint (anteroposterior view).

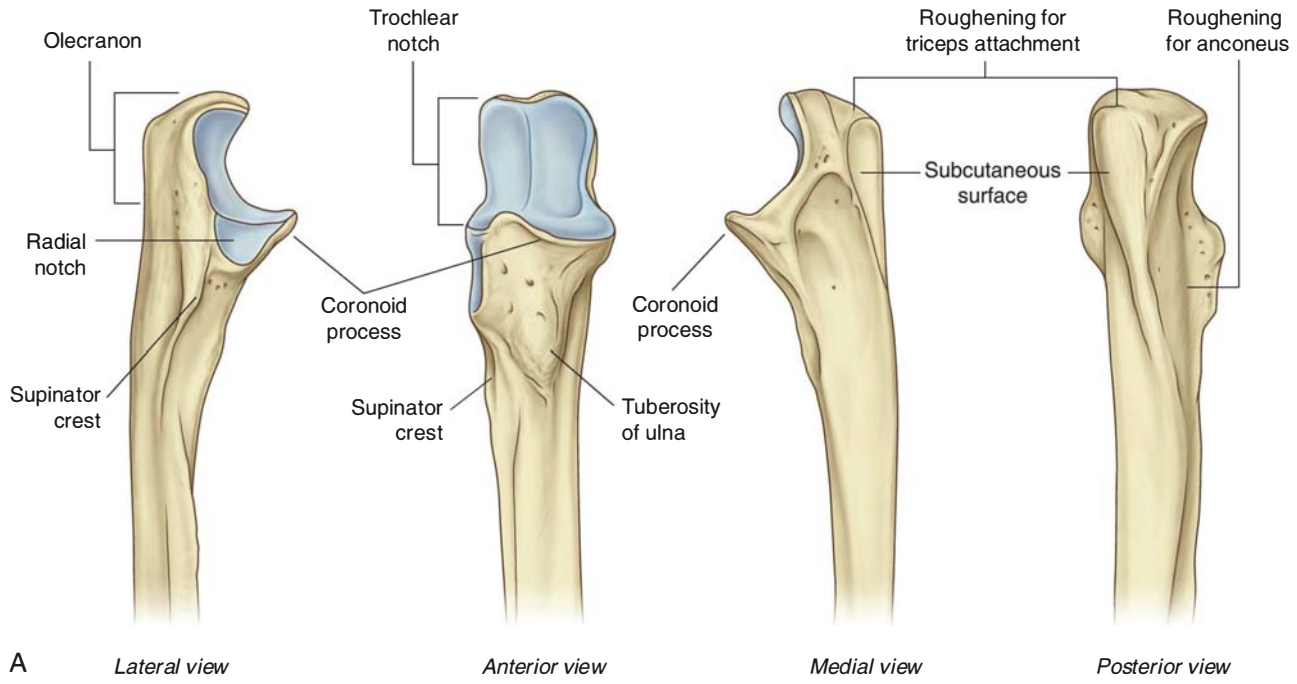


Fig. 7.63 A. Lateral, anterior, medial, and posterior views of the proximal end of the ulna.

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Fig. 7.63, cont'd B. Radiograph of the elbow joint (lateral view).

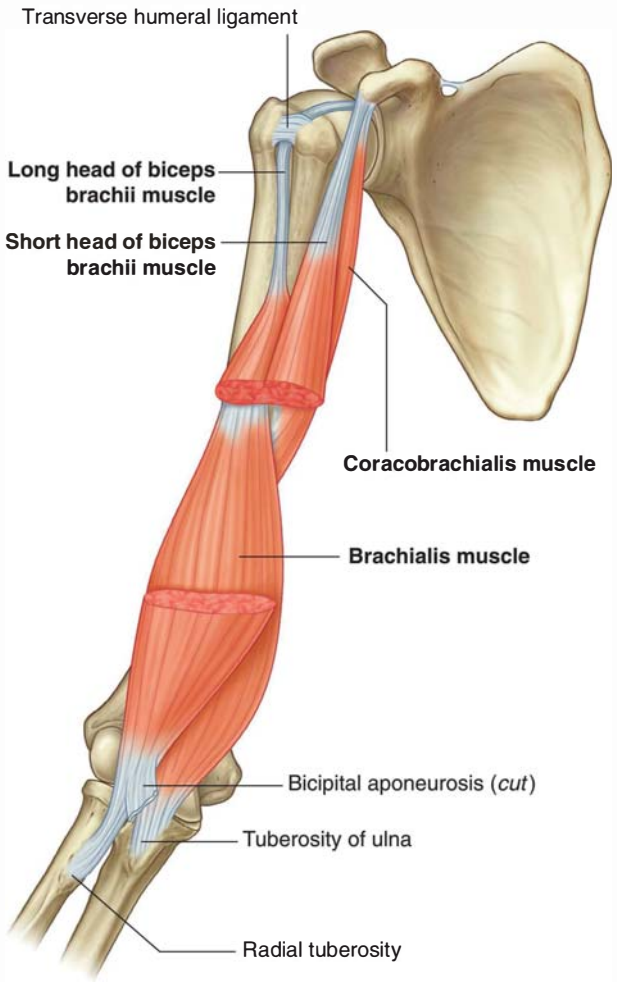


Fig. 7.64 Coracobrachialis, biceps brachii, and brachialis muscles.

Muscles

The anterior compartment of the arm contains three muscles—the coracobrachialis, brachialis, and biceps brachii muscles—which are innervated predominantly by the musculocutaneous nerve.

The posterior compartment contains one muscle—the triceps brachii muscle—which is innervated by the radial nerve.

Coracobrachialis

The **coracobrachialis muscle** extends from the tip of the coracoid process of the scapula to the medial side of the midshaft of the humerus (Fig. 7.64 and Table 7.8). It passes through the axilla and is penetrated and innervated by the musculocutaneous nerve.

The coracobrachialis muscle flexes the arm.

Biceps brachii

The **biceps brachii** muscle has two heads:

- The short head of the muscle originates from the coracoid process in conjunction with the coracobrachialis.
- The long head originates as a tendon from the supraglenoid tubercle of the scapula (Fig. 7.64 and Table 7.8).

The tendon of the long head passes through the glenohumeral joint superior to the head of the humerus and then passes through the intertubercular sulcus and enters the arm. In the arm, the tendon joins with its muscle belly and, together with the muscle belly of the short head, overlies the brachialis muscle.

The long and short heads converge to form a single tendon, which inserts onto the radial tuberosity.

As the tendon enters the forearm, a flat sheet of connective tissue (the **bicipital aponeurosis**) fans out from the medial side of the tendon to blend with deep fascia covering the anterior compartment of the forearm.

The biceps brachii muscle is a powerful flexor of the forearm at the elbow joint; it is also the most powerful supinator of the forearm when the elbow joint is flexed. Because the two heads of the biceps brachii muscle cross the glenohumeral joint, the muscle can also flex the glenohumeral joint.

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Table 7.8 Muscles of the anterior compartment of the arm (spinal segments in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Coracobrachialis	Apex of coracoid process	Linear roughening on midshaft of humerus on medial side	Musculocutaneous nerve (**C5, C6, C7**)	Flexor of the arm at the glenohumeral joint
Biceps brachii	Long head—supraglenoid tubercle of scapula; short head—apex of coracoid process	Radial tuberosity	Musculocutaneous nerve (**C5, C6**)	Powerful flexor of the forearm at the elbow joint and supinator of the forearm; accessory flexor of the arm at the glenohumeral joint
Brachialis	Anterior aspect of humerus (medial and lateral surfaces) and adjacent intermuscular septae	Tuberosity of the ulna	Musculocutaneous nerve (C5, **C6**); small contribution by the radial nerve (C7) to lateral part of muscle	Powerful flexor of the forearm at the elbow joint

The biceps brachii muscle is innervated by the musculocutaneous nerve. A tap on the tendon of the biceps brachii at the elbow is used to test predominantly spinal cord segment C6.

In the clinic

Rupture of biceps tendon

It is relatively unusual for muscles and their tendons to rupture in the upper limb; however, the tendon that most commonly ruptures is the tendon of the long head of the biceps brachii muscle. In isolation, this has relatively little effect on the upper limb, but it does produce a characteristic deformity—on flexing the elbow, there is an extremely prominent bulge of the muscle belly as its unrestrained fibers contract—the "Popeye" sign.

Brachialis

The **brachialis muscle** originates from the distal half of the anterior aspect of the humerus and from adjacent parts of the intermuscular septa, particularly on the medial side (Fig. 7.64 and Table 7.8). It lies beneath the biceps brachii muscle, is flattened dorsoventrally, and converges to form a tendon, which attaches to the tuberosity of the ulna.

The brachialis muscle flexes the forearm at the elbow joint.

Innervation of the brachialis muscle is predominantly by the musculocutaneous nerve. A small component of the lateral part is innervated by the radial nerve.

Posterior compartment

The only muscle of the posterior compartment of the arm is the **triceps brachii muscle** (Fig. 7.65 and Table 7.9). The triceps brachii muscle has three heads:

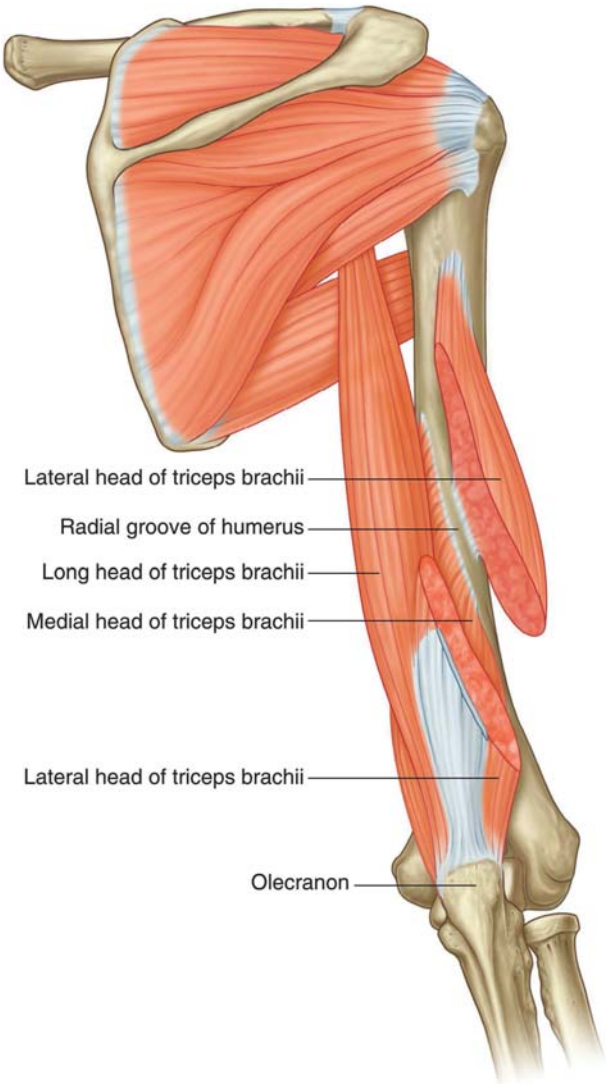


Fig. 7.65 Triceps muscle.

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Table 7.9 Muscle of the posterior compartment of the arm (spinal segment indicated in bold is the major segment innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Triceps brachii	Long head—infraglenoid tubercle of scapula; medial head—posterior surface of humerus; lateral head—posterior surface of humerus	Olecranon	Radial nerve (C6, **C7**, C8)	Extension of the forearm at the elbow joint; long head can also extend and adduct the arm at the shoulder joint

- The long head originates from the infraglenoid tubercle of the scapula.
- The medial head originates from the extensive area on the posterior surface of the shaft of the humerus inferior to the radial groove.
- The lateral head originates from a linear roughening superior to the radial groove of the humerus.

The three heads converge to form a large tendon, which inserts on the superior surface of the olecranon of the ulna.

The triceps brachii muscle extends the forearm at the elbow joint.

Innervation of the triceps brachii is by branches of the radial nerve. A tap on the tendon of the triceps brachii tests predominantly spinal cord segment C7.

Arteries and veins

Brachial artery

The major artery of the arm, the **brachial artery**, is found in the anterior compartment (Fig. 7.66A). Beginning as a continuation of the axillary artery at the lower border of the teres major muscle, it terminates just distal to the elbow joint where it divides into the radial and ulnar arteries.

In the proximal arm, the brachial artery lies on the medial side. In the distal arm, it moves laterally to assume a position midway between the lateral epicondyle and the medial epicondyle of the humerus. It crosses anteriorly to the elbow joint where it lies immediately medial to the tendon of the biceps brachii muscle. The brachial artery is palpable along its length. In proximal regions, the brachial artery can be compressed against the medial side of the humerus.

Branches of the brachial artery in the arm include those to adjacent muscles and two ulnar collateral vessels, which contribute to a network of arteries around the elbow joint (Fig. 7.66B). Additional branches are the profunda brachii artery and nutrient arteries to the humerus, which pass through a foramen in the anteromedial surface of the humeral shaft.

Profunda brachii artery

The **profunda brachii artery,** the largest branch of the brachial artery, passes into and supplies the posterior compartment of the arm (Fig. 7.66A,B). It enters the posterior compartment with the radial nerve and together they pass through the triangular interval, which is formed by the shaft of the humerus, the inferior margin of the teres major muscle, and the lateral margin of the long head of the triceps muscle. They then pass along the radial groove on the posterior surface of the humerus deep to the lateral head of the triceps brachii muscle.

Branches of the profunda brachii artery supply adjacent muscles and anastomose with the posterior circumflex humeral artery. The artery terminates as two collateral vessels, which contribute to an anastomotic network of arteries around the elbow joint (Fig. 7.66B).

In the clinic

Blood pressure measurement

Blood pressure measurement is an extremely important physiological parameter. High blood pressure (hypertension) requires treatment to prevent long-term complications such as stroke. Low blood pressure may be caused by extreme blood loss, widespread infection, or poor cardiac output (e.g., after myocardial infarction). Accurate measurement of blood pressure is essential.

Most clinicians use a sphygmomanometer and a stethoscope. The sphygmomanometer is a device that inflates a cuff around the midportion of the arm to compress the brachial artery against the humerus. The cuff is inflated so it exceeds the systolic blood pressure (greater than 120 mm Hg). The clinician places a stethoscope over the brachial artery in the cubital fossa and listens (auscultates) for the pulse. As the pressure in the arm cuff of the sphygmomanometer is reduced just below the level of the systolic blood pressure, the pulse becomes audible as a regular thumping sound. As the pressure in the sphygmomanometer continues to drop, the regular thumping sound becomes clearer. When the pressure in the sphygmomanometer is less than that of the diastolic blood pressure, the audible thumping sound becomes inaudible. Using the simple scale on the sphygmomanometer, the patient's blood pressure can be determined. The normal range is 120/80 mm Hg (systolic blood pressure/diastolic blood pressure).

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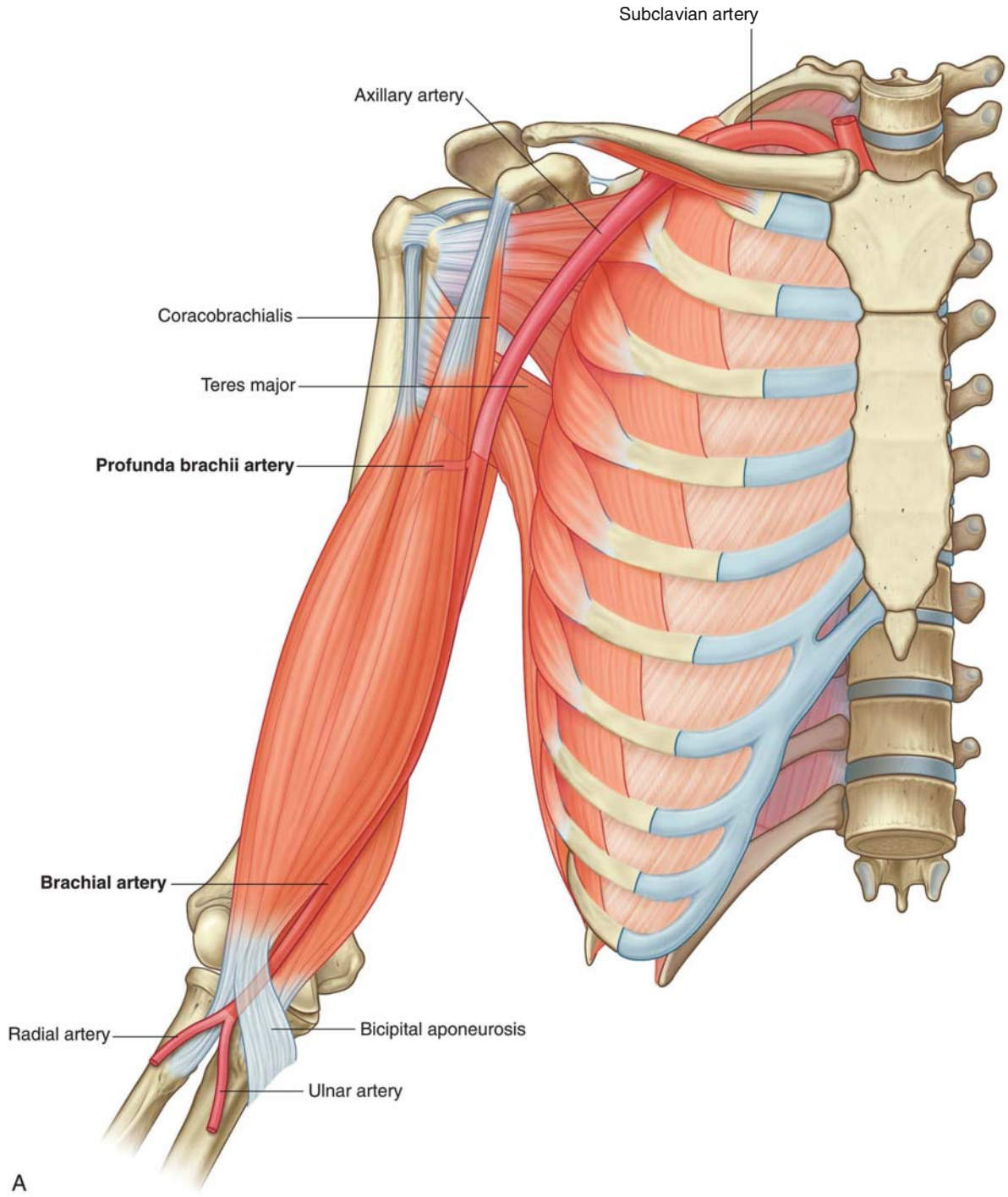


Fig. 7.66 Brachial artery. **A.** In context. *Continued*

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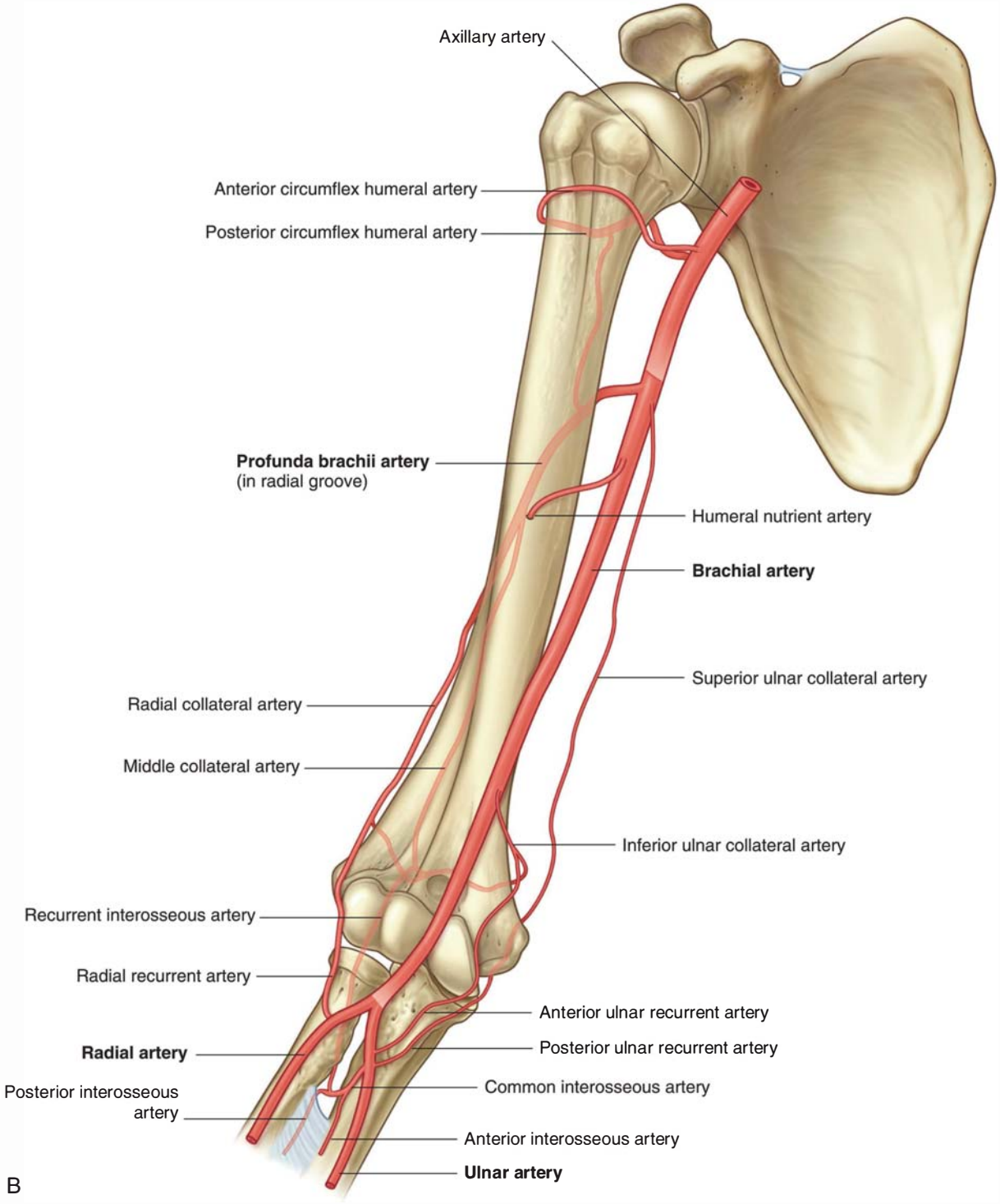


Fig. 7.66, cont'd Brachial artery. **B.** Branches.

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Veins

Paired brachial veins pass along the medial and lateral sides of the brachial artery, receiving tributaries that accompany branches of the artery (Fig. 7.67).

In addition to these deep veins, two large subcutaneous veins, the basilic vein and the cephalic vein, are located in the arm.

The basilic vein passes vertically in the distal half of the arm, penetrates deep fascia to assume a position medial to the brachial artery, and then becomes the axillary vein at the lower border of the teres major muscle. The brachial veins join the basilic, or axillary, vein.

The cephalic vein passes superiorly on the anterolateral aspect of the arm and through the anterior wall of the axilla to reach the axillary vein.

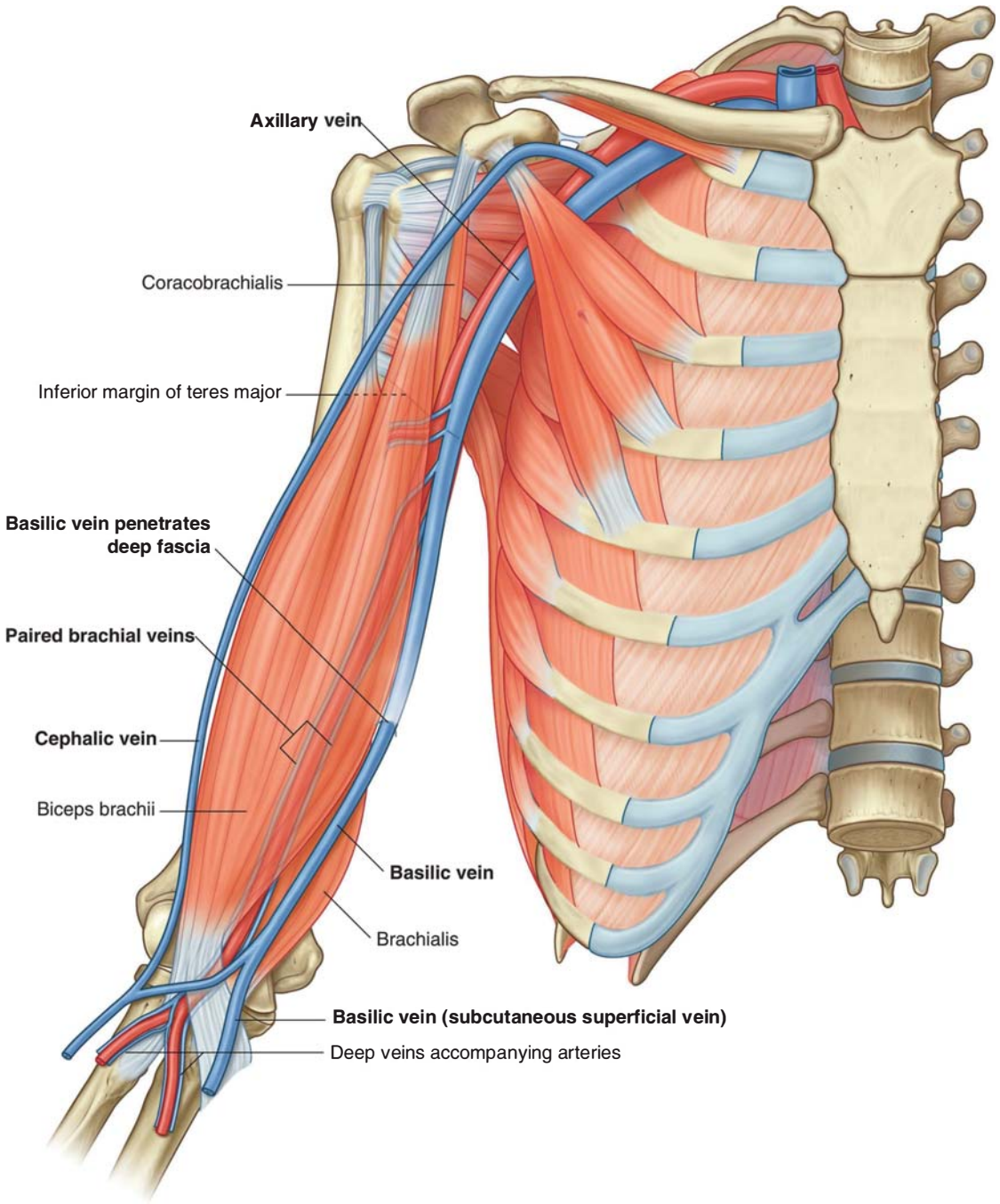


Fig. 7.67 Veins of the arm.

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Nerves

Musculocutaneous nerve

The musculocutaneous nerve leaves the axilla and enters the arm by passing through the coracobrachialis muscle (Fig. 7.68). It passes diagonally down the arm in the plane between the biceps brachii and brachialis muscles. After giving rise to motor branches in the arm, it emerges laterally to the tendon of the biceps brachii muscle at the elbow, penetrates deep fascia, and continues as the **lateral cutaneous nerve of the forearm**.

The musculocutaneous nerve provides:

- motor innervation to all muscles in the anterior compartment of the arm, and
- sensory innervation to skin on the lateral surface of the forearm.

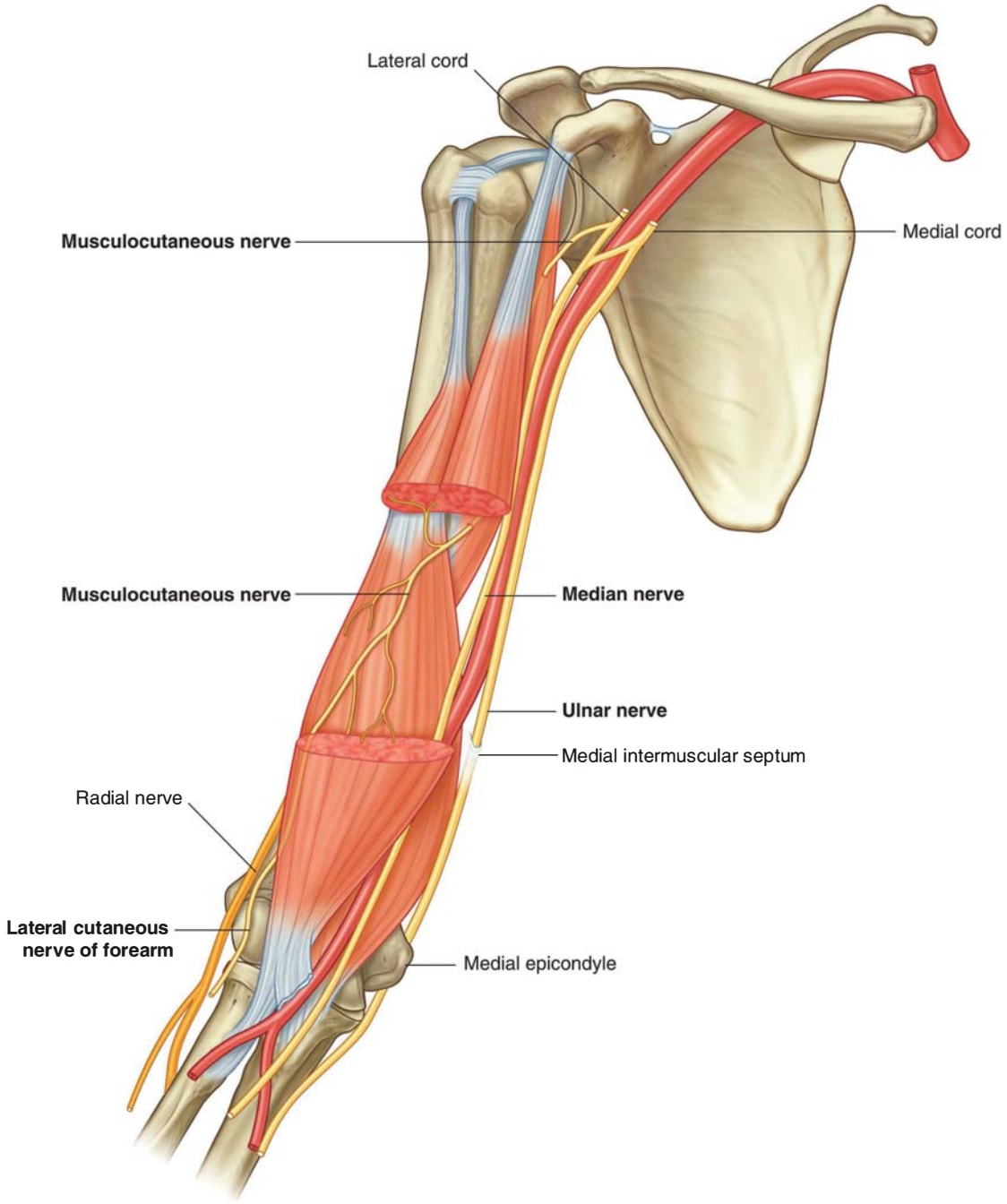


Fig. 7.68 Musculocutaneous, median, and ulnar nerves in the arm.

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Median nerve

The median nerve enters the arm from the axilla at the inferior margin of the teres major muscle (Fig. 7.68). It passes vertically down the medial side of the arm in the anterior compartment and is related to the brachial artery throughout its course:

- In proximal regions, the median nerve is immediately lateral to the brachial artery.
- In more distal regions, the median nerve crosses to the medial side of the brachial artery and lies anterior to the elbow joint.

The median nerve has no major branches in the arm, but a branch to one of the muscles of the forearm, the pronator teres muscle, may originate from the nerve immediately proximal to the elbow joint.

Ulnar nerve

The ulnar nerve enters the arm with the median nerve and axillary artery (Fig. 7.68). It passes through proximal regions medial to the axillary artery. In the middle of the arm, the ulnar nerve penetrates the medial intermuscular septum and enters the posterior compartment where it lies anterior to the medial head of the triceps brachii muscle. It passes posterior to the medial epicondyle of the humerus and then into the anterior compartment of the forearm.

The ulnar nerve has no major branches in the arm.

Radial nerve

The radial nerve originates from the posterior cord of the brachial plexus and enters the arm by crossing the inferior margin of the teres major muscle (Fig. 7.69). As it enters the arm, it lies posterior to the brachial artery. Accompanied by the profunda brachii artery, the radial nerve enters the posterior compartment of the arm by passing through the triangular interval.

As the radial nerve passes diagonally, from medial to lateral, through the posterior compartment, it lies in the radial groove directly on bone. On the lateral side of the arm, it passes anteriorly through the lateral intermuscular septum and enters the anterior compartment where it lies between the brachialis muscle and a muscle of the posterior compartment of the forearm—the brachioradialis muscle, which attaches to the lateral supraepicondylar ridge of the humerus. The radial nerve enters the forearm anterior to the lateral epicondyle of the humerus, just deep to the brachioradialis muscle.

In the arm, the radial nerve has muscular and cutaneous branches (Fig. 7.69).

- Muscular branches include those to the triceps brachii, brachioradialis, and extensor carpi radialis longus muscles. In addition, the radial nerve contributes to the innervation of the lateral part of the brachialis muscle. One of the branches to the medial head of the triceps brachii muscle arises before the radial nerve's entrance into the posterior compartment and passes vertically down the arm in association with the ulnar nerve.
- Cutaneous branches of the radial nerve that originate in the posterior compartment of the arm are the **inferior lateral cutaneous nerve of the arm** and the **posterior cutaneous nerve of the forearm**, both of which penetrate through the lateral head of the triceps brachii muscle and the overlying deep fascia to become subcutaneous.

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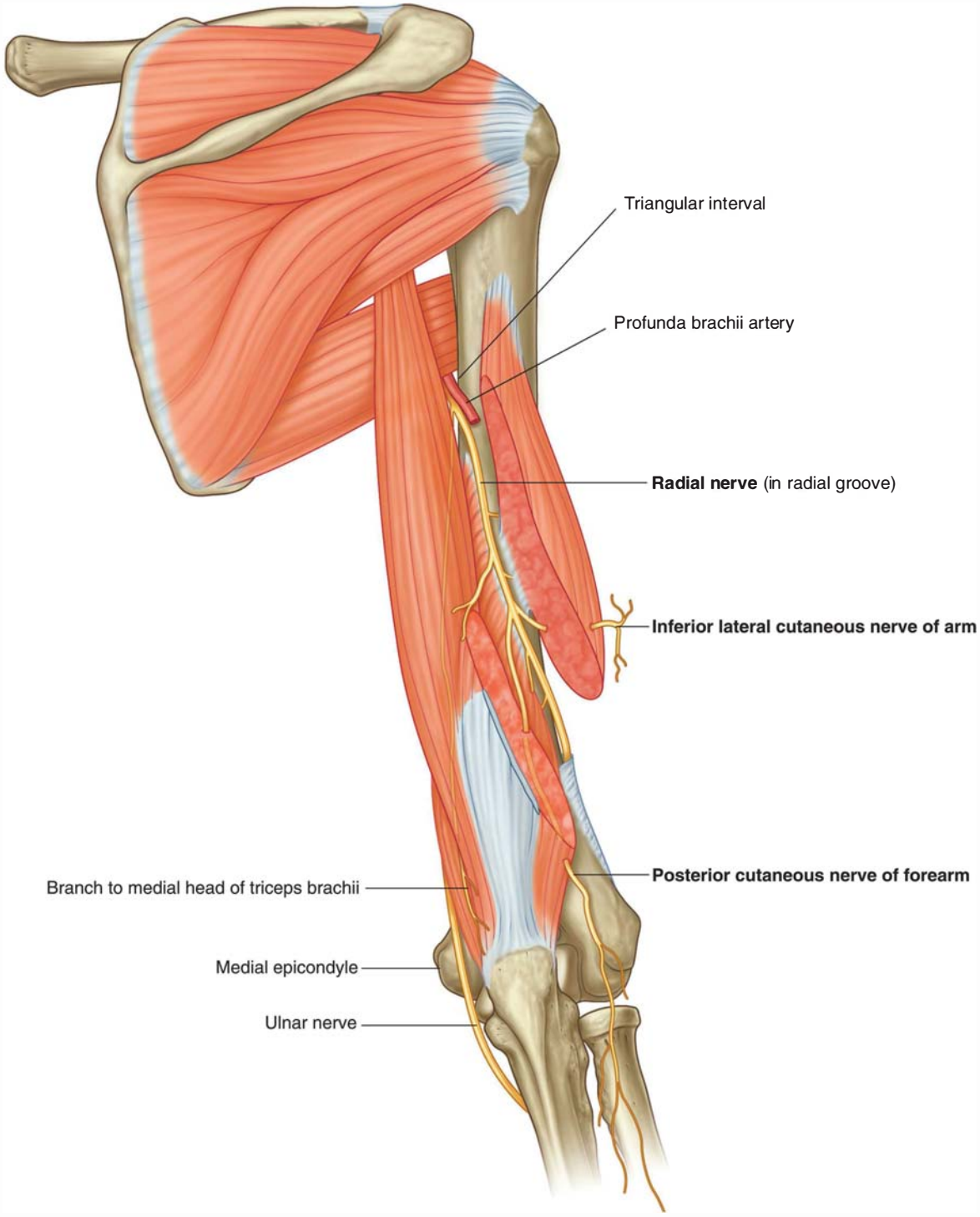


Fig. 7.69 Radial nerve in the arm.

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In the clinic

Radial nerve injury in the arm

The radial nerve is tightly bound with the profunda brachii artery between the medial and lateral heads of the triceps brachii muscle in the radial groove. If the humerus is fractured, the radial nerve may become stretched or transected in this region, leading to permanent damage and loss of function. This injury is typical (Fig. 7.70) and the nerve should always be tested when a fracture of the midshaft of the humerus is suspected. The patient's symptoms usually include wrist drop (due to denervation of the extensor muscles) and sensory changes over the dorsum of the hand.



Fig. 7.70 Radiograph of the humerus demonstrating a midshaft fracture, which may disrupt the radial nerve.

In the clinic

Median nerve injury in the arm

In the arm and forearm the median nerve is usually not injured by trauma because of its relatively deep position. The commonest neurological problem associated with the median nerve is compression beneath the flexor retinaculum at the wrist (carpal tunnel syndrome).

On very rare occasions, a fibrous band may arise from the anterior aspect of the humerus beneath which the median nerve passes. This is an embryological remnant of the coracobrachialis muscle and is sometimes called the ligament of Struthers; occasionally, it may calcify. This band can compress the median nerve, resulting in weakness of the flexor muscles in the forearm and the thenar muscles. Nerve conduction studies will demonstrate the site of nerve compression.

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ELBOW JOINT

The elbow joint is a complex joint involving three separate articulations, which share a common synovial cavity (Fig. 7.71).

- The joints between the trochlear notch of the ulna and the trochlea of the humerus and between the head of the radius and the capitulum of the humerus are primarily involved with hinge-like flexion and extension of the forearm on the arm and, together, are the principal articulations of the elbow joint.
- The joint between the head of the radius and the radial notch of the ulna, the proximal radio-ulnar joint, is involved with pronation and supination of the forearm.

The articular surfaces of the bones are covered with hyaline cartilage.

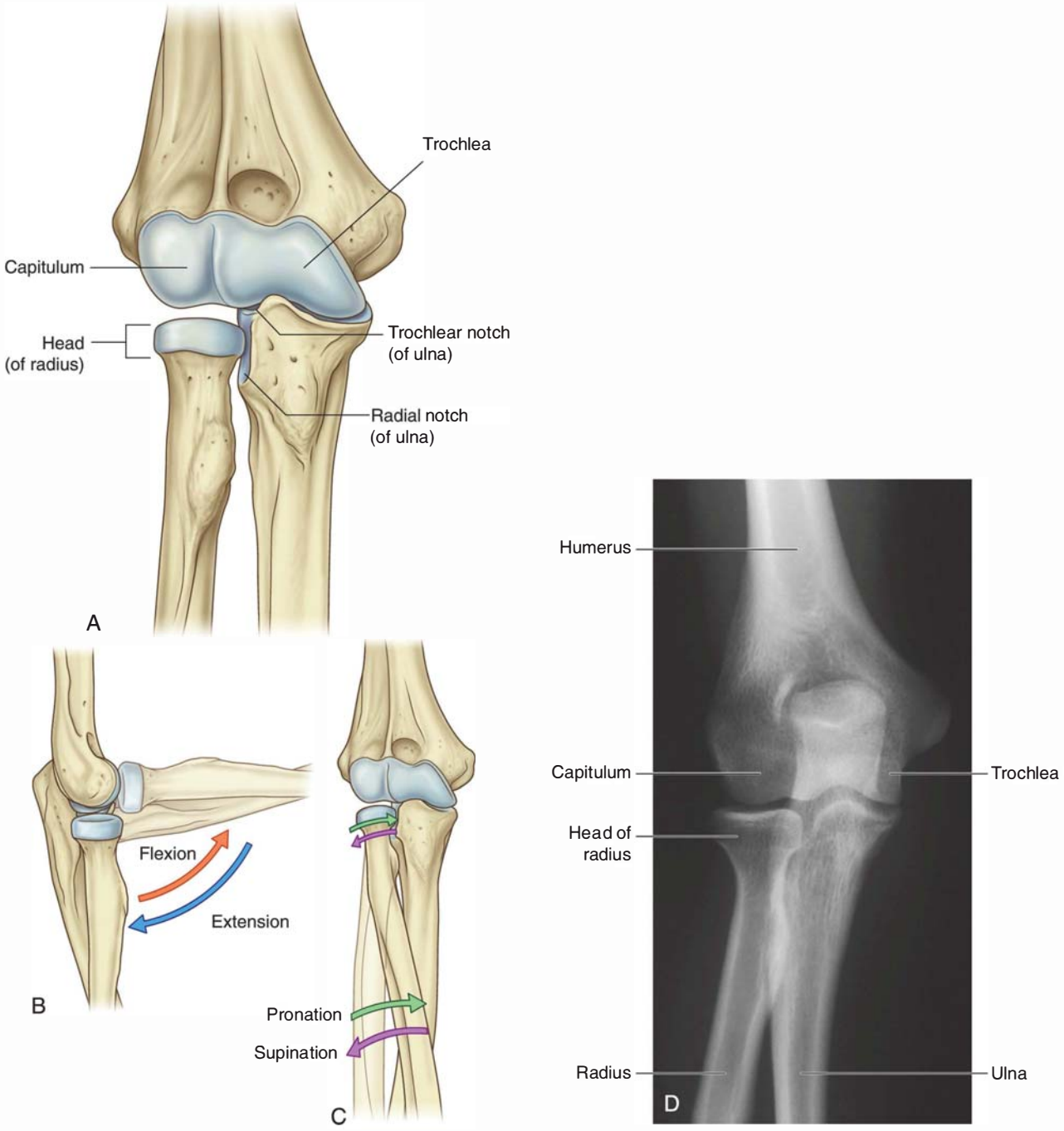


Fig. 7.71 Components and movements of the elbow joint. **A.** Bones and joint surfaces. **B.** Flexion and extension. **C.** Pronation and supination. **D.** Radiograph of a normal elbow joint (anteroposterior view).

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The synovial membrane originates from the edges of the articular cartilage and lines the radial fossa, the coronoid fossa, the olecranon fossa, the deep surface of the joint capsule, and the medial surface of the trochlea (Fig. 7.72).

The synovial membrane is separated from the fibrous membrane of the joint capsule by pads of fat in regions overlying the coronoid fossa, the olecranon fossa, and the radial fossa. These fat pads accommodate the related bony processes during extension and flexion of the elbow. Attachments of the brachialis and triceps brachii muscles to the joint capsule overlying these regions pull the attached fat pads out of the way when the adjacent bony processes are moved into the fossae.

The fibrous membrane of the joint capsule overlies the synovial membrane, encloses the joint, and attaches to the medial epicondyle and the margins of the olecranon, coronoid, and radial fossae of the humerus (Fig. 7.73). It also attaches to the coronoid process and olecranon of the ulna. On the lateral side, the free inferior margin of the joint capsule passes around the neck of the radius from an anterior attachment to the coronoid process of the ulna to a posterior attachment to the base of the olecranon.

The fibrous membrane of the joint capsule is thickened medially and laterally to form collateral ligaments, which

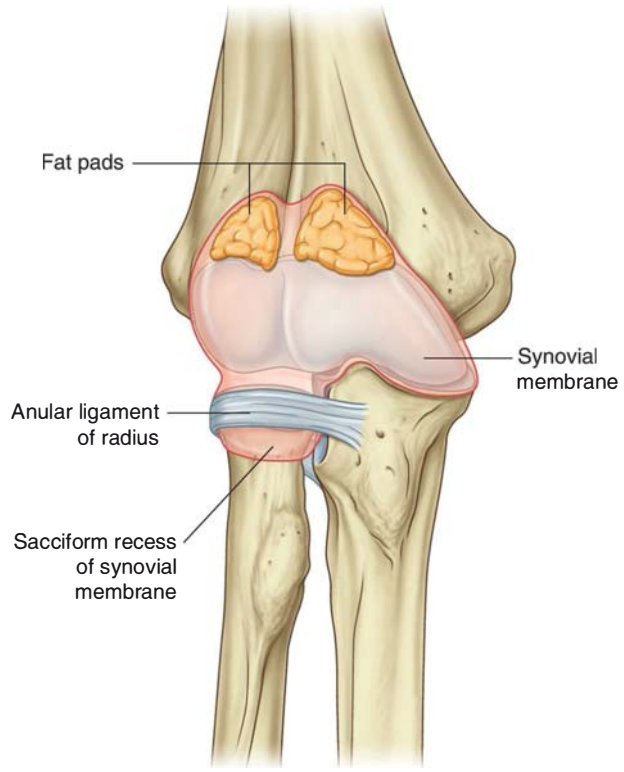


Fig. 7.72 Synovial membrane of elbow joint (anterior view).

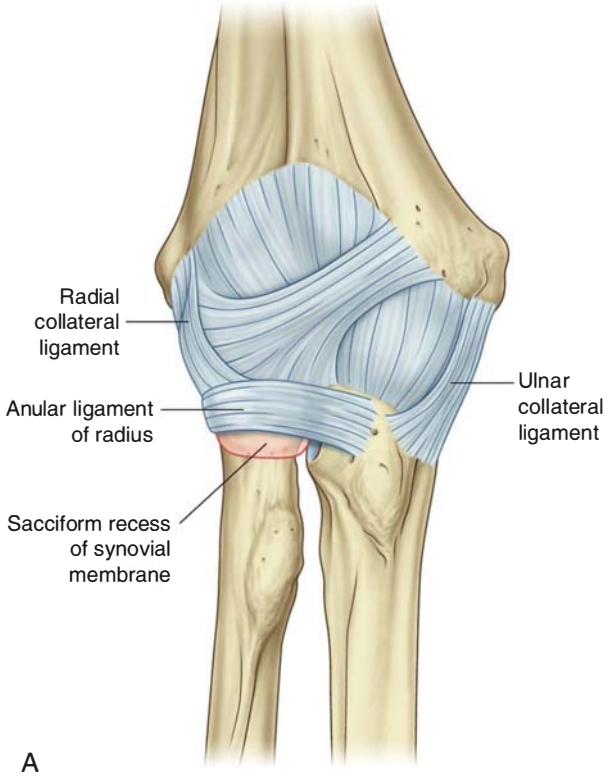




Fig. 7.73 Elbow joint. **A.** Joint capsule and ligaments of the right elbow joint. **B.** Magnetic resonance image of the elbow joint in the coronal plane.

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support the flexion and extension movements of the elbow joint (Fig. 7.73).

In addition, the external surface of the joint capsule is reinforced laterally where it cuffs the head of the radius with a strong **anular ligament of the radius**. Although this ligament blends with the fibrous membrane of the joint capsule in most regions, they are separate posteriorly. The anular ligament of the radius also blends with the **radial collateral ligament**.

The anular ligament of the radius and related joint capsule allow the radial head to slide against the radial notch of the ulna and pivot on the capitulum during pronation and supination of the forearm.

The deep surface of the fibrous membrane of the joint capsule and the related anular ligament of the radius that articulate with the sides of the radial head are lined by cartilage. A pocket of synovial membrane (sacciform recess) protrudes from the inferior free margin of the joint capsule and facilitates rotation of the radial head during pronation and supination.

Vascular supply to the elbow joint is through an anastomotic network of vessels derived from collateral and recurrent branches of the brachial, profunda brachii, radial, and ulnar arteries.

The elbow joint is innervated predominantly by branches of the radial and musculocutaneous nerves, but there may be some innervation by branches of the ulnar and median nerves.

In the clinic

Supracondylar fracture of the humerus

Elbow injuries in children may result in a transverse fracture of the distal end of the humerus, above the level of the epicondyles. This fracture is termed a supracondylar fracture. The distal fragment and its soft tissues are pulled posteriorly by the triceps muscle. This posterior displacement effectively "bowstrings" the brachial artery over the irregular proximal fracture fragment. In children, this is a relatively devastating injury: the muscles of the anterior compartment of the forearm are rendered ischemic and form severe contractions, significantly reducing the function of the anterior compartment and flexor muscles (Volkmann's ischemic contracture).

In the clinic

Pulled elbow

Pulled elbow is a disorder that typically occurs in children under 5 years of age. It is commonly caused by a sharp pull of the child's hand, usually when the child is pulled up a curb. The not-yet-developed head of the radius and the laxity of the anular ligament of the radius allow the head to sublux from this cuff of tissue. Pulled elbow is extremely painful, but can be treated easily by simple supination and compression of the elbow joint by the clinician. When the radial head is relocated the pain subsides immediately and the child can continue with normal activity.

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In the clinic

Developmental changes in the elbow joint

The elbow joint can be injured in many ways; the types of injuries are age dependent. When a fracture or soft tissue trauma is suspected, a plain lateral and an anteroposterior radiograph are obtained. In an adult it is usually not difficult to interpret the radiograph, but in children additional factors require interpretation.

As the elbow develops in children, numerous secondary ossification centers appear before and around puberty. It is easy to mistakenly interpret these as fractures. In addition, it is also possible for the epiphyses and apophyses to be "pulled off" or disrupted. Therefore, when interpreting a child's radiograph of the elbow, the physician must know the child's age (Fig. 7.74). Fusion occurs at around the time of puberty. An understanding of the normal epiphyses and apophyses and their normal relationship to the bones will secure a correct diagnosis. The approximate ages of appearance of the secondary ossification centers around the elbow joint are:

- capitulum—1 year,
- head (of radius)—5 years,
- medial epicondyle—5 years,
- trochlea—11 years,
- olecranon—12 years, and
- lateral epicondyle—13 years.

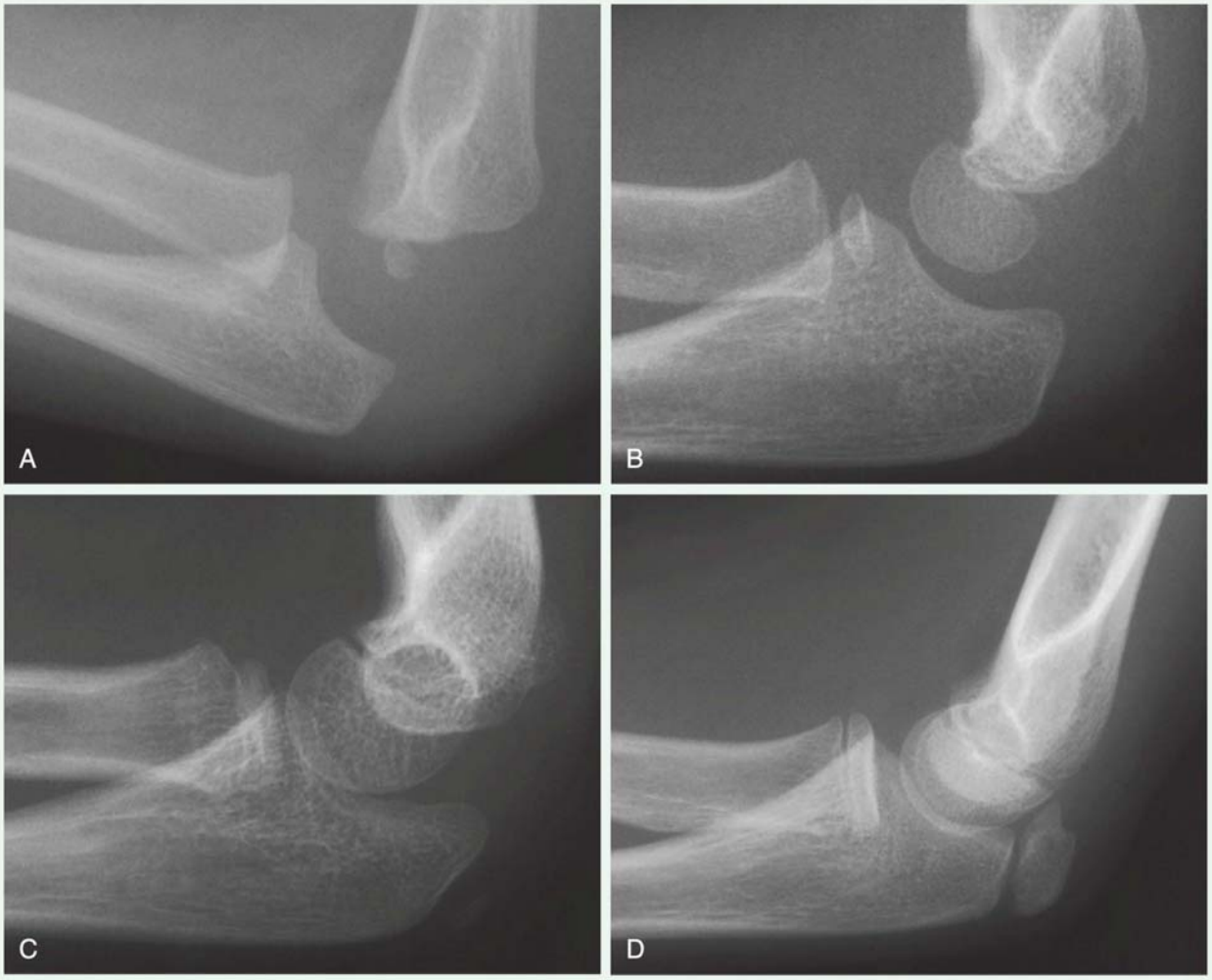


Fig. 7.74 Radiographs of elbow joint development. **A.** At age 2 years. **B.** At age 5 years. **C.** At age 5–6 years. **D.** At age 12 years.

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In the clinic

Fracture of the head of the radius

A fracture of the head of the radius is a common injury and can cause appreciable morbidity. It is one of the typical injuries that occur with a fall on the outstretched hand. On falling, the force is transmitted to the radial head, which fractures. These fractures typically result in loss of full extension, and potential surgical reconstruction may require long periods of physiotherapy to obtain a full range of movement at the elbow joint.

A lateral radiograph of a fracture of the head of the radius typically demonstrates the secondary phenomenon of this injury. When the bone is fractured, fluid fills the synovial cavity, elevating the small pad of fat within the coronoid and olecranon fossae. These fat pads appear as areas of lucency on the lateral radiograph—the "fat pad" sign. This radiological finding is useful because fracture of the head of the radius is not always clearly visible. If there is an appropriate clinical history, tenderness around the head of the radius, and a positive fat pad sign, a fracture can be inferred clinically even if no fracture can be identified on the radiograph, and appropriate treatment can be instituted.

In the clinic

"Tennis" and "golfer's" elbow (epicondylitis)

It is not uncommon for people who are involved in sports such as golf and tennis to develop an overuse strain of the origins of the flexor and extensor muscles of the forearm. The pain is typically around the epicondyles and usually resolves after rest. If pain and inflammation persist, surgical division of the extensor or flexor origin from the bone may be necessary. Typically, in tennis players this pain occurs on the lateral epicondyle and common extensor origin (tennis elbow), whereas in golfers it occurs on the medial epicondyle and common flexor origin.

In the clinic

Elbow arthritis

Osteoarthritis is extremely common and is usually most severe in the dominant limb. From time to time an arthritic elbow may undergo such degenerative change that small bone fragments appear in the articular cavity. Given the relatively small joint space, these fragments can result in an appreciable reduction in flexion and extension, and typically lodge within the olecranon and coronoid fossae.

In the clinic

Ulnar nerve injury at the elbow

Posterior to the medial epicondyle of the humerus the ulnar nerve is bound in a fibro-osseous tunnel (the cubital tunnel) by a retinaculum. Older patients may develop degenerative changes within this tunnel, which compresses the ulnar nerve when flexed. The repeated action of flexion and extension of the elbow may cause local nerve damage, resulting in impaired function of the ulnar nerve. Accessory muscles and localized neuritis in this region secondary to direct trauma may also produce ulnar nerve damage.

CUBITAL FOSSA

The cubital fossa is an important area of transition between the arm and the forearm. It is located anterior to the elbow joint and is a triangular depression formed between two forearm muscles:

- the brachioradialis muscle originating from the lateral supra-epicondylar ridge of the humerus, and
- the pronator teres muscle originating from the medial epicondyle of the humerus (Fig. 7.75A).

The base of the triangle is an imaginary horizontal line between the medial and lateral epicondyles. The bed or floor of the fossa is formed mainly by the brachialis muscle.

The major contents of the cubital fossa, from lateral to medial, are:

- the tendon of the biceps brachii muscle,
- the brachial artery, and
- the median nerve (Fig. 7.75B).

The brachial artery normally bifurcates into the radial and ulnar arteries in the apex of the fossa (Fig. 7.75B), although this bifurcation may occur much higher in the arm, even in the axilla. When taking a blood pressure reading from a patient, the clinician places the stethoscope over the brachial artery in the cubital fossa.

The median nerve lies immediately medial to the brachial artery and leaves the fossa by passing between the ulnar and humeral heads of the pronator teres muscle (Fig. 7.75C).

The brachial artery and the median nerve are covered and protected anteriorly in the distal part of the cubital fossa by the bicipital aponeurosis (Fig. 7.75B). This flat connective tissue membrane passes between the medial side of the tendon of the biceps brachii muscle and deep

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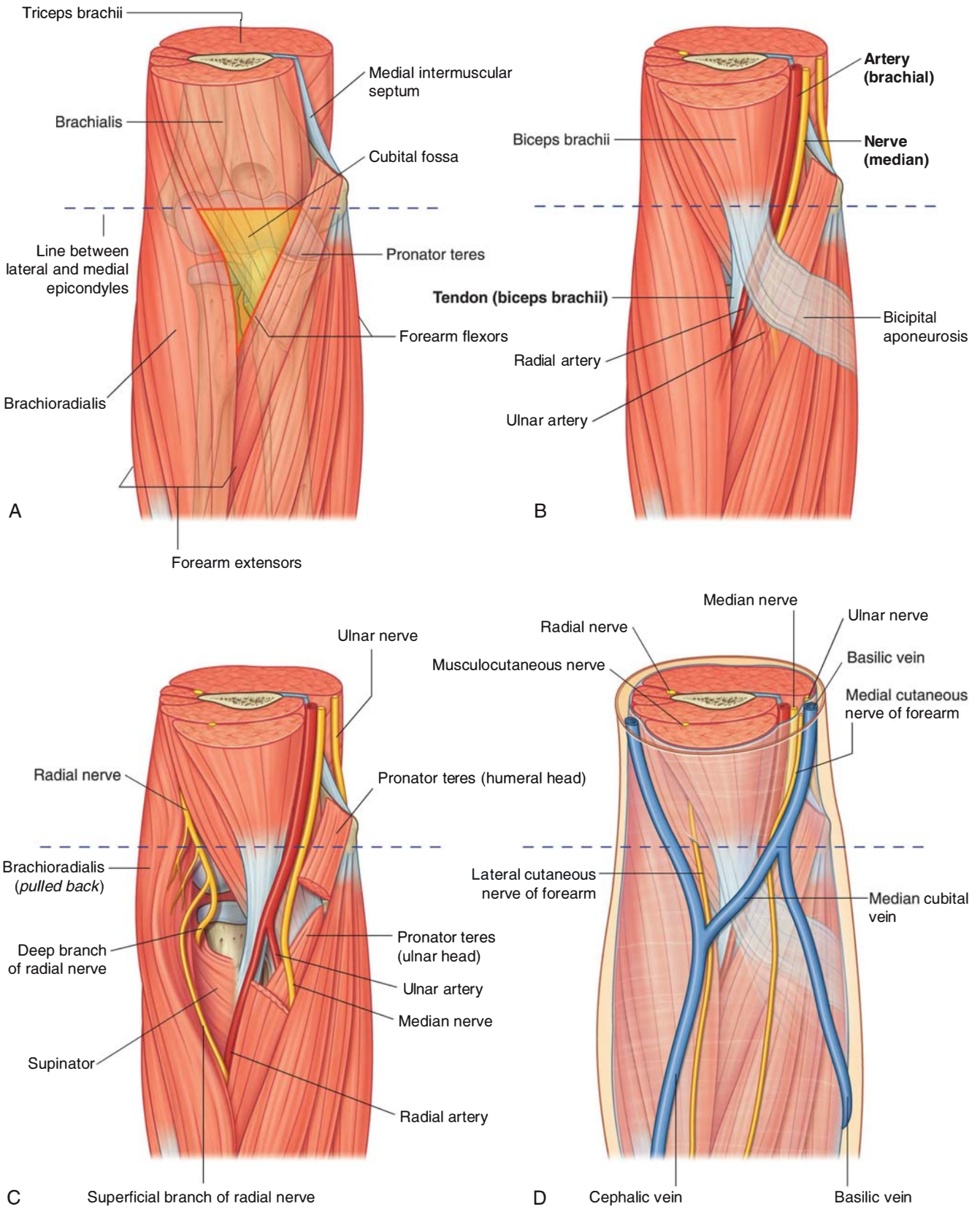


Fig. 7.75 Cubital fossa. **A.** Margins. **B.** Contents. **C.** Position of the radial nerve. **D.** Superficial structures.

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fascia of the forearm. The sharp medial margin of the bicipital aponeurosis can often be felt.

The radial nerve lies just under the lip of the brachioradialis muscle, which forms the lateral margin of the fossa (Fig. 7.75C). In this position, the radial nerve divides into superficial and deep branches:

- The superficial branch continues into the forearm just deep to the brachioradialis muscle.
- The deep branch passes between the two heads of the supinator muscle (see pp. 788–789 and Fig. 7.90) to access the posterior compartment of the forearm.

The ulnar nerve does not pass through the cubital fossa. Instead, it passes posterior to the medial epicondyle.

The roof of the cubital fossa is formed by superficial fascia and skin. The most important structure within the roof is the median cubital vein (Fig. 7.75D), which passes diagonally across the roof and connects the cephalic vein on the lateral side of the upper limb with the basilic vein on the medial side. The bicipital aponeurosis separates the median cubital vein from the brachial artery and median nerve. Other structures within the roof are cutaneous nerves—the medial cutaneous and lateral cutaneous nerves of the forearm.

In the clinic

Construction of a dialysis fistula

Many patients throughout the world require renal dialysis for kidney failure. The patient's blood is filtered and cleaned by the dialysis machine. Blood therefore has to be taken from patients into the filtering device and then returned to them. This process of dialysis occurs over many hours and requires considerable flow rates of 250–500 mL per minute. To enable such large volumes of blood to be removed from and returned to the body, the blood is taken from vessels that have a high flow. As no veins in the peripheral limbs have such high flow, a surgical procedure is necessary to create such a system. In most patients, the radial artery is anastomosed (joined) to the cephalic vein (Fig. 7.76) at the wrist, or the brachial artery is anastomosed to the cephalic vein at the elbow. Some surgeons place an arterial graft between these vessels.

After six weeks, the veins increase in size in response to their arterial blood flow and are amenable to direct cannulation or dialysis.

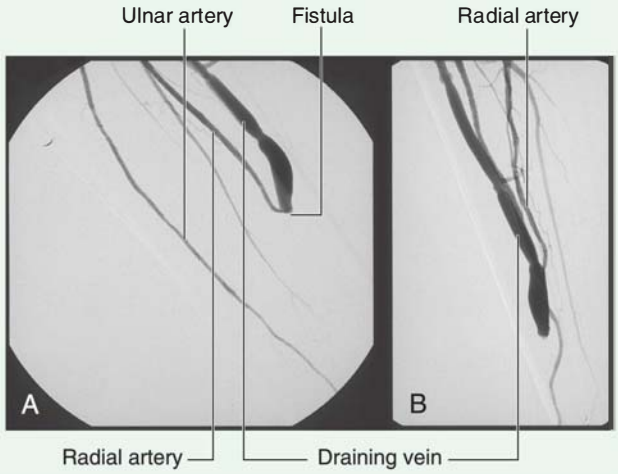


Fig. 7.76 Digital subtraction angiograms of forearm demonstrating a surgically created radiocephalic fistula. **A.** Anteroposterior view. **B.** Lateral view.

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FOREARM

The forearm is the part of the upper limb that extends between the elbow joint and the wrist joint. Proximally, most major structures pass between the arm and forearm through, or in relation to, the cubital fossa, which is anterior to the elbow joint (Fig. 7.77). The exception is the ulnar nerve, which passes posterior to the medial epicondyle of the humerus.

Distally, structures pass between the forearm and the hand through, or anterior to, the carpal tunnel (Fig. 7.77). The major exception is the radial artery, which passes dorsally around the wrist to enter the hand posteriorly.

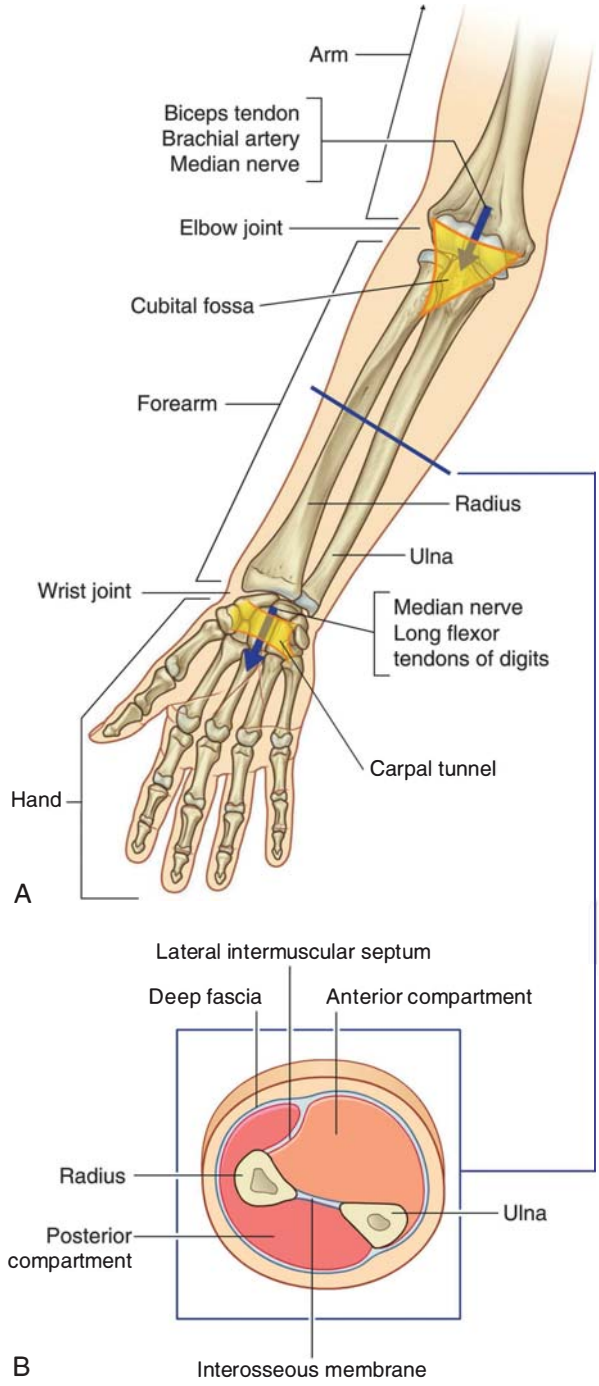


Fig. 7.77 Forearm. **A.** Proximal and distal relationships of the forearm. **B.** Transverse section through the middle of the forearm.

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The bone framework of the forearm consists of two parallel bones, the radius and the ulna (Figs. 7.77 and 7.78B). The radius is lateral in position and is small proximally, where it articulates with the humerus, and large distally, where it forms the wrist joint with the carpal bones of the hand.

The ulna is medial in the forearm, and its proximal and distal dimensions are the reverse of those for the radius: the ulna is large proximally and small distally. Proximal and distal joints between the radius and the ulna allow the distal end of the radius to swing over the adjacent end of the ulna, resulting in pronation and supination of the hand.

As in the arm, the forearm is divided into anterior and posterior compartments (Fig. 7.77). In the forearm, these compartments are separated by:

- a lateral intermuscular septum, which passes from the anterior border of the radius to deep fascia surrounding the limb;
- an interosseous membrane, which links adjacent borders of the radius and ulna along most of their length; and
- the attachment of deep fascia along the posterior border of the ulna.

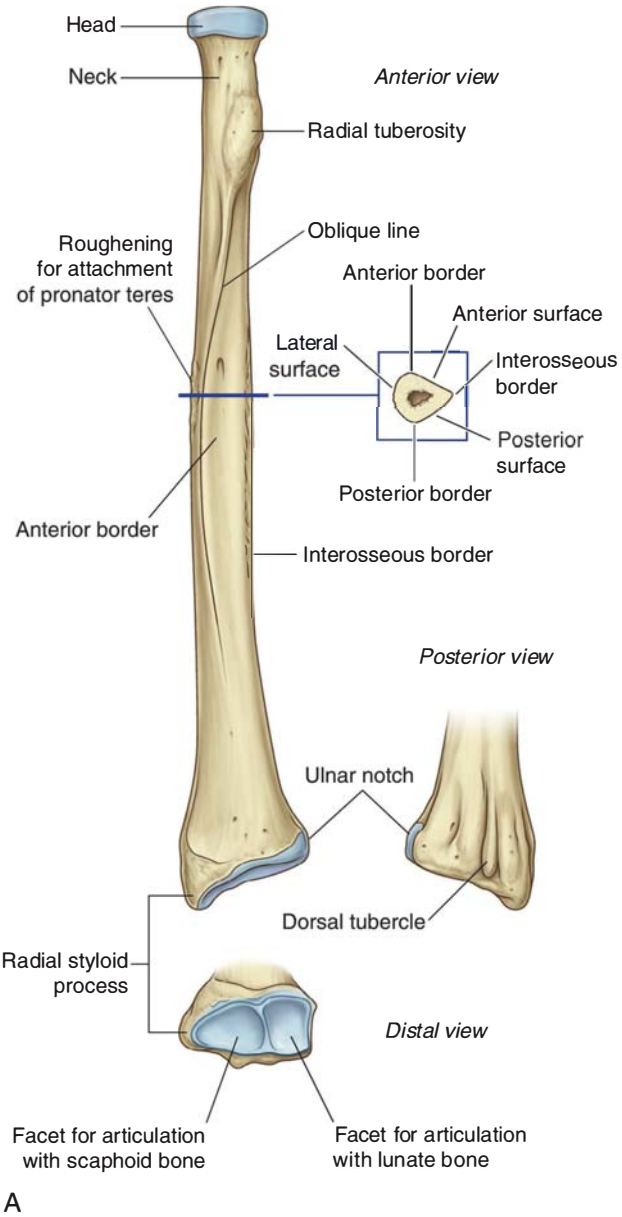




Fig. 7.78 Radius. **A.** Shaft and distal end of the right radius. **B.** Radiograph of the forearm (anteroposterior view).

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